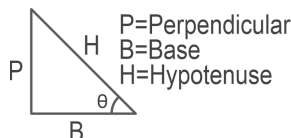


Trigonometry

It is that branch of Mathematics which deals with the measurements of the sides and angles of a triangle and also the problems associated with angles.

Important Formulae

In a right angled triangle, as shown below



$$\bullet \sin \theta = \frac{P}{H} \quad \bullet \operatorname{cosec} \theta = \frac{H}{P}$$

$$\bullet \cos \theta = \frac{B}{H} \quad \bullet \sec \theta = \frac{H}{B}$$

$$\bullet \tan \theta = \frac{P}{B} \quad \bullet \cot \theta = \frac{B}{P}$$

Note that

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}, \therefore \operatorname{cosec} \theta \times \sin \theta = 1$$

$$\text{Similarly, } \sec \theta = \frac{1}{\cos \theta}$$

$$\therefore \sec \theta \times \cos \theta = 1$$

$$\text{And, } \cot \theta = \frac{1}{\tan \theta}$$

$$\therefore \cot \theta \times \tan \theta = 1$$

$$\text{Also, } \tan \theta = \frac{\sin \theta}{\cos \theta}$$

- $\sin \theta = \cos (90^\circ - \theta)$
- $\operatorname{cosec} \theta = \sec (90^\circ - \theta)$
- $\tan \theta = \cot (90^\circ - \theta)$

Conversion of trigonometric ratios

- $\sin (-x) = -\sin x$
- $\operatorname{cosec} (-x) = -\operatorname{cosec} x$
- $\cos (-x) = \cos x$
- $\sec (-x) = \sec x$
- $\tan (-x) = -\tan x$
- $\cot (-x) = -\cot x$

Signs of the trigonometric ratios in different quadrants :

2nd quadrant Sine and cosecant are positive	1st quadrant All positive
3rd quadrant tangent and cotangent are positive	4th quadrant cosine and secant are positive

- x axis $\pm \theta \rightarrow$ no change
 - y axis $\pm \theta \rightarrow$ changes like
- $\sin \theta \leftrightarrow \cos \theta$
 $\tan \theta \leftrightarrow \cot \theta$
 $\operatorname{cosec} \theta \leftrightarrow \sec \theta$

Important Identities

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\sec^2 \theta - \tan^2 \theta = 1$
- $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

We can derive all other identities from these three basic identities.

Values of Trigonometric functions for different values of θ .

	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

The value of $\sin \theta$ or $\cos \theta$ never exceeds 1, whereas the value of $\sec \theta$ or $\operatorname{cosec} \theta$ is always greater than or equal to 1.

Range of all trigonometric functions

	Minimum	Maximum
$\sin x, \cos x$ (odd powers)	-1	+1
$\sin x, \cos x$ (even powers)	0	+1
$\tan x, \cot x$ (odd powers)	$-\infty$	$+\infty$
$\tan x, \cot x$ (even powers)	0	$+\infty$
$\sec x, \operatorname{cosec} x$ (odd powers)	$-\infty$	$+\infty$
$\sec x, \operatorname{cosec} x$ (even powers)	+1	$+\infty$

Maximum and Minimum value of the trigonometric expressions of the different form :

$$\bullet a \sin x \pm b \cos x$$

$$\text{Maximum value} = \sqrt{a^2 + b^2}$$

$$\text{Minimum value} = -\sqrt{a^2 + b^2}$$

$$\bullet a \sin^2 x + b \cos^2 x$$

$$\text{If, } a > b \text{ Maximum value} = a$$

$$\text{Minimum value} = b$$

$$\text{If } a < b \text{ Maximum value} = b$$

$$\text{Minimum value} = a$$

$$\bullet a \operatorname{cosec}^2 x + b \sec^2 x$$

$$\text{Maximum value} = \infty$$

$$\text{Minimum value} = (\sqrt{a} + \sqrt{b})^2$$

$$\bullet a \sin^2 x + b \operatorname{cosec}^2 x$$

$$\bullet a \tan^2 x + b \cot^2 x$$

$$\bullet a \cos^2 x + b \sec^2 x$$

Maximum and Minimum value of these three trigonometric expression is ∞ and $2\sqrt{ab}$ respectively.

$$\bullet \sin^{2n} x + \cos^{2m} x$$

$$\text{Maximum value} = 1$$

$$\text{Minimum value} = (\text{use } x = 45^\circ)$$

Example : Find the minimum and

maximum value of $\sin^2 x + \cos^4 x$?

Maximum value of $\sin^2 x + \cos^4 x = 1$

Minimum value = $\sin^2 45^\circ + \cos^4 45^\circ =$

$$\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\bullet (\sin x \cos x)^n$$

Maximum value = ∞

$$\text{Minimum value} = \left(\frac{1}{2}\right)^n$$

Example : Find the minimum value of $\sin^4 x \cos^4 x$?

$$\text{Minimum value} = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

Compound Formulae

- $\sin (A + B)$
 $= \sin A \cos B + \cos A \sin B$
- $\sin (A - B)$
 $= \sin A \cos B - \cos A \sin B$
- $\cos (A + B)$
 $= \cos A \cos B - \sin A \sin B$
- $\cos (A - B)$
 $= \cos A \cos B + \sin A \sin B$
- $\sin 2A = 2 \sin A \cos A$
 $= \frac{2 \tan A}{1 + \tan^2 A}$
- $\cos 2A = \cos^2 A - \sin^2 A$
 $= 1 - 2 \sin^2 A$
- $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
 $= 2 \cos^2 A - 1$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$
- $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$
- $\sin \theta \cdot \sin 2\theta \cdot \sin 4\theta$
 $= \frac{1}{4} \sin 3\theta$
- $\cos \theta \cdot \cos 2\theta \cdot \cos 4\theta$
 $= \frac{1}{4} \cos 3\theta$
- $\tan \theta \cdot \tan 2\theta \cdot \tan 4\theta = \tan 3\theta$
- $\tan (A + B) = \frac{\tan A + \tan B}{1 - (\tan A \times \tan B)}$
- $\tan (A - B) = \frac{\tan A - \tan B}{1 + (\tan A \times \tan B)}$
- $\cot (A + B) = \frac{\cot B \cot A - 1}{\cot B + \cot A}$
- $\cot (A - B) = \frac{\cot B \cot A + 1}{\cot B - \cot A}$

Transformations of sums or differences into products

- $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$

$$\bullet \cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$\bullet \cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$$

Concepts

$$\bullet a \sin X + b \cos X = m$$

$$a \cos X - b \sin X = n$$

$$\text{Then, } a^2 + b^2 = m^2 + n^2$$

Example : if $3 \sin \theta + 5 \cos \theta = 5$. Find the value of $3 \cos \theta - 5 \sin \theta$?

$$3 \cos \theta - 5 \sin \theta = \sqrt{5^2 + 3^2 - 5^2} = \pm 3$$

$$\bullet a \sec X + b \tan X = m$$

$$a \tan X - b \sec X = n$$

$$\text{Then, } a^2 - b^2 = m^2 - n^2$$

If $\sec x$ and $\tan x$ are replaced by $\operatorname{cosec} x$ and $\cot x$ respectively. Then the outcome will be the same.

Example : if $29 \sec \theta - 21 \tan \theta = 20$ Find the value of $29 \tan \theta - 21 \sec \theta$?

$$\text{Sol:- } 29 \tan \theta - 21 \sec \theta$$

$$= \sqrt{20^2 - 29^2 + 21^2} = 0$$

$$\bullet \sec x + \tan x = \frac{1}{\sec x - \tan x}$$

$$\bullet \operatorname{cosec} x + \cot x = \frac{1}{\operatorname{cosec} x - \cot x}$$

Some Important results

(i). If $A + B + C = 180^\circ$. then,

$$\bullet \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\bullet \cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

(ii). If $A + B + C = 90^\circ$. then,

$$\bullet \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$$

$$\bullet \cot A + \cot B + \cot C = \cot A \cot B \cot C$$

(iii). If $A + B = 45^\circ$. then

$$\bullet (1 + \tan A)(1 + \tan B) = 2$$

$$\bullet (1 - \cot A)(1 - \cot B) = 2$$

Variety Questions

Q.1. If $\cot^4 A + \cot^2 A = 1$, then find the value of $\cos^4 A - 3 \cos^2 A$.

SSC CHSL Tier II 02/11/2023

$$(a) -1 \quad (b) -2 \quad (c) 2 \quad (d) 1$$

Q.2 If $\cot^2 \theta = 1 - e^2$, then the value of $\operatorname{cosec} \theta + \cot^3 \theta \sec \theta$ is:

SSC CPO 05/10/2023 (3rd Shift)

$$(a) (2 - e^2)^{\frac{3}{2}} \quad (b) (1 - e^2)$$

$$(c) (1 - e^2)^{\frac{3}{2}} \quad (d) (2 - e^2)^{\frac{1}{2}}$$

Q.3. If $\operatorname{cosec} A + \cot A = a\sqrt{b}$, then find

$$\text{the value of } \frac{(a^2 b - 1)}{(a^2 b + 1)}.$$

SSC CHSL 17/08/2023 (2nd Shift)

$$(a) \cos A \quad (b) \tan A \quad (c) \frac{1}{\sin A} \quad (d) \frac{1}{\cot A}$$

Q.4. If $\sin \theta + \tan \theta = p$ and $\tan \theta - \sin \theta = q$, then which of the relation satisfies for given values of p and q ?

SSC CHSL 11/08/2023 (4th Shift)

$$(a) p^2 - q^2 = 4\sqrt{pq} \quad (b) p^2 - q^2 = 8\sqrt{pq}$$

$$(c) p^2 + q^2 = 4\sqrt{pq} \quad (d) p^2 + q^2 = 4pq$$

Q.5. If θ is an acute angle and $\sin \theta \cos \theta = 2 \cos^3 \theta - \frac{1}{4} \cos \theta$, then the value of $\sin \theta$ is:

SSC CHSL 10/08/2023 (2nd Shift)

$$(a) \frac{\sqrt{15} - 1}{8} \quad (b) \frac{\sqrt{15} - 1}{4}$$

$$(c) \frac{\sqrt{15} + 1}{4} \quad (d) \frac{\sqrt{15} - 1}{2}$$

Q.6. Simplify the following: $7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ)$

SSC CHSL 09/08/2023 (3rd Shift)

$$(a) 7 \quad (b) -7 \quad (c) -1 \quad (d) 1$$

Q.7. The given expression is equal to :

$$\frac{\sin^4 A + \cos^4 A}{1 - 2 \sin^2 A \cos^2 A}$$

SSC CHSL 09/08/2023 (2nd Shift)

$$(a) 1 \quad (b) -1 \quad (c) 0 \quad (d) 2$$

Q.8. If $\cot A + \cos A = p$ and $\cot A - \cos A = q$, then which of the following relation is correct?

SSC CHSL 03/08/2023 (4th Shift)

$$(a) \sqrt{\frac{1}{16pq}} = p^2 + q^2 \quad (b) \frac{1}{4pq} = p^2 + q^2$$

$$(c) \sqrt{16pq} = p^2 - q^2 \quad (d) 4pq = p^2 + q^2$$

Q.9. The value of the expression $4(\sin^6 A + \cos^6 A) - 6(\sin^4 A + \cos^4 A) + 8$ is :

SSC CHSL 02/08/2023 (4th Shift)

$$(a) 4 \quad (b) 8 \quad (c) 7 \quad (d) 6$$

Q.10. Evaluate the following:

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos 8\theta}}}$$

SSC CGL 27/07/2023 (3rd shift)

$$(a) 2 \cos \theta \quad (b) 2 \cos 2\theta \quad (c) \sin 2\theta \quad (d) \cos 2\theta$$

Q.11. Find the value of $\sqrt{\frac{1 - \sin 3\theta}{1 + \sin 3\theta}}$

SSC CGL 21/07/2023 (2nd shift)

$$(a) \sec 3\theta - \tan 3\theta \quad (b) (\sec 3\theta - \tan 3\theta)^3$$

$$(c) (\sec 3\theta - \tan 3\theta)^2 \quad (d) \sec 3\theta + \tan 3\theta$$

Q.12. The solution of the equation $\cos^2 \theta + \sin \theta + 1 = 0$ lies between :

Graduate Level 30/06/2023 (shift 1)

$$(a) \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right) \quad (b) \left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$$

$$(c) \left(\frac{\pi}{4}, \frac{3\pi}{4}\right) \quad (d) \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$$

Q.13. What is the value of $(1 + \cot A +$

$$\tan A)(\sin A - \cos A) \left(\frac{\sin A \cos A}{\sin^3 A - \cos^3 A}\right) ?$$

SSC CHSL Tier II (26/06/2023)

$$(a) 1 \quad (b) -1 \quad (c) 0 \quad (d) 2$$

Q.14. If $\sin m + \sin n = p$, $\cos m + \cos n = q$, then find the value of $\sin m \times \sin n + \cos m \times \cos n$.

SSC CHSL 20/03/2023 (2nd Shift)

$$(a) p^2 + q^2 - 2 \quad (b) \frac{(p^2 + q^2 - 2)}{2}$$

$$(c) p + q - pq \quad (d) p + q + pq$$

Q.15. If $a \cot \theta + b \operatorname{cosec} \theta = p$ and $b \cot \theta + a \operatorname{cosec} \theta = q$ then $p^2 - q^2$ is equal to _____.

SSC CGL Tier II (07/03/2023)

$$(a) b^2 - a^2 \quad (b) a^2 - b^2 \quad (c) b - a \quad (d) a^2 + b^2$$

Q.16. If A is an acute angle, the simplified form of

$$\cos(\pi - A) \cdot \cot\left(\frac{\pi}{2} + A\right) \cos(-A)$$

$$\tan(\pi + A) \tan\left(\frac{3\pi}{2} + A\right) \sin(2\pi - A)$$

SSC CGL 13/12/2022 (4th Shift)

$$(a) \cos^2 A \quad (b) \sin A \quad (c) \sin^2 A \quad (d) \cos A$$

Q.17. What is the value of the expression $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$?

SSC CGL 12/12/2022 (1st Shift)

$$(a) \sin(2A - 2B) \quad (b) \sin(2A + 2B)$$

$$(c) \cos(2A + 2B) \quad (d) \cos(2A - 2B)$$

Q.18. If $\frac{k - k \cot^2 30^\circ}{1 + \cot^2 30^\circ} = \sin^2 60^\circ +$

$4 \tan^2 45^\circ - \operatorname{cosec}^2 60^\circ$, then the value of k (correct to two decimal places) is:

SSC CGL 03/12/2022 (3rd Shift)

$$(a) 5.55 \quad (b) -6.83 \quad (c) -5.58 \quad (d) 6.83$$

Q.19. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, find the value of $(\cos \theta - \sin \theta)$.

SSC CPO 10/11/2022 (Afternoon)

$$(a) \sqrt{2} \sin \theta \quad (b) \sqrt{2} \cos \theta$$

$$(c) \frac{1}{\sqrt{2}} \sin \theta \quad (d) \frac{1}{2} \cos \theta$$

Q.20. The value of $\sin 73^\circ + \cos 137^\circ$ is:

SSC CHSL 10/06/2022 (Evening)

$$(a) \sin 13^\circ \quad (b) \cos 13^\circ$$

$$(c) \cos 18^\circ \quad (d) \sin 18^\circ$$

Q.21. The value of $\sin 18^\circ$ is given as $\frac{\sqrt{5} - 1}{4}$. Using this value, find the value

of $\cos 18^\circ$.

SSC CHSL 30/05/2022 (Afternoon)

$$(a) \frac{\sqrt{10 + 2\sqrt{5}}}{4} \quad (b) \frac{\sqrt{5} + 1}{4}$$

$$(c) \frac{\sqrt{6} + \sqrt{2}}{4} \quad (d) \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

Q.22. The value of $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$ is:

SSC CHSL 27/05/2022 (Morning)

$$(a) \tan 3\theta \quad (b) \cot 3\theta \quad (c) \tan \theta \quad (d) \cot \theta$$

Q.23. The value of $\cot^2 46^\circ - \sec^2 44^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ)$
SSC CHSL 10/08/2021 (Evening)
(a) 23.5 (b) 20.5 (c) 22.5 (d) 21.5

Q.24. If A lies between 45° and 540° , and $\sin A = 0.5$, what is the value of $\frac{A}{3}$ in degree?
SSC CHSL 15/04/2021 (Morning)
(a) 170° (b) 175° (c) 165° (d) 160°

Q.25. If $\cos 27^\circ = x$, then the value of $\tan 63^\circ$ is:
SSC CHSL 26/10/2020 (Evening)

- (a) $\frac{\sqrt{1+x^2}}{x}$ (b) $\frac{x}{\sqrt{1+x^2}}$
(c) $\frac{\sqrt{1-x^2}}{x}$ (d) $\frac{x}{\sqrt{1-x^2}}$

Q.26. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta = y \cos \theta$, then the value of $x^2 + y^2$ is:
SSC CHSL 20/10/2020 (Morning)
(a) 0 (b) 4 (c) 1 (d) 2

Q.27. If $\cos^2 \theta + \cos^4 \theta = 1$, then the value of $\sin \theta + \sin^2 \theta$ is:
SSC CHSL 16/10/2020 (Morning)
(a) 0 (b) $\frac{1}{2}$ (c) 2 (d) 1

Q.28. If $\sin \theta - \cos \theta = \frac{7}{17}$, then find the value of $\sin \theta + \cos \theta$.
SSC CHSL 14/10/2020 (Evening)
(a) $\frac{8}{17}$ (b) $\frac{23}{13}$ (c) $\frac{23}{17}$ (d) $\frac{8}{13}$

Q.29. If $\sec \theta$ and $\sin \theta$ ($0^\circ < \theta < 90^\circ$) are the roots of the equation $\sqrt{6}x^2 - kx + \sqrt{6} = 0$, then the value of k is:
SSC CHSL 14/10/2020 (Morning)
(a) $\sqrt{3}$ (b) $3\sqrt{2}$ (c) $2\sqrt{3}$ (d) $3\sqrt{3}$

Q.30. If $\cos A, \sin A, \cot A$ are in geometric progression, then the value of $\tan^6 A - \tan^2 A$ is:
SSC CHSL 13/10/2020 (Afternoon)
(a) $\frac{1}{2}$ (b) 3 (c) $\frac{1}{3}$ (d) 1

Q.31. Find the value of $\cos 225^\circ$
SSC CHSL 17/03/2020 (Afternoon)
(a) -0.7071 (b) -0.866
(c) 0.7071 (d) 0.866

Q.32. If $0^\circ < A, B < 45^\circ$, $\cos(A+B) = \frac{24}{25}$ and $\sin(A-B) = \frac{15}{17}$, then $\tan 2A = ?$
SSC CGL 06/03/2020 (Evening)
(a) 0 (b) 1 (c) $\frac{416}{87}$ (d) $\frac{213}{4}$

Q.33. If $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$, then which of the following is true?
SSC CPO 13/12/2019 (Evening)
(a) $\cos \theta \cos \theta = 1$
(b) $\cos \theta \cos \theta = \sqrt{2}$
(c) $\cos^2 \theta - \sin^2 \theta = \cot^2 \theta$
(d) $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$

Q.34. everything between 0° and 90° For $\alpha'_i s$, ($i = 1, 2, 3, \dots, 20$) it is given that $\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3 + \dots + \sin \alpha_{20} = 20$. Find the value (in degrees) of $(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{20})$.
SSC CHSL 11/07/2019 (Morning)
(a) 1800° (b) 900° (c) 0° (d) 20°

Q.35. For α and β both being acute angles, it is given that $\sin(\alpha + \beta) = 1$, $\cos(\alpha - \beta) = \frac{1}{2}$. The values of α and β are:
SSC CHSL 10/07/2019 (Evening)
(a) $75^\circ, 15^\circ$ (b) $45^\circ, 15^\circ$
(c) $75^\circ, 45^\circ$ (d) $60^\circ, 30^\circ$

Q.36. If $\theta = 9^\circ$, then what is the value of $\cot \theta \cdot \cot 2\theta \cdot \cot 3\theta \cdot \cot 4\theta \cdot \cot 5\theta \cdot \cot 6\theta \cdot \cot 7\theta \cdot \cot 8\theta \cdot \cot 9\theta$?
SSC CHSL 10/07/2019 (Evening)
(a) $\sqrt{3} - 1$ (b) 1 (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

Q.37. Which among the following is an irrational quantity?
SSC CHSL 10/07/2019 (Afternoon)
(a) $\tan 30^\circ \cdot \tan 60^\circ$ (b) $\sin 30^\circ$
(c) $\tan 45^\circ$ (d) $\cos 30^\circ$

Q.38. Which among the following increases continuously in the range $0^\circ < \theta < 90^\circ$?
SSC CHSL 09/07/2019 (Evening)
(a) $\cot \theta$ (b) $\operatorname{cosec} \theta$ (c) $\tan \theta$ (d) $\cos \theta$

Q.39. If $\sec \theta = 4x$ and $\tan \theta = \frac{4}{x}$, ($x \neq 0$) then the value of $8(x^2 - \frac{1}{x^2})$ is:
SSC CHSL 05/07/2019 (Afternoon)
(a) $\frac{1}{16}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{8}$

Q.40. If $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1-r}{1+r}$, then the value of r is:
SSC CHSL 03/07/2019 (Afternoon)
(a) $\sin \theta$ (b) $\operatorname{Cosec} \theta$ (c) $\sec \theta$ (d) $\cos \theta$

Q.41. If $\frac{(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)}{\tan^2 \theta - \sin^2 \theta} = r^3$, then $r = ?$

SSC CHSL 03/07/2019 (Morning)
(a) $\sin \theta \cos \theta$ (b) $\tan \theta$
(c) $\cot \theta$ (d) $\operatorname{Cosec} \theta \sec \theta$

Q.42. If $3 - 2\sin^2 \theta - 3\cos \theta = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of $(2\operatorname{Cosec} \theta + \tan \theta)$?
SSC CHSL 02/07/2019 (Evening)

- (a) $7\sqrt{3}$ (b) $5\sqrt{3}$ (c) $\frac{5\sqrt{3}}{3}$ (d) $\frac{7\sqrt{3}}{3}$

Q.43. A simplified of $(\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta})(\frac{1}{\tan \theta + \cot \theta})$ is:
SSC CHSL 02/07/2019 (Evening)
(a) $\cos \theta$ (b) $2\sin \theta$ (c) $\sin \theta$ (d) $2\cos \theta$

Q.44. The value of $\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \cdot \sec 34^\circ + \cos 25^\circ \cdot \operatorname{cosec} 65^\circ}$ is:
SSC CHSL 02/07/2019 (Evening)
(a) $\frac{1}{2}$ (b) 4 (c) 2 (d) 1

Q.45. $\frac{(\sec \theta + \tan \theta)(1 - \sin \theta)}{\operatorname{cosec} \theta(1 + \cos \theta)(\operatorname{cosec} \theta - \cot \theta)}$ is equal to:
SSC CHSL 02/07/2019 (Morning)
(a) $\sin \theta$ (b) $\sec \theta$ (c) $\cos \theta$ (d) $\operatorname{Cosec} \theta$

Q.46. If $3\cos^2 A + 7\sin^2 A = 4$, then what is the value of $\cot A$, given that A is an acute angle?
SSC CGL 13/06/2019 (Evening)
(a) 1 (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$

Q.47. If $\operatorname{Cosec} \theta = \frac{13}{12}$, then $\sin \theta + \cos \theta - \tan \theta$ is equal to:
SSC CGL 11/06/2019 (Afternoon)
(a) $\frac{91}{65}$ (b) $\frac{139}{65}$ (c) $\frac{71}{65}$ (d) $-\frac{71}{65}$

Q.48. The value of $\sin^2 38^\circ + \sin^2 52^\circ + \sin^2 30^\circ - \tan^2 45^\circ$ is equal to:
SSC CGL 10/06/2019 (Afternoon)
(a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{1}{2}$

Q.49. If $\sec 4\theta = \operatorname{Cosec}(\theta + 20^\circ)$, then θ is equal to:
SSC CGL 10/06/2019 (Afternoon)
(a) 22° (b) 18° (c) 14° (d) 20°

Q.50. If $3\sin \theta = 2\cos^2 \theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan^2 \theta + \sec^2 \theta - \operatorname{cosec}^2 \theta)$ is:
SSC CGL 10/06/2019 (Morning)
(a) -2 (b) $-\frac{7}{3}$ (c) $\frac{7}{3}$ (d) 2

Q.51. If $\frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} = \frac{k+1}{k-1}$, then $k = ?$
SSC CGL 10/06/2019 (Morning)
(a) $\operatorname{cosec} \theta$ (b) $\sec \theta$ (c) $\cos \theta$ (d) $\sin \theta$

Q.52. If $\frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1} = 2$
 $\sec \theta$, $0^\circ < \theta < 90^\circ$, then the value of
 $(\cot \theta + \cos \theta)$ is:

SSC CGL 07/06/2019 (Evening)

- (a) $\frac{1+\sqrt{2}}{2}$ (b) $\frac{2+\sqrt{2}}{2}$
 (c) $\frac{2+\sqrt{3}}{\sqrt{2}}$ (d) $1 + \sqrt{2}$

Q.53. If $\sin \theta = \frac{a}{\sqrt{a^2 + b^2}}$, $0^\circ < \theta < 90^\circ$,

then the value of $\sec \theta + \tan \theta$ is :

SSC CGL 07/06/2019 (Afternoon)

- (a) $\frac{\sqrt{a^2 + b^2} + a}{b}$ (b) $\frac{\sqrt{a^2 + b^2} + b}{2a}$
 (c) $\frac{\sqrt{a^2 + b^2} + a}{2b}$ (d) $\frac{\sqrt{a^2 + b^2} + b}{a}$

Q.54. If $\cos^2 \theta - 3 \cos \theta + 2 = \sin^2 \theta$, $0^\circ < \theta < 90^\circ$, then the value of $2 \operatorname{cosec} \theta + 4 \cot \theta$ is :

SSC CGL 07/06/2019 (Afternoon)

- (a) $\frac{8\sqrt{3}}{3}$ (b) $\frac{4\sqrt{3}}{4}$ (c) $2\sqrt{3}$ (d) $4\sqrt{3}$

Q.55. The value of $\frac{1}{\sec x - \tan x} - \frac{1}{\cos x}$,

$0^\circ < x < 90^\circ$, is equal to :

SSC CGL 07/06/2019 (Morning)

- (a) $2 \sec x$ (b) $\tan x$ (c) $2 \cos x$ (d) $\cot x$

Q.56. If $\tan^2 \theta - 3 \sec \theta + 3 = 0$, $0^\circ < x < 90^\circ$, then the value of $\sin \theta + \cot \theta$ is :

SSC CGL 07/06/2019 (Morning)

- (a) $\frac{5\sqrt{3}}{6}$ (b) $2\sqrt{3}$ (c) $\frac{5\sqrt{3}}{3}$ (d) $3\sqrt{3}$

Q.57. If $(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$, then $\sqrt{k} = ?$

SSC CGL 06/06/2019 (Evening)

- (a) $\operatorname{cosec} \theta \cdot \sec \theta$ (b) $\operatorname{cosec} \theta \cdot \cos \theta$
 (c) $\sin \theta \cdot \cos \theta$ (d) $\sin \theta \cdot \sec \theta$

Q.58. The value of $\frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta}$ is

SSC CGL 06/06/2019 (Evening)

- (a) 0 (b) 1 (c) 2 (d) -1

Q.59. $(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \operatorname{cosec}^2 \theta}) = ?$

SSC CGL 06/06/2019 (Evening)

- (a) $\sin^2 \theta$ (b) 1 (c) $\operatorname{cosec}^2 \theta$ (d) 2

Q.60. If $\frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan \theta + \operatorname{cosec} \theta)$ is :

SSC CGL 06/06/2019 (Evening)

- (a) $\frac{5\sqrt{2}}{2}$ (b) $\frac{5\sqrt{3}}{3}$ (c) $\frac{4\sqrt{3}}{3}$ (d) $\frac{5\sqrt{2}}{3}$

Q.61. $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = ?$

SSC CGL 06/06/2019 (Morning)

- (a) $\sec \theta \sin \theta$ (b) $\sec \theta \tan \theta$
 (c) $\sec \theta + \tan \theta$ (d) $\sec \theta - \tan \theta$

Q.62. The value of

$\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \times \sqrt{\tan^2 \theta - \sin^2 \theta}$ is equal to :

SSC CGL 06/06/2019 (Morning)

- (a) $\operatorname{cosec} \theta \sec^2 \theta$ (b) $\sin \theta \sec^2 \theta$
 (c) $\sin \theta \cos^2 \theta$ (d) $\operatorname{cosec} \theta \cos^2 \theta$

Q.63. If $\sec \theta - \tan \theta = P$, then $\operatorname{cosec} \theta = ?$

SSC CGL 04/06/2019 (Evening)

- (a) $\frac{2P}{1 - P^2}$ (b) $\frac{1 - P^2}{1 + P^2}$
 (c) $\frac{P^2 + 1}{1 - P^2}$ (d) $\frac{2P}{1 + P^2}$

Q.64. If $0^\circ < \theta < 90^\circ$ and $\cos^2 \theta = 3(\cot^2 \theta - \cos^2 \theta)$ then the value of $(\frac{1}{2} \sec \theta + \sin \theta)^{-1}$ is :

SSC CGL 04/06/2019 (Afternoon)

- (a) $\sqrt{3} + 2$ (b) $2(2 - \sqrt{3})$
 (c) $2(\sqrt{3} - 1)$ (d) $\sqrt{3} + 1$

Q.65. $(\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta})^2 + 1$, $\theta \neq 45^\circ$ is equal to.

SSC CGL 04/06/2019 (Afternoon)

- (a) $\operatorname{cosec}^2 \theta$ (b) $\sec^2 \theta$ (c) $\cot^2 \theta$ (d) $2 \tan^2 \theta$

Q.66. $(\operatorname{cosec} A - \sin A)^2 + (\sec A - \cos A)^2 - (\cot A - \tan A)^2$ is equal to:

SSC CPO 16/03/2019 (Evening)

- (a) 2 (b) 0 (c) 1 (d) -1

Q.67. Find the value of $1 + \frac{\tan^2 A}{1 + \sec A}$

SSC CPO 15/03/2019 (Evening)

- (a) $\operatorname{cosec} A$ (b) $\cos A$ (c) $\sec A$ (d) $\sin A$

Q.68. If $x = a \cos \theta + b \sin \theta$ and $y = a \sin \theta - b \cos \theta$, the value of $x^2 + y^2$ is:

SSC CPO 15/03/2019 (Morning)

- (a) $a^2 - b^2$ (b) $a - b$ (c) $a^2 + b^2$ (d) $a + b$

Practice Questions

SSC CPO 2023 Tier - 1

Q.69. Find the value of $(\tan^2 \theta + \tan^4 \theta)$.

SSC CPO 03/10/2023 (1st Shift)

- (a) $\sec^2 \theta - \sec^4 \theta$ (b) $\cot^2 \theta - \tan^2 \theta$
 (c) $\sec^4 \theta + \sec^2 \theta$ (d) $\sec^4 \theta - \sec^2 \theta$

Q.70. If θ is an acute angle and $\sin \theta + \operatorname{cosec} \theta = 2$, then the value of $\sin^5 \theta + \operatorname{cosec}^5 \theta$ is:

SSC CPO 03/10/2023 (1st Shift)

- (a) 2 (b) 10 (c) 4 (d) 5

Q.71. If $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$, then $(\cos \theta + \sin \theta)$ is:

SSC CPO 03/10/2023 (1st Shift)

- (a) $-\sqrt{2} \sin \theta$ (b) $\sqrt{2} \tan \theta$
 (c) $\sqrt{2} \cos \theta$ (d) $-\sqrt{2} \cos \theta$

Q.72. The value of

$(1 + \sin^4 A - \cos^4 A) \operatorname{cosec}^2 A$ is:

SSC CPO 03/10/2023 (2nd Shift)

- (a) -1 (b) 1 (c) -2 (d) 2

Q.73. If $\cot \theta = \frac{4}{3}$, $0 < \theta < \frac{\pi}{2}$, and

$5p \cos^2 \theta \sin \theta = \cot^2 \theta$, then the value of p is :

SSC CPO 03/10/2023 (2nd Shift)

- (a) $\frac{7}{27}$ (b) $\frac{5}{27}$ (c) $\frac{25}{27}$ (d) $\frac{125}{27}$

Q.74. If $\cos A = \frac{15}{17}$, $0 \leq A \leq 90^\circ$, then

the value of $\cot(90^\circ - A)$ is :

SSC CPO 03/10/2023 (2nd Shift)

- (a) $\frac{8}{15}$ (b) $\frac{7}{15}$ (c) $\frac{\sqrt{2}}{15}$ (d) $\frac{2\sqrt{2}}{15}$

Q.75. If $\sin 2A = \cos 15^\circ$, then the smallest positive value of A is:

SSC CPO 03/10/2023 (3rd Shift)

- (a) 7.5° (b) 47.5° (c) 75° (d) 37.5°

Q.76. If $A + B = 90^\circ$, then the expression

$\frac{\cot A}{\cot B} + \cos^2 A + \cos^2 B$ is equal to:

SSC CPO 03/10/2023 (3rd Shift)

- (a) $\cot^2 B$ (b) $\operatorname{cosec}^2 A$
 (c) $\operatorname{cosec}^2 B$ (d) $\cos^2 A$

Q.77. In $\triangle ABC$, $\angle B = 90^\circ$ and $AB : BC = 1 :$

2. The value of $\cos A + \tan C$ is:

SSC CPO 04/10/2023 (1st Shift)

- (a) $\frac{1 + \sqrt{5}}{2\sqrt{5}}$ (b) $\frac{2\sqrt{5}}{2 + \sqrt{5}}$
 (c) $\frac{5 + \sqrt{5}}{2\sqrt{5}}$ (d) $\frac{2 + \sqrt{5}}{2\sqrt{5}}$

Q.78. If $\sin A + \sin^2 A = 1$, then the value of $\cos^4 A + \cos^6 A$ is :

SSC CPO 04/10/2023 (1st Shift)

- (a) 1 (b) $\cos A$ (c) $\sin A$ (d) 0

Q.79. If $\sec A - \tan A = p$, then find the value of $\sec A$.

SSC CPO 04/10/2023 (2nd Shift)

- (a) $\frac{p^2 - 1}{p^2 + 1}$ (b) $\frac{p^2 + 1}{p^2 - 1}$
 (c) $\frac{p^2 + 1}{2p}$ (d) $\frac{p^2 + 1}{p}$

Q.80. $2(\sin 1^\circ \times \sec 89^\circ) + 3(\cos 11^\circ \times \operatorname{cosec} 79^\circ) + 5(\tan 21^\circ \times \tan 69^\circ) = ?$

SSC CPO 04/10/2023 (2nd Shift)

- (a) 20 (b) 12 (c) 11 (d) 10

Q.81. Find the value of $\sqrt{\frac{1 - \tan A}{1 + \tan A}}$.

SSC CPO 04/10/2023 (2nd Shift)

- (a) $\sqrt{\frac{1 + \sin A}{\cos A}}$ (b) $\sqrt{\frac{1 + \sin 2A}{\cos 2A}}$
(c) $\sqrt{\frac{1 - \sin A}{\cos A}}$ (d) $\sqrt{\frac{1 - \sin 2A}{\cos 2A}}$

Q.82. $7 \sin^2 A + 3 \cos^2 A = 4$, then find $\cot A$:

SSC CPO 04/10/2023 (2nd Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\sqrt{3}$ (d) $\sqrt{2}$

Q.83. Find the least value of $16 \operatorname{cosec}^2 \theta + 25 \sin^2 \theta$.

SSC CPO 04/10/2023 (3rd Shift)

- (a) 35 (b) 38 (c) 42 (d) 40

Q.84. If $\operatorname{cosec} A + \cot A = 3$, $0 \leq A \leq 90^\circ$, then find the value of $\cos A$.

SSC CPO 04/10/2023 (3rd Shift)

- (a) $\frac{2}{5}$ (b) $\frac{3}{4}$ (c) $\frac{3}{5}$ (d) $\frac{4}{5}$

Q.85. The value of $\sqrt{\frac{1 + \sin A}{1 - \sin A}}$ is :

SSC CPO 04/10/2023 (3rd Shift)

- (a) $\operatorname{cosec} A + \cot A$ (b) $\operatorname{cosec} A - \cot A$
(c) $\sec A - \tan A$ (d) $\sec A + \tan A$

Q.86. If $\sin A = \frac{2}{3}$, then find the value of $(7 - \tan A)(3 + \cos A)$.

SSC CPO 05/10/2023 (1st Shift)

- (a) $\frac{61}{3} + \frac{17}{\sqrt{5}}$ (b) $\frac{61}{3} - \frac{17}{3\sqrt{5}}$
(c) $\frac{61}{3} + \frac{17}{3\sqrt{5}}$ (d) $\frac{61}{3\sqrt{5}} + \frac{17}{3}$

Q.87. If $\sin A = \frac{\sqrt{3}}{2}$, $0 < A < 90^\circ$, then find

the value of $2(\operatorname{cosec} A + \cot A)$.

SSC CPO 05/10/2023 (1st Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) $\frac{2}{\sqrt{3}}$ (d) $2\sqrt{3}$

Q.88. If $\sin \beta = \frac{1}{3}$, $(\sec \beta - \tan \beta)^2$ is

equal to :

SSC CPO 05/10/2023 (1st Shift)

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

Q.89. Find the value of

$$\frac{\sin^2 39^\circ + \sin^2 (90^\circ - 39^\circ)}{\cos^2 35^\circ + \cos^2 (90^\circ - 35^\circ)} + 3 \tan 25^\circ \tan 65^\circ$$

SSC CPO 05/10/2023 (2nd Shift)

- (a) 1 (b) 3 (c) 4 (d) 2

Q.90. If $\tan \theta + \cot \theta = 2$, θ is an acute angle, then find the value of $2 \tan^{25} \theta + 3 \cot^{20} \theta + 5 \tan^{30} \theta \cot^{15} \theta$

SSC CPO 05/10/2023 (2nd Shift)

- (a) 12 (b) 10 (c) 8 (d) 6

Q.91. Find the value of $\sin (50^\circ + A) - \cos (40^\circ - A)$.

SSC CPO 05/10/2023 (2nd Shift)

- (a) 1 (b) 0 (c) -1 (d) 2

Q.92. If $\sin \theta + \cos \theta = \frac{\sqrt{11}}{3}$, then what is

$\sin \theta - \cos \theta$?

SSC CPO 05/10/2023 (2nd Shift)

- (a) $\frac{\sqrt{5}}{3}$ (b) $\frac{\sqrt{7}}{3}$ (c) $\frac{\sqrt{7}}{4}$ (d) $\frac{\sqrt{5}}{2}$

Q.93. If $\sin \theta + \cos \theta = \sqrt{2} \cos \theta$, then find $\frac{\sin \theta - \cos \theta}{\sin \theta}$.

SSC CPO 05/10/2023 (3rd Shift)

- (a) $-\sqrt{2}$ (b) -1 (c) 1 (d) $\sqrt{2}$

SSC CHSL 2023 Tier - 1

Q.94. If $\sin A + \sin^2 A = 1$, then the value of the expression $(\cos^2 A + \cos^4 A)$ is

SSC CHSL 02/08/2023 (1st Shift)

- (a) $\frac{3}{2}$ (b) 1 (c) 2 (d) $\frac{1}{2}$

Q.95. If $\cot \theta = \frac{9}{17}$, find the value of

$\operatorname{cosec}^2 \theta$.

SSC CHSL 02/08/2023 (1st Shift)

- (a) $1\frac{9}{17}$ (b) $\frac{208}{289}$ (c) $\frac{8}{17}$ (d) $1\frac{81}{289}$

Q.96. $\frac{\cos \theta}{\sec \theta - 1} + \frac{\cos \theta}{\sec \theta + 1}$ is equal to :

SSC CHSL 02/08/2023 (1st Shift)

- (a) $2 \sec^2 \theta$ (b) $2 \cot^2 \theta$
(c) $2 \cos^2 \theta$ (d) $2 \sin^2 \theta$

Q.97. If $A = 60^\circ$, $B = 30^\circ$, then find the value of $\sin A \cos B + \cos A \sin B$.

SSC CHSL 02/08/2023 (2nd Shift)

- (a) 0 (b) 12 (c) 13 (d) 1

Q.98. If $(\sin \theta - \cos \theta) = 0$, then the value

of $\sin(\pi - \theta) + \sin(\frac{\pi}{2} - \theta)$ is:

SSC CHSL 02/08/2023 (2nd Shift)

- (a) 1 (b) 0 (c) $\sqrt{3}$ (d) $\sqrt{2}$

Q.99. The value of $\cot 13^\circ \cot 27^\circ \cot 60^\circ \cot 63^\circ \cot 77^\circ$ is:

SSC CHSL 02/08/2023 (2nd Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) 0 (c) $\sqrt{3}$ (d) 1

Q.100. If $\cos \theta + \sec \theta = \sqrt{3}$, then the

value of $\cos^3 \theta + \sec^3 \theta$ is :

SSC CHSL 02/08/2023 (3rd Shift)

- (a) 0 (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) $2\sqrt{3}$

Q.101. If $\cos(3\alpha) = \sin(\alpha - 22^\circ)$, where $3\alpha < 90^\circ$, then what is the value of α ?

SSC CHSL 02/08/2023 (3rd Shift)

- (a) 29° (b) 26° (c) 27° (d) 28°

Q.102. If $\sec \theta - \tan \theta = m$, then $2 \tan \theta = ?$

SSC CHSL 02/08/2023 (3rd Shift)

- (a) $\frac{1-m}{m}$ (b) $\frac{1-m^2}{m}$
(c) $\frac{1+m^2}{m}$ (d) $\frac{m^2-1}{m}$

Q.103. If $\cos A = \frac{63}{65}$, then find the value

of $\tan A + \cot A$ (up to two places of decimal).

SSC CHSL 02/08/2023 (4th Shift)

- (a) 4.19 (b) 2.76 (c) 3.19 (d) 5.23

Q.104. If $\frac{\cos \theta}{(1 + \sin \theta)} + \frac{\cos \theta}{(1 - \sin \theta)} = 4$

and θ is acute, then the value of θ is :

SSC CHSL 03/08/2023 (1st Shift)

- (a) 60° (b) 15° (c) 45° (d) 30°

Q.105. If $\sin \alpha + \cos \alpha = \frac{2}{\sqrt{3}}$ then what

is $(\tan \alpha + \cot \alpha)$ equal to?

SSC CHSL 03/08/2023 (1st Shift)

- (a) 1 (b) 2 (c) 6 (d) 0

Q.106. Find the value of

$\cos 0^\circ + \cos 30^\circ - \tan 45^\circ + \operatorname{cosec} 60^\circ + \cot 90^\circ$.

SSC CHSL 03/08/2023 (1st Shift)

- (a) $\frac{7}{6\sqrt{3}}$ (b) $\frac{\sqrt{3}}{6}$ (c) $\frac{7}{6}$ (d) $\frac{7}{2\sqrt{3}}$

Q.107. The value of

$(\operatorname{cosec}^2 B - 1)(\sec^2 B - 1)$ is :

SSC CHSL 03/08/2023 (2nd Shift)

- (a) 4 (b) 3 (c) 2 (d) 1

Q.108. If $\sin \theta + \cos \theta = \frac{1}{29}$, find the

value of $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$.

SSC CHSL 03/08/2023 (2nd Shift)

- (a) $\frac{1}{41}$ (b) $\frac{43}{29}$ (c) $\frac{41}{29}$ (d) $\frac{1}{43}$

Q.109. $\sin^2 35^\circ + \sin^2 55^\circ =$ _____.

SSC CHSL 03/08/2023 (3rd Shift)

- (a) 0 (b) 1 (c) $2 \sin 35^\circ$ (d) $2 \sin 55^\circ$

Q.110. The given expression is equal to :

$$\frac{(1 + \tan^2 A)}{\operatorname{cosec}^2 A \tan A}$$

SSC CHSL 03/08/2023 (3rd Shift)

- (a) $\sec^2 A$ (b) $\sec A$ (c) $\tan A$ (d) $\tan^2 A$

Q.111. $\cos^4 A - \sin^4 A$ is equal to :

SSC CHSL 03/08/2023 (3rd Shift)

- (a) $2 \cos^2 A + 1$ (b) $1 - 2 \sin^2 A$
(c) $2 \sin^2 A - 1$ (d) $-2 \sin^2 A + 1$

Q.112. If $\sin(p + q) = 1$ and $\cos(p - q) = \frac{\sqrt{3}}{2}$, find p .

SSC CHSL 03/08/2023 (4th Shift)

- (a) 90° (b) 80° (c) 60° (d) 120°

Q.113. If $\operatorname{cosec}\theta = 1\frac{7}{22}$, find the value of $\cot^2\theta$.

SSC CHSL 03/08/2023 (4th Shift)

- (a) $\frac{49}{484}$ (b) $\frac{357}{484}$ (c) $\frac{225}{484}$ (d) $\frac{7}{22}$

Q.114. What is the value of the expression

$$\sin A \left(1 + \frac{\sin A}{\cos A}\right) + \cos A \left(1 + \frac{\cos A}{\sin A}\right) ?$$

SSC CHSL 04/08/2023 (1st Shift)

- (a) $\sec A + \operatorname{cosec} A$ (b) $\sin A + \cos A$
(c) $\sin A - \cos A$ (d) $\sec A - \operatorname{cosec} A$

Q.115. The value of the expression $[\cot 1^\circ \cdot \cot 2^\circ \cdot \cot 3^\circ \cdot \cot 4^\circ \cdot \cot 5^\circ \dots \cot 178^\circ \cot 179^\circ]$ is:

SSC CHSL 04/08/2023 (1st Shift)

- (a) 1235 (b) $\frac{1}{2}$ (c) 1 (d) 0

Q.116. Simplify the expression:

$$\frac{3 - \sin^2 A + \cos^2 A}{2 + 2\cos^2 A}$$

SSC CHSL 04/08/2023 (1st Shift)

- (a) 1 (b) 0 (c) -1 (d) 2

Q.117. If $\tan\theta - \cot\theta = 4$ then find the value of $\tan^2\theta + \cot^2\theta$.

SSC CHSL 04/08/2023 (2nd Shift)

- (a) 16 (b) 20 (c) 14 (d) 18

Q.118. If $A = 30^\circ$ then find the value of

$$\frac{(2\tan A)}{(1 - \tan^2 A)}$$

SSC CHSL 04/08/2023 (2nd Shift)

- (a) $4\sqrt{3}$ (b) $\frac{3}{\sqrt{3}}$ (c) 3 (d) $2\sqrt{3}$

Q.119. If $\sec\theta + \tan\theta = \sqrt{3}$, then the positive value of $\sin\theta + \cos\theta$ is:

SSC CHSL 04/08/2023 (3rd Shift)

- (a) $\frac{(1+\sqrt{3})}{2}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2}{(1+\sqrt{3})}$

Q.120. If $\sin(A - B) = \frac{1}{2}$, and $\sin(A + B) = 1$, where A and B are positive acute angles and $A \geq B$, then A and B are respectively:

SSC CHSL 04/08/2023 (4th Shift)

- (a) $80^\circ, 10^\circ$ (b) $60^\circ, 30^\circ$
(c) $70^\circ, 20^\circ$ (d) $50^\circ, 40^\circ$

Q.121. If $2\tan^2 A + 4\cos^4 A = 3$, then the possible value of A is:

SSC CHSL 04/08/2023 (4th Shift)

- (a) 45° (b) 60° (c) 0° (d) 30°

Q.122. If $a = \cot A + \cos A$ and $b = \cot A - \cos A$, then find the value of $a^2 - b^2 - 4\sqrt{ab}$.

SSC CHSL 04/08/2023 (4th Shift)

- (a) 0 (b) -1 (c) 1 (d) -4

Q.123. The given expression is equal to:

$$1 - \frac{\tan^2 \phi}{\sec^2 \phi}$$

SSC CHSL 07/08/2023 (1st Shift)

- (a) $\sin^2 \phi \cos^2 \phi$ (b) $\sin^2 \phi \cot^2 \phi$
(c) $\cot^2 \phi \cos^2 \phi$ (d) $\tan^2 \phi \cos^2 \phi$

Q.124. If $\alpha \sin 45^\circ = \beta \operatorname{cosec} 30^\circ$, then

$$\frac{\alpha^4}{\beta^4} \text{ is :}$$

SSC CHSL 07/08/2023 (1st Shift)

- (a) 4^4 (b) 3^3 (c) 2^3 (d) 4^3

Q.125. Find the value of

$$\cot^2 B - \operatorname{cosec}^2 B \text{ for } 0 < B < 90^\circ.$$

SSC CHSL 07/08/2023 (1st Shift)

- (a) 1 (b) 2 (c) -1 (d) -2

Q.126. Find the value of $\sin 60^\circ \sec 30^\circ + \tan 45^\circ \cos 0^\circ + \sec 60^\circ \sin 30^\circ$.

SSC CHSL 07/08/2023 (2nd Shift)

- (a) 0 (b) 3 (c) 1 (d) 2

Q.127. If $\sin\theta \cdot \sin 45^\circ = \sin 30^\circ$ and θ is an acute angle, then 2θ is equal to:

SSC CHSL 07/08/2023 (2nd Shift)

- (a) 45° (b) 0° (c) 30° (d) 90°

Q.128. In ΔOPQ , right-angled at P , $OP = 7$ cm and $OQ - PQ = 1$ cm. Determine the values of $\sin Q + \cos Q$.

SSC CHSL 07/08/2023 (2nd Shift)

- (a) $\frac{13}{25}$ (b) $\frac{21}{25}$ (c) $\frac{31}{25}$ (d) $\frac{33}{25}$

Q.129. The given expression is equal to:

$$\frac{2 \sec A}{\sec^2 A - 1}$$

SSC CHSL 07/08/2023 (3rd Shift)

- (a) $2 \operatorname{cosec} A \cot A$ (b) $2 \cos A \cot A$
(c) $2 \operatorname{cosec} A \tan A$ (d) $2 \cos A \sin A$

Q.130. The given expression is equal to:

$$\frac{\cos^2 A}{(1 + \sin A)}$$

SSC CHSL 07/08/2023 (3rd Shift)

- (a) $1 - \sin A$ (b) $1 + \cos A$
(c) $\sin A$ (d) $\cos A$

Q.131. The value of $\sin^2 30^\circ + \sin^2 40^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 60^\circ + \sin^2 70^\circ + \sin^2 80^\circ$ is:

SSC CHSL 07/08/2023 (3rd Shift)

- (a) 2 (b) 4 (c) 3 (d) 1

Q.132. Simplify the given expression.

$$\frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos 53^\circ \sin 37^\circ$$

SSC CHSL 07/08/2023 (4th Shift)

- (a) 2 (b) -1 (c) -2 (d) 1

Q.133. The value of $\frac{1}{\sin\theta} - \frac{\cot^2\theta}{1 + \operatorname{cosec}\theta}$ is

SSC CHSL 07/08/2023 (4th Shift)

- (a) 2 (b) 1 (c) -1 (d) 0

Q.134. The value of $\frac{\sin 58^\circ}{\cos 32^\circ} +$

$$\frac{\sin 55^\circ \sec 35^\circ}{\tan 5^\circ \tan 45^\circ \tan 85^\circ}$$
 is equal to:

SSC CHSL 07/08/2023 (4th Shift)

- (a) 2 (b) 1 (c) 0 (d) 3

Q.135. If $\sec\theta = 2\frac{4}{23}$ find the value of $\tan^2\theta$

SSC CHSL 08/08/2023 (1st Shift)

- (a) $2\frac{16}{529}$ (b) $1\frac{200}{529}$ (c) $3\frac{384}{529}$ (d) $3\frac{177}{529}$

Q.136. Find the value of

$$\frac{\cos^2 25^\circ - \sin^2 65^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} :$$

SSC CHSL 08/08/2023 (1st Shift)

- (a) $\frac{1}{2}$ (b) 1 (c) -1 (d) 0

Q.137. Simplify the given equation.

$$\frac{\cot^3 A - 1}{\cot A - 1} = ?$$

SSC CHSL 08/08/2023 (2nd Shift)

- (a) $\operatorname{cosec}^2 A - \cot A$ (b) $\operatorname{cosec}^2 A + \cot A$
(c) $\cot^2 A + \operatorname{cosec} A$ (d) $\cot^2 A - \operatorname{cosec} A$

Q.138. If $\tan\theta + \sec\theta = a$, then what is $\sec\theta$ equal to:

SSC CHSL 08/08/2023 (2nd Shift)

- (a) $\frac{a+1}{2a^2}$ (b) $\frac{(a-1)}{2a^2}$
(c) $\frac{a^2-1}{2a}$ (d) $\frac{a^2+1}{2a}$

Q.139. Find the value of

$$\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} + 2.$$

SSC CHSL 08/08/2023 (3rd Shift)

- (a) $1 + 2\sec A$ (b) $2 - \sec A$
(c) $2 + \sec A$ (d) $2(1 + \sec A)$

Q.140. If θ is an acute angle and $\sin\theta = \frac{13}{19}$, what is the value of $\cos\theta$?

SSC CHSL 08/08/2023 (3rd Shift)

- (a) $\frac{6}{19}$ (b) $\frac{10\sqrt{2}}{19}$ (c) $\frac{14}{19}$ (d) $\frac{8\sqrt{3}}{19}$

Q.141. If $\sec\theta + \tan\theta = \frac{1}{\sqrt{3}}$, then the

positive value of $\cot\theta + \cos\theta$ is:

SSC CHSL 08/08/2023 (4th Shift)

(a) $\frac{3\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{2}{3\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$

Q.142. If $\frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta} = \frac{4}{5}$, then the value of $\frac{\operatorname{cosec}^2\theta}{2 - \operatorname{cosec}^2\theta}$ is:

SSC CHSL 09/08/2023 (1st Shift)

(a) $\frac{16}{25}$ (b) $\frac{40}{41}$ (c) $\frac{41}{40}$ (d) $\frac{31}{30}$

Q.143. ΔPQR is right angled at Q. If PQ = 12 cm PR = 13 cm, find $\tan P + \cot R$.

SSC CHSL 09/08/2023 (1st Shift)

(a) $\frac{9}{10}$ (b) 0 (c) $\frac{10}{12}$ (d) $\frac{12}{10}$

Q.144. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then

the value of $A + B$ is:

SSC CHSL 09/08/2023 (1st Shift)

(a) 60° (b) 15° (c) 30° (d) 45°

Q.145. If $\cot A = \frac{5}{12}$, find the value of the following expression.

$$\frac{5(1 - \cos^2 A)}{6(1 - \sin^2 A)}$$

SSC CHSL 09/08/2023 (2nd Shift)

(a) $\frac{144}{5}$ (b) $\frac{24}{25}$ (c) $\frac{144}{25}$ (d) $\frac{24}{5}$

Q.146. In a right-angled triangle XYZ, if $X = 60^\circ$ and $Y = 30^\circ$, then find the value of $\sin(X - Y)$.

SSC CHSL 09/08/2023 (2nd Shift)

(a) $\frac{3}{5}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{2}{3}$

Q.147. $\cos 2\theta =$ _____.

SSC CHSL 09/08/2023 (3rd Shift)

(a) $1 - 2 \cos^2 \theta$ (b) $2 \cos^2 \theta - 1$

(c) $2 \sin^2 \theta - 1$ (d) $1 - \sin^2 \theta$

Q.148. If $\tan\theta \cdot \tan 2\theta = 1$ then the value of $\cot 5\theta$ is:

SSC CHSL 09/08/2023 (4th Shift)

(a) -1 (b) 1 (c) $-\sqrt{3}$ (d) $\sqrt{3}$

Q.149. If $4 \sin^2 A - 3 = 0$ and $0 \leq A \leq 90^\circ$, then $3 \sin A - 4 \sin^3 A$ is:

SSC CHSL 10/08/2023 (1st Shift)

(a) $\frac{1}{2}$ (b) $\sqrt{\frac{3}{2}}$ (c) 0 (d) 1

Q.150. If $\sin A = \frac{2}{5}$, where A is an acute angle, what is the value of $\frac{5 \sin A + 2 \operatorname{cosec} A}{\sqrt{21 \sec A}}$?

SSC CHSL 10/08/2023 (1st Shift)

(a) $\frac{5}{4}$ (b) $\frac{7}{5}$ (c) $\frac{4}{5}$ (d) $\frac{5}{7}$

Q.151. If $\tan(5\theta - 10^\circ) = \cot(5\phi + 20^\circ)$, then the value of $\theta + \phi$ is:

SSC CHSL 10/08/2023 (2nd Shift)

(a) 16° (b) 18° (c) 15° (d) 20°

Q.152. Evaluate the given expression:

$$\cos^2 36^\circ + \cos 54^\circ \cdot \sin 36^\circ + \left(\frac{\tan 26^\circ}{\cot 64^\circ}\right)$$

SSC CHSL 10/08/2023 (3rd Shift)

(a) 4 (b) 3 (c) 1 (d) 2

Q.153. ΔRMS is right-angled at M. The length of the base, RM = 4 cm and the length of perpendicular MS = 3 cm. Find the value of $\sec R$.

SSC CHSL 10/08/2023 (3rd Shift)

(a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) $\frac{4}{5}$ (d) $\frac{2}{5}$

Q.154. If $A = 60^\circ$ and $B = 30^\circ$, find the

value of $\frac{(\tan A - \tan B)}{(1 + \tan A \tan B)}$

SSC CHSL 10/08/2023 (3rd Shift)

(a) 3 (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) 1

Q.155. Find the value of the following expression.

$$2(\tan^2 A \cos^2 A + \cos^2 A)$$

SSC CHSL 10/08/2023 (4th Shift)

(a) 4 (b) 1 (c) 3 (d) 2

Q.156. If A, B, C are the acute angles, such that $\tan(A + B - C) = \frac{1}{\sqrt{3}}$, $\cos(B + C$

$$- A) = \frac{1}{2}$$
 and $\sin(C + A - B) = \frac{1}{\sqrt{2}}$. The

value of $A + B + C$ will be equal to:

SSC CHSL 10/08/2023 (4th Shift)

(a) 90° (b) 110° (c) 170° (d) 135°

Q.157. Find the value of the following expression.

$$\frac{(1 + \sec \theta)}{(\sec \theta)} (1 - \cos \theta)$$

SSC CHSL 10/08/2023 (4th Shift)

(a) $\cos^2 \theta$ (b) $\sec^2 \theta$ (c) $\cot^2 \theta$ (d) $\sin^2 \theta$

Q.158. Let ABC be a right-angled triangle where $\angle A = 90^\circ$ and $\angle C = 45^\circ$. Find the value of $\sec C + \sin C \sec C$.

SSC CHSL 11/08/2023 (1st Shift)

(a) 1 (b) $1 - \sqrt{2}$ (c) $1 + \sqrt{2}$ (d) $\sqrt{2} - 1$

Q.159. If $(4 \sin \theta + 5 \cos \theta) = 3$, then the value of $(4 \cos \theta - 5 \sin \theta)$ is:

SSC CHSL 11/08/2023 (1st Shift)

(a) $3\sqrt{2}$ (b) $4\sqrt{2}$ (c) $2\sqrt{3}$ (d) $2\sqrt{5}$

Q.160. If $\tan 2\theta = \cot(\theta - 36^\circ)$, where 2θ is an actual angle, then the value of θ is:

SSC CHSL 11/08/2023 (2nd Shift)

(a) 18° (b) 36° (c) 30° (d) 42°

Q.161. If $\tan(90^\circ - \theta) = \frac{2}{\sqrt{3}}$, then the

value of $2\sqrt{3} \tan \theta + 1$ is:

SSC CHSL 11/08/2023 (2nd Shift)

(a) 4 (b) 5 (c) 3 (d) 6

Q.162. If $\sec \theta + \tan \theta = 5$, then $\sin \theta =$ _____.

SSC CHSL 11/08/2023 (2nd Shift)

(a) 0 (b) $\frac{13}{12}$ (c) $\frac{12}{13}$ (d) $\frac{1}{5}$

Q.163. If $\sin \theta = \frac{6}{10}$ and $\cos \theta = \frac{8}{10}$ then

$\sin 2\theta =$ _____.

SSC CHSL 11/08/2023 (3rd Shift)

(a) $\frac{10}{8}$ (b) $\frac{12}{10}$ (c) $\frac{16}{10}$ (d) $\frac{24}{25}$

Q.164. Let ABC be a triangle right angled

at B. If $\tan A = \frac{12}{5}$, then find the values

of $\operatorname{cosec} A$ and $\sec A$, respectively.

SSC CHSL 11/08/2023 (3rd Shift)

(a) $\frac{13}{10}, \frac{5}{13}$ (b) $\frac{13}{12}, \frac{13}{5}$

(c) $\frac{10}{13}, \frac{5}{13}$ (d) $\frac{12}{13}, \frac{5}{13}$

Q.165. If $\cos \theta - \sin \theta = 0$, then

$(\sin^8 \theta + \cos^8 \theta)$ is:

SSC CHSL 11/08/2023 (3rd Shift)

(a) $\frac{1}{8}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) $\frac{1}{2}$

Q.166. Simplify the given expression

$$\frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z}$$

SSC CHSL 11/08/2023 (4th Shift)

(a) 9 (b) 2 (c) 4 (d) 6

Q.167. If θ be an acute angle and $\tan \theta + \cot \theta = 2$, then the value of $2 \tan^2 \theta + \cot^2 \theta + \tan^4 \theta \cot^4 \theta$ is:

SSC CHSL 14/08/2023 (1st Shift)

(a) 4 (b) 2 (c) 3 (d) 6

Q.168. In right angled triangle ABC, if $\angle A = 30^\circ$, find the value of $3 \sin A - 4 \sin^3 A$.

SSC CHSL 14/08/2023 (1st Shift)

(a) 2 (b) -1 (c) -2 (d) 1

Q.169. Find the value of $\sin 47^\circ \cos 33^\circ - \cos 43^\circ \sin 57^\circ$.

SSC CHSL 14/08/2023 (2nd Shift)

(a) -1 (b) 1 (c) 0 (d) $\frac{1}{2}$

Q.170. If $\cos 5\alpha = \sin \alpha$ and $5\alpha < 90^\circ$, then the value of $\tan 2\alpha$ is:

SSC CHSL 14/08/2023 (3rd Shift)

(a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

Q.171. The value of $\frac{1 + \sin A}{\cos A} + \frac{\cos A}{1 + \sin A}$ is:

SSC CHSL 14/08/2023 (3rd Shift)

(a) $2 \sec A$ (b) $2 \operatorname{cosec} A$

(c) $\sec A$ (d) $\operatorname{cosec} A$

Q.172. Simplify the given equation. $(1 + \tan^2 A)(1 + \cot^2 A) = ?$

SSC CHSL 14/08/2023 (3rd Shift)

- (a) $\frac{1}{\cos^2 A(1 + \sin^2 A)}$
 (b) $\frac{1}{\sin^2 A(1 - \sin^2 A)}$
 (c) $\frac{1}{\sin^2 A + 1 + \operatorname{cosec}^2 A}$
 (d) $\frac{1}{\sin^2 A(1 + \cos^2 A)}$

Q.173. The given expression is equal to:

$(\cot B - \tan B) \cdot \sin B \cdot \cos B$

SSC CHSL 14/08/2023 (4th Shift)

- (a) $1 - 2\sec^2 B$ (b) $1 - 2\cos^2 B$
 (c) $2\cos^2 B - 1$ (d) $2\sec^2 B - 1$

Q.174. $\cot^2 \theta - \operatorname{cosec}^2 \theta$ equals to :

SSC CHSL 17/08/2023 (1st Shift)

- (a) 0 (b) $\tan \theta$ (c) 1 (d) -1

Q.175. If $\tan A = \frac{3}{4}$, find the value of the following expression.

$$\frac{6 \sin A}{1 - \sin A}$$

SSC CHSL 17/08/2023 (2nd Shift)

- (a) 18 (b) 9 (c) 24 (d) 12

Q.176. In a right-angled triangle PQR, PQ = 5 cm, QR = 13 cm and $\angle P = 90^\circ$. Find the value of $\tan Q - \tan R$.

SSC CHSL 17/08/2023 (2nd Shift)

- (a) $\frac{5}{14}$ (b) $\frac{119}{60}$ (c) $\frac{60}{119}$ (d) $\frac{14}{5}$

Q.177. If $\sin \theta \cos \theta = \frac{1}{\sqrt{3}}$ then the value

of $(\sin^4 \theta + \cos^4 \theta)$ is:

SSC CHSL 17/08/2023 (3rd Shift)

- (a) 1 (b) $\frac{5}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

Q.178. Simplify the given expression.

$$\sqrt{\frac{1 + \cos P}{1 - \cos P}}$$

SSC CHSL 17/08/2023 (3rd Shift)

- (a) $\operatorname{cosec} P - \cot P$ (b) $\sec P - \tan P$
 (c) $\sec P + \tan P$ (d) $\operatorname{cosec} P + \cot P$

Q.179. Find the value of $\tan(50^\circ + \theta) - \cot(40^\circ - \theta)$.

SSC CHSL 17/08/2023 (4th Shift)

- (a) $\frac{1}{2}$ (b) -1 (c) 0 (d) 1

Q.180. What is the value of $(1 - \sin^2 A) \operatorname{cosec}^2 A$?

Given A is an acute angle.

SSC CHSL 17/08/2023 (4th Shift)

- (a) $\cot^2 A$ (b) $\cos^2 A$ (c) $\tan^2 A$ (d) $\sin^2 A$

Q.181. If $\tan 15^\circ = 2 - \sqrt{3}$, then the value of $\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$ is:

SSC CHSL 17/08/2023 (4th Shift)

- (a) 6 (b) 10 (c) 8 (d) 14

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Q.182. If $\sin(a + b) = 1$ and $\cos(a - b) = \frac{1}{2}$, then find a.

SSC CGL 14/07/2023 (1st shift)

- (a) 75° (b) 30° (c) 15° (d) 45°

Q.183. If $\left\{ \left(\frac{3\sin\theta - \cos\theta}{\cos\theta + \sin\theta} \right) \right\} = 1$, then the value of $\cot\theta$ is:

SSC CGL 14/07/2023 (1st shift)

- (a) 3 (b) 0 (c) 1 (d) 2

Q.184. Evaluate the following: $\cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos(54^\circ + A) \cdot \cos(54^\circ - A)$

SSC CGL 14/07/2023 (2nd shift)

- (a) $\sin 2A$ (b) $\cos A$ (c) $\sin A$ (d) $\cos 2A$

Q.185. What is the value of $\sec^2 54^\circ - \cot^2 36^\circ + \frac{3}{2} \sin^2 37^\circ \times \sec^2 53^\circ + \frac{2}{\sqrt{3}}$

$\tan 60^\circ$?

SSC CGL 14/07/2023 (2nd shift)

- (a) $\frac{5}{2}$ (b) $\frac{9}{2}$ (c) $\frac{3}{2}$ (d) $\frac{7}{2}$

Q.186. If $\sin 23^\circ = \frac{a}{b}$, then the value of

$\sec 23^\circ - \sin 67^\circ$ is _____

SSC CGL 14/07/2023 (3rd shift)

- (a) $\frac{a^2}{\sqrt{b^2 - a^2}}$ (b) $\frac{b^2 - a^2}{ab}$
 (c) $\frac{a^2}{b\sqrt{b^2 + a^2}}$ (d) $\frac{a^2}{b\sqrt{b^2 - a^2}}$

Q.187. If $\frac{\cos\beta}{\sec\alpha} = 15$ and $\frac{\sin\beta}{\sec\alpha} = 16$, then

the value of $\sin^2\beta$ is _____.

SSC CGL 14/07/2023 (3rd shift)

- (a) $\frac{256}{481}$ (b) $-\frac{256}{481}$ (c) $\frac{481}{256}$ (d) $-\frac{481}{256}$

Q.188. If $\sec^2\theta + \tan^2\theta = \frac{25}{18}$, then the

value of $\sec^4\theta - \tan^4\theta$ is:

SSC CGL 14/07/2023 (3rd shift)

- (a) $\frac{18}{25}$ (b) $\frac{25}{12}$ (c) $\frac{25}{9}$ (d) $\frac{25}{18}$

Q.189. Simplify the following.

$$\frac{\sin^3 \alpha + \cos^3 \alpha}{\sin \alpha + \cos \alpha}$$

SSC CGL 14/07/2023 (4th shift)

- (a) $1 + \sin \alpha \cos \alpha$ (b) $\tan \alpha$
 (c) $1 - \sin \alpha \cos \alpha$ (d) $\sec \alpha$

Q.190. If $\tan \frac{\pi}{6} + \sec \frac{\pi}{6} = x$, then find x.

SSC CGL 17/07/2023 (2nd shift)

- (a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$ (c) $-\frac{1}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$

Q.191. If $\cos x = -\frac{1}{2}$, x lies in third

quadrant, then $\tan x = ?$

SSC CGL 17/07/2023 (3rd shift)

- (a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{2}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{3}}$

Q.192. Using $\cos(A + B) = \cos A \cos B - \sin A \sin B$, find the value of $\cos 75^\circ$.

SSC CGL 17/07/2023 (3rd shift)

- (a) $\frac{\sqrt{5} - 1}{4}$ (b) $\frac{\sqrt{5} + 1}{4}$
 (c) $\frac{\sqrt{6} - \sqrt{2}}{4}$ (d) $\frac{\sqrt{6} + \sqrt{2}}{4}$

Q.193. $\frac{\sin^4 \theta + \cos^4 \theta}{1 - 2\sin^2 \theta \cdot \cos^2 \theta} = \underline{\hspace{2cm}}$

SSC CGL 17/07/2023 (4th shift)

- (a) 1 (b) 2 (c) -1 (d) 0

Q.194. Using trigonometric formulas, find

the value of $\left(\frac{\sin(x - y)}{\sin(x + y)} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right)$

SSC CGL 18/07/2023 (1st shift)

- (a) -2 (b) 2 (c) 0 (d) 1

Q.195. If $\tan \frac{A}{2} = x$, then find x.

SSC CGL 18/07/2023 (1st shift)

- (a) $\frac{\sqrt{1 + \cos A}}{\sqrt{1 - \cos A}}$ (b) $\frac{\sqrt{1 - \sin A}}{\sqrt{1 + \cos A}}$
 (c) $\frac{\sqrt{1 - \cos A}}{\sqrt{1 + \cos A}}$ (d) $\frac{\sqrt{\cos A - 1}}{\sqrt{1 + \cos A}}$

Q.196. The value of

$\left(\frac{1}{\sin \theta} + \frac{1}{\tan \theta} \right) \left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta} \right)$ is :

SSC CGL 18/07/2023 (2nd shift)

- (a) 0 (b) 2 (c) 3 (d) 1

Q.197. Simplify the following:

$\sin 2x + 2 \sin 4x + \sin 6x$

SSC CGL 18/07/2023 (2nd shift)

- (a) $4 \cos^2 x \sin 4x$ (b) $4 \cos^2 x \sin x$
 (c) $2 \cos^2 x \sin 4x$ (d) $4 \sin^2 x \sin 4x$

Q.198. If $\sin A = -\frac{3}{5}$, A lies in III

quadrant, the value of $\sec A$ is:

SSC CGL 18/07/2023 (2nd shift)

- (a) $-\frac{4}{5}$ (b) $-\frac{5}{4}$ (c) $-\frac{3}{4}$ (d) $\frac{3}{4}$

Q.199. If $\sin(5x - 25^\circ) = \cos(5y + 25^\circ)$, where $5x - 25^\circ$ and $5y + 25^\circ$ are acute

angles, then the value of $(x + y)$ is :

SSC CGL 18/07/2023 (3rd shift)

- (a) 50° (b) 40° (c) 18° (d) 16°

Q.200. Find the value of the following expression. $12(\sin^4 \theta + \cos^4 \theta) + 18$

$(\sin^6 \theta + \cos^6 \theta) + 78 \sin^2 \theta \cos^2 \theta$

SSC CGL 18/07/2023 (3rd shift)

- (a) 30 (b) 40 (c) 10 (d) 20

Q.201. Using $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

find the value of $\tan 15^\circ$.

SSC CGL 18/07/2023 (4th shift)

- (a) $\sqrt{3} + 1$ (b) $\sqrt{3} - 1$ (c) $2 - \sqrt{3}$ (d) $2 + \sqrt{3}$

Q.202. If $\frac{\sin x - \cos x}{\sin x + \cos x} = \frac{2}{5}$, then the

value of $\frac{1 + \cot^2 x}{1 - \cot^2 x}$ is:

SSC CGL 19/07/2023 (1st shift)

- (a) 2.25 (b) 1.45 (c) 3.75 (d) 5.25

Q.203. Find the value of $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$

SSC CGL 19/07/2023 (1st shift)

- (a) 4 (b) 0 (c) 2 (d) 1

Q.204. If $\left(\frac{\cos A}{1 - \sin A}\right) + \left(\frac{\cos A}{1 + \sin A}\right) = 4$

then what will be the value of A? ($0^\circ < \theta < 90^\circ$)

SSC CGL 19/07/2023 (1st shift)

- (a) 90° (b) 45° (c) 60° (d) 30°

Q.205. Using $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$,

find the value of $\cot 15^\circ$.

SSC CGL 19/07/2023 (2nd shift)

- (a) $2 + \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) $\sqrt{3} - 1$ (d) $\sqrt{3} + 1$

Q.206. If $P \cos \alpha = 3$ and $4 \tan \alpha = Q$, then what is the relation between P and Q, which is independent of α ?

SSC CGL 19/07/2023 (2nd shift)

- (a) $\frac{9}{P^2} + \frac{16}{Q^2} = 1$ (b) $\frac{9}{P^2} - \frac{16}{Q^2} = 1$
(c) $\frac{P^2}{9} - \frac{Q^2}{16} = 1$ (d) $\frac{P^2}{9} + \frac{Q^2}{16} = 1$

Q.207. For any acute angle θ , $\sin \theta + \sin^2 \theta = 1$, then the value of $\cos^2 \theta + \cos^4 \theta =$ _____.

SSC CGL 19/07/2023 (3rd shift)

- (a) 0 (b) 1 (c) 2 (d) -1

Q.208. If $\cos x + \sin x = \sqrt{2} \cos x$, what is the value of $(\cos x - \sin x)^2 + (\cos x + \sin x)^2$?

SSC CGL 19/07/2023 (4th shift)

- (a) 2 (b) 1 (c) 0 (d) $\frac{1}{\sqrt{2}}$

Q.209. Solve the given expression.

$$\frac{\sec^2(90^\circ - \theta) - \cot^2 \theta}{2(\sin^2 35^\circ + \sin^2 55^\circ)}$$

SSC CGL 20/07/2023 (1st shift)

- (a) 0.5 (b) 2 (c) 1.5 (d) 1

Q.210. $\operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ + \cot 1755^\circ = ?$

SSC CGL 20/07/2023 (1st shift)

- (a) 3 (b) 1 (c) 4 (d) 0

Q.211. The maximum value of $(\sin^{12} \theta + \cos^{20} \theta)$ for all the real values of θ is:

SSC CGL 20/07/2023 (3rd shift)

- (a) 2 (b) 3 (c) 1 (d) 0

Q.212. Find the value of $\frac{2}{3} \tan^2 60^\circ + 3$

$$\cos^2 30^\circ - 2 \sec^2 30^\circ - \frac{3}{4} \cot^2 60^\circ.$$

SSC CGL 20/07/2023 (3rd shift)

- (a) 2.33 (b) 2.01 (c) 1.33 (d) 3.33

Q.213. What is the value of $\sin(-405^\circ)$?

SSC CGL 20/07/2023 (3rd shift)

- (a) $\frac{\sqrt{5}}{2}$ (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) $-\frac{1}{\sqrt{2}}$

Q.214. In a right-angle triangle, the hypotenuse is 5 cm and the base is 3 cm. If one angle is θ , then $\tan \theta$ is equal to:

SSC CGL 20/07/2023 (4th shift)

- (a) $\frac{4}{3}$ (b) $\frac{5}{3}$ (c) $\frac{3}{5}$ (d) $\frac{5}{4}$

Q.215. Simplify the following expression

$$\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta}$$

SSC CGL 21/07/2023 (3rd shift)

- (a) $\tan \theta$ (b) $\sin \theta$ (c) $\sec \theta$ (d) $\cos \theta$

Q.216. Simplify $\frac{1 + \sin t}{4 - 4 \sin t} - \frac{1 - \sin t}{4 + 4 \sin t}$

SSC CGL 21/07/2023 (4th shift)

- (a) $4 \tan t \cdot \sin t$ (b) $\tan t \cdot \sec t$
(c) $\tan t - \sin t$ (d) $\tan t + \sin t$

Q.217. PQR is a triangle right angled at Q and PQ : QR = 3 : 4. What is the value of $\sin P + \sin Q + \sin R$?

SSC CGL 21/07/2023 (4th shift)

- (a) $\frac{4}{5}$ (b) $\frac{3}{5}$ (c) $\frac{12}{5}$ (d) $\frac{2}{5}$

Q.218. If $2 \cot \theta = 3$, then find the value

$$\text{of } \frac{\sqrt{13} \sin \theta - 3 \tan \theta}{3 \tan \theta + \sqrt{13} \cos \theta}$$

SSC CGL 24/07/2023 (1st shift)

- (a) $\frac{1}{\sqrt{13}}$ (b) $\frac{2}{\sqrt{13}}$ (c) 0 (d) $\frac{2}{3}$

Q.219. Using the identity $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$ find the value of $\tan 15^\circ$

correct to three decimal places. [Use $\sqrt{3} = 1.732$]

SSC CGL 24/07/2023 (1st shift)

- (a) 0.268 (b) 0.27 (c) 0.267 (d) 0.269

Q.220. If $\operatorname{Cosec} \theta + \cot \theta = m$, find the

$$\text{value of } \frac{m^2 - 1}{m^2 + 1}$$

SSC CGL 24/07/2023 (2nd shift)

- (a) 1 (b) 0 (c) $\cos \theta$ (d) $-\sin \theta$

Q.221. The value of $\tan 5^\circ \tan 25^\circ \tan 45^\circ$

$\tan 65^\circ \tan 85^\circ$ is equal to

SSC CGL 24/07/2023 (2nd shift)

- (a) 4 (b) 3 (c) 1 (d) 2

Q.222. If $\tan x = \frac{7}{5}$, then the value of

$$\frac{9 \sin x - \frac{42}{5} \cos x}{15 \sin x + 21 \cos x}$$
 is:

SSC CGL 24/07/2023 (3rd shift)

- (a) 0 (b) 1 (c) 0.1 (d) 0.5

Q.223. If $\sin \theta + \operatorname{cosec} \theta = \sqrt{5}$, then the value of $\sin^3 \theta + \operatorname{cosec}^3 \theta =$ _____.

SSC CGL 24/07/2023 (3rd shift)

- (a) 0 (b) $3\sqrt{5}$ (c) $\frac{1}{\sqrt{5}}$ (d) $2\sqrt{5}$

Q.224. If $a \cot \theta = b$, then what will be the value of $\frac{b \cos \theta - a \sin \theta}{b \cos \theta + a \sin \theta}$?

SSC CGL 24/07/2023 (4th shift)

- (a) $\frac{b^2 + a^2}{b^2 - a^2}$ (b) $b^2 + a^2$
(c) $\frac{b^2 - a^2}{b^2 + a^2}$ (d) 0

Q.225. Simplify the given expression.

$$\frac{1 + \sin^4 \theta + \cos^4 \theta}{\cos^2 \theta + \sin^4 \theta}$$

SSC CGL 24/07/2023 (4th shift)

- (a) 3 (b) 1 (c) 2 (d) 4

Q.226. Using $\operatorname{cosec}(\alpha + \beta) = \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$, find the value of $\operatorname{cosec} 75^\circ$.

SSC CGL 25/07/2023 (1st shift)

- (a) $\frac{\sqrt{6} + \sqrt{2}}{4}$ (b) $\frac{\sqrt{6} - \sqrt{2}}{4}$
(c) $\sqrt{6} - \sqrt{2}$ (d) $\sqrt{6} + \sqrt{2}$

Q.227. If $\cot A = 1$, $\sin B = \frac{1}{\sqrt{2}}$ then find

the value of $\sin(A + B) - \cot(A + B)$.

SSC CGL 25/07/2023 (1st shift)

- (a) $1 - \sqrt{2}$ (b) $\frac{1}{2}$ (c) 0 (d) 1

Q.228. The value of $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$ is:

SSC CGL 25/07/2023 (3rd shift)

- (a) $\sec \theta + \tan \theta$ (b) $\operatorname{cosec} \theta - \cot \theta$
(c) $\operatorname{cosec} \theta + \cot \theta$ (d) $\sec \theta - \tan \theta$

Q.229. What is the value of $5\sqrt{3} \cos 60^\circ \tan 30^\circ - 3 \cos 0^\circ + 3 \cos^2 45^\circ + 2 \sin^2 60^\circ$?

SSC CGL 25/07/2023 (3rd shift)

- (a) $2\sqrt{3}$ (b) 2.5 (c) 3.5 (d) $3\sqrt{2}$

Q.230. If $\cos^2 \theta = \frac{3}{4}$, where θ is an acute angle, then the value of $\sin(\theta + 30^\circ)$ is:

SSC CGL 25/07/2023 (3rd shift)

(a) 1 (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

Q.231. Find $(\frac{1}{\sin\theta}) - \sin\theta$.

SSC CGL 25/07/2023 (4th shift)

- (a) $\cos\theta \cot\theta$ (b) $\cos\theta \sec\theta$
(c) $\cos\theta \operatorname{cosec}\theta$ (d) $\cos\theta \tan\theta$

Q.232. The value of $\operatorname{cosec} 30^\circ - \cos 60^\circ$ is :
SSC CGL 26/07/2023 (1st shift)

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{3}{2}$ (d) $\frac{2\sqrt{2}-1}{\sqrt{2}}$

Q.233. $\triangle ABC$ is a right triangle. If $\angle B = 90^\circ$ and $\tan A = \frac{1}{\sqrt{3}}$ then the value of $\sin$

$A \cos C + \cos A \sin C$ is:

SSC CGL 26/07/2023 (2nd shift)

- (a) 1 (b) 2 (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

Q.234. If $\frac{21 \cos A + 3 \sin A}{3 \cos A + 4 \sin A} = 2$, then find the value of $\cot A$

SSC CGL 26/07/2023 (2nd shift)

- (a) $\frac{9}{11}$ (b) $\frac{11}{9}$ (c) $\frac{1}{3}$ (d) $\frac{11}{10}$

Q.235. If $\cos x + \sec x = \frac{7}{2\sqrt{3}}$ then the

value of $\cos^2 x + \sec^2 x$ will be _____

SSC CGL 26/07/2023 (3rd shift)

- (a) $\frac{15}{12}$ (b) $\frac{10}{12}$ (c) $\frac{25}{10}$ (d) $\frac{25}{12}$

Q.236. $\tan 4384^\circ + \cot 6814^\circ = ?$

SSC CGL 26/07/2023 (3rd shift)

- (a) -1 (b) 2 (c) 0 (d) 1

Q.237. If $A = 22.5^\circ$ what is the value of $(10\sqrt{2} \sin 2A - 7\sqrt{2} \cos 2A + 9 \tan 2A)$?

SSC CGL 26/07/2023 (4th shift)

- (a) 12 (b) 15 (c) 10 (d) 6

Q.238. $\sin 600^\circ \cos 750^\circ + \sin 150^\circ \cos 240^\circ = ?$

SSC CGL 26/07/2023 (4th shift)

- (a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) -1

Q.239. What is the values of $\tan \frac{\theta}{2}$?

SSC CGL 26/07/2023 (4th shift)

- (a) $\frac{\cos\theta}{1 - \sin\theta}$ (b) $\frac{\sin\theta}{1 - \cos\theta}$
(c) $\frac{\cos\theta}{1 - \cos\theta}$ (d) $\frac{\sin\theta}{1 + \cos\theta}$

Q.240. Given $\theta_1 + \theta_2 = \frac{\pi}{2}$ and $\sin\theta_1$

$= \frac{1}{2}$, find the value of θ_2 .

SSC CGL 27/07/2023 (1st shift)

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) π

Q.241. If $3x = \sec A$ and $\frac{3}{x} = \tan A$, then

$9(x^2 - \frac{1}{x^2})$ is ____.

SSC CGL 27/07/2023 (2nd shift)

- (a) 3 (b) 9 (c) 1 (d) $\frac{1}{9}$

Q.242. If $\tan^4\theta + \tan^2\theta = 1$, what is the value of $11(\cos^4\theta + \cos^2\theta)$?

SSC CGL 27/07/2023 (3rd shift)

- (a) -11 (b) 8 (c) 0 (d) 11

SSC Selection Post (Phase - XI)

Q.243. The value of

$$\frac{32}{\sec^2\theta} - \frac{20}{1 + \tan^2\theta} + 12\sin^2\theta$$

Graduate Level 27/06/2023 (Shift - 4)

- (a) 20 (b) 32 (c) 24 (d) 12

Q.244. If $\sin\theta + \cos\theta = \frac{\sqrt{11}}{3}$, then what is $\tan\theta + \cot\theta$ equal to?

Graduate Level 27/06/2023 (Shift - 4)

- (a) 5 (b) 9 (c) 7 (d) 11

Q.245. If $\tan\theta + \sec\theta = a$, then what is $\sec\theta$?

Graduate Level 28/06/2023 (Shift - 3)

- (a) $\frac{a-1}{2a}$ (b) $\frac{a^2+1}{2a}$
(c) $\frac{1}{2a}$ (d) $\frac{a^2-1}{2a}$

Q.246. What is the value of $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 177^\circ \cdot \cos 178^\circ \cdot \cos 179^\circ$

Graduate Level 28/06/2023 (Shift - 3)

- (a) 0 (b) 1 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{2}$

Q.247. If $1 + \sin^2\theta = a \cos^2\theta$, then find $\sec^4\theta - \tan^4\theta$.

Graduate Level 30/06/2023 (Shift - 1)

- (a) a^{-2} (b) a^{-1} (c) a (d) a^2

Q.248. In a triangle PQR, if $QR = 3$ cm, $PR = 5$ cm and $\sin P = \frac{3}{5}$, then $\angle Q$ equals:

Higher Secondary 28/06/2023 (Shift - 2)

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

Q.249. The value of

$$\operatorname{cosec} 41^\circ - \sec 49^\circ$$
 is :

Higher Secondary 30/06/2023 (Shift - 2)

- (a) 1 (b) 0 (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

SSC CHSL 2022 Tier - 2

Q.250. If $\sin x + \sin^2 x = 1$, then what is the value of $\cos^8 x + 2 \cos^6 x + \cos^4 x$?

SSC CHSL Tier II (26/06/2023)

- (a) 2 (b) 0 (c) 4 (d) 1

SSC CHSL 2022 Tier - 1

Q.251. If $\sec\theta + \tan\theta = 5$, then what is the value of $\sec\theta - \tan\theta$?

SSC CHSL 09/03/2023 (1st Shift)

- (a) $\frac{1}{5}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{1}{2}$

Q.252. What is the value of

$$\left(\frac{1 + \sec^2 A}{1 + \cos^2 A}\right) \left(\frac{1 + \sin^2 A}{1 + \operatorname{cosec}^2 A}\right) ?$$

SSC CHSL 09/03/2023 (1st Shift)

- (a) $\cos^2 A$ (b) $\operatorname{cosec}^2 A$
(c) $\sec^2 A$ (d) $\tan^2 A$

Q.253. What is $\frac{\sqrt{(1 - \sin A)}}{\sqrt{(1 + \sin A)}}$?

SSC CHSL 09/03/2023 (2nd Shift)

- (a) $\sec^2 A$ (b) $\sec A + \tan A$
(c) $\tan^2 A$ (d) $\sec A - \tan A$

Q.254. What is the value of

$$(1 + \tan\theta - \sec\theta)(1 + \cot\theta + \operatorname{cosec}\theta) ?$$

SSC CHSL 09/03/2023 (3rd Shift)

- (a) -1 (b) 0 (c) 1 (d) 2

Q.255. What is the value of

$$\frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \sec 40^\circ} ?$$

SSC CHSL 10/03/2023 (1st Shift)

- (a) 1 (b) 2 (c) 0.5 (d) 0

Q.256. If $\tan A + \sec A = K$, then what is the value of $\sin A$?

SSC CHSL 10/03/2023 (1st Shift)

- (a) $\frac{2K}{K^2 + 1}$ (b) $\frac{K^2 - 1}{2K}$
(c) $\frac{K^2 + 1}{k^2 - 1}$ (d) $\frac{K^2 - 1}{K^2 + 1}$

Q.257. If $4 \cot A = 5$, then what is the value of $6 \sec A \tan A$?

SSC CHSL 10/03/2023 (1st Shift)

- (a) $\frac{20\sqrt{41}}{21}$ (b) $\frac{25\sqrt{41}}{24}$
(c) $\frac{24\sqrt{41}}{25}$ (d) $\frac{3}{2}\sqrt{41}$

Q.258. What is $\frac{\operatorname{cosec} 90^\circ}{(1 - \cos^2 60^\circ)}$?

SSC CHSL 10/03/2023 (2nd Shift)

- (a) 0 (b) $\frac{4}{3}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

Q.259. If $\theta = 45^\circ$ degree, then what is the value of $2 \operatorname{cosec}^2\theta + 3 \sec^2\theta$?

SSC CHSL 10/03/2023 (3rd Shift)

- (a) 10 (b) 5 (c) 9 (d) 8

Q.260. If $\sin\theta = \sqrt{3} \cos\theta$, then what is the value of $\frac{3 \sin^2\theta + \cos\theta}{2 \cos\theta + 5}$?

SSC CHSL 10/03/2023 (3rd Shift)

- (a) $\frac{19}{25}$ (b) $\frac{18}{5}$ (c) $\frac{11}{24}$ (d) $\frac{3}{2}$

Q.261. If $\sin(C + D) = \frac{\sqrt{3}}{2}$ and $\sec(C - D)$

$= \frac{2}{\sqrt{3}}$ then what is the value of C and D?

SSC CHSL 10/03/2023 (3rd Shift)

- (a) 45 degree and 15 degree
(b) 30 degree and 30 degree
(c) 15 degree and 30 degree
(d) 60 degree and 30 degree

Q.262. If $\tan 2A = \cot(46^\circ - A)$, then what is the value of A?

SSC CHSL 10/03/2023 (4th Shift)

- (a) 2 degree (b) 16 degree
(c) 8 degree (d) 44 degree

Q.263. What is the value of

$$\sqrt{\frac{\operatorname{cosec} A + 1}{\operatorname{cosec} A - 1}} + \sqrt{\frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}} ?$$

SSC CHSL 13/03/2023 (1st Shift)

- (a) $2 \cos A$ (b) $\sec A$
(c) $\cos A$ (d) $2 \sec A$

Q.264. What is the value of $\frac{1 + \sin A}{\cos^2 A}$?

SSC CHSL 13/03/2023 (1st Shift)

- (a) $\frac{1}{1 - \sin A}$ (b) $\frac{1}{1 + \sin A}$
(c) $\frac{1}{1 + \cos A}$ (d) $\frac{1}{1 - \cos A}$

Q.265. If $m = \sec \theta - \tan \theta$ and $n = \operatorname{cosec} \theta + \cot \theta$, then what is the value of $m + n$ ($m - 1$) ?

SSC CHSL 13/03/2023 (2nd Shift)

- (a) 0 (b) 2 (c) 1 (d) - 1

Q.266. $\triangle ABC$ is a right angled triangle. $\angle A = 90^\circ$, $AB = 4$ cm, $BC = 5$ cm. What is the value of $\cos B + \cot C$?

SSC CHSL 13/03/2023 (3rd Shift)

- (a) $\frac{17}{20}$ (b) $\frac{29}{20}$ (c) $\frac{23}{20}$ (d) $\frac{31}{20}$

Q.267. If $\triangle PQR$ is right angled at Q, then what is the value of $\cos(P + R)$?

SSC CHSL 13/03/2023 (4th Shift)

- (a) 0.5 (b) 1 (c) 0.2 (d) 0

Q.268. If $\sqrt{3} \tan^2 \theta - 4 \tan \theta + \sqrt{3} = 0$, then what is the value of $\tan^2 \theta + \cot^2 \theta$?

SSC CHSL 13/03/2023 (4th Shift)

- (a) $\frac{4}{3}$ (b) $\frac{10}{3}$ (c) 3 (d) $\frac{6}{5}$

Q.269. If $\tan \theta + \cot \theta = 8$, then what is the value of $\tan^2 \theta + \cot^2 \theta$?

SSC CHSL 14/03/2023 (1st Shift)

- (a) 36 (b) 64 (c) 62 (d) 16

Q.270. If $6 \cot \theta = 5$, then find the value of $\frac{(6 \cos \theta + \sin \theta)}{(6 \cos \theta - 4 \sin \theta)}$?

SSC CHSL 14/03/2023 (2nd Shift)

- (a) 5 (b) 1 (c) 6 (d) 0

Q.271. $\triangle EFG$ is a right angled triangle. $\angle F = 90^\circ$, $EF = 10$ cm and $FG = 15$ cm. What is the value of $\operatorname{cosec} G$?

SSC CHSL 14/03/2023 (2nd Shift)

- (a) $\frac{\sqrt{13}}{2}$ (b) $\frac{2}{\sqrt{3}}$ (c) $\frac{\sqrt{13}}{10}$ (d) $\frac{10}{\sqrt{3}}$

Q.272. If $-\sin \theta + \operatorname{cosec} \theta = 6$, then what is the value of $\sin \theta + \operatorname{cosec} \theta$?

SSC CHSL 14/03/2023 (2nd Shift)

- (a) 6 (b) $\sqrt{40}$ (c) $\sqrt{34}$ (d) $\sqrt{38}$

Q.273. If $\sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$, then what is the value $\sin^2 \theta + \tan^2 \theta$?

SSC CHSL 14/03/2023 (3rd Shift)

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{5}{2}$ (d) $\frac{3}{2}$

Q.274. What is the value of

$$\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} ?$$

SSC CHSL 14/03/2023 (4th Shift)

- (a) $\frac{1}{(\sin^2 \theta - \cos^2 \theta)}$
(b) $2(\sin^2 \theta - \cos^2 \theta)$
(c) $\frac{2}{(\sin^2 \theta - \cos^2 \theta)}$
(d) $\sin^2 \theta - \cos^2 \theta$

Q.275. If $\cot 3A = \tan(A - 36^\circ)$, then what is the value of A ?

SSC CHSL 14/03/2023 (4th Shift)

- (a) 33.5 degree (b) 25 degree
(c) 30 degree (d) 31.5 degree

Q.276. Simplify the following expression:- $\frac{1 - \sin A}{\cos A} + \frac{\cos A}{1 - \sin A}$?

SSC CHSL 15/03/2023 (1st Shift)

- (a) $2 \sin A$ (b) $2 \cos A$ (c) $2 \sec A$ (d) $2 \tan A$

Q.277. If $\cos(A + B) = 0$ and $\sin(A - B) = 0$, then what is the value of angle A and angle B? ($0^\circ < A, B \leq 90^\circ$)

SSC CHSL 15/03/2023 (1st Shift)

- (a) 20 degree, 70 degree
(b) 45 degree, 45 degree
(c) 15 degree, 75 degree
(d) 60 degree, 30 degree

Q.278. If $\sec x + \tan x = 5$ and $\operatorname{cosec} y - \cot y = \frac{1}{3}$ then find the value of $(\sec x + \operatorname{cosec} y) - (\tan x - \cot y)$.

SSC CHSL 15/03/2023 (1st Shift)

- (a) 2.2 (b) 3.2 (c) 4.2 (d) 3.1

Q.279. If $\sec 3A = \operatorname{cosec}(A - 72^\circ)$, then what is the value of A ?

SSC CHSL 15/03/2023 (2nd Shift)

- (a) 34.5 degree (b) 30 degree
(c) 40.5 degree (d) 25 degree

Q.280. If $\sin(x - y) = \frac{1}{2}$ and $\cos(x + y) = \frac{1}{2}$, then what is the value of $\sin x \cos x +$

$2 \sin^2 x + \cos^3 x \sec x$?

SSC CHSL 15/03/2023 (3rd Shift)

- (a) 2 (b) $\frac{3}{4}$ (c) $\sqrt{2} + 1$ (d) 1

Q.281. What is the value of $\frac{1 + \sin^2 45^\circ}{1 - \cos^2 45^\circ}$?

SSC CHSL 15/03/2023 (3rd Shift)

- (a) 2 (b) 3 (c) $\frac{3}{2}$ (d) $\frac{1}{2}$

Q.282. What is the value of

$$\frac{3 \operatorname{cosec} 42^\circ}{\sec 48^\circ} - \frac{5 \cos 32^\circ}{\sin 58^\circ} ?$$

SSC CHSL 15/03/2023 (4th Shift)

- (a) - 1 (b) 5 (c) 0 (d) - 2

Q.283. If $A = a \operatorname{cosec} \theta + b \cot \theta$ and $B = a \cot \theta + b \operatorname{cosec} \theta$, then what is the value of $A^2 - B^2$?

SSC CHSL 15/03/2023 (4th Shift)

- (a) ab (b) $(a + b)$ (c) $(a^2 - b^2)$ (d) $(a - b)$

Q.284. If $\cos \theta + \sin \theta = \sqrt{2}$, then what is the value of $\sec \theta \operatorname{cosec} \theta$?

SSC CHSL 16/03/2023 (1st Shift)

- (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 0

Q.285. If $\operatorname{cosec} \theta + \cot \theta = s$, then what is the value of $\cos \theta$?

SSC CHSL 16/03/2023 (2nd Shift)

- (a) $\frac{s^2 - 1}{s^2 + 1}$ (b) $\frac{s^2 + 1}{s^2 - 1}$
(c) $\frac{s^2 - 1}{2s}$ (d) $\frac{2s}{s^2 + 1}$

Q.286. Find the value of $\tan x$, if $\sec x + \tan x = \sqrt{3}$ and x lies between 0° and 90° .

SSC CHSL 16/03/2023 (3rd Shift)

- (a) $\sqrt{3}$ (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) 0

Q.287. Tan of complementary of an angle is same as which of the angles?

SSC CHSL 17/03/2023 (1st Shift)

- (a) $\sin$ of that angle
(b) $\cot$ of that angle
(c) $\cos$ of that angle
(d) cosec of that angle

Q.288. If $\sin^4 x + \sin^2 x = 1$, then what is the value of $\cot^4 x + \cot^2 x$?

SSC CHSL 17/03/2023 (1st Shift)

- (a) 1 (b) 2 (c) - 1 (d) - 2

Q.289. If $\sec \theta + \tan \theta = 2$, then what is the value of $\cot \theta$?

SSC CHSL 17/03/2023 (2nd Shift)

- (a) $\frac{5}{3}$ (b) $\frac{3}{5}$ (c) $\frac{4}{3}$ (d) $\frac{3}{4}$

Q.290. What is the value of $(\tan 10^\circ \tan 80^\circ + \tan 20^\circ \tan 70^\circ + \tan 30^\circ \tan 60^\circ + \tan 40^\circ \tan 50^\circ)$?

SSC CHSL 17/03/2023 (4th Shift)

- (a) 4 (b) 3 (c) 5 (d) 2

Q.291. If $3 \operatorname{cosec} A = 7$, then what is the value of $\cos A \tan A$?

SSC CHSL 17/03/2023 (4th Shift)

- (a)
- $\frac{2\sqrt{10}}{7}$
- (b)
- $\frac{3}{7}$
- (c)
- $\frac{4}{7}$
- (d)
- $\frac{\sqrt{10}}{7}$

Q.292. If $2 \sec^2 \theta - 9 \sec \theta + 7 = 0$, then what is the value of $\cos^2 \theta + \sec^2 \theta$?

SSC CHSL 17/03/2023 (4th Shift)

- (a) 2 (b) 3 (c) 5 (d) 1

Q.293. What is the value of $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1}$?

SSC CHSL 20/03/2023 (2nd Shift)

- (a)
- $2 \sec \theta$
- (b)
- $2 \operatorname{cosec} \theta$
-
- (c)
- $2 \cot \theta$
- (d)
- $\cot \theta + \operatorname{cosec} \theta$

Q.294. If $6 \sec \theta = 10$, then find the value of $\frac{(5 \operatorname{cosec} \theta - 3 \cot \theta)}{(4 \cos \theta + 3 \sin \theta)}$?

SSC CHSL 20/03/2023 (3rd Shift)

- (a)
- $\frac{2}{3}$
- (b)
- $\frac{3}{2}$
- (c)
- $\frac{5}{6}$
- (d)
- $\frac{6}{5}$

Q.295. What is the value of $\sec 60^\circ + \tan 45^\circ - \sin^2 90^\circ$?

SSC CHSL 20/03/2023 (4th Shift)

- (a)
- $\frac{2}{\sqrt{3}}$
- (b)
- $\frac{1}{\sqrt{3}}$
- (c) 1 (d) 2

Q.296. If $\sin \theta + \cos \theta = \sqrt{2}$, then find the value of $\operatorname{cosec} \theta + \cot \theta$.

SSC CHSL 21/03/2023 (1st Shift)

- (a)
- $\sqrt{2} - 1$
- (b)
- $\sqrt{2} + 5$
- (c)
- $\sqrt{3} - 1$
- (d)
- $\sqrt{2} + 1$

Q.297. If $\cot(P + Q) = 0$ and $\tan(P - Q) = \frac{1}{\sqrt{3}}$, then what is the value of $\angle P$ and $\angle Q$?

(0 degree < P, Q ≤ 90 degree)

SSC CHSL 21/03/2023 (1st Shift)

- (a) 25 degree, 65 degree
-
- (b) 45 degree, 45 degree
-
- (c) 60 degree, 30 degree
-
- (d) 15 degree, 75 degree

Q.298. What is the value of $\operatorname{cosec} A(1 - \cos A)(\operatorname{cosec} A + \cot A)$?

SSC CHSL 21/03/2023 (2nd Shift)

- (a) 2 (b) 1 (c) 0 (d) 3

Q.299. If $\frac{(2 \cos^3 A - \cos A)}{(\sin A - 2 \sin^3 A)} = 1$, then what is the value of A ?

SSC CHSL 21/03/2023 (2nd Shift)

- (a)
- 90°
- (b)
- 30°
- (c)
- 45°
- (d)
- 60°

Q.300. What is the value of $\frac{\cot 60^\circ - \cot 30^\circ}{1 + \cot 60^\circ \cot 30^\circ}$?

SSC CHSL 21/03/2023 (3rd Shift)

- (a)
- $\frac{1}{\sqrt{3}}$
- (b)
- $\sqrt{3}$
- (c)
- $-\sqrt{3}$
- (d)
- $-\frac{1}{\sqrt{3}}$

Q.301. If $A = \cot 30^\circ \tan 60^\circ + \cot 60^\circ \tan 30^\circ$, then what is the value of A ?

SSC CHSL 21/03/2023 (4th Shift)

- (a)
- $\frac{1}{13}$
- (b)
- $\frac{10}{3}$
- (c)
- $\frac{10}{\sqrt{3}}$
- (d)
- $\frac{1}{3}$

Q.302. If $\tan \theta + \sin \theta = A$ and $\tan \theta - \sin \theta = B$, then what is the value of $A^2 - B^2$?

SSC CHSL 21/03/2023 (4th Shift)

- (a)
- $\tan \theta \sin \theta$
- (b)
- $4 \cot \theta$
-
- (c)
- $4 \tan \theta \sin \theta$
- (d)
- $2 \tan \theta \sin \theta$

SSC CGL 2022 Tier - 2**Q.303.** $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = ?$
SSC CGL Tier II (07/03/2023)

- (a)
- $5 + \tan^2 \theta + \cot^2 \theta$
-
- (b)
- $7 + \tan^2 \theta - \cot^2 \theta$
-
- (c)
- $7 + \tan^2 \theta + \cot^2 \theta$
-
- (d)
- $5 + \tan^2 \theta - \cot^2 \theta$

SSC CGL 2022 Tier - 1**Q.304.** The value of $\tan^2 \theta + \cot^2 \theta - \sec^2 \theta \operatorname{cosec}^2 \theta$ is:

SSC CGL 01/12/2022 (1st Shift)

- (a) 2 (b) -2 (c) 0 (d)
- $\frac{1}{2}$

Q.305. What will be the value of $\frac{\sin 30^\circ \sin 40^\circ \sin 50^\circ \sin 60^\circ}{\cos 30^\circ \cos 40^\circ \cos 50^\circ \cos 60^\circ}$?

SSC CGL 01/12/2022 (2nd Shift)

- (a)
- $\frac{1}{\sqrt{2}}$
- (b)
- $\sqrt{3}$
- (c) 1 (d)
- $\frac{1}{\sqrt{3}}$

Q.306. What is $\sin \alpha - \sin \beta$?

SSC CGL 01/12/2022 (2nd Shift)

- (a)
- $2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
-
- (b)
- $2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
-
- (c)
- $2 \cos \frac{\alpha - \beta}{2} \sin \frac{\alpha + \beta}{2}$
-
- (d)
- $2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$

Q.307. What is the value of $\tan 240^\circ$?

SSC CGL 01/12/2022 (3rd Shift)

- (a)
- $\sqrt{2}$
- (b)
- $(-\sqrt{3})$
- (c)
- $\sqrt{3}$
- (d) 3

Q.308. If $\tan^2 \theta = 1 - a^2$, then the value of $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta$ is:

SSC CGL 01/12/2022 (4th Shift)

- (a)
- $(2 - a)^{\frac{3}{2}}$
- (b)
- $(a^2 - 1)^{\frac{3}{2}}$
-
- (c)
- $(2 - a^2)^{\frac{3}{2}}$
- (d)
- $a^{\frac{3}{2}}$

Q.309. If $\sin(A + B) = \cos(A + B)$, what is the value of $\tan A$?

SSC CGL 02/12/2022 (1st Shift)

- (a)
- $\frac{1 - \tan B}{1 + \tan B}$
- (b)
- $\frac{1 + \tan B}{1 - \tan B}$

- (c)
- $\frac{1 + \sec B}{1 - \sec B}$
- (d)
- $\frac{1 - \operatorname{cosec} B}{1 + \operatorname{cosec} B}$

Q.310. In ΔABC , right-angled at B, $AB = 7$ cm and $AC - BC = 1$ cm. Find the value of $\sin C$.

SSC CGL 02/12/2022 (2nd Shift)

- (a)
- $\frac{3}{7}$
- (b)
- $\frac{12}{13}$
- (c)
- $\frac{3}{25}$
- (d)
- $\frac{7}{25}$

Q.311. If $\tan A + \cot A = 2$, then the value of $2(\tan^2 A + \cot^2 A)$ is:

SSC CGL 02/12/2022 (4th Shift)

- (a) 1 (b) 4 (c) 2 (d) 3

Q.312. If $\frac{(1 - \cos \theta)}{\sin \theta} = \frac{1}{5}$, then what will be the value of $\frac{(1 + \cos \theta)}{\sin \theta}$?

SSC CGL 02/12/2022 (4th Shift)

- (a) 5 (b)
- $\frac{2}{5}$
- (c)
- $\frac{4}{5}$
- (d)
- $\frac{1}{5}$

Q.313. Simplify: $\frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$

SSC CGL 03/12/2022 (1st Shift)

- (a)
- $\tan A$
- (b)
- $\cos A - \sin A$
-
- (c)
- $\cos A \sin A$
- (d)
- $\cos A + \sin A$

Q.314. $\frac{1}{1 + \cos(90^\circ - \theta)} + \frac{1}{1 - \cos(90^\circ - \theta)} = \underline{\hspace{2cm}}$

SSC CGL 03/12/2022 (1st Shift)

- (a)
- $2 \sec^2 \theta$
- (b) 1 (c) 0 (d)
- $2 \tan^2 \theta$

Q.315. $\sin^4 \theta + \cos^4 \theta$ in terms of $\sin \theta$ can be written as:

SSC CGL 03/12/2022 (2nd Shift)

- (a)
- $2 \sin^4 \theta + 2 \sin^2 \theta - 1$
-
- (b)
- $2 \sin^4 \theta - 2 \sin^2 \theta$
-
- (c)
- $2 \sin^4 \theta - 2 \sin^2 \theta - 1$
-
- (d)
- $2 \sin^4 \theta - 2 \sin^2 \theta + 1$

Q.316. Simplify the following :

$$\frac{\cos x - \sqrt{3} \sin x}{2}$$

SSC CGL 03/12/2022 (3rd Shift)

- (a)
- $\cos(\frac{\pi}{3} - x)$
- (b)
- $\sin(\frac{\pi}{3} + x)$
-
- (c)
- $\cos(\frac{\pi}{3} + x)$
- (d)
- $\sin(\frac{\pi}{3} - x)$

Q.317. The value of $\sqrt{\frac{1 + \cos A}{1 - \cos A}}$ is:

SSC CGL 03/12/2022 (4th Shift)

- (a)
- $\sec A - \tan A$
- (b)
- $\operatorname{cosec} A + \cot A$
-
- (c)
- $\sec A + \tan A$
- (d)
- $\operatorname{cosec} A - \cot A$

Q.318. If $\sin \theta + \cos \theta = \frac{\sqrt{3} - 1}{2\sqrt{2}}$, then what is the value of $\tan \theta + \cot \theta$?

SSC CGL 05/12/2022 (1st Shift)

- (a)
- $8(\sqrt{3} - 2)$
- (b)
- $12(\sqrt{3} - 2)$
-
- (c)
- $12(\sqrt{3} + 2)$
- (d)
- $8(\sqrt{3} + 2)$

Q.319. If $\sec 2\theta = \operatorname{cosec}(\theta - 36^\circ)$, where 2θ is an acute angle, find the value of θ .
SSC CGL 05/12/2022 (1st Shift)
(a) 32° (b) 46° (c) 20° (d) 42°

Q.320. $\cos^2 35^\circ + \cos 55^\circ \cdot \sin 35^\circ + \frac{\tan 34^\circ}{\cot 56^\circ}$
= _____.
SSC CGL 05/12/2022 (2nd Shift)
(a) 2 (b) 3 (c) 4 (d) 1

Q.321. What will be the value of $\cos x \operatorname{cosec} x - \sin x \sec x$?

SSC CGL 05/12/2022 (2nd Shift)

- (a) $\cot \frac{2x}{2}$ (b) $\tan 2x$
(c) $\cot 2x$ (d) $2\cot 2x$

Q.322. $\cot^2 A - \cos^2 A$ is equal to :
SSC CGL 05/12/2022 (3rd Shift)
(a) $\cos^2 A \cdot \sin^2 A$ (b) $\cos^2 A \cdot \cot^2 A$
(c) 1 (d) 0

Q.323. If $\operatorname{bsin} \theta = a$, then $\sec \theta + \tan \theta =$?
SSC CGL 06/12/2022 (1st Shift)

- (a) $\sqrt{\frac{b+a}{b-a}}$ (b) $\sqrt{\frac{1}{b+a}}$
(c) $\sqrt{\frac{1}{b-a}}$ (d) $\sqrt{\frac{b-a}{b+a}}$

Q.324. $\frac{\cos 20^\circ}{\sin 70^\circ} + \frac{\cos \theta}{\sin(90^\circ - \theta)} =$ _____.

SSC CGL 06/12/2022 (1st Shift)

- (a) -2 (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) 2

Q.325. What will be the value of $\sin 10^\circ - \frac{4}{3} \sin^3 10^\circ$?

SSC CGL 06/12/2022 (1st Shift)

- (a) $\frac{1}{3\sqrt{3}}$ (b) $\frac{1}{6}$ (c) $\frac{1}{2\sqrt{3}}$ (d) $\frac{\sqrt{3}}{6}$

Q.326. What is the value of $\cos(-\frac{17\pi}{3})$?

SSC CGL 06/12/2022 (2nd Shift)

- (a) 1 (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) 0

Q.327. Solve the following to find its value in terms of trigonometric ratios.

$$(\sin A + \cos A)(1 - \sin A \cos A)$$

SSC CGL 06/12/2022 (3rd Shift)

- (a) $\sin^3 A + \cos^3 A$
(b) $\sin^2 A - \cos^2 A$
(c) $[\cos A - \sin A][\sin^2 A + \cos^2 A]$
(d) $\sin^3 A - \cos^3 A$

Q.328. If $\sec \theta - 2\cos \theta = \frac{7}{2}$, where θ is a positive acute angle, then the value of $\sec \theta$ is:

SSC CGL 06/12/2022 (4th Shift)

- (a) 6 (b) 8 (c) 5 (d) 4

Q.329. Find the value of $\tan 3\theta$ if $\sec 3\theta$

= $\operatorname{cosec}(4\theta - 15^\circ)$.

SSC CGL 06/12/2022 (4th Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) -1 (d) 1

Q.330. If $\tan A - \tan B - \tan C = \tan A \tan B \tan C$, what is the value of A in terms of B and C ?

SSC CGL 07/12/2022 (1st Shift)

- (a) $A = B + C$ (b) $A = 2B - 2C$
(c) $A = B - C$ (d) $A = \frac{B-C}{2}$

Q.331. If $\tan^2 \alpha = 3 + Q^2$, then $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha =$?

SSC CGL 07/12/2022 (1st Shift)

- (a) $(3 + Q^2)^{\frac{3}{2}}$ (b) $(7 + Q^2)^{\frac{3}{2}}$
(c) $(5 - Q^2)^{\frac{3}{2}}$ (d) $(4 + Q^2)^{\frac{3}{2}}$

Q.332. If $1 + \sin \theta = m \cos \theta$, then what is the value of $\sin \theta$?

SSC CGL 07/12/2022 (2nd Shift)

- (a) $\frac{2m^2 - 1}{m^2 + 1}$ (b) $\frac{m^2 - 1}{m^2 + 1}$
(c) $\frac{m^2 + 1}{2m^2 - 1}$ (d) $\frac{m^2 + 1}{m^2 - 1}$

Q.333. $\cos(30^\circ + \theta) - \sin(60^\circ - \theta) =$ _____

SSC CGL 07/12/2022 (3rd Shift)

- (a) $\frac{\sqrt{3}}{2}$ (b) 0 (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$

Q.334. Find the value of the following expression.

$$5(\sin^4 \theta + \cos^4 \theta) + 3(\sin^6 \theta + \cos^6 \theta) + 19\sin^2 \theta \cos^2 \theta$$

SSC CGL 07/12/2022 (3rd Shift)

- (a) 8 (b) 5 (c) 6 (d) 7

Q.335. If $\tan \theta + \cot \theta = 12$, then $\tan^2 \theta + \cot^2 \theta$ is:

SSC CGL 07/12/2022 (4th Shift)

- (a) 66 (b) 128 (c) 142 (d) 132

Q.336. If $\sin Y = x$, then what will be the value of $\cos 2Y$ (where $0 \leq Y \leq 90^\circ$)?

SSC CGL 08/12/2022 (1st Shift)

- (a) $(\sqrt{2} - 1)x$ (b) $\sqrt{2}x$ (c) $1 - 2x$ (d) $1 - 2x^2$

Q.337. If $A + B = C$, then $\tan A \tan B \tan C =$?

SSC CGL 08/12/2022 (1st Shift)

- (a) $\tan C + \tan A - \tan B$
(b) $\tan C + \tan A + \tan B$
(c) $\tan A - \tan B - \tan C$
(d) $\tan C - \tan A - \tan B$

Q.338. If $x = 8(\sin \theta + \cos \theta)$ and $y = 9(\sin \theta - \cos \theta)$, then the value of

$$\frac{x^2}{8^2} + \frac{y^2}{9^2}$$

is:
SSC CGL 08/12/2022 (1st Shift)

- (a) 4 (b) 6 (c) 8 (d) 2

Q.339. What is the value of $\tan 6^\circ \times \tan 45^\circ \times \tan 84^\circ$?

SSC CGL 08/12/2022 (2nd Shift)

- (a) 1 (b) $\tan 6^\circ \times \tan 39^\circ$
(c) 3 (d) $\tan 6^\circ + \tan 45^\circ + \tan 84^\circ$

Q.340. $\theta = 135^\circ$; $y = 15^\circ$. What is the value of $2\cos(\theta) \sin(y)$?

SSC CGL 08/12/2022 (3rd Shift)

- (a) $\frac{\sqrt{3}-1}{2}$ (b) $\frac{1-\sqrt{3}}{2}$
(c) $2-\sqrt{3}$ (d) $\sqrt{3}-2$

Q.341. If $\sec x - \cos x = 4$, then what will

be the value of $\frac{(1 + \cos^2 x)}{\cos x}$?

SSC CGL 08/12/2022 (3rd Shift)

- (a) $\frac{9}{4}$ (b) $\frac{1}{4}$ (c) $2\sqrt{5}$ (d) $\sqrt{5}$

Q.342. If α is an acute angle, $\tan(4\alpha - 50^\circ) = \cot(50^\circ - \alpha)$, then find the value of α (in degrees).

SSC CGL 08/12/2022 (4th Shift)

- (a) 60° (b) 45° (c) 30° (d) 90°

Q.343. If $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$, then $x =$?

SSC CGL 08/12/2022 (4th Shift)

- (a) B (b) A (c) $A + B$ (d) $A - B$

Q.344. $\cos^3 60^\circ - \cos^3 240^\circ - \cos^3 360^\circ =$

SSC CGL 09/12/2022 (1st Shift)

- (a) $-\frac{3}{4}$ (b) $-\frac{7}{5}$ (c) $-\frac{3}{5}$ (d) $-\frac{9}{7}$

Q.345. If $5 \sin^2 A + 3 \cos^2 A = 4$, $0 < A < 90^\circ$, then what is the value of $\tan A$?

SSC CGL 09/12/2022 (1st Shift)

- (a) 0 (b) 3 (c) 1 (d) 2

Q.346. If $\cos(A - B) = \frac{\sqrt{3}}{2}$ and $\cos(A + B) = 0$, where A and B are positive acute angles and $A \geq B$ then the measures of A and B are:

SSC CGL 09/12/2022 (2nd Shift)

- (a) 80° and 10° (b) 60° and 30°
(c) 70° and 20° (d) 50° and 40°

Q.347. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{3}{2}$, then the value of $\sin^4 \theta - \cos^4 \theta$ is

SSC CGL 09/12/2022 (2nd Shift)

- (a) $\frac{5}{12}$ (b) $\frac{12}{13}$ (c) $\frac{11}{12}$ (d) $\frac{5}{13}$

Q.348. If $\sin^2 \theta \cos^2 \theta = \frac{2}{9}$, then what will be the value of $\operatorname{cosec}^2 \theta + \sec^2 \theta$?

SSC CGL 09/12/2022 (3rd Shift)

- (a) $\frac{7}{2}$ (b) $\frac{5}{2}$ (c) $\frac{9}{2}$ (d) $9\sqrt{2}$

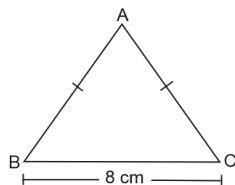
Q.349. If $2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$,

then $a, b = ?$

SSC CGL 09/12/2022 (3rd Shift)

- (a) $\frac{-3}{2}, \frac{-1}{2}$ (b) $\frac{3}{2}, \frac{1}{2}$ (c) -3, -1 (d) 3, 1

Q.350. In the given figure, ABC is an isosceles triangle with $BC = 8$ cm and $AB = AC = 5$ cm. The value of $\tan C - \cot B$ is _____.



SSC CGL 09/12/2022 (4th Shift)

- (a) $-\frac{5}{12}$ (b) $-\frac{7}{12}$ (c) $\frac{7}{12}$ (d) $\frac{5}{12}$

Q.351. If $b \cos \theta = a$, then $\operatorname{cosec} \theta + \cot \theta =$ _____

SSC CGL 09/12/2022 (4th Shift)

- (a) $\sqrt{\frac{1}{b+a}}$ (b) $\sqrt{\frac{b-a}{b+a}}$
(c) $\sqrt{\frac{b+a}{b-a}}$ (d) $\sqrt{\frac{1}{b-a}}$

Q.352. If $\operatorname{cosec} \theta + \cot \theta = p$, then the value of $\frac{p^2 - 1}{p^2 + 1}$ is:

SSC CGL 12/12/2022 (1st Shift)

- (a) $\cos \theta$ (b) $\sin \theta$ (c) $\cot \theta$ (d) $\operatorname{cosec} \theta$

Q.353. If $\sin^2 \theta - 3 \sin \theta + 2 = 0$, then find the value of θ ($0^\circ \leq \theta \leq 90^\circ$).

SSC CGL 12/12/2022 (2nd Shift)

- (a) 45° (b) 0° (c) 60° (d) 90°

Q.354. $\frac{1 + \sin \theta}{\cos \theta}$ is equal to which of

the following (where $\theta \neq \frac{\pi}{2}$) ?

SSC CGL 12/12/2022 (2nd Shift)

- (a) $\frac{1 + \cos \theta}{\sin \theta}$ (b) $\frac{\tan \theta + 1}{\tan \theta - 1}$
(c) $\frac{\tan \theta - 1}{\tan \theta + 1}$ (d) $\frac{\cos \theta}{1 - \sin \theta}$

Q.355. If $\tan 3\theta \cdot \tan 7\theta = 1$, where 7θ is an acute angle, then find the value of $\cot 15\theta$.

SSC CGL 12/12/2022 (3rd Shift)

- (a) 1 (b) -1 (c) $-\sqrt{3}$ (d) $\sqrt{3}$

Q.356. If $2 \sin \theta + 2 \sin^2 \theta = 2$, then the value of $2 \cos^4 \theta + 2 \cos^2 \theta$ is:

SSC CGL 12/12/2022 (3rd Shift)

- (a) 4 (b) 2 (c) 1 (d) 0

Q.357. If $\cos \theta + \cos^2 \theta = 1$, find the value of $\sqrt{\sin^4 \theta + \cos^2 \theta}$

SSC CGL 12/12/2022 (4th Shift)

- (a) $\sqrt{2} \cos \theta$ (b) $2 \cos \theta$
(c) $\sqrt{2} \sin \theta$ (d) $2 \sin \theta$

Q.358. If $\sec^2 A + \tan^2 A = 3$, then what is the value of $\cot A$?

SSC CGL 13/12/2022 (1st Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) 0 (c) 1 (d) $\sqrt{3}$

Q.359. What is the value of $\operatorname{cosec} 15^\circ \sec 15^\circ$?

SSC CGL 13/12/2022 (2nd Shift)

- (a) 0.5 (b) 4 (c) 2 (d) 1

Q.360. If $a = 45^\circ$ and $b = 15^\circ$, what is the value of $\frac{\cos(a-b) - \cos(a+b)}{\cos(a-b) + \cos(a+b)}$?

SSC CGL 13/12/2022 (2nd Shift)

- (a) $2 - 2\sqrt{2}$ (b) $3 - \sqrt{6}$ (c) $3 - \sqrt{2}$ (d) $2 - \sqrt{3}$

Q.361. $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} = ?$

SSC CGL 13/12/2022 (2nd Shift)

- (a) $\tan A - \cot A$ (b) $\tan A + \cot A$
(c) $\sin A - \cos A$ (d) $\sin A + \cos A$

Q.362. If $\sec \theta + \frac{1}{\cos \theta} = 2$, find the value

of $\sec^{55} \theta + \frac{1}{\sec^{55} \theta}$

SSC CGL 13/12/2022 (3rd Shift)

- (a) 2 (b) 0 (c) 1 (d) 55

Q.363. If $\tan 40^\circ = a$, then find

$\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ}$

SSC CGL 13/12/2022 (3rd Shift)

- (a) $\frac{1 - a^2}{a}$ (b) $\frac{1 + a^2}{2a}$
(c) $\frac{1 - a^2}{2a}$ (d) $\frac{1 + a^2}{a}$

Q.364. If $\tan(\alpha + \beta) = a$, $\tan(\alpha - \beta) = b$ then the value of $\tan 2\alpha$ is:

SSC CGL 13/12/2022 (4th Shift)

- (a) $\frac{a+b}{1-ab}$ (b) $\frac{a+b}{1+ab}$
(c) $\frac{a-b}{1+ab}$ (d) $\frac{a-b}{1-ab}$

SSC CPO 2022 Tier - 1

Q.365. If $x = \frac{2 \sin \theta}{(1 + \cos \theta + \sin \theta)}$ then the

value of $\frac{1 - \cos \theta + \sin \theta}{(1 + \sin \theta)}$ is:

SSC CPO 09/11/2022 (Morning)

- (a) $\frac{x}{(1+x)}$ (b) x (c) $\frac{1}{x}$ (d) $\frac{(1+x)}{x}$

Q.366. If $\left\{ \left(\frac{\sec \theta - 1}{\sec \theta + 1} \right)^n = \operatorname{cosec} \theta - \cot \theta \right\}$, then $n = ?$

SSC CPO 09/11/2022 (Morning)

- (a) 1 (b) 0.5 (c) -1 (d) -0.5

Q.367. The value of $(2 \cos^2 \theta - 1)$

$\left[\frac{1 + \tan \theta}{1 - \tan \theta} + \frac{1 - \tan \theta}{1 + \tan \theta} \right]$ is;

SSC CPO 09/11/2022 (Morning)

- (a) 2 (b) 0 (c) $\frac{\sqrt{3}}{2}$ (d) 1

Q.368. Simplify.

$\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$

SSC CPO 09/11/2022 (Afternoon)

- (a) $\cot \theta$ (b) $\cot 2\theta$ (c) $\tan \theta$ (d) $\tan 2\theta$

Q.369. The value of $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 1$ is:

SSC CPO 09/11/2022 (Afternoon)

- (a) 1 (b) 2 (c) 3 (d) 0

Q.370. Given that A and B are second quadrant angles, $\sin A = \frac{1}{3}$ and $\sin B = \frac{1}{5}$,

then find the value of $\cos(A - B)$.

SSC CPO 09/11/2022 (Evening)

- (a) $\frac{4\sqrt{3} + 1}{15}$ (b) $\frac{8\sqrt{3} - 1}{15}$
(c) $\frac{8\sqrt{3} + 1}{15}$ (d) $\frac{4\sqrt{3} - 1}{15}$

Q.371. What is the value of $\frac{\sin(A+B)}{\sin A \cos B}$?

SSC CPO 09/11/2022 (Evening)

- (a) $1 + \cot A \tan B$ (b) $1 + \tan A \cot B$
(c) $1 - \sin A \cos B$ (d) $1 - \cot A \tan B$

Q.372. The value of the expression $\cos^2 45^\circ + \cos^2 135^\circ + \cos^2 225^\circ + \cos^2 315^\circ$ is:

SSC CPO 10/11/2022 (Morning)

- (a) 2 (b) $\frac{1}{2}$ (c) $\frac{3}{2}$ (d) 1

Q.373. The value of $\frac{2 \cos^3 \theta - \cos \theta}{\sin \theta - 2 \sin^3 \theta}$ is:

SSC CPO 10/11/2022 (Morning)

- (a) $\sec \theta$ (b) $\sin \theta$ (c) $\cot \theta$ (d) $\tan \theta$

Q.374. If $\cos 48^\circ = \frac{m}{n}$, then $\sec 48^\circ - \cot 42^\circ$ is equal to:

SSC CPO 10/11/2022 (Morning)

- (a) $\frac{m - \sqrt{n^2 - m^2}}{m}$ (b) $\frac{m - \sqrt{n^2 - m^2}}{n}$
(c) $\frac{n - \sqrt{n^2 - m^2}}{n}$ (d) $\frac{n - \sqrt{n^2 - m^2}}{m}$

Q.375. Solve the following equation.

$\sec^2 \theta \sqrt{1 - \sin^2 \theta} =$ _____.

SSC CPO 10/11/2022 (Morning)

- (a) $\tan \theta$ (b) $\operatorname{cosec} \theta$ (c) $\sec \theta$ (d) 1

Q.376. If the value of $\operatorname{cosec} A + \cot A = m$, then the value of $\operatorname{cosec} A - \cot A$ is:

SSC CPO 10/11/2022 (Afternoon)

- (a) $\frac{1}{m}$ (b) m (c) $\sqrt{m}$ (d) m^2

Q.377. If $\cos \theta + \sec \theta = 2$, then the value of $\sin^6 \theta + \cos^6 \theta$ is:

SSC CPO 10/11/2022 (Evening)

(a) $\frac{1}{3}$ (b) 0 (c) 1 (d) $\frac{1}{2}$

Q.378. The value of $(\operatorname{cosec} A + \cot A)(1 - \cos A)$ is:

SSC CPO 10/11/2022 (Evening)

(a) $\cos A$ (b) $\tan A$ (c) $\cot A$ (d) $\sin A$

Q.379. If $\cos A = \sin^2 A$, and $a \sin^{12} A + b \sin^{10} A + c \sin^8 A + \sin^6 A = 1$, then $a + b + c = ?$

SSC CPO 11/11/2022 (Morning)

(a) 7 (b) 8 (c) 9 (d) 6

Q.380. If $4 \cos \theta + 3 \sin \theta = x$ and $4 \sin \theta - 3 \cos \theta = y$, find the value of $x^2 + y^2$.

SSC CPO 11/11/2022 (Morning)

(a) 16 (b) 9 (c) 25 (d) 1

Q.381. Simplify. $\frac{\cos^4 \theta - \sin^4 \theta}{\sin^2 \theta}$

SSC CPO 11/11/2022 (Afternoon)

(a) $1 - \tan^2 \theta$ (b) $\tan^2 \theta - 1$
(c) $\cot^2 \theta - 1$ (d) $1 - \cot^2 \theta$

Q.382. The value of $\frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} + 1$ is?

SSC CPO 11/11/2022 (Afternoon)

(a) $\frac{1}{2}$ (b) 3 (c) 0 (d) 2

Q.383. If $1 + \sin^2 \theta - 3 \sin \theta \cos \theta = 0$, then the value of $\cot \theta$ is:

SSC CPO 11/11/2022 (Evening)

(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

Q.384. If $\cos^4 \alpha - \sin^4 \alpha = \frac{5}{6}$, then the value of $2 \cos^2 \alpha - 1 =$ _____.

SSC CPO 11/11/2022 (Evening)

(a) $\frac{11}{6}$ (b) $\frac{5}{6}$ (c) $\frac{6}{11}$ (d) $\frac{6}{5}$

SSC CGL 2021 Tier - 2

Q.385. Which of the following is equal to

$$\left[\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \right] ?$$

SSC CGL Tier II (08/08/2022)

(a) $\operatorname{cosec} \theta \cdot \sec \theta$ (b) $\sec \theta \cdot \tan \theta$
(c) $\operatorname{cosec} \theta \cdot \tan \theta$ (d) $\cot \theta \cdot \sec \theta$

Q.386. What is the value of $\frac{\cos 50^\circ}{\sin 40^\circ} + \frac{3 \operatorname{cosec} 80^\circ}{\sec 10^\circ} - 2 \cos 50^\circ \cdot \operatorname{cosec} 40^\circ$?

SSC CGL Tier II (08/08/2022)

(a) 3 (b) 2 (c) 5 (d) 4

Q.387. If $\sqrt{3} \tan \theta = 3 \sin \theta$, then what is the value of $\sin^2 \theta - \cos^2 \theta$?

SSC CGL Tier II (08/08/2022)

(a) $\frac{1}{5}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

Q.388. Which of the following is equal to

$$\frac{1}{\tan \theta} + \tan \theta ?$$

SSC CGL Tier II (08/08/2022)

(a) $\sec \theta \cdot \operatorname{cosec} \theta$ (b) 1
(c) $\frac{\operatorname{cosec} \theta}{\sec \theta}$ (d) $\tan^2 \theta$

SSC Selection Post (Phase - X)

Q.389. If $\operatorname{cosec}(A - B) = 2$ and $\cot(A + B) = \frac{1}{\sqrt{3}}$, then what is the value of A?

Graduate Level 01/08/2022 (Shift - 4)

(a) 90° (b) 60° (c) 30° (d) 45°

Q.390. $[(\sin \theta \tan \theta + \cos \theta)^2 - 1]$ is equal to:

Graduate Level 02/08/2022 (Shift - 3)

(a) $\tan^2 \theta$ (b) $\operatorname{cosec} \theta$ (c) $\sec^2 \theta$ (d) $\sec \theta$

Q.391. The value of

$$\sin^2 \frac{2\pi}{3} + \cos^2 \frac{5\pi}{6} - \tan^2 \frac{3\pi}{4}$$
 is :

Graduate Level 03/08/2022 (Shift - 3)

(a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) 4

Q.392. $\frac{2 \sin \theta}{\sqrt{4 - 4 \sin^2 \theta}} = ?$

Graduate Level 04/08/2022 (Shift - 4)

(a) $\tan \theta$ (b) $2 \tan \theta$ (c) $\cot \theta$ (d) $2 \cot \theta$

Q.393. Simplify.

$$\frac{1}{\cos x} \sqrt{\frac{\cos(\pi + x) \cos(-x)}{\sin(\pi - x) \cos(\frac{\pi}{2} + x)}}$$

Graduate Level 04/08/2022 (Shift - 4)

(a) $\sec x$ (b) $\tan x$ (c) $\operatorname{cosec} x$ (d) $\cot x$

Q.394. If $\cos \theta = \sin(2\theta) \neq 0$, what is the value of $\cos^4 \theta + \sin^4 \theta + \cos^3 \theta + \sin^3 \theta + \sin^2 \theta + \cos^2 \theta + \sin \theta + \cos \theta$?

Graduate Level 05/08/2022 (Shift - 3)

(a) $\frac{8 + 7\sqrt{3}}{18}$ (b) $\frac{18 + 7\sqrt{3}}{8}$
(c) $\frac{18 + 8\sqrt{3}}{7}$ (d) $\frac{7 + 18\sqrt{3}}{8}$

Q.395. If $\tan 3\theta = \cot(\theta - 22^\circ)$, where 3θ is an acute angle, find the value of θ .

Higher Secondary 01/08/2022 (Shift - 3)

(a) 26° (b) 29° (c) 28° (d) 27°

SSC CHSL 2021 Tier - 1

Q.396. If $\frac{x - x \operatorname{cosec}^2 30^\circ}{1 + \operatorname{cosec}^2 30^\circ} = \cos^2 60^\circ$

+ $4 \cot^2 45^\circ - \sec^2 60^\circ$, then the value of x is _____.

SSC CHSL 25/05/2022 (Morning)

(a) $\frac{5}{12}$ (b) $-\frac{5}{12}$ (c) $\frac{1}{5}$ (d) $-\frac{1}{12}$

Q.397. If $\sec \theta - \cos \theta = 14$ and $14 \sec \theta = x$, then the value of x is _____.

SSC CHSL 25/05/2022 (Afternoon)

(a) $\tan^2 \theta$ (b) $\sec^2 \theta$ (c) $2 \sec \theta$ (d) $2 \tan \theta$

Q.398. The value of $(1 + \tan 10^\circ)(1 + \tan 35^\circ)$ is:

SSC CHSL 25/05/2022 (Evening)

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) 1 (d) 2

Q.399. The value of $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$ is:

SSC CHSL 27/05/2022 (Morning)

(a) $\tan 3\theta$ (b) $\cot 3\theta$ (c) $\tan \theta$ (d) $\cot \theta$

Q.400. If $\cos 2A = \sin 75^\circ$, then the smallest positive value of A is :

SSC CHSL 27/05/2022 (Afternoon)

(a) 15° (b) 7.5° (c) 30° (d) 37.5°

Q.401. If $\sin x - 3 \cos x = \sqrt{3} \cos x$, then the value of $\cot x$ is _____.

SSC CHSL 31/05/2022 (Morning)

(a) $3 - \sqrt{3}$ (b) $3 + \sqrt{3}$
(c) $\sqrt{3}$ (d) $\frac{3 - \sqrt{3}}{6}$

Q.402. What is the value of

$$(4 \sin^3 x - 3 \sin x + \sin 3x)?$$

SSC CHSL 31/05/2022 (Afternoon)

(a) 1 (b) 0 (c) 4 (d) 3

Q.403. What is the simplified value of

$$\cos^2(90^\circ - \theta) - \left[\frac{\{\cos(90^\circ - \theta) \cos \theta\}}{\cot \theta} \right]$$

SSC CHSL 31/05/2022 (Evening)

(a) 4 (b) 2 (c) 0 (d) 1

Q.404. The value of $\sin 2\theta (\tan \theta + \cot \theta)$ is :

SSC CHSL 01/06/2022 (Morning)

(a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) 2

Q.405. Using $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$, find the value of $\cos 75^\circ \cos 15^\circ$.

SSC CHSL 01/06/2022 (Afternoon)

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{4}$

Q.406. If $\operatorname{cosec} \theta + \cot \theta = 5$, then $\operatorname{cosec} \theta$ is equal to _____.

SSC CHSL 03/06/2022 (Morning)

(a) $(5 + \frac{1}{5})$ (b) $\frac{1}{2}(3 + \frac{1}{3})$
(c) $\frac{1}{2}(5 + \frac{1}{5})$ (d) $(3 + \frac{1}{3})$

Q.407. If $1 + \tan \theta = \sqrt{3}$, then

$$\sqrt{3} \cot \theta - 1 =$$

SSC CHSL 03/06/2022 (Afternoon)

(a) $\frac{\sqrt{3} + 1}{2}$ (b) $\frac{\sqrt{3} - 1}{2}$

(c) $\frac{2\sqrt{3}-1}{2}$ (d) $\frac{2\sqrt{3}+1}{2}$

Q.408. What is the value of the expression $100(\sin 15^\circ \cos 15^\circ)$?
SSC CHSL 03/06/2022 (Evening)
(a) 50 (b) 75 (c) 100 (d) 25

Q.409. If $\operatorname{cosec}^2 \theta + \cot^2 \theta = \frac{5}{3}$, then

what is the value of $\cot 2\theta$

SSC CHSL 07/06/2022 (Morning)

(a) $\frac{1}{\sqrt{3}}$ (b) $-\frac{1}{\sqrt{3}}$ (c) $\frac{2}{\sqrt{3}}$ (d) $-\frac{2}{\sqrt{3}}$

Q.410. $(\operatorname{cosec} \theta - \cot \theta)^2 = ?$, $0^\circ < \theta < 90^\circ$
SSC CHSL 09/06/2022 (Evening)

(a) $\frac{1 - \sin \theta}{1 + \cos \theta}$ (b) $\frac{1 - \cos \theta}{1 + \cos \theta}$

(c) $\frac{1 + \cos \theta}{1 - \cos \theta}$ (d) $\frac{1 + \cos \theta}{1 - \sin \theta}$

Q.411. Using the formula $\tan \frac{x}{2}$

$= \frac{1 - \cos x}{\sin x}$, find the value of $\tan 22.5^\circ$.

SSC CHSL 10/06/2022 (Morning)

(a) $\sqrt{2} - 1$ (b) $\sqrt{2} + 1$

(c) $\frac{\sqrt{3} + \sqrt{2}}{2}$ (d) $\frac{\sqrt{3} - \sqrt{2}}{2}$

SSC CGL 2021 Tier - 1

Q.412. The value of

$\left(\frac{\sin 23^\circ \cos 67^\circ + \sec 52^\circ \sin 38^\circ}{\operatorname{cosec}^2 20^\circ - \tan^2 70^\circ} \right) + \left(\frac{\cos 23^\circ \sin 67^\circ + \operatorname{cosec} 52^\circ \cos 38^\circ}{\operatorname{cosec}^2 20^\circ - \tan^2 70^\circ} \right)$ is:

SSC CGL 11/04/2022 (Morning)

(a) 3 (b) 4 (c) 2 (d) 0

Q.413. If $4\sin^2 \theta = 3(1 + \cos \theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $(2\tan \theta + 4\sin \theta - \sec \theta)$?

SSC CGL 11/04/2022 (Morning)

(a) $3\sqrt{15} - 4$ (b) $15\sqrt{3} - 4$

(c) $15\sqrt{3} + 3$ (d) $4\sqrt{15} - 3$

Q.414. If $3 \sec^2 \theta + \tan \theta - 7 = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of

$\left(\frac{2 \sin \theta + 3 \cos \theta}{\operatorname{cosec} \theta + \sec \theta} \right)$?

SSC CGL 12/04/2022 (Morning)

(a) 10 (b) $\frac{5}{2}$ (c) $\frac{5}{4}$ (d) $4\sqrt{2}$

Q.415. The value of $1 + \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$,

if $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 12/04/2022 (Afternoon)

(a) $1 - \sec \theta + \tan \theta$ (b) $1 - \sec \theta - \tan \theta$

(c) $1 + \sec \theta - \tan \theta$ (d) $1 + \sec \theta + \tan \theta$

Q.416. The value of $\left(\frac{1 - \cot \theta}{1 - \tan \theta} \right)^2 - 1$

when $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 12/04/2022 (Afternoon)

(a) $\cos^2 \theta - 1$ (b) $\sec^2 \theta + 1$

(c) $\cot^2 \theta - 1$ (d) $\sin^2 \theta - 1$

Q.417. The expression $(\cos^6 \theta + \sin^6 \theta - 1)(\tan^2 \theta + \cot^2 \theta + 2) + 3$ is equal to:
SSC CGL 13/04/2022 (Afternoon)

(a) 0 (b) 1 (c) 2 (d) -1

Q.418. If $\sin^2 \theta - \cos^2 \theta - 3\sin \theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of $1 + \sec \theta + \tan \theta$?

SSC CGL 13/04/2022 (Evening)

(a) $-1 + \sqrt{3}$ (b) $-1 - \sqrt{3}$

(c) $1 + \sqrt{3}$ (d) $1 - \sqrt{3}$

Q.419. $(\sec \theta - \tan \theta)^2 (1 + \sin \theta)^2 \div \cos^2 \theta =$
SSC CGL 13/04/2022 (Evening)

(a) $\cos^2 \theta$ (b) 1 (c) $\cot^2 \theta$ (d) -1

Q.420. If $2 \sin^2 \theta + 3 \cos \theta = 3$, $0^\circ < \theta < 90^\circ$, then the value of $(\sec^2 \theta + \cot^2 \theta)$ is
SSC CGL 18/04/2022 (Evening)

(a) $3\frac{2}{3}$ (b) $3\frac{1}{3}$ (c) $4\frac{1}{3}$ (d) $4\frac{1}{2}$

Q.421. Simplify the following expression:

$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} - \sin A$

SSC CGL 19/04/2022 (Evening)

(a) $1 + \cos A$ (b) $(1 + \sin A) \cos A$

(c) $1 + \sin A$ (d) $\cos A$

Q.422. Simplify the following expression:
 $\operatorname{cosec}^4 A (1 - \cos^4 A) - 2\cot^2 A - 1$

SSC CGL 20/04/2022 (Morning)

(a) $\sin^2 A$ (b) $\operatorname{cosec}^2 A$ (c) 1 (d) 0

Q.423. If $6 \tan A (\tan A + 1) = 5 - \tan A$,

Given that $0 < A < \frac{\pi}{2}$ What is the value of

$(\sin A + \cos A)$?

SSC CGL 20/04/2022 (Morning)

(a) $3\sqrt{5}$ (b) $\frac{5}{\sqrt{3}}$ (c) $5\sqrt{3}$ (d) $\frac{3}{\sqrt{5}}$

Q.424. The value of $\left(\frac{1 - \cot \theta}{1 - \tan \theta} \right)^2 + 1$, if

$0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 20/04/2022 (Evening)

(a) $\operatorname{cosec}^2 \theta$ (b) $\sin^2 \theta$ (c) $\cos^2 \theta$ (d) $\sec^2 \theta$

Q.425. The value of

$\frac{5 \cos^2 62^\circ + 5 \cos^2 28^\circ - 21}{7 \sin^2 35^\circ + 7 \sin^2 55^\circ + 1}$ is:

SSC CGL 20/04/2022 (Evening)

(a) 2 (b) -2 (c) 3 (d) -3

Q.426. If $\frac{\sin^2 \theta - 3 \sin \theta + 2}{\cos^2 \theta} = 1$, where

$0^\circ < \theta < 90^\circ$, then what is the value of $(\cos 2\theta - \sin 3\theta + \operatorname{cosec} 2\theta)$?

SSC CGL 21/04/2022 (Afternoon)

(a) $\frac{-3 - 4\sqrt{3}}{6}$ (b) $\frac{-3 + 4\sqrt{3}}{6}$

(c) $\frac{3 + 4\sqrt{3}}{6}$

(d) $\frac{3 - 4\sqrt{3}}{6}$

SSC CGL 2020 Tier - 2

Q.427. The value of

$\frac{4 \tan^2 30^\circ + \sin^2 30^\circ \cos^2 45^\circ + \sec^2 48^\circ - \cot^2 42^\circ}{\cos 37^\circ \sin 53^\circ + \sin 37^\circ \cos 53^\circ + \tan 18^\circ \tan 72^\circ}$

SSC CGL Tier II (29/01/2022)

(a) $\frac{59}{48}$ (b) $\frac{35}{24}$ (c) $\frac{49}{24}$ (d) $\frac{35}{48}$

Q.428. $\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$,

$0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (29/01/2022)

(a) $\tan \theta$ (b) $\sec \theta$ (c) $\operatorname{cosec} \theta$ (d) $\cot \theta$

Q.429. If $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$,

$0^\circ < \theta < 90^\circ$, then $\frac{a+b}{a-b} = ?$

SSC CGL Tier II (29/01/2022)

(a) $\sec \theta$ (b) $\sin \theta$ (c) $\cos \theta$ (d) $\operatorname{cosec} \theta$

Q.430.

$\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2}$

, $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (29/01/2022)

(a) $\cos \theta$ (b) $1 - \cos \theta$ (c) $\sin \theta$ (d) $1 - \sin \theta$

Q.431. $\left(\frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta \right)$

$\div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta)$, $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (29/01/2022)

(a) $\operatorname{cosec} \theta \sec \theta$ (b) $\operatorname{cosec} \theta$

(c) $\sin \theta \cos \theta$ (d) $\sec \theta$

Q.432. If $3 \tan \theta = 2\sqrt{3} \sin \theta$, $0^\circ < \theta < 90^\circ$,

then the value of $\frac{\operatorname{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2 \theta + \tan^2 \theta}$ is:

SSC CGL Tier II (29/01/2022)

(a) $\frac{4}{13}$ (b) $\frac{20}{27}$ (c) $\frac{20}{39}$ (d) $\frac{4}{3}$

Q.433. The expression

$\frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 2}$, $0^\circ < \theta < 90^\circ$, is equal to:

90° , is equal to:

SSC CGL Tier II (03/02/2022)

(a) $\sec^2 \theta \operatorname{cosec}^2 \theta$ (b) $-\sec^2 \theta \operatorname{cosec}^2 \theta$

(c) $\cos^2 \theta \sin^2 \theta$ (d) $-\cos^2 \theta \sin^2 \theta$

Q.434. The expression

$\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}$

, $0^\circ < \theta < 90^\circ$, is equal to

SSC CGL Tier II (03/02/2022)

(a) $\frac{1}{2} \sin \theta$ (b) $2 \sin \theta$ (c) $\sec \theta$ (d) $2 \operatorname{cosec} \theta$

Q.435. The expression

$$\frac{(1 - \sin\theta + \cos\theta)^2(1 - \cos\theta)\sec^3\theta \operatorname{cosec}^2\theta}{(\sec\theta - \tan\theta)(\tan\theta + \cot\theta)}$$

, ($0^\circ < \theta < 90^\circ$), is equal to

SSC CGL Tier II (03/02/2022)

(a) $\sin\theta$ (b) $2\cos\theta$ (c) $\cot\theta$ (d) $2\tan\theta$

Q.436. If $\frac{\sin^2\theta}{\cos^2\theta - 3\cos\theta + 2} = 1$, θ lies in the first quadrant, then the value of $\frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta}$ is:

SSC CGL Tier II (03/02/2022)

(a) $\frac{2\sqrt{3}}{27}$ (b) $\frac{5\sqrt{3}}{27}$ (c) $\frac{2\sqrt{3}}{9}$ (d) $\frac{7\sqrt{3}}{54}$

Q.437. The expression

$$\frac{(1 - 2\sin^2\theta \cos^2\theta)(\cot\theta + 1) \cos\theta}{(\sin^4\theta + \cos^4\theta)(1 + \tan\theta) \operatorname{cosec}\theta} - 1,$$

$0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (03/02/2022)

(a) $\cos^2\theta$ (b) $-\sin^2\theta$ (c) $\sec^2\theta$ (d) $-\sec^2\theta$

SSC MTS 2020 Tier - 1

Q.438. What is the average of the sixty terms given below?

$\cos^2 x, \cos^2 2x, \cos^2 3x, \dots, \cos^2 30x, \sin^2 x, \sin^2 2x, \sin^2 3x, \dots, \sin^2 30x$

SSC MTS 12/10/2021 (Afternoon)

(a) $\cos^2 x$ (b) 0.5 (c) 1 (d) $\cos^2 x \sin^2 x$

SSC CGL 2020 Tier - 1

Q.439. $\frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - 1} + \frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta + 1} - \tan^2\theta$, $0^\circ < \theta < 90^\circ$

SSC CGL 13/08/2021 (Morning)

(a) $2\sec^2\theta$ (b) $\sec^2\theta + 1$
(c) $\sec^2\theta$ (d) $1 - \tan^2\theta$

Q.440. If $2\cos^2\theta = 3\sin\theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\sec^2\theta - \tan^2\theta + \cos^2\theta)$ is:

SSC CGL 13/08/2021 (Afternoon)

(a) $\frac{7}{4}$ (b) $\frac{5}{4}$ (c) $\frac{9}{4}$ (d) $\frac{3}{4}$

Q.441. If $3\tan\theta = 2\sqrt{3}\sin\theta$, $0^\circ < \theta < 90^\circ$ then find the value of $2\sin^2\theta - 3\cos^2\theta$.

SSC CGL 13/08/2021 (Evening)

(a) 1 (b) $\frac{3}{2}$ (c) $\frac{1}{2}$ (d) $-\frac{3}{2}$

Q.442. The value of $\frac{\tan(45^\circ - \alpha)}{\cot(45^\circ + \alpha)} -$

$$\frac{(\cos 19^\circ + \sin 71^\circ)(\sec 19^\circ + \operatorname{cosec} 71^\circ)}{\tan 12^\circ \tan 24^\circ \tan 66^\circ \tan 78^\circ}$$

SSC CGL 13/08/2021 (Evening)

(a) -3 (b) 0 (c) -2 (d) 2

Q.443. $1 + 2\tan^2\theta + 2\sin\theta \sec^2\theta$, $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 16/08/2021 (Morning)

(a) $\frac{1 - \sin\theta}{1 + \sin\theta}$ (b) $\frac{1 + \cos\theta}{1 - \cos\theta}$
(c) $\frac{1 - \cos\theta}{1 + \cos\theta}$ (d) $\frac{1 + \sin\theta}{1 - \sin\theta}$

Q.444. If $2\sin(3x - 15)^\circ = 1$, $0^\circ < (3x - 15) < 90^\circ$, then find the value of $\cos^2(2x + 15)^\circ + \cot^2(x + 15)^\circ$.

SSC CGL 16/08/2021 (Afternoon)

(a) 1 (b) $\frac{5}{2}$ (c) $-\frac{7}{2}$ (d) $\frac{7}{2}$

Q.445. If $\sin\left(\frac{2A+B}{2}\right) = \cos\left(\frac{2A-B}{2}\right) =$

$$= \frac{\sqrt{3}}{2}, 0^\circ < \frac{2A+B}{2} \text{ and } \frac{2A-B}{2} < 90^\circ$$

then find the value of $\sin[3(A - B)]$.

SSC CGL 16/08/2021 (Evening)

(a) 1 (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

Q.446. $(\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta})$

$$\left(\frac{\sin\theta(1 + \cos\theta)}{1 + \cos\theta - \sin^2\theta}\right), 0^\circ < \theta < 90^\circ \text{ is}$$

equal to:

SSC CGL 17/08/2021 (Morning)

(a) $\cot\theta$ (b) $\operatorname{cosec}^2\theta$ (c) $\sec^2\theta$ (d) $\tan\theta$

Q.447. For $0^\circ < \theta < 90^\circ$,

$$\frac{1}{\cos\theta} + \frac{1}{\tan\theta - \sec\theta} \text{ is equal to:}$$

SSC CGL 17/08/2021 (Afternoon)

(a) $-\sec\theta$ (b) $\tan\theta$ (c) $\sec\theta$ (d) $-\tan\theta$

Q.448. If $4\sin^2(2x - 10)^\circ = 3$, $0 \leq (2x - 10) \leq 90$, then find the value of

$$\frac{\sin^4(x - 5)^\circ + \cos^4(x - 5)^\circ}{1 - 2\sin^2(3x - 15)^\circ \cos^2(3x - 15)^\circ}$$

SSC CGL 17/08/2021 (Evening)

(a) 1 (b) $\frac{5}{8}$ (c) $-\frac{5}{8}$ (d) -1

Q.449. The value of

$$\frac{\sqrt{2}\tan(60^\circ - \theta) \tan(30^\circ + \theta)}{\sin^2(45^\circ + \theta) + \sin^2(45^\circ - \theta)} \text{ is:}$$

SSC CGL 17/08/2021 (Evening)

(a) $\frac{1}{\sqrt{2}}$ (b) 1 (c) 2 (d) $\sqrt{2}$

Q.450. In ΔPQR , $\angle Q = 90^\circ$. If $\tan R = \frac{1}{3}$,

then what is the value of

$$\frac{\sec P(\cos R + \sin P)}{\operatorname{cosec} R(\sin R - \operatorname{cosec} P)}?$$

SSC CGL 18/08/2021 (Afternoon)

(a) $-\frac{2}{7}$ (b) $\frac{18}{7}$ (c) $\frac{2}{7}$ (d) $-\frac{18}{7}$

Q.451. Find the value of

$$(\sec\theta + \tan\theta)^2 + \frac{1 + \operatorname{cosec}\theta}{1 - \operatorname{cosec}\theta}$$

where $0^\circ < \theta < 90^\circ$

SSC CGL 18/08/2021 (Evening)

(a) 0 (b) -2 (c) 1 (d) 2

Q.452. If $\sin^6\theta + \cos^6\theta = \frac{1}{3}$, $0^\circ < \theta$

$< 90^\circ$, then what is the value of $\sin\theta \cdot \cos\theta$

SSC CGL 18/08/2021 (Evening)

(a) $\frac{\sqrt{6}}{6}$ (b) $\frac{\sqrt{2}}{3}$ (c) $\frac{\sqrt{2}}{\sqrt{3}}$ (d) $\frac{2}{9}$

Q.453. If $2\cos^2\theta - 5\cos\theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then the value of $(\sec\theta + \tan\theta)$ is:

SSC CGL 20/08/2021 (Afternoon)

(a) $1 - \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) $1 + \sqrt{3}$ (d) $2 + \sqrt{3}$

Q.454. The value of

$$\frac{\tan 13^\circ \tan 36^\circ \tan 45^\circ \tan 54^\circ \tan 77^\circ}{2 \sec^2 60^\circ (\sin^2 60^\circ - 3 \cos 60^\circ + 2)} \text{ is:}$$

SSC CGL 20/08/2021 (Evening)

(a) $-\frac{1}{4}$ (b) $-\frac{1}{10}$ (c) $\frac{1}{10}$ (d) $\frac{1}{4}$

Q.455. The value of

$$\sec^4\theta(1 - \sin^4\theta) - 2\tan^2\theta \text{ is:}$$

SSC CGL 20/08/2021 (Evening)

(a) 1 (b) 0 (c) -1 (d) $-\frac{1}{2}$

Q.456. If $\sin^2\theta - \cos^2\theta - 3\sin\theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of

$$\frac{1}{\sqrt{\sec\theta - \tan\theta}} \text{ is:}$$

SSC CGL 23/08/2021 (Morning)

(a) $\sqrt[4]{3}$ (b) $\sqrt[2]{2}$ (c) $\sqrt[3]{3}$ (d) $\sqrt[4]{2}$

Q.457. If $0^\circ < \theta < 90^\circ$,

$$\sqrt{\frac{\sec^2\theta + \operatorname{cosec}^2\theta}{\tan^2\theta - \sin^2\theta}} \text{ is equal to:}$$

SSC CGL 23/08/2021 (Afternoon)

(a) $\sec^3\theta$ (b) $\sin^2\theta$ (c) $\operatorname{cosec}^3\theta$ (d) $\sec^2\theta$

Q.458. If $\frac{1}{1 - \sin\theta} + \frac{1}{1 + \sin\theta} = 4 \sec\theta$,

$0^\circ < \theta < 90^\circ$, then the value of $\cot\theta + \operatorname{cosec}\theta$ is:

SSC CGL 23/08/2021 (Evening)

(a) $\frac{5\sqrt{3}}{3}$ (b) $3\sqrt{3}$ (c) $\sqrt{3}$ (d) $\frac{4\sqrt{3}}{3}$

Q.459. If $\sin^2\theta = 2\sin\theta - 1$, $0^\circ \leq \theta \leq 90^\circ$,

then find the value of $\frac{1 + \operatorname{cosec}\theta}{1 - \cos\theta}$.

SSC CGL 24/08/2021 (Afternoon)

(a) -1 (b) 1 (c) 2 (d) -2

Q.460. If $\sec 31^\circ = x$, then $\sin^2 59^\circ +$

$$\frac{1}{\sec^2 31^\circ} - \frac{1}{\sin^2 59^\circ \operatorname{cosec}^2 59^\circ} \text{ is equal to:}$$

SSC CGL 24/08/2021 (Evening)

(a) $\frac{x^2 - 2}{x}$ (b) $\frac{2 - x^2}{x^2}$
(c) $\frac{x^2 - 2}{x^2}$ (d) $\frac{2 - x^2}{x}$

SSC CHSL 2020 Tier - 1

Q.461. If $2\tan x + 3\cot x = 5$, then the

value of $4\tan^2 x + 9\cot^2 x$ is:

SSC CHSL 12/04/2021 (Afternoon)

(a) 13 (b) 11 (c) 17 (d) 15

Q.462. If $2\cos\theta + \sec\theta - 2\sqrt{2} = 0$, where θ is an acute angle. Find the value of θ .

SSC CHSL 12/04/2021 (Evening)

(a) 60° (b) 30° (c) 15° (d) 45°

Q.463. If $\cot\theta = \sqrt{2} + 1$, then $\operatorname{cosec}\theta \sec\theta = ?$

SSC CHSL 13/04/2021 (Morning)

(a) $\frac{\sqrt{2}}{2}$ (b) $4\sqrt{2}$ (c) $\frac{\sqrt{2}}{4}$ (d) $2\sqrt{2}$

Q.464. If $0^\circ < \theta < 90^\circ$, then $\frac{(1 - \sin\theta)(\sec\theta + \tan\theta)\tan\theta}{(\sec\theta + \tan\theta + 1)(\cot\theta - \operatorname{cosec}\theta + 1)} = ?$

SSC CHSL 13/04/2021 (Afternoon)

(a) $\sin\theta$ (b) $\frac{\sin\theta}{2}$ (c) $\cos\theta$ (d) $\frac{\cos\theta}{2}$

Q.465. If $\frac{3\sqrt{3}\sec\theta + 4\tan\theta}{3\tan\theta + \sqrt{3}\sec\theta} = 2$, $0^\circ < \theta < 90^\circ$ then the value of $\cos\theta$ will be:

SSC CHSL 13/04/2021 (Evening)

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

Q.466. What number should be subtracted from $4(\sin^4 60^\circ + \cos^4 30^\circ) - (\tan^2 45^\circ - \cot^2 30^\circ) + \cos^2 45^\circ - \operatorname{cosec}^2 45^\circ + \sec^2 60^\circ$ to get 2?

SSC CHSL 13/04/2021 (Evening)

(a) 5 (b) 4 (c) 7 (d) 3

Q.467. If $Y = \tan 35^\circ$, then the value of $(2\tan 55^\circ + \cot 55^\circ)$ is:

SSC CHSL 15/04/2021 (Evening)

(a) $\frac{2+Y^2}{Y}$ (b) $\frac{2-Y}{Y^2}$ (c) $\frac{2}{Y^2}$ (d) $\frac{2-Y^2}{Y}$

Q.468. If $\tan\theta + \cot\theta = 4$, then the ratio of $3(\tan^2\theta + \cot^2\theta)$ to $(2\operatorname{cosec}^2\theta \sec^2\theta - 4)$ will be:

SSC CHSL 15/04/2021 (Evening)

(a) 4:3 (b) 3:4 (c) 5:4 (d) 3:2

Q.469. If $3(\sec^2\theta + \tan^2\theta) = 5$, $0^\circ < \theta < 90^\circ$, then the value of $\operatorname{cosec}\theta$ is:

SSC CHSL 16/04/2021 (Afternoon)

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{2\sqrt{3}}{3}$ (c) 2 (d) $\frac{1}{2}$

Q.470. If $(\cos^2\theta - 1)(2\sec^2\theta) + \sec^2\theta + 2\tan^2\theta = 2$, $0^\circ < \theta < 90^\circ$, then the value of $\frac{(\sec\theta + \sin\theta)}{(\operatorname{cosec}\theta - \cos\theta)}$ will be:

SSC CHSL 16/04/2021 (Afternoon)

(a) -1 (b) -3 (c) 3 (d) 2

Q.471. $\frac{(1 + \cos\theta)(\operatorname{cosec}\theta - \cot\theta)\sec\theta}{\sin\theta(1 - \sin\theta)(\sec\theta + \tan\theta)} =$

SSC CHSL 19/04/2021 (Morning)

(a) $\sec^2\theta$ (b) $\sin^2\theta$ (c) $\cos^2\theta$ (d) $\operatorname{Cosec}^2\theta$

Q.472. If $\sin\theta(2\sin\theta + 3) = 2$, $0^\circ < \theta < 90^\circ$, then what is the value of $(\sec^2\theta + \cot^2\theta - \cos^2\theta)$?

SSC CHSL 19/04/2021 (Morning)

(a) $\frac{13}{3}$ (b) $\frac{31}{12}$ (c) $\frac{7}{2}$ (d) $\frac{43}{12}$

Q.473. If $\sin\theta = \frac{2\sqrt{ab}}{a+b}$, $a > b > 0$, then the value of $\frac{\cos\theta + 1}{\cos\theta - 1}$ will be:

SSC CHSL 19/04/2021 (Afternoon)

(a) $-\frac{b}{a}$ (b) $-\frac{a}{b}$ (c) $\frac{a}{b}$ (d) $\frac{b}{a}$

Q.474. The value of

$$\frac{3\cos^2 27^\circ - 5 + 3\cos^2 63^\circ}{\tan^2 32^\circ + 4 - \operatorname{cosec}^2 58^\circ} + \sin 35^\circ \cos 55^\circ$$

+ $\cos 35^\circ \sin 55^\circ$ is:

SSC CHSL 19/04/2021 (Afternoon)

(a) $-\frac{1}{4}$ (b) $-\frac{1}{3}$ (c) $\frac{1}{3}$ (d) $1\frac{2}{3}$

Q.475. $2\cos^2\theta = 3(1 - \sin\theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $(\tan 2\theta + \operatorname{cosec} 3\theta - \sec 2\theta)$?

SSC CHSL 04/08/2021 (Evening)

(a) $\sqrt{3} + 1$ (b) $\sqrt{3} - 1$
(c) $\frac{1 + \sqrt{3}}{\sqrt{3}}$ (d) $\frac{1 - \sqrt{3}}{\sqrt{3}}$

Q.476. If $\cos\theta = \frac{\sqrt{3}}{2}$, then the value of

$$\frac{2 - \sin^2\theta}{1 - \cot^2\theta} + (\sec^2\theta + \operatorname{cosec}\theta)$$
 is:

SSC CHSL 05/08/2021 (Morning)

(a) $\frac{25}{12}$ (b) $\frac{59}{24}$ (c) $-\frac{25}{12}$ (d) $-\frac{59}{24}$

Q.477. Find the value of θ , if $\sec^2\theta + (1 - \sqrt{3})\tan\theta - (1 + \sqrt{3}) = 0$, where θ is an acute angle.

SSC CHSL 05/08/2021 (Evening)

(a) 60° (b) 30° (c) 45° (d) 15°

Q.478. If $\operatorname{cosec}^2\theta(\cos\theta - 1)(1 + \cos\theta) = k$, then what is the value of k ?

SSC CHSL 06/08/2021 (Morning)

(a) 1 (b) -1 (c) $\frac{1}{2}$ (d) 0

Q.479. If $\frac{\cot\theta + \cos\theta}{\cot\theta - \cos\theta} = \frac{k+1}{1-k}$, $k \neq 1$, then k is equal to:

SSC CHSL 09/08/2021 (Morning)

(a) $\sin\theta$ (b) $\operatorname{cosec}\theta$ (c) $\cos\theta$ (d) $\sec\theta$

Q.480. If $\cos^2\theta - \sin^2\theta - 3\cos\theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then what will be the value of $\sec\theta - \cos\theta$?

SSC CHSL 09/08/2021 (Morning)

(a) $\frac{4}{3}$ (b) $\frac{1}{2}$ (c) $\frac{3}{2}$ (d) $\frac{2}{3}$

Q.481. If $\frac{\operatorname{cosec}^2\theta}{\operatorname{cosec}^2\theta - \cot^2\theta} = \frac{13}{4}$,

$0^\circ < \theta < 90^\circ$, then the value of

$$\frac{52\cos^2\theta - 9\tan^2\theta}{18\sec^2\theta + 8\cot^2\theta}$$
 will be:

SSC CHSL 09/08/2021 (Afternoon)

(a) $\frac{4}{11}$ (b) $\frac{8}{11}$ (c) $\frac{8}{13}$ (d) $\frac{5}{11}$

Q.482. Evaluate the following expression

$$\frac{3(\cot^2 46^\circ - \sec^2 44^\circ)}{2(\sin^2 28^\circ + \sin^2 62^\circ)} + \frac{2\cos^2 60^\circ \tan^2 33^\circ \tan^2 57^\circ}{\sec^2(90^\circ - \theta) - \cot^2\theta}$$

SSC CHSL 10/08/2021 (Morning)

(a) -1 (b) 1 (c) -2 (d) 2

Q.483. For $\theta: 0^\circ < \theta < 90^\circ$, $3\sec\theta + 4\cos\theta = 4\sqrt{3}$, find the value of $(1 - \sin\theta + \cos\theta)$

SSC CHSL 10/08/2021 (Morning)

(a) $\frac{1 + 2\sqrt{3}}{2}$ (b) $\frac{1 + \sqrt{3}}{2}$
(c) $\frac{1 - \sqrt{3}}{2}$ (d) $\frac{1 - 2\sqrt{3}}{2}$

Q.484. If $\frac{5\cot\theta + \sqrt{3}\operatorname{cosec}\theta}{2\sqrt{3}\operatorname{cosec}\theta + 3\cot\theta} = 1$,

$0^\circ < \theta < 90^\circ$, then the value of

$$\frac{\frac{7}{2}\cot^2\theta - \frac{3}{4}\operatorname{cosec}^2\theta}{4\sin^2\theta + \frac{3}{2}\tan^2\theta}$$
 will be:

SSC CHSL 10/08/2021 (Afternoon)

(a) 7 (b) 2 (c) 3 (d) 5

Q.485. If $(1 + \cot^2\theta) + (1 + (\cot^2\theta)^{-1})$ is equal to k , then $\sqrt{k} = ?$

SSC CHSL 10/08/2021 (Evening)

(a) $\sin\theta \sec\theta$ (b) $\operatorname{cosec}\theta \cos\theta$
(c) $\operatorname{cosec}\theta \sec\theta$ (d) $\sin\theta \cos\theta$

Q.486. The value of $\frac{\sec\theta \operatorname{cosec}\theta}{2 + \tan^2\theta + \cot^2\theta}$ is

equal to:

SSC CHSL 11/08/2021 (Morning)

(a) $\sec\theta \operatorname{cosec}\theta$ (b) $\sec\theta \sin\theta$
(c) $\sin\theta \cos\theta$ (d) $\cos\theta \operatorname{cosec}\theta$

Q.487. If $\cot A + \operatorname{cosec} A = 2$ and A is an acute angle, then the value of $\frac{9\tan A + 16\operatorname{cosec} A}{5\sin A + 3\tan A}$ is:

SSC CHSL 11/08/2021 (Afternoon)

(a) 3 (b) 6 (c) 4 (d) 8

Q.488. If $\cot^2\theta + \cot^4\theta = 2$, then the value of $2\sin^4\theta + \sin^2\theta$ is:

SSC CHSL 11/08/2021 (Evening)

(a) 3 (b) 5 (c) 1 (d) 2

Q.489. If $\sin A + \operatorname{cosec} A = \tan \frac{\pi}{3}$, then

the value of $(\sin^3 \alpha + \operatorname{cosec}^3 \alpha)$ is equal to:

SSC CHSL 11/08/2021 (Evening)

- (a) $\frac{1}{2}$ (b) 1 (c) 0 (d) $\frac{3}{2}$

Q.490. The value of $\operatorname{cosec}(58^\circ + \theta) - \sec(32^\circ - \theta) + \sin 15^\circ \cdot \sin 35^\circ \cdot \sec 55^\circ \cdot \sin 30^\circ \cdot \sec 75^\circ$ is:

SSC CHSL 12/08/2021 (Morning)

- (a) 1 (b) 2 (c) 0.5 (d) 0

SSC CPO 2020 Tier - 1

Q.491. value of $\frac{\sin^2 52^\circ + 2 + \sin^2 38^\circ}{4\cos^2 43^\circ - 5 + 4\cos^2 47^\circ}$ is

SSC CPO 24/11/2020 (Evening)

- (a) 3 (b) $\frac{1}{3}$ (c) -3 (d) $-\frac{1}{3}$

Q.492. If $4(\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ) - \cos 90^\circ + y \times \tan^2 66^\circ \times \tan^2 24^\circ = \frac{y}{2}$

then the value of y is:

SSC CPO 25/11/2020 (Morning)

- (a) 4 (b) 8 (c) -4 (d) -8

SSC CGL 2019 Tier - 2

Q.493

$\frac{\sin \theta [(1 - \tan \theta) \tan \theta + \sec^2 \theta]}{(1 - \sin \theta) \tan \theta (1 + \tan \theta) (\sec \theta + \tan \theta)}$ is equal to

SSC CGL Tier II (15/11/2020)

- (a) 1 (b) $\operatorname{cosec} \theta \sec \theta$ (c) $\sin \theta \cos \theta$ (d) -1

Q.494. $\left(\frac{1}{\cos \theta} - \frac{1}{\sin \theta}\right) + \frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sec \theta + \tan \theta} = ?$

SSC CGL Tier II (15/11/2020)

- (a) $\sec \theta \operatorname{cosec} \theta$ (b) $\sin \theta \tan \theta$
(c) $\operatorname{cosec} \theta \cot \theta$ (d) $\sin \theta \cos \theta$

Q.495. If $\sec \theta = \frac{a}{b}$, $b \neq 0$, then

$$\frac{1 - \tan^2 \theta}{2 - \sin^2 \theta} = ?$$

SSC CGL Tier II (15/11/2020)

- (a) $\frac{a^2(2b^2 + a^2)}{b^2(a^2 - b^2)}$ (b) $\frac{a^2(2b^2 + a^2)}{b^2(a^2 + b^2)}$
(c) $\frac{a^2(2b^2 - a^2)}{b^2(a^2 + b^2)}$ (d) $\frac{a^2(2b^2 - a^2)}{a^2(a^2 + b^2)}$

Q.496. The value of

$$\frac{\sec^2 \theta (2 + \tan^2 \theta + \cot^2 \theta) \div (\sin^2 \theta - \tan^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)(1 + \cot^2 \theta)^2}$$

SSC CGL Tier II (15/11/2020)

- (a) -1 (b) 1 (c) -2 (d) 2

Q.497. The value of

$$\frac{2 \sin^2 38^\circ \sec^2 52^\circ + \cos 64^\circ \sin 26^\circ + \sin^2 64^\circ}{\tan^2 23^\circ + \cot^2 23^\circ - \sec^2 67^\circ - \operatorname{cosec}^2 67^\circ}$$

SSC CGL Tier II (15/11/2020)

- (a) -2 (b) $\frac{3}{2}$ (c) 2 (d) $-\frac{3}{2}$

Q.498. The value of

$$\frac{\cos^6 \theta + \sin^6 \theta + 3 \sin^2 \theta \cos^2 \theta}{\operatorname{cosec} \theta \sec \theta (\sin \theta + \cos \theta - 1)(\sin \theta + \cos \theta + 1)}$$

SSC CGL Tier II (15/11/2020)

- (a) 1 (b) 2 (c) $\frac{1}{2}$ (d) 3

Q.499. $\frac{\sin^2 \theta}{\cos \theta (1 + \cos \theta)} + \frac{1 + \cos \theta}{\cos \theta} = ?$

SSC CGL Tier II (16/11/2020)

- (a) $2 \cos \theta$ (b) $\sec \theta$ (c) $2 \sec \theta$ (d) $\operatorname{cosec} \theta$

Q.500. The value of

$$\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$$

SSC CGL Tier II (16/11/2020)

- (a) 1 (b) -1 (c) -2 (d) 2

Q.501. $\frac{\sec A (\sec A + \tan A)(1 - \sin A)}{(\operatorname{cosec}^2 A - 1) \sin^2 A}$ is

equal to.

SSC CGL Tier II (16/11/2020)

- (a) $\sec^2 A$ (b) $\cos^2 A$ (c) $\cot A$ (d) $\cos A$

Q.502. If $(\sin \theta + \operatorname{cosec} \theta)^2 +$

$(\cos \theta + \sec \theta)^2 = k + \tan^2 \theta + \cot^2 \theta$, then the value of k is equal to:

SSC CGL Tier II (18/11/2020)

- (a) 7 (b) 2 (c) 9 (d) 5

Q.503. If $3 \sin x + 4 \cos x = 2$, then the value of $3 \cos x - 4 \sin x$ is equal to:

SSC CGL Tier II (18/11/2020)

- (a) $\sqrt{21}$ (b) $\sqrt{23}$ (c) 21 (d) $\sqrt{29}$

SSC CHSL 2019 Tier - 1

Q.504. The least value of $8 \operatorname{cosec}^2 \theta + 25 \sin^2 \theta$ is:

SSC CHSL 12/10/2020 (Morning)

- (a) $20\sqrt{2}$ (b) $10\sqrt{2}$ (c) $40\sqrt{2}$ (d) $30\sqrt{2}$

Q.505. If $A = 2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta)$ then the value of 3α such that $\cos$

$$\alpha = \sqrt{\frac{3+A}{5+A}}$$
 is:

SSC CHSL 13/10/2020 (Morning)

- (a) 45° (b) 135° (c) 180° (d) 90°

Q.506. If $117 \cos^2 A + 129 \sin^2 A = 120$

and $170 \cos^2 B + 158 \sin^2 B = 161$, then

the value of $\operatorname{Cosec}^2 A \sec^2 B$ is:

SSC CHSL 13/10/2020 (Morning)

- (a) 1 (b) 9 (c) 4 (d) 16

Q.507. $\frac{\cot x}{1 + \operatorname{cosec} x} + \frac{1 + \operatorname{cosec} x}{\cot x}$ is equal

to:

SSC CHSL 13/10/2020 (Evening)

(a) $2 \sec x$ (b) $2 \cos x$ (c) $2 \operatorname{cosec} x$ (d) $2 \sin x$

Q.508. Which of the following values suits for A to make the equation

$$\frac{A \tan 62^\circ \sec 28^\circ \cot 38^\circ}{\operatorname{Cosec} 62^\circ \tan 11^\circ} = 1 \text{ true?}$$

SSC CHSL 14/10/2020 (Afternoon)

- (a) $\frac{\tan 38^\circ}{\tan 79^\circ \tan 28^\circ}$ (b) $\frac{\tan 28^\circ \tan 79^\circ}{\tan 38^\circ}$
(c) $\frac{\tan 28^\circ \tan 38^\circ}{\tan 79^\circ}$ (d) $\frac{\tan 38^\circ \tan 79^\circ}{\tan 28^\circ}$

Q.509. $\frac{\cos x}{1 + \sin x} + \frac{1 + \sin x}{\cos x}$ is equal to:

SSC CHSL 15/10/2020 (Morning)

- (a) $2 \sin x$ (b) $2 \cos x$ (c) $2 \operatorname{cosec} x$ (d) $2 \sec x$

Q.510. Find the smallest positive angle which satisfies the given trigonometric

equation:- $2 \sin^2 x + \sqrt{3} \cos x + 1 = 0$

SSC CHSL 19/10/2020 (Morning)

- (a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{5\pi}{6}$ (d) $\frac{\pi}{6}$

Q.511. The value of $\tan(63^\circ - \theta) - \cot(27^\circ$

$+ \theta) + \frac{\operatorname{cosec}^2 70^\circ - \tan^2 20^\circ}{\sec^2 37^\circ - \cot^2 53^\circ}$ is:

SSC CHSL 19/10/2020 (Morning)

- (a) 2 (b) 3 (c) 0 (d) 1

Q.512. If $x = \operatorname{cosec} A + \cos A$ and $y = \operatorname{cosec} A - \cos A$, then find the value of

$$\left(\frac{2}{x+y}\right)^2 + \left(\frac{x-y}{2}\right)^2 - 1$$

SSC CHSL 19/10/2020 (Morning)

- (a) 3 (b) 0 (c) 2 (d) 1

Q.513. If $0 < \theta < 90^\circ$, $3b \operatorname{cosec} \theta = a \sec \theta$ and $3a \sec \theta - b \operatorname{cosec} \theta = 8$ then the value of $9b^2 + a^2$ is:

SSC CHSL 19/10/2020 (Afternoon)

- (a) 6 (b) 8 (c) 9 (d) 7

Q.514. If $\sqrt{3} \cos \theta = \sin \theta$, then the value

of $\frac{4 \sin^2 \theta - 5 \cos \theta}{3 \cos \theta + 1}$ is:

SSC CHSL 20/10/2020 (Morning)

- (a) $\frac{1}{4}$ (b) $\frac{1}{5}$ (c) 5 (d) $\frac{2}{5}$

Q.515. If θ is a positive acute angle and $\tan 2\theta \cdot \tan 3\theta = 1$, then the value of θ is:

SSC CHSL 20/10/2020 (Afternoon)

- (a) 45° (b) 18° (c) 60° (d) 36°

Q.516. If $\sin 7x = \cos 11x$, $0^\circ < x < 90^\circ$, then the value of $\tan 9x$ is:

SSC CHSL 21/10/2020 (Morning)

- (a) $-\frac{\sqrt{3}}{2}$ (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

Q.517. If $\sin(\theta + 30^\circ) = \frac{3}{\sqrt{12}}$, then the

value of θ is equal to:

SSC CHSL 21/10/2020 (Afternoon)

- (a) 60° (b) 15° (c) 45° (d) 30°

Q.518. $(\sin\theta + \cos\theta)^2 = 2$, $0^\circ < \theta < 90^\circ$, then the value of θ is:

SSC CHSL 17/03/2020 (Afternoon)

- (a) 0 (b) $\frac{\pi}{2}$ (c) π (d) $\frac{\pi}{4}$

Q.519. If $a \sin A + b \cos A = c$, then $a \cos A - b \sin A$ is equal to:

SSC CHSL 18/03/2020 (Evening)

- (a) $\sqrt{a^2 - b^2 - c^2}$ (b) $\sqrt{a^2 + b^2 - c^2}$
(c) $\sqrt{a^2 + b^2 + c^2}$ (d) $\sqrt{a^2 - b^2 + c^2}$

Q.520. $\frac{1 - \tan A}{1 + \tan A} =$

$$\frac{\tan 3^\circ \tan 15^\circ \tan 30^\circ \tan 75^\circ \tan 87^\circ}{\tan 27^\circ \tan 39^\circ \tan 51^\circ \tan 60^\circ \tan 63^\circ},$$

then the value of $\cot A$ is:

SSC CHSL 18/03/2020 (Evening)

- (a) 3 (b) 2 (c) 1 (d) 4

Q.521. $\frac{\sqrt{\csc x - 1}}{\sqrt{\csc x + 1}}$ is equal to:

SSC CHSL 19/03/2020 (Evening)

- (a) $\sec x - \tan x$ (b) $\tan x - \sec x$
(c) $\sec x \cdot \tan x$ (d) $\tan x + \sec x$

Q.522. The elimination of θ from $x \cos\theta - y \sin\theta = 2$ and $x \sin\theta + y \cos\theta = 4$ will give:

SSC CHSL 21/10/2020 (Evening)

- (a) $3x^2 - y^2 = 20$ (b) $x^2 + y^2 = 20$
(c) $3x^2 + y^2 = 20$ (d) $x^2 - y^2 = 20$

SSC CGL 2019 Tier - 1

Q.523. If $A + B = 45^\circ$, then the value of $2(1 + \tan A)(1 + \tan B)$ is:

SSC CGL 03/03/2020 (Morning)

- (a) 2 (b) 4 (c) 1 (d) 0

Q.524. The value of the expression $\operatorname{cosec}(85^\circ + \theta) - \sec(5^\circ - \theta) - \tan(55^\circ + \theta) + \cot(35^\circ - \theta)$ is:

SSC CGL 03/03/2020 (Afternoon)

- (a) $\frac{3}{2}$ (b) 0 (c) -1 (d) 1

Q.525. The expression $3\sec^2\theta \tan^2\theta + \tan^6\theta - \sec^6\theta$ is equal to:

SSC CGL 04/03/2020 (Afternoon)

- (a) -2 (b) 1 (c) 2 (d) -1

Q.526. The value of $\frac{\tan^2\theta - \sin^2\theta}{2 + \tan^2\theta + \cot^2\theta}$ is:

SSC CGL 04/03/2020 (Afternoon)

- (a) $\operatorname{cosec}^6\theta$ (b) $\cos^4\theta$ (c) $\sin^6\theta$ (d) $\sec^4\theta$

Q.527. If $x \cos A - y \sin A = 1$ and $x \sin A + y \cos A = 4$, then the value of $17x^2 + 17y^2$ is:

SSC CGL 06/03/2020 (Morning)

- (a) 0 (b) 49 (c) 7 (d) 289

Q.528. If $(\cos^2\theta - 1)(1 + \tan^2\theta) + 2 \tan^2\theta$

$= 1$, $0^\circ < \theta < 90^\circ$, then θ is

SSC CGL 06/03/2020 (Afternoon)

- (a) 45° (b) 60° (c) 30° (d) 90°

Q.529. If A lies in the third quadrant, and $20 \tan A = 21$, then the value of

$$\frac{5 \sin A - 2 \cos A}{4 \cos A - \frac{5}{7} \sin A}$$
 is:

SSC CGL 06/03/2020 (Evening)

- (a) $\frac{13}{12}$ (b) 1 (c) $-\frac{65}{29}$ (d) $\frac{5}{29}$

Q.530. The value of

$$4 \left[\frac{(1 - \sec A)^2 + (1 + \sec A)^2}{1 + \sec^2 A} \right]$$
 is:

SSC CGL 06/03/2020 (Evening)

- (a) 2 (b) 4 (c) 8 (d) 1

Q.531. If $\frac{\sin A + \cos A}{\cos A} = \frac{17}{12}$, then the

value of $\frac{1 - \cos A}{\sin A}$ is:

SSC CGL 07/03/2020 (Afternoon)

- (a) -5 (b) 1 (c) $\frac{5}{12}$ (d) $\frac{1}{5}$

Q.532. If $6 \tan\theta - 5\sqrt{3}\sec\theta + 12 \cot\theta = 0$, $0^\circ < \theta < 90^\circ$, then the value of $(\operatorname{cosec}\theta + \sec\theta)$ is:

SSC CGL 09/03/2020 (Morning)

- (a) $\frac{3 + 2\sqrt{3}}{2}$ (b) $\frac{2}{3}(3 + \sqrt{3})$
(c) $\frac{3 + \sqrt{3}}{2}$ (d) $\frac{2(3 + 2\sqrt{3})}{3}$

Q.533. The value of $(\operatorname{cosec} A + \cot A + 1)(\operatorname{cosec} A - \cot A + 1) - 2 \operatorname{cosec} A$ is:

SSC CGL 09/03/2020 (Afternoon)

- (a) $4 \operatorname{cosec} A$ (b) 2 (c) $2 \operatorname{cosec} A$ (d) 0

SSC CPO 2019 Tier - 1

Q.534. If $\frac{1}{\operatorname{Cosec}\theta + 1} + \frac{1}{\operatorname{Cosec}\theta - 1} =$

$2 \sec\theta$, $0^\circ < \theta < 90^\circ$, then value of

$$\frac{\tan\theta + 2 \sec\theta}{\operatorname{cosec}\theta}$$
 is:

SSC CPO 09/12/2019 (Evening)

- (a) $\frac{2 + \sqrt{3}}{2}$ (b) $\frac{4 + \sqrt{2}}{2}$
(c) $\frac{2 + \sqrt{2}}{2}$ (d) $\frac{4 + \sqrt{3}}{2}$

Q.535. If $(\cos\theta + \sin\theta) :$

$(\cos\theta - \sin\theta) = (\sqrt{3} + 1) : (\sqrt{3} - 1)$, $0^\circ < \theta < 90^\circ$, then what is the value of $\sec\theta$?

SSC CPO 09/12/2019 (Evening)

- (a) 1 (b) 2 (c) $\sqrt{2}$ (d) $\frac{2\sqrt{3}}{3}$

SSC CGL 2018 Tier - 2

Q.536. $\frac{(2 \sin A)(1 + \sin A)}{1 + \sin A + \cos A}$ is equal to:

SSC CGL Tier II (11/09/2019)

- (a) $1 + \sin A - \cos A$ (b) $1 - \sin A \cos A$
(c) $1 + \cos A - \sin A$ (d) $1 + \sin A \cos A$

Q.537. $\frac{(1 + \cos\theta)^2 + \sin^2\theta}{(\operatorname{cosec}^2\theta - 1)\sin^2\theta} = ?$

SSC CGL Tier II (11/09/2019)

- (a) $\cos\theta(1 + \sin\theta)$ (b) $2 \cos\theta(1 + \sec\theta)$
(c) $\sec\theta(1 + \sin\theta)$ (d) $2 \sec\theta(1 + \sec\theta)$

Q.538. $\sqrt{\frac{\cot\theta + \cos\theta}{\cot\theta - \cos\theta}}$ is equal to:

SSC CGL Tier II (11/09/2019)

- (a) $\sec\theta + \tan\theta$ (b) $1 + \sec\theta \cdot \tan\theta$
(c) $1 - \sec\theta \cdot \tan\theta$ (d) $\sec\theta - \tan\theta$

Q.539. If $\frac{1 + \sin\theta}{1 - \sin\theta} = \frac{p^2}{q^2}$, then $\sec\theta$ is

equal to:

SSC CGL Tier II (12/09/2019)

- (a) $\frac{2p^2 q^2}{p^2 + q^2}$ (b) $\frac{1}{2} \left(\frac{q}{p} + \frac{p}{q} \right)$
(c) $\frac{1}{p^2} + \frac{1}{q^2}$ (d) $\frac{p^2 q^2}{p^2 + q^2}$

Q. 540. The value of

$$\frac{\sec\theta(1 - \sin\theta)(\sin\theta + \cos\theta)(\sec\theta + \tan\theta)}{\sin\theta(1 + \tan\theta) + \cos\theta(1 + \cot\theta)}$$

is equal to:

SSC CGL Tier II (12/09/2019)

- (a) $2 \cos\theta$ (b) $\operatorname{cosec}\theta \sec\theta$
(c) $2 \sin\theta$ (d) $\sin\theta \cos\theta$

Q.541. $(\sec\theta - \tan\theta)^2(1 + \sin\theta)^2 \div \sin^2\theta = ?$

SSC CGL Tier II (12/09/2019)

- (a) $\cos\theta$ (b) $\cot^2\theta$ (c) $\sec\theta$ (d) $\cos^2\theta$

Q.542. The value of $\sqrt{\frac{\operatorname{cosec}\theta - \cot\theta}{\operatorname{cosec}\theta + \cot\theta}}$

$\div \frac{\sin\theta}{1 + \cos\theta}$ is equal to:

SSC CGL Tier II (12/09/2019)

- (a) $\operatorname{cosec}\theta$ (b) $\frac{1}{2}$ (c) $\sec\theta$ (d) 1

Q.543. If $\sec\theta + \tan\theta = p$, ($p > 1$) then

$$\frac{\operatorname{cosec}\theta + 1}{\operatorname{cosec}\theta - 1} = ?$$

SSC CGL Tier II (13/09/2019)

- (a) $\frac{p+1}{p-1}$ (b) p^2 (c) $\frac{p-1}{p+1}$ (d) $2p^2$

Q.544. The value of $\frac{\sec^2\theta}{\operatorname{cosec}^2\theta} + \frac{\operatorname{cosec}^2\theta}{\sec^2\theta} -$

$(\sec^2\theta + \operatorname{cosec}^2\theta)$ is:

SSC CGL Tier II (13/09/2019)

- (a) 0 (b) -2 (c) 2 (d) 1

Q.545. Find the value

$$\frac{(\sin\theta - \cos\theta)(1 + \tan\theta + \cot\theta)}{1 + \sin\theta \cos\theta} = ?$$

SSC CGL Tier II (13/09/2019)

- (a) $\sec\theta - \operatorname{cosec}\theta$ (b) $\operatorname{cosec}\theta - \sec\theta$
 (c) $\sin\theta + \cos\theta$ (d) $\tan\theta - \cot\theta$

Q.546. If $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = \frac{4}{\sqrt{3}}$

$0^\circ < \theta < 90^\circ$, then the value of

$(\tan\theta + \sec\theta)^{-1}$ is:

SSC CGL Tier II (13/09/2019)

- (a) $2 - \sqrt{3}$ (b) $3 - \sqrt{2}$ (c) $2 + \sqrt{3}$ (b) $3 + \sqrt{2}$

SSC CHSL 2018 Tier - 1

Q.547. If $\cot\theta = \frac{1}{\sqrt{3}}$, then the value of

$\frac{2 - \sin^2\theta}{1 - \cos^2\theta} + \operatorname{cosec}^2\theta + \sec\theta$ is:

SSC CHSL 02/07/2019 (Morning)

- (a) 4 (b) 6 (c) 7 (d) 5

Q.548. If $\operatorname{Cosec} 31^\circ = x$, then $\sin^2 59^\circ + \frac{1}{\operatorname{cosec}^2 31^\circ} + \tan^2 59^\circ - \frac{1}{\sin^2 59^\circ \cdot \operatorname{cosec}^2 59^\circ}$

is equal to:

SSC CHSL 02/07/2019 (Afternoon)

- (a) $x + 1$ (b) $x^2 - 1$ (c) $x - 1$ (d) $x^2 + 1$

Q.549. If $\sin\theta \cdot \sec^2\theta = \frac{2}{3}$, $0^\circ < \theta < 90^\circ$,

then the value of $(\tan^2\theta + \cos^2\theta)$ is:

SSC CHSL 03/07/2019 (Morning)

- (a) $\frac{7}{6}$ (b) $\frac{11}{12}$ (c) $\frac{13}{12}$ (d) $\frac{5}{4}$

Q.550. If $\cos x = \frac{-\sqrt{3}}{2}$ and $\pi < x < \frac{3\pi}{2}$,

then the value of $4\cot^2 x - 3\operatorname{cosec}^2 x$ is:

SSC CHSL 09/07/2019 (Afternoon)

- (a) 8 (b) 0 (c) 2 (d) 1

Q.551. If θ is an acute angle, and it is given that $5\sin\theta + 12\cos\theta = 13$, then what is the value of $\tan\theta$?

SSC CHSL 09/07/2019 (Evening)

- (a) $\frac{5}{13}$ (b) $\frac{13}{12}$ (c) $\frac{12}{13}$ (d) $\frac{5}{12}$

Q.552. If $\sin\theta + \operatorname{cosec}\theta = 2$, then what is the value of $(\sin^{153}\theta + \operatorname{cosec}^{253}\theta)$?

SSC CHSL 10/07/2019 (Morning)

- (a) $\frac{1}{153 \times 253}$ (b) $\frac{253}{123}$ (c) 2 (d) $\frac{153}{253}$

SSC CGL 2018 Tier - 1

Q.553. $\frac{2 + \tan^2\theta + \cot^2\theta}{\sec\theta \operatorname{cosec}\theta}$ is equal to:

SSC CGL 04/06/2019 (Morning)

- (a) $\cot\theta$ (b) $\cos\theta \sin\theta$
 (c) $\sec\theta \operatorname{cosec}\theta$ (d) $\tan\theta$

Q.554. If $\cos\theta = \frac{2p}{(1+p^2)}$, then $\tan\theta$ is equal to:

SSC CGL 04/06/2019 (Afternoon)

- (a) $\frac{p^2}{1+p^2}$ (b) $\frac{2p}{1-p^2}$
 (c) $\frac{1-p^2}{1+p^2}$ (d) $\frac{1-p^2}{2p}$

Q.555. The value of θ when $\sqrt{3}\cos\theta + \sin\theta = 1$ ($0^\circ \leq \theta \leq 90^\circ$), is:

SSC CGL 04/06/2019 (Evening)

- (a) 90° (b) 30° (c) 60° (d) 0°

Q.556. If $\sin\theta = 4\cos\theta$, then what is the value of $\sin\theta \cos\theta$?

SSC CGL 06/06/2019 (Afternoon)

- (a) $\frac{2}{9}$ (b) $\frac{3}{10}$ (c) $\frac{4}{17}$ (d) $\frac{3}{4}$

Q.557. The value of $\sin^2 30^\circ \cos^2 45^\circ + 4\tan^2 30^\circ + \frac{1}{2}\sin^2 90^\circ + 2\cos 90^\circ$ is:

SSC 07/06/2019 (Evening)

- (a) $\frac{15}{8}$ (b) $\frac{47}{24}$ (c) $\frac{23}{12}$ (d) 2

Q.558. If $\tan\theta = \frac{2}{3}$, then

$\frac{3\sin\theta - 4\cos\theta}{3\sin\theta + 4\cos\theta}$ is equal to:

SSC CGL 10/06/2019 (Afternoon)

- (a) $-\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $-\frac{2}{3}$ (d) $\frac{1}{3}$

Q.559. The value of $\frac{\sin 44^\circ}{\cos 46^\circ} + \sin^2 60^\circ$

$- \cos^2 45^\circ + \sec 60^\circ$ is equal to:

SSC CGL 12/06/2019 (Morning)

- (a) $\frac{13}{4}$ (b) $\frac{11}{4}$ (c) $\frac{11}{3}$ (d) $\frac{7}{4}$

SSC CPO 2018 Tier - 1

Q.560. If $\sin(A+B) = \frac{\sqrt{3}}{2}$ and $\tan(A-B)$

$= \frac{1}{\sqrt{3}}$, then $(2A+3B)$ is equal to:

SSC CPO 13/03/2019 (Morning)

- (a) 120° (b) 135° (c) 130° (d) 125°

Answer Key :-

1.(a)	2.(a)	3.(a)	4.(a)
5.(b)	6.(a)	7.(a)	8.(c)
9.(d)	10.(a)	11.(a)	12.(b)
13.(a)	14.(b)	15.(a)	16.(d)
17.(c)	18.(b)	19.(a)	20.(a)
21.(a)	22.(a)	23.(d)	24.(a)
25.(d)	26.(c)	27.(d)	28.(c)
29.(d)	30.(d)	31.(a)	32.(c)
33.(d)	34.(a)	35.(a)	36.(b)
37.(d)	38.(c)	39.(c)	40.(d)

41.(c)	42.(d)	43.(d)	44.(c)
45.(c)	46.(b)	47.(d)	48.(b)
49.(c)	50.(b)	51.(b)	52.(b)
53.(a)	54.(a)	55.(b)	56.(a)
57.(a)	58.(b)	59.(b)	60.(b)
61.(c)	62.(b)	63.(c)	64.(b)
65.(b)	66.(c)	67.(c)	68.(c)
69.(d)	70.(a)	71.(c)	72.(d)
73.(c)	74.(a)	75.(d)	76.(b)
77.(d)	78.(c)	79.(c)	80.(d)
81.(d)	82.(c)	83.(d)	84.(d)
85.(d)	86.(c)	87.(d)	88.(b)
89.(c)	90.(b)	91.(b)	92.(b)
93.(a)	94.(b)	95.(d)	96.(b)
97.(d)	98.(d)	99.(a)	100.(a)
101.(d)	102.(b)	103.(a)	104.(a)
105.(c)	106.(d)	107.(d)	108.(a)
109.(b)	110.(c)	111.(b)	112.(c)
113.(b)	114.(a)	115.(d)	116.(a)
117.(d)	118.(b)	119.(a)	120.(b)
121.(a)	122.(a)	123.(b)	124.(d)
125.(c)	126.(b)	127.(d)	128.(c)
129.(a)	130.(a)	131.(b)	132.(a)
133.(b)	134.(a)	135.(c)	136.(d)
137.(b)	138.(d)	139.(d)	140.(d)
141.(a)	142.(c)	143.(c)	144.(d)
145.(d)	146.(b)	147.(b)	148.(c)
149.(c)	150.(b)	151.(a)	152.(d)
153.(b)	154.(b)	155.(d)	156.(d)
157.(d)	158.(c)	159.(b)	160.(d)
161.(a)	162.(c)	163.(d)	164.(b)
165.(a)	166.(d)	167.(a)	168.(d)
169.(c)	170.(c)	171.(a)	172.(b)
173.(c)	174.(d)	175.(b)	176.(b)
177.(d)	178.(d)	179.(c)	180.(a)
181.(d)	182.(a)	183.(c)	184.(d)
185.(b)	186.(d)	187.(a)	188.(d)
189.(c)	190.(a)	191.(a)	192.(c)
193.(a)	194.(d)	195.(c)	196.(d)
197.(a)	198.(b)	199.(c)	200.(a)
201.(c)	202.(b)	203.(c)	204.(c)
205.(a)	206.(c)	207.(b)	208.(a)
209.(a)	210.(c)	211.(c)	212.(c)
213.(d)	214.(a)	215.(a)	216.(b)
217.(c)	218.(c)	219.(a)	220.(c)
221.(c)	222.(c)	223.(d)	224.(c)

225.(c)	226.(c)	227.(d)	228.(c)
229.(b)	230.(d)	231.(a)	232.(c)
233.(a)	234.(c)	235.(d)	236.(c)
237.(a)	238.(d)	239.(d)	240.(c)
241.(c)	242.(d)	243.(d)	244.(b)
245.(b)	246.(a)	247.(c)	248.(b)
249.(b)	250.(d)	251.(a)	252.(d)
253.(d)	254.(d)	255.(d)	256.(d)
257.(c)	258.(b)	259.(a)	260.(c)
261.(a)	262.(d)	263.(d)	264.(a)
265.(d)	266.(d)	267.(d)	268.(b)
269.(c)	270.(c)	271.(a)	272.(b)
273.(d)	274.(c)	275.(d)	276.(c)
277.(b)	278.(b)	279.(c)	280.(a)
281.(b)	282.(d)	283.(c)	284.(c)
285.(a)	286.(c)	287.(b)	288.(a)
289.(c)	290.(a)	291.(b)	292.(a)
293.(d)	294.(c)	295.(d)	296.(d)
297.(c)	298.(b)	299.(c)	300.(d)
301.(b)	302.(c)	303.(c)	304.(b)
305.(c)	306.(a)	307.(c)	308.(c)
309.(a)	310.(d)	311.(b)	312.(a)
313.(b)	314.(a)	315.(d)	316.(c)
317.(b)	318.(a)	319.(d)	320.(a)
321.(d)	322.(b)	323.(a)	324.(d)
325.(b)	326.(c)	327.(a)	328.(d)
329.(d)	330.(a)	331.(d)	332.(b)
333.(b)	334.(a)	335.(c)	336.(d)
337.(d)	338.(d)	339.(a)	340.(b)
341.(c)	342.(c)	343.(c)	344.(a)
345.(c)	346.(b)	347.(b)	348.(c)
349.(c)	350.(b)	351.(c)	352.(a)
353.(d)	354.(d)	355.(b)	356.(b)
357.(a)	358.(c)	359.(b)	360.(d)
361.(d)	362.(a)	363.(c)	364.(a)
365.(b)	366.(b)	367.(a)	368.(d)
369.(a)	370.(c)	371.(a)	372.(a)
373.(c)	374.(d)	375.(c)	376.(a)
377.(c)	378.(d)	379.(a)	380.(c)
381.(c)	382.(b)	383.(b)	384.(b)
385.(a)	386.(b)	387.(d)	388.(a)
389.(d)	390.(a)	391.(b)	392.(a)
393.(c)	394.(b)	395.(c)	396.(b)
397.(a)	398.(d)	399.(a)	400.(b)
401.(d)	402.(b)	403.(c)	404.(d)
405.(b)	406.(c)	407.(a)	408.(d)

409.(b)	410.(b)	411.(a)	412.(a)
413.(a)	414.(c)	415.(d)	416.(c)
417.(a)	418.(c)	419.(b)	420.(c)
421.(d)	422.(d)	423.(d)	424.(a)
425.(b)	426.(b)	427.(a)	428.(d)
429.(d)	430.(d)	431.(c)	432.(c)
433.(d)	434.(b)	435.(d)	436.(d)
437.(b)	438.(b)	439.(b)	440.(a)
441.(c)	442.(a)	443.(d)	444.(d)
445.(b)	446.(c)	447.(d)	448.(b)
449.(d)	450.(d)	451.(a)	452.(b)
453.(d)	454.(c)	455.(a)	456.(a)
457.(c)	458.(c)	459.(c)	460.(b)
461.(a)	462.(d)	463.(d)	464.(b)
465.(d)	466.(c)	467.(a)	468.(d)
469.(c)	470.(c)	471.(a)	472.(d)
473.(b)	474.(c)	475.(b)	476.(b)
477.(a)	478.(b)	479.(a)	480.(c)
481.(b)	482.(a)	483.(b)	484.(d)
485.(c)	486.(c)	487.(c)	488.(c)
489.(c)	490.(c)	491.(c)	492.(d)
493.(a)	494.(a)	495.(c)	496.(a)
497.(d)	498.(c)	499.(c)	500.(a)
501.(a)	502.(a)	503.(a)	504.(a)
505.(b)	506.(d)	507.(a)	508.(c)
509.(d)	510.(c)	511.(d)	512.(b)
513.(c)	514.(b)	515.(b)	516.(b)
517.(d)	518.(d)	519.(b)	520.(b)
521.(a)	522.(b)	523.(b)	524.(b)
525.(d)	526.(c)	527.(d)	528.(a)
529.(b)	530.(c)	531.(d)	532.(b)
533.(b)	534.(b)	535.(d)	536.(a)
537.(d)	538.(a)	539.(b)	540.(d)
541.(b)	542.(d)	543.(b)	544.(b)
545.(a)	546.(a)	547.(d)	548.(b)
549.(c)	550.(b)	551.(d)	552.(c)
553.(c)	554.(d)	555.(a)	556.(c)
557.(b)	558.(a)	559.(a)	560.(b)

Solutions:-

Sol.1.(a) $\cot^4 A + \cot^2 A = 1$
 $\cot^2 A (\cot^2 A + 1) = 1$
 $\cot A = \sin A$ or $\sin^2 A = \cos A$
Now,
 $\cos^4 A - 3\cos^2 A =$
 $\left\{ (1 - \sin^2 A)^2 - 3\sin^4 A \right\}$

$$1 + \sin^4 A - 2\sin A - 3\sin^4 A$$

$$1 - 2(\sin^4 A + \sin^2 A) =$$

$$1 - 2(\cot^4 A + \cot^2 A)$$

$$1 - 2 = -1$$

Sol.2.(a) According to the question,

$$\operatorname{cosec} \theta + \cot^3 \theta \sec \theta$$

$$\frac{1}{\sin \theta} + \frac{\cos^3 \theta}{\sin^3 \theta} \times \frac{1}{\cos \theta}$$

$$\frac{\sin^2 \theta}{\sin^3 \theta} + \frac{\cos^2 \theta}{\sin^3 \theta} = \frac{1}{\sin^3 \theta} = \operatorname{cosec}^3 \theta$$

now,

$$\cot^2 \theta = 1 - e^2 \quad (\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta)$$

$$\operatorname{cosec}^2 \theta = 2 - e^2$$

$$\operatorname{cosec} \theta = (2 - e^2)^{\frac{1}{2}} \quad (\text{on cubing both sides})$$

$$\operatorname{cosec}^3 \theta = (2 - e^2)^{\frac{3}{2}}$$

Sol.3.(a) $\operatorname{cosec} A + \cot A = a\sqrt{b}$

$$\left(\frac{1}{\sin A} + \frac{\cos A}{\sin A} \right)^2 = \left(\frac{1 + \cos A}{\sin A} \right)^2 = a^2 b$$

$$\frac{a^2 b - 1}{a^2 b + 1} = \frac{\left(\frac{1 + \cos A}{\sin A} \right)^2 - 1}{\left(\frac{1 + \cos A}{\sin A} \right)^2 + 1}$$

$$= \frac{1 + \cos^2 A + 2\cos A - \sin^2 A}{1 + \cos^2 A + 2\cos A + \sin^2 A}$$

$$= \frac{2\cos^2 A + 2\cos A}{2 + 2\cos A} = \frac{2\cos A (\cos A + 1)}{2 (\cos A + 1)} = \cos A$$

Sol.4.(a) Put $\theta = 45^\circ$

Now, $\sin \theta + \tan \theta = p$ (1)

$$\Rightarrow \frac{1}{\sqrt{2}} + 1 = p \Rightarrow \frac{1 + \sqrt{2}}{\sqrt{2}} = p$$

And, $\tan \theta - \sin \theta = q$ (2)

$$\Rightarrow 1 - \frac{1}{\sqrt{2}} = q \Rightarrow \frac{\sqrt{2} - 1}{\sqrt{2}} = q$$

Now, checking through options, (a) satisfies the value of p and q.

$$p^2 - q^2 = 4\sqrt{pq} \Rightarrow \left(\frac{1 + \sqrt{2}}{\sqrt{2}} \right)^2 - \left(\frac{\sqrt{2} - 1}{\sqrt{2}} \right)^2$$

$$= 4 \sqrt{\frac{1 + \sqrt{2}}{\sqrt{2}} \times \frac{\sqrt{2} - 1}{\sqrt{2}}}$$

$$= \frac{(1 + \sqrt{2})^2 - (\sqrt{2} - 1)^2}{2} = 4 \times \sqrt{\frac{1}{2}}$$

$$\Rightarrow 2\sqrt{2} = 2\sqrt{2}$$

Sol.5.(b)

$$\sin \theta \cdot \cos \theta = 2\cos^3 \theta - \frac{1}{4} \cos \theta$$

$$4\sin \theta \cdot \cos \theta = 8\cos^3 \theta - \cos \theta$$

$$4\sin \theta = 8\cos^2 \theta - 1$$

$$4\sin \theta = 8 - 8\sin^2 \theta - 1$$

$$8\sin^2 \theta + 4\sin \theta - 7 = 0$$

$$\sin \theta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-4 \pm \sqrt{4^2 - 4 \times 8 \times (-7)}}{2 \times 8}$$

$$= \frac{-4 \pm \sqrt{240}}{2 \times 8} = \frac{-1 \pm \sqrt{15}}{4}$$

Here θ is an acute angle, so

$$\Rightarrow \sin \theta = \frac{\sqrt{15} - 1}{4}$$

Sol.6.(a)

If $A + B = 90^\circ$, then $\cot A \times \cot B = 1$
 $= 7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ)$
 $= 7(\cot 11^\circ \times \cot 79^\circ)(\cot 13^\circ \times \cot 77^\circ)$
 $(\cot 19^\circ \times \cot 71^\circ)(\cot 29^\circ \times \cot 61^\circ)$
 $= 7(1 \times 1 \times 1 \times 1) = 7$

Sol.7.(a) $\sin^2 A + \cos^2 A = 1$

$$\sin^4 A + \cos^4 A + 2\sin^2 A \cos^2 A = 1$$

$$\sin^4 A + \cos^4 A = 1 - 2\sin^2 A \cos^2 A$$

$$\frac{\sin^4 A + \cos^4 A}{1 - 2\sin^2 A \cos^2 A} = 1$$

Sol.8.(c) $pq = \cot^2 A - \cos^2 A = \cot^2 A \times \cos^2 A$... (identity)

$$\sqrt{pq} = \cot A \cos A$$

$$p^2 = \cot^2 A + \cos^2 A + 2 \times \cot A \cos A$$

$$q^2 = \cot^2 A + \cos^2 A - 2 \times \cot A \cos A$$

$$p^2 - q^2 = 4 \times \cot A \cos A$$

Now,

$$p^2 - q^2 = 4 \times \cot A \cos A = \sqrt{16pq}$$

Sol.9.(d) $4(\sin^6 A + \cos^6 A) - 6$

$$(\sin^4 A + \cos^4 A) + 8$$

$$[\because a^3 + b^3 = (a + b)^3 - 3ab(a + b) \text{ and } a^2 + b^2 = (a + b)^2 - 2ab]$$

$$= 4(1 - 3\sin^2 A \cos^2 A) - 6(1 - 2\sin^2 A \cos^2 A) + 8$$

$$= 4 - 12\sin^2 A \cos^2 A - 6 + 12\sin^2 A \cos^2 A$$

$$\cos^2 A + 8 = 6$$

Sol.10.(a) Concept used :

$$(1 + \cos 2\theta) = 2\cos^2 \theta$$

$$\sqrt{2 + \sqrt{2} + \sqrt{2} + 2\cos 8\theta}$$

$$= \sqrt{2 + \sqrt{2} + \sqrt{2}(1 + \cos 8\theta)}$$

$$= \sqrt{2 + \sqrt{2} + \sqrt{2} \times 2\cos^2 4\theta}$$

$$= \sqrt{2 + \sqrt{2} + 2\cos 4\theta}$$

$$= \sqrt{2 + \sqrt{2}(1 + \cos 4\theta)}$$

$$= \sqrt{2 + \sqrt{2} \times 2\cos^2 2\theta} = \sqrt{2 + 2\cos 2\theta}$$

$$= \sqrt{2(1 + \cos 2\theta)} = \sqrt{2 \times 2\cos^2 \theta} = 2\cos \theta$$

Short trick -

$$2\cos \left[\frac{8\theta}{2^3} \right] = 2\cos 2\theta$$

Sol.11.(a) $\sqrt{\frac{1 - \sin 3\theta}{1 + \sin 3\theta}}$

After rationalizing,

$$= \sqrt{\frac{(1 - \sin 3\theta) \times (1 - \sin 3\theta)}{(1 + \sin 3\theta) \times (1 - \sin 3\theta)}}$$

$$= \sqrt{\frac{(1 - \sin 3\theta) \times (1 - \sin 3\theta)}{(1 - \sin^2 3\theta)}}$$

$$= \sqrt{\left(\frac{1 - \sin 3\theta}{\cos 3\theta} \right)^2}$$

$$= \left(\frac{1 - \sin 3\theta}{\cos 3\theta} \right) = \sec 3\theta - \tan 3\theta$$

Sol.12.(b) $\cos^2 \theta + \sin \theta + 1 = 0$

$$1 - \sin^2 \theta + \sin \theta + 1 = 0$$

$$\sin^2 \theta - \sin \theta - 2 = 0$$

$$\sin^2 \theta - 2\sin \theta + \sin \theta - 2 = 0$$

$$\sin \theta (\sin \theta - 2) + 1(\sin \theta - 2) = 0$$

$$(\sin \theta - 2)(\sin \theta + 1) = 0$$

$$\sin \theta = -1 \text{ or } +2$$

Since $\sin \theta$ has a range of -1 to +1.

$$\text{So, } \sin \theta = -1 \text{ or } \sin \theta = \sin \frac{3\pi}{2}$$

$$\text{Comparing both sides } \Rightarrow \theta = \frac{3\pi}{2}$$

Therefore solution of $\cos^2 \theta + \sin \theta + 1 = 0$ lies between $(\frac{5\pi}{4}, \frac{7\pi}{4})$.

Sol.13.(a) $(1 + \cot A + \tan A)$

$$\frac{\sin A \cos A}{(\sin^3 A - \cos^3 A)}$$

$$= (1 + \cot A + \tan A)(\sin A - \cos A)$$

$$\frac{\sin A \cos A}{(\sin A - \cos A)(\sin^2 A + \cos^2 A + \sin A \cos A)}$$

$$= (1 + \cot A + \tan A) \frac{\sin A \cos A}{(1 + \sin A \cos A)}$$

Now, put $A = 45^\circ$

$$(3) \times \frac{\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}}{\left(1 + \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right)} = 3 \times \frac{1}{2} \times \frac{2}{3} = 1$$

Sol.14.(b) $\sin m + \sin n = p$

$$\Rightarrow (\sin m + \sin n)^2 = p^2$$

$$\Rightarrow \sin^2 m + \sin^2 n + 2 \sin m \sin n = p^2$$

$$\text{----- (eq. 1)}$$

And $\cos m + \cos n = q$

$$\Rightarrow (\cos m + \cos n)^2 = q^2$$

$$\Rightarrow \cos^2 m + \cos^2 n + 2 \cos m \cos n = q^2$$

$$\text{----- (eq. 2)}$$

After adding (eq. 1 and 2)

Then,

$$(\sin^2 m + \cos^2 m) + (\sin^2 n + \cos^2 n) +$$

$$2(\sin m \sin n + \cos m \cos n) = p^2 + q^2$$

$$\Rightarrow 2 + 2(\sin m \sin n + \cos m \cos n) = p^2 + q^2$$

$$\Rightarrow (\sin m \sin n + \cos m \cos n)$$

$$= \frac{(p^2 + q^2 - 2)}{2}$$

Sol.15.(a)

$$b \operatorname{cosec} \theta + a \cot \theta = p$$

$$b \cot \theta + a \operatorname{cosec} \theta = q$$

$$b^2 - a^2 = p^2 - q^2$$

Sol.16.(d)

$$\frac{\cos(\pi - A) \cot(\frac{\pi}{2} + A) \cos(-A)}{\tan(\pi + A) \tan(\frac{3\pi}{2} + A) \sin(2\pi - A)}$$

$$= \frac{-\cos A \cdot (-\tan A) \cos A}{\tan A \cdot (-\cot A) \cdot (-\sin A)} = \frac{\cos^2 A}{\sin A \cot A}$$

$$= \cos A$$

Sol.17.(c) $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B) = \cos 2A \cos 2B +$
 $\{ \sin(A - B) + \sin(A + B) \} \times$
 $\{ \sin(A - B) - \sin(A + B) \}$
 $= \cos 2A \cos 2B + \{ 2 \sin A \cos B \times (-2 \cos A \sin B) \}$
 $= \cos 2A \cos 2B - \sin 2A \sin 2B$
 $= \cos(2A + 2B)$

Sol.18.(b) $\frac{k - k \cot^2 30^\circ}{1 + \cot^2 30^\circ}$

$$= \sin^2 60^\circ + 4 \tan^2 45^\circ - \operatorname{cosec}^2 60^\circ$$

$$\frac{k(1 - 3)}{(1 + 3)} = \frac{3}{4} + 4 \times 1 - \frac{4}{3}$$

$$k = -2 \left\{ \frac{3}{4} + 4 - \frac{4}{3} \right\} = -2 \times \left(\frac{41}{12} \right) = -6.83$$

Sol.19.(a) $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$

$$\sin \theta = \sqrt{2} \cos \theta - \cos \theta$$

$$\sin \theta = \cos \theta (\sqrt{2} - 1)$$

$$\sin \theta \times \frac{1}{\sqrt{2} - 1} = \cos \theta$$

$$(\sqrt{2} + 1) \sin \theta = \cos \theta$$

Subtracting $\sin \theta$ from it

$$\sin \theta (\sqrt{2} + 1 - 1) = \cos \theta - \sin \theta$$

$$\sqrt{2} \sin \theta = \cos \theta - \sin \theta$$

Sol.20.(a) $\sin 73^\circ + \cos 137^\circ$

$$\cos 137^\circ = \cos(90^\circ + 47^\circ) = -\sin 47^\circ$$

$$\sin 73^\circ - \sin 47^\circ = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$= 2 \cos \left(\frac{73 + 47}{2} \right) \sin \left(\frac{73 - 47}{2} \right)$$

$$= 2 \cos \left(\frac{120^\circ}{2} \right) \sin \left(\frac{26^\circ}{2} \right)$$

$$= 2 \cos 60^\circ \sin 13^\circ = 2 \times \frac{1}{2} \times \sin 13^\circ = \sin 13^\circ$$

Sol.21.(a) As we know that,

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 18^\circ = 1 - \left(\frac{\sqrt{5} - 1}{4} \right)^2$$

$$= \frac{16 - (5 + 1 - 2\sqrt{5})}{16} = \frac{10 + 2\sqrt{5}}{16}$$

$$\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}$$

Sol.22.(a) $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$

$$= \frac{2 \sin \left(\frac{4\theta + 2\theta}{2} \right) \cos \left(\frac{4\theta - 2\theta}{2} \right)}{2 \cos \left(\frac{4\theta + 2\theta}{2} \right) \cos \left(\frac{4\theta - 2\theta}{2} \right)}$$

$$= \frac{\sin 3\theta \cos \theta}{\cos 3\theta \cos \theta} = \frac{\sin 3\theta}{\cos 3\theta} = \tan 3\theta$$

Sol.23.(d)

$$\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ$$

$$\text{No. of terms} = \frac{\text{Last term} - \text{1st term}}{\text{common difference}} + 1$$

$$= \frac{89 - 1}{2} + 1 = 45$$

$$\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ$$

$$= \frac{45}{2} = 22.5$$

$$\cot^2 46^\circ - \sec^2 44^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ)$$

$$= \tan^2 44^\circ - \sec^2 44^\circ + (22.5) = -1 + 22.5 = 21.5$$

Sol.24.(a) $\sin A = 0.5$ is true for all the values of $(2\pi n + 30^\circ)$ and $(2n + 1)\pi - 30^\circ$
For the $n = 0$

$$2\pi n + 30^\circ = 30^\circ \text{ and } (2n + 1)\pi - 30^\circ = 150^\circ$$

And for $n = 1$

$$2\pi n + 30^\circ = 390^\circ \text{ and } (2n + 1)\pi - 30^\circ = 510^\circ$$

$\sin A = 0.5$ is true for $150^\circ, 390^\circ$ and 510°

$$\text{So, } \frac{A}{3} = 50^\circ, 130^\circ \text{ and } 170^\circ$$

Only 170° is given in the answer so it is the correct answer.

Sol.25.(d) $\cos 27^\circ = x$

$$\sin 63^\circ = \cos 27^\circ = x$$

$$\cos 63^\circ = \sqrt{1 - x^2}$$

$$\Rightarrow \tan 63^\circ = \frac{x}{\sqrt{1 - x^2}}$$

Sol.26.(c) $x \sin \theta = y \cos \theta$

$$= x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$$

$$= x \sin^3 \theta + \frac{x \sin \theta}{\cos \theta} \times \cos^3 \theta = \sin \theta \cos \theta$$

$$= x \sin^3 \theta + x \sin \theta \times \cos^2 \theta = \sin \theta \cos \theta$$

$$= x \sin \theta (\sin^2 \theta + \cos^2 \theta) = \sin \theta \cos \theta$$

$$= x \sin \theta = \sin \theta \cos \theta \Rightarrow x = \cos \theta$$

$$= x \sin \theta = y \cos \theta$$

$$= \cos \theta \cdot \sin \theta = y \cos \theta \Rightarrow y = \sin \theta$$

$$\Rightarrow x^2 + y^2 = \sin^2 \theta + \cos^2 \theta = 1$$

Short trick:- In such type of questions, value of $x^2 + y^2$ is the square of constant term before $\sin \theta \cos \theta$

Sol.27.(d) $\cos^2 \theta + \cos^4 \theta = 1$

$$\cos^4 \theta = 1 - \cos^2 \theta \Rightarrow \cos^4 \theta = \sin^2 \theta \text{ Or } \cos^2 \theta = \sin \theta$$

$$\sin \theta + \sin^2 \theta = \cos^2 \theta + \sin^2 \theta = 1$$

Sol.28.(c) $\sin \theta - \cos \theta = \frac{7}{17}$

$$\frac{p}{h} - \frac{b}{h} = \frac{7}{17} \text{ (here, } h = 17 \text{ or its multiple) \& We know,}$$

15, 8 and 17 make right angle triangles.
Put, $p = 15$ and $b = 8$

$$\sin \theta + \cos \theta = \frac{p}{h} + \frac{b}{h} = \frac{15 + 8}{17} = \frac{23}{17}$$

Short trick :- $\sin \theta - \cos \theta = \frac{7}{17}$

$$\begin{aligned} \sin \theta + \cos \theta &= \sqrt{a^2 + b^2 - m^2} \\ &= \sqrt{1^2 + 1^2 - \left(\frac{7}{17}\right)^2} = \sqrt{\frac{529}{289}} = \frac{23}{17} \end{aligned}$$

Sol.29.(d) sum of roots

$$= \sec \theta + \sin \theta = \frac{-b}{a} = \frac{k}{\sqrt{6}}$$

$$\text{product of roots} = \sec \theta \times \sin \theta$$

$$= \tan \theta = \frac{c}{a} = \frac{\sqrt{6}}{\sqrt{6}} = 1 \Rightarrow \tan \theta = 1$$

it means, $\theta = 45^\circ$

$$\sec \theta + \sin \theta = \frac{k}{\sqrt{6}} \Rightarrow \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{k}{\sqrt{6}}$$

$$\frac{3}{\sqrt{2}} = \frac{k}{\sqrt{6}} \Rightarrow k = 3\sqrt{3}$$

Sol.30.(d) If $\cos A$, $\sin A$ and $\cot A$ are in geometric progression, then

$$\sin^2 A = \cos A \times \cot A$$

$$\sin^2 A = \cos A \times \frac{\cos A}{\sin A} \Rightarrow \sin^3 A = \cos^2 A$$

After dividing by $\cos^3 A$

$$\frac{\sin^3 A}{\cos^3 A} = \frac{1}{\cos A} \Rightarrow \tan^3 A = \sec A$$

$$\tan^6 A - \tan^2 A = \sec^2 A - \tan^2 A = 1$$

Sol.31.(a) $\cos 225^\circ = \cos (180 + 45)^\circ$

$$= -\cos 45^\circ = -\frac{1}{\sqrt{2}} = -\frac{1}{1.414} = -0.7071$$

Sol.32.(c) $\tan 2A = \tan[(A + B) + (A - B)]$

$$= \frac{\tan(A + B) + \tan(A - B)}{1 - \tan(A + B) \tan(A - B)}$$

$$\cos(A + B) = \frac{24}{25} \Rightarrow \tan(A + B) = \frac{7}{24}$$

$$\sin(A - B) = \frac{15}{17} \Rightarrow \tan(A - B) = \frac{15}{8}$$

$$\tan[(A + B) + (A - B)]$$

$$= \frac{\frac{7}{24} + \frac{15}{8}}{1 - \frac{7}{24} \times \frac{15}{8}} = \frac{416}{87}$$

Sol.33.(d) $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$

$$\Rightarrow \frac{\cos^2 \theta - \sin^2 \theta}{1} = \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$\Rightarrow \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta}$$

By using componendo and Dividendo rule,

$$\Rightarrow \frac{\cos^2 \theta}{-\sin^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\Rightarrow \frac{-\sin^2 \theta}{\cos^2 \theta} = \frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta + \cos^2 \theta}$$

$$\Rightarrow \tan^2 \theta = \cos^2 \theta - \sin^2 \theta$$

Sol.34.(a) There are a total 20 terms and the value of $\sin \theta$ can't be greater than 1. Clearly for the sum of 20 terms being 20 values of each term must be 1. So each angle must be 90°

$$(\alpha_1 + \alpha_2 + \alpha_3 \dots \alpha_{20}) = 90 \times 20 = 1800^\circ$$

Sol.35.(a) Given, $\sin(\alpha + \beta) = 1$

$$\sin(\alpha + \beta) = \sin 90^\circ \Rightarrow \alpha + \beta = 90^\circ \dots (1)$$

$$\cos(\alpha - \beta) = \frac{1}{2}$$

Similarly, $\alpha - \beta = 60^\circ \dots (2)$

Adding (1) and (2)

$$2\alpha = 150^\circ \Rightarrow \alpha = 75^\circ$$

$$\text{From eq (1)} \Rightarrow \beta = 90^\circ - 75^\circ = 15^\circ$$

Sol.36.(b) Given $\theta = 9^\circ$

$$\cot \theta \cdot \cot 2\theta \cdot \cot 3\theta \cdot \cot 4\theta \cdot \cot 5\theta \cdot \cot 6\theta \cdot \cot 7\theta \cdot \cot 8\theta \cdot \cot 9\theta$$

$$= \cot 9 \cdot \cot 18 \cdot \cot 27 \cdot \cot 36 \cdot \cot 45 \cdot \cot 54 \cdot \cot 63 \cdot \cot 72 \cdot \cot 81 = 1 \times 1 \times 1 \times 1 \times 1$$

$$\dots (\cot A \cot B = 1, \text{ if } A + B = 90^\circ) = 1$$

Sol.37.(d)

$$\tan 30^\circ \cdot \tan 60^\circ \Rightarrow \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

$$\sin 30^\circ \Rightarrow \frac{1}{2}; \tan 45^\circ \Rightarrow 1 \Rightarrow \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Clearly $\cos 30^\circ$ is an irrational quantity.

Sol.38.(c) Values of $\cos \theta$, $\cot \theta$ and $\operatorname{cosec} \theta$ continuously decreases. Only value of $\tan \theta$ is continuously increases.

Sol.39.(c) Given,

$$\sec \theta = 4x \text{ and } \tan \theta = \frac{4}{x}$$

Squaring both sides

$$\sec^2 \theta = 16x^2 \dots (1) \text{ and } \tan^2 \theta = \frac{16}{x^2} \dots (2)$$

Subtracting (2) from (1)

$$\sec^2 \theta - \tan^2 \theta = 16x^2 - \frac{16}{x^2}$$

$$\Rightarrow 1 = 2(8x^2 - \frac{8}{x^2}) \Rightarrow 8(x^2 - \frac{1}{x^2}) = \frac{1}{2}$$

Sol.40.(d)

$$\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1 - r}{1 + r}$$

$$\Rightarrow \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta}}}$$

$$= \frac{1 - r}{1 + r} \Rightarrow \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$= \frac{1 - r}{1 + r} \Rightarrow \frac{(1 - \cos \theta)}{(1 + \cos \theta)} = \frac{1 - r}{1 + r}$$

Comparing both the terms $\Rightarrow r = \cos \theta$

Sol.41.(c)

$$\frac{(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)}{\tan^2 \theta - \sin^2 \theta} = r^3$$

$$\text{Put, } \theta = 30^\circ \Rightarrow \frac{(\frac{1}{2} - 2)(\frac{\sqrt{3}}{2} - \frac{2}{\sqrt{3}})}{(\frac{1}{\sqrt{3}})^2 - (\frac{1}{2})^2} = r^3$$

$$\Rightarrow \frac{(\frac{-3}{2})(\frac{3-4}{2\sqrt{3}})}{\frac{1}{3} - \frac{1}{4}} = r^3 \Rightarrow \frac{(\frac{3}{4\sqrt{3}})}{\frac{1}{12}} = r^3$$

$$\Rightarrow r^3 = 3\sqrt{3} \Rightarrow r = \sqrt{3}$$

$$\text{through options} \Rightarrow \cot\theta \Rightarrow \cot 30^\circ = \sqrt{3}$$

Clearly $r = \cot\theta$

$$\text{Sol.42.(d)} \quad 3 - 2\sin^2\theta - 3\cos\theta = 0$$

$$= 3 - 2(1 - \cos^2\theta) - 3\cos\theta = 0$$

$$= 2\cos^2\theta - 3\cos\theta + 1 = 0$$

$$= 2\cos^2\theta - 2\cos\theta - \cos\theta + 1 = 0$$

$$= 2\cos\theta(\cos\theta - 1) - 1(\cos\theta - 1) = 0$$

$$= \cos\theta = 1 \text{ or } \frac{1}{2} \Rightarrow \text{Clearly } \theta = 60^\circ$$

$$(2\operatorname{cosec}\theta + \tan\theta) \Rightarrow 2\left(\frac{2}{\sqrt{3}}\right) + \sqrt{3}$$

$$\Rightarrow \frac{4}{\sqrt{3}} + \sqrt{3} = \frac{7\sqrt{3}}{3}$$

Short trick:- put $\theta = 60^\circ$

$$3 - 2\sin^2\theta - 3\cos\theta = 3 - 2 \times \frac{3}{4} - 3 \times \frac{1}{2}$$

$$= 3 - \frac{3}{2} - \frac{3}{2} = 0$$

Condition satisfied. Clearly $\theta = 60^\circ$

$$\text{So, } (2\operatorname{cosec}\theta + \tan\theta) = 2\left(\frac{2}{\sqrt{3}}\right) + \sqrt{3}$$

$$= \frac{4}{\sqrt{3}} + \sqrt{3} = \frac{7\sqrt{3}}{3}$$

Sol.43.(d)

$$\left(\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}\right)\left(\frac{1}{\tan\theta + \cot\theta}\right)$$

$$= \left(\frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}\right)\left(\frac{1}{\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}}\right)$$

$$= \left(\frac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)}\right)\left(\frac{1}{\frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}}\right)$$

$$= \frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)} \left(\frac{\sin\theta\cos\theta}{1}\right)$$

$$= \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)} \times \sin\theta\cos\theta = 2\cos\theta$$

Sol.44.(c)

$$\frac{(\cos 90^\circ + \sin 81^\circ)(\sec 90^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \sec 34^\circ + \cos 25^\circ \operatorname{cosec} 65^\circ}$$

$$\frac{[\cos(90^\circ - 81^\circ) + \sin 81^\circ][\sec(90^\circ - 81^\circ) + \operatorname{cosec} 81^\circ]}{\sin 56^\circ \sec(90^\circ - 56^\circ) + \cos(90^\circ - 65^\circ) \operatorname{cosec} 65^\circ}$$

$$= \frac{(2\sin 81^\circ)(2\operatorname{cosec} 81^\circ)}{\sin 56^\circ \operatorname{cosec} 56^\circ + \sin 65^\circ \operatorname{cosec} 65^\circ}$$

$$= \frac{4 \times 1}{1+1} = 2$$

Sol.45.(c)

$$\frac{(\sec\theta + \tan\theta)(1 - \sin\theta)}{\operatorname{cosec}\theta(1 + \cos\theta)(\operatorname{cosec}\theta - \cot\theta)}$$

$$= \frac{\left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right)(1 - \sin\theta)}{(\operatorname{cosec}\theta + \operatorname{cosec}\theta \cos\theta)(\operatorname{cosec}\theta - \cot\theta)}$$

$$= \frac{\left(\frac{1}{\cos\theta}\right)(1 + \sin\theta)(1 - \sin\theta)}{(\operatorname{cosec}\theta + \cot\theta)(\operatorname{cosec}\theta - \cot\theta)}$$

$$= \frac{\frac{\cos^2\theta}{\cos\theta}}{\operatorname{cosec}^2\theta - \cot^2\theta} = \cos\theta$$

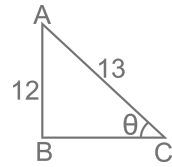
$$\text{Sol.46.(b)} \quad 3\cos^2 A + 7\sin^2 A = 4$$

$$= 3\cos^2 A + 7\sin^2 A = 4(\cos^2 A + \sin^2 A)$$

$$= 3\sin^2 A + \cos^2 A$$

$$= \cot^2 A = 3 \Rightarrow \cot A = \sqrt{3}$$

Sol.47.(d)



$$\operatorname{Cosec}\theta = \frac{AC}{AB} = \frac{13}{12}$$

$$AB^2 + BC^2 = AC^2 \Rightarrow BC = \sqrt{13^2 - 12^2} = 5$$

$$\sin\theta = \frac{12}{13}, \cos\theta = \frac{5}{13} \text{ and } \tan\theta = \frac{12}{5}$$

$$\sin\theta + \cos\theta - \tan\theta$$

$$= \frac{12}{13} + \frac{5}{13} - \frac{12}{5} = -\frac{71}{65}$$

Sol.48.(b)

$$\sin^2 38^\circ + \sin^2 52^\circ + \sin^2 30^\circ - \tan^2 45^\circ$$

$$\Rightarrow 1 + \left(\frac{1}{2}\right)^2 - (1)^2 \dots\dots (\sin^2 A + \sin^2 B = 1,$$

$$\text{if } A + B = 90^\circ) \Rightarrow 1 + \frac{1}{4} - 1 = \frac{1}{4}$$

Sol.49.(c) Given, $\sec 4\theta = \operatorname{cosec}(\theta + 20^\circ)$

$$= \operatorname{cosec}(90^\circ - 4\theta) = \operatorname{cosec}(\theta + 20^\circ)$$

$$\text{Clearly, } (4\theta + \theta + 20^\circ) = 90^\circ$$

$$5\theta = 70^\circ \Rightarrow \theta = 14^\circ$$

Sol.50.(b) Put $\theta = 30^\circ$

$$3\sin\theta = 2\cos^2\theta = 3 \times \frac{1}{2} = 2 \times \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{3}{2} = \frac{3}{2} \text{ (Condition satisfied)}$$

$$(\tan^2\theta + \sec^2\theta - \operatorname{cosec}^2\theta)$$

$$= \left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{2}{\sqrt{3}}\right)^2 - (2)^2$$

$$\Rightarrow \frac{1}{3} + \frac{4}{3} - 4 = -\frac{7}{3}$$

Sol.51.(b)

Applying componendo and dividendo

$$\frac{\tan\theta + \sin\theta}{\tan\theta - \sin\theta} = \frac{k+1}{k-1} \Rightarrow \frac{\tan\theta}{\sin\theta} = \frac{k}{1}$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta \sin\theta} = k \Rightarrow \sec\theta = k$$

Sol.52.(b)

$$\frac{1}{\operatorname{cosec}\theta - 1} + \frac{1}{\operatorname{cosec}\theta + 1} = 2\sec\theta$$

$$= \frac{(\operatorname{cosec}\theta + 1) + (\operatorname{cosec}\theta - 1)}{(\operatorname{cosec}\theta + 1)(\operatorname{cosec}\theta - 1)} = 2\sec\theta$$

$$= \frac{2\operatorname{cosec}\theta}{(\operatorname{cosec}^2\theta - 1)} = 2\sec\theta$$

$$= \frac{2\operatorname{cosec}\theta}{\cot^2\theta} = 2\sec\theta$$

$$= \frac{2\sin^2\theta}{\sin\theta \cos^2\theta} = 2 \times \frac{1}{\cos\theta}$$

$$= \tan\theta = 1 \Rightarrow \theta = 45^\circ$$

$$(\cot\theta + \cos\theta) \Rightarrow 1 + \frac{1}{\sqrt{2}} = \frac{\sqrt{2}+1}{\sqrt{2}}$$

$$= \frac{2+\sqrt{2}}{2}$$

Short trick :- Put $\theta = 45^\circ$

$$\text{LHS} \Rightarrow \frac{1}{\operatorname{cosec}\theta - 1} + \frac{1}{\operatorname{cosec}\theta + 1}$$

$$= \frac{1}{\sqrt{2}-1} + \frac{1}{\sqrt{2}+1}$$

$$= \sqrt{2} + 1 + \sqrt{2} - 1 = 2\sqrt{2}$$

$$\text{RHS} = 2\sec\theta = 2\sqrt{2}$$

$$\cot\theta + \cos\theta \Rightarrow 1 + \frac{1}{\sqrt{2}} = \frac{\sqrt{2}+1}{\sqrt{2}}$$

$$= \frac{2+\sqrt{2}}{2}$$

$$\text{Sol.53.(a)} \quad \sin\theta = \frac{a}{\sqrt{a^2+b^2}}$$

$$\cos\theta = \sqrt{1 - \left(\frac{a}{\sqrt{a^2+b^2}}\right)^2} = \frac{b}{\sqrt{a^2+b^2}}$$

$$\sec\theta + \tan\theta$$

$$= \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} \Rightarrow \frac{1+\sin\theta}{\cos\theta}$$

$$= \frac{1 + \frac{a}{\sqrt{a^2+b^2}}}{\frac{b}{\sqrt{a^2+b^2}}} = \frac{\sqrt{a^2+b^2} + a}{b}$$

Sol.54.(a) $\cos^2\theta - 3\cos\theta + 2 = \sin^2\theta$

$$\Rightarrow \cos^2\theta - 3\cos\theta + 2 = 1 - \cos^2\theta$$

$$= 2\cos^2\theta - 3\cos\theta + 1 = 0$$

$$= 2\cos^2\theta - 2\cos\theta - \cos\theta + 1 = 0$$

$$= 2\cos\theta(\cos\theta - 1) - 1(\cos\theta - 1) = 0$$

$$\Rightarrow \cos\theta = 1 \text{ or } \frac{1}{2}$$

Clearly $\theta = 60^\circ$, (as $0^\circ < \theta < 90^\circ$)

$$2\operatorname{cosec}\theta + 4\cot\theta$$

$$= 2\left(\frac{2}{\sqrt{3}}\right) + 4\left(\frac{1}{\sqrt{3}}\right) = \frac{8\sqrt{3}}{3}$$

$$\text{Sol.55.(b)} \quad \frac{1}{\sec x - \tan x} - \frac{1}{\cos x}$$

$$= \sec x + \tan x - \sec x \dots\dots \{ \because \sec x + \tan x$$

$$= \frac{1}{\sec x - \tan x} \} = \tan x$$

Sol.56.(a) $\tan^2\theta - 3\sec\theta + 3 = 0$

$$\Rightarrow \sec^2\theta - 1 - 3\sec\theta + 3 = 0$$

$$\Rightarrow \sec^2\theta - 3\sec\theta + 2 = 0$$

$$\Rightarrow \sec^2\theta - 2\sec\theta - \sec\theta + 2 = 0$$

$$\Rightarrow \sec\theta(\sec\theta - 2) - 1(\sec\theta - 2) = 0$$

$$\Rightarrow (\sec\theta - 1)(\sec\theta - 2) = 0$$

$$\Rightarrow \sec\theta = 1 \text{ or } 2, \text{ Clearly } \theta = 60^\circ \text{ or } 0^\circ$$

$$\Rightarrow \text{But } \theta \neq 0, \text{ so } \theta = 60^\circ$$

$$\sin\theta + \cot\theta = \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}} = \frac{3+2}{2\sqrt{3}} = \frac{5\sqrt{3}}{6}$$

Sol.57.(a)

$$(1 + \tan^2\theta) + (1 + (\tan^2\theta)^{-1}) = k$$

$$\Rightarrow (1 + \tan^2\theta) + (1 + \cot^2\theta) = k$$

$$\Rightarrow \sec^2\theta + \operatorname{cosec}^2\theta = k$$

$$\Rightarrow \frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta} = k$$

$$\Rightarrow \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = k \Rightarrow \sec^2 \theta \cdot \operatorname{cosec}^2 \theta = k$$

$$\Rightarrow \sec \theta \cdot \operatorname{cosec} \theta = \sqrt{k}$$

Short-Trick: Put $\theta = 45^\circ$

$$(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$$

$$\Rightarrow (1 + 1) + (1 + 1) = k$$

$$\Rightarrow 4 = k \Rightarrow 2 = \sqrt{k}$$

Going through options, Option A :

$$\operatorname{cosec} \theta \sec \theta = \sqrt{2} \cdot \sqrt{2} = 2$$

Clearly option (a) is the right answer.

$$\text{Sol.58.(b)} \quad \frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \frac{(\operatorname{cosec}^2 \theta - 1)}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \frac{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \operatorname{cosec} \theta + 1 = 1$$

$$\text{Sol.59.(b)} \quad \left(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \operatorname{cosec}^2 \theta} \right)$$

$$= \left(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \frac{1}{\sin^2 \theta}} \right)$$

$$\Rightarrow \left(\frac{1}{1 + \sin^2 \theta} + \frac{\sin^2 \theta}{1 + \sin^2 \theta} \right) = 1$$

$$\text{Sol.60.(b)} \quad \frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$$

$$= \cos \theta \left(\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} \right) = 4$$

$$= \cos \theta \left(\frac{1 + \sin \theta + 1 - \sin \theta}{1 - \sin^2 \theta} \right) = 4$$

$$\Rightarrow \cos \theta = \frac{1}{2}, \text{ Clearly } \theta = 60^\circ$$

$$\text{So, } \tan \theta + \operatorname{cosec} \theta \Rightarrow \sqrt{3} + \frac{2}{\sqrt{3}}$$

$$= \frac{3 + 2}{\sqrt{3}} = \frac{5\sqrt{3}}{3}$$

Sol.61.(c) Put $\theta = 45^\circ$,

$$\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1}$$

$$= \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1$$

Going through the options

$$\sec \theta + \tan \theta = \sqrt{2} + 1$$

Sol.62.(b) Put $\theta = 45^\circ$

$$\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \times \sqrt{\tan^2 \theta - \sin^2 \theta}$$

$$\Rightarrow \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} \times \sqrt{1 - \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{2}$$

On checking all the options one by one, option (b) satisfies.

$$\sin \theta \sec^2 \theta \Rightarrow \frac{1}{\sqrt{2}} \times (\sqrt{2})^2 = \sqrt{2}$$

Sol.63.(c) $\sec \theta - \tan \theta = P$

$$\sec \theta + \tan \theta = \frac{1}{P}$$

$$2 \sec \theta = P + \frac{1}{P} \Rightarrow \cos \theta = \frac{2P}{P^2 + 1}$$

$$\cos^2 \theta = \left[\frac{2P}{P^2 + 1} \right]^2$$

$$1 - \sin^2 \theta = \frac{4P^2}{2P^2 + 1 + P^4}$$

$$\sin^2 \theta = 1 - \frac{4P^2}{2P^2 + 1 + P^4}$$

$$= \frac{2P^2 + 1 + P^4 - 4P^2}{2P^2 + 1 + P^4}$$

$$\sin \theta = \frac{1 - P^2}{P^2 + 1}, \operatorname{cosec} \theta = \frac{P^2 + 1}{1 - P^2}$$

Alternate :-

$$\sec \theta - \tan \theta = P \Rightarrow \sec \theta + \tan \theta = \frac{1}{P}$$

$$2 \sec \theta = P + \frac{1}{P}$$

$$\cos \theta = \frac{2P}{P^2 + 1} = \frac{\text{base}}{\text{hypotenuse}}$$

$$\text{Perpendicular} = 1 - P^2$$

$$\operatorname{cosec} \theta = \frac{\text{hypotenuse}}{\text{Perpendicular}} = \frac{P^2 + 1}{1 - P^2}$$

$$\text{Sol.64.(b)} \quad \cos^2 \theta = 3 \cot^2 \theta - 3 \cos^2 \theta$$

$$\Rightarrow 4 \cos^2 \theta = 3 \cot^2 \theta \Rightarrow \sin^2 \theta = \frac{3}{4}$$

$$\Rightarrow \sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ$$

$$\left(\frac{1}{2} \sec \theta + \sin \theta \right)^{-1} = \frac{1}{\left(\frac{1}{2} \times 2 + \frac{\sqrt{3}}{2} \right)}$$

$$= \frac{2}{\sqrt{3} + 2} = 2(2 - \sqrt{3})$$

$$\text{Sol.65.(b)} \quad \left[\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} \right]^2 + 1$$

$$= \left[\frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)} \right]^2 + 1$$

$$= \left[\frac{\sin \theta \cdot \cos 2\theta}{\cos \theta \cdot \cos 2\theta} \right]^2 + 1$$

$$= \tan^2 \theta + 1 = \sec^2 \theta$$

Sol.66.(c) Let $A = 45^\circ$

$$(\operatorname{cosec} A - \sin A)^2 + (\sec A - \cos A)^2 -$$

$$(\cot A - \tan A)^2$$

$$= (\operatorname{cosec} 45^\circ - \sin 45^\circ)^2 + (\sec 45^\circ - \cos 45^\circ)^2 - (\cot 45^\circ - \tan 45^\circ)^2$$

$$= \left(\sqrt{2} - \frac{1}{\sqrt{2}} \right)^2 + \left(\sqrt{2} - \frac{1}{\sqrt{2}} \right)^2 - (1 - 1)^2$$

$$= \frac{1}{2} + \frac{1}{2} - 0 = 1$$

$$\text{Sol.67.(c)} \quad 1 + \frac{\tan^2 A}{1 + \sec A} = 1 + \frac{\sec^2 A - 1}{1 + \sec A}$$

$$= 1 + \sec A - 1 = \sec A$$

Sol.68.(c) $x = a \cos \theta + b \sin \theta$

Squaring both sides

$$x^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta + 2ab \cdot \sin \theta \cdot \cos \theta$$

.....(1)

$y = a \sin \theta - b \cos \theta$, Squaring both sides

$$y^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta - 2ab \cdot \sin \theta \cdot \cos \theta$$

.....(2)

Add both the equations

$$x^2 + y^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta + 2ab$$

$$\cdot \sin \theta \cdot \cos \theta + a^2 \sin^2 \theta + b^2 \cos^2 \theta - 2ab$$

$$\cdot \sin \theta \cdot \cos \theta$$

$$\Rightarrow a^2 (\cos^2 \theta + \sin^2 \theta) + b^2 (\cos^2 \theta + \sin^2 \theta)$$

$$= a^2 + b^2$$

Short trick :- It is very useful to solve questions directly. $x^2 + y^2 = a^2 + b^2$

Sol.69.(d)

$$\tan^2 \theta + \tan^4 \theta = \tan^2 \theta (1 + \tan^2 \theta)$$

$$= \tan^2 \theta \cdot \sec^2 \theta = (\sec^2 \theta - 1) \cdot \sec^2 \theta$$

$$= \sec^4 \theta - \sec^2 \theta$$

Sol.70.(a) Put $\theta = 90^\circ$,

$$\text{Given, } \sin \theta + \operatorname{cosec} \theta = 2$$

$$\Rightarrow 1 + 1 = 2$$

$$\text{Now, } \sin^5 \theta + \operatorname{cosec}^5 \theta = 1 + 1 = 2$$

$$\text{Sol.71.(c)} \quad \cos \theta - \sin \theta = \sqrt{2} \sin \theta$$

Squaring both sides we get,

$$\Rightarrow (\cos \theta - \sin \theta)^2 = 2 \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta - 2 \cos \theta \sin \theta = 2 \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta - 2 \sin^2 \theta = 2 \cos \theta \sin \theta$$

$$\Rightarrow \cos^2 \theta - \sin^2 \theta = 2 \cos \theta \sin \theta$$

$$\Rightarrow (\cos \theta - \sin \theta)(\cos \theta + \sin \theta)$$

$$= 2 \cos \theta \sin \theta$$

$$\Rightarrow \sqrt{2} \sin \theta (\cos \theta + \sin \theta) = 2 \cos \theta \sin \theta$$

$$\Rightarrow (\cos \theta + \sin \theta) = \sqrt{2} \cos \theta$$

Sol.72.(d) According to the question,

$$(1 + \sin^4 A - \cos^4 A) \operatorname{cosec}^2 A$$

On Putting $A = 90^\circ$

$$(1 + 1^4 - 0^4) 1^2 = 2$$

Sol.73.(c) According to the question,

$$\cot \theta = \frac{4}{3} = \frac{\text{base}}{\text{perpendicular}} \quad (3, 4, 5)$$

pythagorean triplets)

So the value of hypotenuse = 5

$$5 \operatorname{cosec}^2 \theta \sin \theta = \cot^2 \theta$$

$$5 \times p \times \frac{16}{25} \times \frac{3}{5} = \frac{16}{9} \Rightarrow p = \frac{25}{27}$$

Sol.74.(a) According to the question,

$$\cos A = \frac{\text{base}}{\text{hypotenuse}} = \frac{15}{17}$$

So the perpendicular = 8 (8, 15, 17 pythagorean triples)

$$\cot (90^\circ - A) = \tan A$$

$$\tan A = \frac{\text{perpendicular}}{\text{base}} = \frac{8}{15}$$

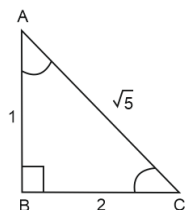
Sol.75.(d) If $\sin A = \cos B$ then, $A + B = 90^\circ$

$$\sin 2A = \cos 15^\circ$$

$$\Rightarrow 2A + 15^\circ = 90^\circ \Rightarrow A = 37.5^\circ$$

Sol.76.(b) Here $A + B = 90^\circ$

$$\begin{aligned} \frac{\cot A}{\cot B} + \cos^2 A + \cos^2 B \\ = \frac{\cot A}{\cot(90-A)} + \cos^2 A + \cos^2(90-A) \\ = \frac{\cot A}{\tan A} + \cos^2 A + \sin^2 A \\ = \cot^2 A + \cos^2 A + \sin^2 A \\ = \cot^2 A + 1 = \operatorname{cosec}^2 A \end{aligned}$$

Sol.77.(d)

$$\cos A + \tan C = \frac{1}{\sqrt{5}} + \frac{1}{2} = \frac{2+\sqrt{5}}{2\sqrt{5}}$$

Sol.78.(c) $\sin A + \sin^2 A = 1$,

$$\Rightarrow \sin A = 1 - \sin^2 A$$

$$\Rightarrow \sin A = \cos^2 A$$

$$\begin{aligned} \text{Now, } \cos^4 A + \cos^6 A &= \sin^2 A + \sin^3 A \\ &= \sin A(\sin A + \sin^2 A) = \sin A \end{aligned}$$

Sol.79.(c) According to the question, $\sec A - \tan A = p$

$$\sec A + \tan A = \frac{1}{p} \Rightarrow \sec A = \frac{p^2 + 1}{2p}$$

$$\begin{aligned} \text{Sol.80.(d)} \quad & 2(\sin 1^\circ \times \sec 89^\circ) + 3(\cos 11^\circ \times \operatorname{cosec} 79^\circ) + 5(\tan 21^\circ \times \tan 69^\circ) \\ &= 2(\sin 1^\circ \times \operatorname{cosec} 1^\circ) + 3(\cos 11^\circ \times \sec 11^\circ) + 5(\tan 21^\circ \times \cot 21^\circ) \\ &= 2 + 3 + 5 = 10 \end{aligned}$$

$$\text{Sol.81.(d)} \sqrt{\frac{1-\tan A}{1+\tan A}}$$

$$\begin{aligned} &= \sqrt{\frac{1-\frac{\sin A}{\cos A}}{1+\frac{\sin A}{\cos A}}} = \sqrt{\frac{\frac{\cos A - \sin A}{\cos A}}{\frac{\cos A + \sin A}{\cos A}}} \\ &= \sqrt{\frac{\cos A - \sin A}{\cos A + \sin A}} \times \frac{\cos A - \sin A}{\cos A - \sin A} \\ &= \sqrt{\frac{(\cos A - \sin A)^2}{\cos^2 A - \sin^2 A}} \\ &= \sqrt{\frac{1 - 2\sin A \cos A}{\cos^2 A - \sin^2 A}} = \sqrt{\frac{1 - \sin 2A}{\cos 2A}} \end{aligned}$$

$$\begin{aligned} \text{Sol.82.(c)} \quad & 7\sin^2 A + 3\cos^2 A = 4 \\ & 4\sin^2 A + [3(\sin^2 A + \cos^2 A)] = 4 \end{aligned}$$

$$4\sin^2 A = 1$$

$$\sin A = \frac{1}{2}, A = 30^\circ \Rightarrow \cot A = \cot 30^\circ = \sqrt{3}$$

Sol.83.(d) Minimum value of $a\sin^2 \theta + b \operatorname{cosec}^2 \theta = 2\sqrt{ab}$ {when, $a > b$ }
Hence, minimum value of $16\operatorname{cosec}^2 \theta + 25\sin^2 \theta = 2\sqrt{16 \times 25} = 40$

Sol.84.(d)

$$\operatorname{cosec} A + \cot A = 3 \text{ -----(1)}$$

$$\operatorname{cosec} A - \cot A = \frac{1}{3} \text{ -----(2)}$$

With the help of equation (1) and (2) we get;

$$2\operatorname{cosec} A = \frac{10}{3} \Rightarrow \operatorname{cosec} A = \frac{5}{3}$$

$$\text{Now, } \operatorname{cosec} A = \frac{\text{hypotenuse}}{\text{perpendicular}} = \frac{5}{3}$$

$$\text{Then, base} = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$$

$$\text{Hence, } \cos A = \frac{\text{base}}{\text{hypotenuse}} = \frac{4}{5}$$

Sol.85.(d)

$$\begin{aligned} \sqrt{\frac{1+\sin A}{1-\sin A}} &= \sqrt{\frac{1+\sin A}{1-\sin A}} \times \frac{1+\sin A}{1+\sin A} \\ &= \sqrt{\frac{(1+\sin A)^2}{(1-\sin A)(1+\sin A)}} \\ &= \frac{(1+\sin A)}{\sqrt{1-\sin^2 A}} = \frac{(1+\sin A)}{\cos A} = \sec A + \tan A \end{aligned}$$

$$\text{Sol.86.(c)} \sin A = \frac{2}{3} = \frac{\text{perpendicular}}{\text{hypotenuse}}$$

$$\text{Then, base} = \sqrt{9 - 4} = \sqrt{5}$$

$$\text{Now, } (7 - \tan A)(3 + \cos A)$$

$$= (7 - \frac{2}{\sqrt{5}})(3 + \frac{\sqrt{5}}{3})$$

$$= (21 - \frac{2}{3}) + (-\frac{7\sqrt{5}}{3} - \frac{6}{\sqrt{5}})$$

$$= \frac{61}{3} + \frac{17}{3\sqrt{5}}$$

$$\text{Sol.87.(d)} \sin A = \frac{\sqrt{3}}{2}, \text{ then } A = 60^\circ$$

$$\text{Now, } 2(\operatorname{cosec} A + \cot A)$$

$$= 2(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}) = 2\sqrt{3}$$

$$\text{Sol.88.(b)} \sin \beta = \frac{1}{3}$$

$$\text{Then, } (\sec \beta - \tan \beta)^2$$

$$= (\frac{1}{\cos \beta} - \frac{\sin \beta}{\cos \beta})^2 = (\frac{1-\sin \beta}{\cos \beta})^2$$

$$= \frac{(1-\sin \beta)^2}{\cos^2 \beta} = \frac{(1-\sin \beta)^2}{1-\sin^2 \beta} = \frac{(1-\frac{1}{3})^2}{1-\frac{1}{9}}$$

$$= \frac{1}{2}$$

Sol.89.(c)

$$\frac{\sin^2 39^\circ + \sin^2(90^\circ - 39^\circ)}{\cos^2 35^\circ + \cos^2(90^\circ - 35^\circ)} + 3\tan 25^\circ \tan 65^\circ$$

$$= \frac{\sin^2 39^\circ + \cos^2 39^\circ}{\cos^2 35^\circ + \sin^2 35^\circ} + 3\tan 25^\circ \times \frac{1}{\tan 25^\circ} = 1 + 3 = 4$$

Sol.90.(b) According to the question,

$$\tan \theta + \cot \theta = 2$$

$$\text{On Putting } \theta = 45^\circ$$

$$1 + 1 = 2 \text{ (satisfied)}$$

Now,

$$2\tan^{25}\theta + 3\cot^{20}\theta + 5\tan^{30}\theta \cot^{15}\theta$$

$$2 \times 1 + 3 \times 1 + 5 \times 1 \times 1 = 10$$

Sol.91.(b) According to the question,

$$\sin(50^\circ + A) - \cos(40^\circ - A)$$

$$\text{Let } A = 10^\circ$$

$$\text{Then } \sin 60^\circ - \cos 30^\circ = \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = 0$$

Sol.92.(b) According to the question,

$$\sin \theta + \cos \theta = \frac{\sqrt{11}}{3} \text{ (on squaring both sides)}$$

$$\sin^2 \theta + \cos^2 \theta + 2\sin \theta \times \cos \theta = \frac{11}{9}$$

$$2\sin \theta \times \cos \theta = \frac{11}{9} - 1 = \frac{2}{9}$$

$$\begin{aligned} \text{Now, } (\sin \theta - \cos \theta)^2 &= \sin^2 \theta + \cos^2 \theta - 2\sin \theta \times \cos \theta \\ (\sin \theta - \cos \theta)^2 &= 1 - \frac{2}{9} = \frac{7}{9} \end{aligned}$$

$$\sin \theta - \cos \theta = \frac{\sqrt{7}}{3}$$

Sol.93.(a) According to the question,

$$\sin \theta + \cos \theta = \sqrt{2} \cos \theta$$

$$\sin \theta = \cos \theta(\sqrt{2} - 1)$$

$$\text{now, } \frac{\sin \theta - \cos \theta}{\sin \theta} = \frac{\sin \theta}{\sin \theta} - \frac{\cos \theta}{\sin \theta}$$

$$1 - \frac{\cos \theta}{\cos \theta(\sqrt{2}-1)}$$

$$= 1 - \frac{1}{\sqrt{2}-1} \times \frac{(\sqrt{2}+1)}{(\sqrt{2}+1)} = -\sqrt{2}$$

$$\text{Sol.94.(b)} \sin A + \sin^2 A = 1$$

$$\sin A = 1 - \sin^2 A = \cos^2 A$$

$$\Rightarrow \sin^2 A = \cos^4 A$$

$$\text{Now, } \cos^2 A + \cos^4 A = \cos^2 A + \sin^2 A = 1$$

$$\text{Sol.95.(d)} \cot \theta = \frac{9}{17} = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\text{Hypotenuse} = \sqrt{9^2 + 17^2} = \sqrt{370}$$

$$\operatorname{cosec}^2 \theta = \left(\frac{\text{Hypotenuse}}{\text{perpendicular}}\right)^2 = \left(\frac{\sqrt{370}}{17}\right)^2$$

$$\operatorname{cosec}^2 \theta = \frac{370}{289} = 1\frac{81}{289}$$

$$\text{Sol.96.(b)} \frac{\cos \theta}{\sec \theta - 1} + \frac{\cos \theta}{\sec \theta + 1}$$

$$= \cos \theta \left(\frac{\sec \theta + 1 + \sec \theta - 1}{\tan^2 \theta}\right)$$

$$= \cos \theta \left(\frac{2\sec \theta}{\tan^2 \theta}\right) = 2\cot^2 \theta$$

Sol.97.(d) Formula used:

$$\sin A \cdot \cos B + \cos A \cdot \sin B = \sin(A + B)$$

$$= \sin(60^\circ + 30^\circ) = \sin(90^\circ) = 1$$

Sol.98.(d) $\sin \theta - \cos \theta = 0$

$$\sin \theta = \cos \theta \Rightarrow \theta = 45^\circ$$

$$\text{Now, } \sin(\pi - \theta) + \sin\left(\frac{\pi}{2} - \theta\right)$$

$$= \sin(\theta) + \cos(\theta) = \sin(45^\circ) + \cos(45^\circ)$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

Sol.99.(a)

If $A + B = 90^\circ$, then $\cot A^\circ \cot B^\circ = 1$
 $\cot 13^\circ \cot 27^\circ \cot 60^\circ \cot 63^\circ \cot 77^\circ$
 $= \cot 13^\circ \cot 77^\circ \cot 27^\circ \cot 63^\circ \cot 60^\circ$
 $= 1 \times 1 \times \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}$

Sol.100.(a) $\cos \theta + \sec \theta = \sqrt{3}$

$$\cos^3 \theta + \sec^3 \theta = (\sqrt{3})^3 - 3(\sqrt{3}) = 0$$

Sol.101.(d)

If $A + B = 90^\circ$, then $\cos(A) = \sin(B)$
 $\cos(3\alpha) = \sin(\alpha - 22^\circ)$
 $(3\alpha) + (\alpha - 22^\circ) = 90^\circ$
 $(4\alpha) = 112^\circ \Rightarrow \alpha = 28^\circ$

Sol.102.(b) If, $\sec \theta - \tan \theta = m$...e.q.(1)

Then, $\sec \theta + \tan \theta = \frac{1}{m}$ e.q.(2)

Now, $2\tan \theta = \frac{1}{m} - m = \frac{1 - m^2}{m}$

Sol.103.(a)

Pythagoras triplets :- (16, 63, 65)

$$\cos A = \frac{63}{65} \Rightarrow \sin A = \frac{16}{65}$$

Now, $\tan A + \cot A = \frac{1}{\sin A \cos A}$

$$= \frac{65 \times 65}{16 \times 63} = 4.19$$

Sol.104.(a) $\frac{\cos \theta}{(1 + \sin \theta)} + \frac{\cos \theta}{(1 - \sin \theta)} = 4$

$$\frac{2}{(\cos^2 \theta)} = \frac{4}{\cos \theta} \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

Sol.105.(c) $\sin \alpha + \cos \alpha = \frac{2}{\sqrt{3}}$

$$1 + 2\sin \alpha \cos \alpha = \frac{4}{3}$$

$$\sin 2\alpha = \frac{4}{3} - 1 = \frac{1}{3}$$

Now, $\tan \alpha + \cot \alpha = \frac{2}{\sin 2\alpha} = 6$

Sol.106.(d) $\cos 0^\circ + \cos 30^\circ - \tan 45^\circ$
 $+ \operatorname{cosec} 60^\circ + \cot 90^\circ$

$$= 1 + \frac{\sqrt{3}}{2} - 1 + \frac{2}{\sqrt{3}} + 0$$

$$= \frac{\sqrt{3}}{2} + \frac{2}{\sqrt{3}} = \frac{7}{2\sqrt{3}}$$

Sol.107.(d) $(\operatorname{cosec}^2 B - 1)(\sec^2 B - 1)$
 $(\cot^2 B)(\tan^2 B) = 1$

Sol.108.(a) $\sin \theta + \cos \theta = \frac{1}{29} \Rightarrow 2\sin \theta$

$$\cos \theta = \left(\frac{1}{29}\right)^2 - 1 = \left(\frac{-840}{841}\right)$$

$$\sin \theta - \cos \theta = \sqrt{1 - \left(\frac{-840}{841}\right)^2}$$

$$= \frac{41}{29}$$

Now, $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{1}{29} \times \frac{29}{41} = \frac{1}{41}$

Sol.109.(b)

If $A + B = 90^\circ$, then $\sin^2 A + \sin^2 B = 1$
 $\sin^2 35^\circ + \sin^2 55^\circ = 1$

Sol.110.(c)

$$\frac{(1 + \tan^2 A)}{\operatorname{cosec}^2 A \times \tan A} = \frac{(\sec^2 A)}{\operatorname{cosec}^2 A \times \tan A}$$

$$= \frac{(\sin^2 A)}{\cos^2 A \times \tan A} = \tan A$$

Sol.111.(b) $\cos^4 A - \sin^4 A$

$$= (\cos^2 A - \sin^2 A)(\cos^2 A + \sin^2 A)$$

$$= 1 - 2\sin^2 A$$

Sol.112.(c) $\sin(p + q) = 1 \Rightarrow (p + q) = 90^\circ$

$$\cos(p - q) = \frac{\sqrt{3}}{2} \Rightarrow (p - q) = 30^\circ$$

$$p = \frac{90^\circ + 30^\circ}{2} = 60^\circ$$

Sol.113.(b) $\operatorname{cosec} \theta = \frac{29}{22}$

$$\cot^2 \theta = \operatorname{cosec}^2 \theta - 1 = \frac{841}{484} - 1 = \frac{357}{484}$$

Sol.114.(a)

$$\sin A \left(1 + \frac{\sin A}{\cos A}\right) + \cos A \left(1 + \frac{\cos A}{\sin A}\right)$$

$$= \sin A + \frac{\sin^2 A}{\cos A} + \cos A + \frac{\cos^2 A}{\sin A}$$

$$= \frac{\sin^2 A + \cos^2 A}{\cos A} + \frac{\sin^2 A + \cos^2 A}{\sin A}$$

$$= \frac{1}{\cos A} + \frac{1}{\sin A} = \sec A + \operatorname{cosec} A$$

Sol.115.(d) $\cot 90^\circ = 0$

So, $\cot 1^\circ \cdot \cot 2^\circ \cdot \cot 3^\circ \dots \cot 90^\circ \dots$
 $\cot 178^\circ \cdot \cot 179^\circ = 0$

Sol.116.(a) $\frac{3 - \sin^2 A + \cos^2 A}{2 + 2\cos^2 A}$

$$= \frac{\sin^2 A + \cos^2 A + 2 - \sin^2 A + \cos^2 A}{2 + 2\cos^2 A}$$

$$= \frac{\cos^2 A + 2 + \cos^2 A}{2 + 2\cos^2 A} = 1$$

Sol.117.(d) $\tan \theta - \cot \theta = 4$

$$\tan^2 \theta + \cot^2 \theta - 2 \times \tan \theta \cdot \cot \theta = 16$$

$$\tan^2 \theta + \cot^2 \theta = 18$$

Sol.118.(b) $\frac{2 \tan A}{(1 - \tan^2 A)} = \tan 2A$

$$\tan 60^\circ = \sqrt{3} = \frac{3}{\sqrt{3}}$$

Sol.119.(a) $\sec \theta + \tan \theta = \sqrt{3}$

$$\sec \theta - \tan \theta = \frac{1}{\sqrt{3}}$$

$$\sec \theta = \frac{\sqrt{3} + \frac{1}{\sqrt{3}}}{2} = \frac{2}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$

Now, $\sin \theta + \cos \theta = \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1 + \sqrt{3}}{2}$

Sol.120.(b)

$$\sin(A - B) = \frac{1}{2} \Rightarrow (A - B) = 30^\circ$$

$$\sin(A + B) = 1 \Rightarrow (A + B) = 90^\circ$$

$$A = \frac{30^\circ + 90^\circ}{2} = 60^\circ \text{ and } B = 30^\circ$$

Sol.121.(a) $2\tan^2 A + 4\cos^4 A = 3$

Put $A = 45^\circ$

$$2\tan^2 45^\circ + 4\cos^4 45^\circ$$

$$= 2 \times 1 + 4 \times \left(\frac{1}{\sqrt{2}}\right)^4 = 2 + 1 = 3$$

Sol.122.(a) $\cot^2 A - \cos^2 A =$

$$\cot^2 A \cdot \cos^2 A \dots (\text{identity})$$

$$a^2 = \cot^2 A + \cos^2 A + 2 \cdot \cot A \cdot \cos A$$

$$b^2 = \cot^2 A + \cos^2 A - 2 \cdot \cot A \cdot \cos A$$

$$ab = \cot^2 A - \cos^2 A$$

Now,

$$a^2 - b^2 - 4\sqrt{ab} = 4 \cdot \cot A \cdot \cos A - 4$$

$$\sqrt{\cot^2 A - \cos^2 A}$$

$$= 4 \cdot \cot A \cdot \cos A - 4 \cdot \cot A \cdot \cos A = 0$$

Sol.123.(b) $1 - \frac{\tan^2 \phi}{\sec^2 \phi} = \frac{\sec^2 \phi - \tan^2 \phi}{\sec^2 \phi}$

$$= \frac{1}{\sec^2 \phi} = \cos^2 \phi = \frac{\sin^2 \phi \cdot \cos^2 \phi}{\sin^2 \phi}$$

$$= \sin^2 \phi \cdot \cot^2 \phi$$

Sol.124.(d) $\alpha \sin 45^\circ = \beta \operatorname{cosec} 30^\circ$

So that, $\frac{\alpha}{\beta} = 2\sqrt{2}$

$$\frac{\alpha^4}{\beta^4} = (2\sqrt{2})^4 = 64 = (4)^3$$

Sol.125.(c) $1 + \cot^2 B = \operatorname{cosec}^2 B$

$$\cot^2 B - \operatorname{cosec}^2 B = -1$$

Sol.126.(b) $\sin 60^\circ \sec 30^\circ + \tan 45^\circ$
 $\cos 0^\circ + \sec 60^\circ \sin 30^\circ$

$$= \frac{\sqrt{3}}{2} \times \frac{2}{\sqrt{3}} + 1 \times 1 + 2 \times \frac{1}{2} = 1 + 1 + 1 = 3$$

Sol.127.(d) $\sin \theta \cdot \sin 45^\circ = \sin 30^\circ$

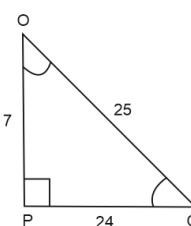
$$\sin \theta \times \frac{1}{\sqrt{2}} = \frac{1}{2}$$

$$\sin \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

So that, $2\theta = 90^\circ$

Sol.128.(c)

Pythagoras triplets:- (7, 24, 25)



$$\sin Q + \cos Q = \frac{7}{25} + \frac{24}{25} = \frac{31}{25}$$

$$\begin{aligned} \text{Sol.129.(a)} \quad \frac{2 \sec A}{\sec^2 A - 1} &= \frac{2 \cot^2 A}{\cos A} \\ &= \frac{2 \cos A}{\sin^2 A} = \frac{2 \cot A}{\sin A} = 2 \operatorname{cosec} A \cdot \cot A \end{aligned}$$

$$\begin{aligned} \text{Sol.130.(a)} \quad \frac{\cos^2 A}{(1+\sin A)} &= \frac{1-\sin^2 A}{(1+\sin A)} \\ &= \frac{(1+\sin A)(1-\sin A)}{(1+\sin A)} = (1-\sin A) \end{aligned}$$

Sol.131.(b)

$$\begin{aligned} \text{If } A + B = 90^\circ \text{ then } \sin^2 A + \sin^2 B &= 1 \\ \sin^2 30^\circ + \sin^2 40^\circ + \sin^2 45^\circ + \sin^2 55^\circ &+ \\ \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 50^\circ + \sin^2 60^\circ &= \\ \sin^2 30^\circ + \sin^2 60^\circ + \sin^2 40^\circ + \sin^2 50^\circ &+ \\ \sin^2 45^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 35^\circ &= \\ 1 + 1 + 1 + 1 &= 4 \end{aligned}$$

Sol.132.(a) If $A + B = 90^\circ$, Then

$$\begin{aligned} \sin^2 A + \sin^2 B &= 1 \\ \text{and } \cos^2 A + \cos^2 B &= 1 \\ \frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos^2 53^\circ &= \\ \sin^2 37^\circ &= \\ \frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos^2 53^\circ &= \\ \frac{1}{1} + 1 &= 2 \end{aligned}$$

$$\begin{aligned} \text{Sol.133.(b)} \quad \frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta} &= \\ \operatorname{cosec} \theta - \frac{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)}{1 + \operatorname{cosec} \theta} &= \\ \operatorname{cosec} \theta - \operatorname{cosec} \theta + 1 &= 1 \end{aligned}$$

Sol.134.(a)

$$\begin{aligned} \frac{\sin 58^\circ}{\cos 32^\circ} + \frac{\sin 55^\circ \cdot \sec 35^\circ}{\tan 5^\circ \cdot \tan 45^\circ \cdot \tan 85^\circ} &= \\ \frac{\cos 32^\circ}{\cos 32^\circ} + \frac{\sin 55^\circ \cdot \operatorname{cosec} 55^\circ}{\tan 5^\circ \cdot \tan 85^\circ} &= \\ 1 + \frac{1}{\tan 5^\circ \cdot \cot 5^\circ} &= 1 + 1 = 2 \end{aligned}$$

$$\text{Sol.135.(c)} \quad \sec \theta = \frac{50}{23} \Rightarrow \sec^2 \theta = \frac{2500}{529}$$

$$\begin{aligned} \tan^2 \theta &= \sec^2 \theta - 1 = \frac{2500 - 529}{529} \\ &= \frac{1971}{529} = 3 \frac{384}{529} \end{aligned}$$

$$\begin{aligned} \text{Sol.136.(d)} \quad \frac{\cos^2 25^\circ - \sin^2 65^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} &= \\ \frac{\cos^2 25^\circ - \cos^2 25^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} &= 0 \end{aligned}$$

$$\begin{aligned} \text{Sol.137.(b)} \quad \frac{\cot^3 A - 1}{\cot A - 1} &= \\ = \frac{(\cot A - 1)(\cot^2 A + \cot A + 1)}{\cot A - 1} &= \\ = (\operatorname{cosec}^2 A + \cot A) & \end{aligned}$$

Sol.138.(d) If $\sec \theta + \tan \theta = a$

$$\text{Then, } \sec \theta - \tan \theta = \frac{1}{a}$$

on solving above equation, we get

$$\sec \theta = \frac{a + \frac{1}{a}}{2} = \frac{a^2 + 1}{2a}$$

$$\begin{aligned} \text{Sol.139.(d)} \quad \frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} + 2 &= \\ = \frac{1 - \sin A}{\cos A} + \frac{1 + \sin A}{\cos A} + 2 &= \\ = \frac{1 - \sin A + 1 + \sin A}{\cos A} + 2 &= \\ = 2 \sec A + 2 = 2(1 + \sec A) & \end{aligned}$$

Sol.140.(d)

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{13}{19}$$

$$\text{Base} = \sqrt{19^2 - 13^2} = 8\sqrt{3}$$

$$\text{Now, } \cos \theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{8\sqrt{3}}{19}$$

$$\text{Sol.141.(a)} \quad \sec \theta + \tan \theta = \frac{1}{\sqrt{3}}$$

$$\sec \theta - \tan \theta = \sqrt{3}$$

$$\sec \theta = \frac{\frac{1}{\sqrt{3}} + \sqrt{3}}{2} = \frac{2}{\sqrt{3}} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$$

Therefore, $\theta = 30^\circ$

$$\text{Now, } \cot \theta + \cos \theta = \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

$$\text{Sol.142.(c)} \quad \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} = \frac{4}{5}$$

$$\frac{\sin \theta - \cos \theta + \sin \theta + \cos \theta}{\sin \theta - \cos \theta - \sin \theta - \cos \theta} = \frac{4 + 5}{4 - 5}$$

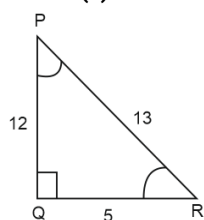
$$\frac{2 \sin \theta}{-2 \cos \theta} = \frac{9}{-1} \Rightarrow \cot \theta = \frac{1}{9}$$

$$\operatorname{cosec}^2 \theta = 1 + \left(\frac{1}{9}\right)^2 = \left(\frac{82}{81}\right)$$

$$\text{Now, } \frac{\operatorname{cosec}^2 \theta}{2 - \operatorname{cosec}^2 \theta} = \frac{\frac{82}{81}}{2 - \frac{82}{81}}$$

$$= \frac{82}{80} = \frac{41}{40}$$

Sol.143.(c)



$$\tan P + \cot R = \frac{5}{12} + \frac{5}{12} = \frac{10}{12}$$

$$\text{Sol.144.(d)} \quad \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} = \frac{\frac{5}{6}}{1 - \frac{1}{6}} = 1$$

$$\Rightarrow A + B = 45^\circ$$

$$\text{Sol.145.(d)} \quad \tan A = \frac{12}{5}$$

$$\begin{aligned} \frac{5(1 - \cos^2 A)}{6(1 - \sin^2 A)} &= \frac{5(\sin^2 A)}{6(\cos^2 A)} = \frac{5}{6} \times \tan^2 A \\ &= \frac{5}{6} \times \frac{144}{25} = \frac{24}{5} \end{aligned}$$

$$\begin{aligned} \text{Sol.146.(b)} \quad \sin(X - Y) &= \sin(60^\circ - 30^\circ) \\ &= \sin(30^\circ) = \frac{1}{2} \end{aligned}$$

Sol.147.(b)

$$\cos 2\theta = 1 - 2\sin^2 \theta = 2\cos^2 \theta - 1$$

Sol.148.(c) $\tan \theta \cdot \tan 2\theta = 1$

Put $\theta = 30^\circ$, Then

$$\tan 30^\circ \cdot \tan 60^\circ = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

....(satisfied)

$$\cot 5\theta = \cot(150^\circ) = \cot(90^\circ + 60^\circ)$$

$$= -\tan 60^\circ = -\sqrt{3}$$

$$\text{Sol.149.(c)} \quad 4\sin^2 A - 3 = 0$$

Multiply by $\sin A$, both sides

$$4\sin^3 A - 3\sin A = 0 \text{ or}$$

$$3\sin A - 4\sin^3 A = 0$$

$$\text{Sol.150.(b)} \quad \sin A = \frac{2}{5} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\Rightarrow \text{Base} = \sqrt{25 - 4} = \sqrt{21}$$

$$\text{Now, } \frac{5 \sin A + 2 \operatorname{cosec} A}{\sqrt{21} \sec A}$$

$$= \frac{5 \times \frac{2}{5} + 2 \times \frac{5}{2}}{\sqrt{21} \times \frac{5}{\sqrt{21}}} = \frac{7}{5}$$

$$\text{Sol.151.(a)} \quad \tan \alpha = \cot \beta \Rightarrow \alpha = \beta = 45^\circ$$

Here,

$$\tan(5\theta - 10^\circ) = \cot(5\varphi + 20^\circ)$$

$$(5\theta - 10^\circ) = 45^\circ \Rightarrow \theta = 11^\circ$$

$$(5\varphi + 20^\circ) = 45^\circ \Rightarrow \varphi = 5^\circ$$

$$\text{Now, } \theta + \varphi = 16^\circ$$

Sol.152.(d)

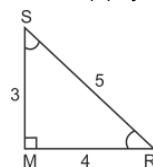
$$\cos^2 36^\circ + \cos 54^\circ \cdot \sin 36^\circ + \left(\frac{\tan 26^\circ}{\cot 64^\circ}\right)$$

$$= \cos^2 36^\circ + \sin 36^\circ \cdot \sin 36^\circ + \left(\frac{\tan 26^\circ}{\tan 26^\circ}\right)$$

$$= \cos^2 36^\circ + \sin^2 36^\circ + (1)$$

$$= (1) + (1) = 2$$

Sol.153.(b) Pythagoras triplets :- (3, 4, 5)



$$\sec R = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{5}{4}$$

$$\text{Sol.154.(b)} \quad \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$= \tan(A - B)$$

$$= \tan(60^\circ - 30^\circ) = \tan(30^\circ) = \frac{1}{\sqrt{3}}$$

$$\text{Sol.155.(d)} \quad 2(\tan^2 A \cdot \cos^2 A + \cos^2 A)$$

$$= 2\cos^2 A (\tan^2 A + 1)$$

$$= 2\cos^2 A (\sec^2 A) = 2$$

Sol.156.(d)

$$\tan(A + B - C) = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\Rightarrow (A + B - C) = 30^\circ \dots \text{e.q. (1)}$$

$$\cos(B + C - A) = \frac{1}{2} = \cos 60^\circ$$

$$\Rightarrow (B + C - A) = 60^\circ \dots \text{e.q. (2)}$$

$$\sin(C + A - B) = \frac{1}{\sqrt{2}} = \sin 45^\circ$$

$$\Rightarrow (C + A - B) = 45^\circ \dots \text{e.q. (3)}$$

Adding e.q. (1), e.q. (2) and e.q. (3), we get

$$A + B + C = 30^\circ + 60^\circ + 45^\circ = 135^\circ$$

$$\text{Sol.157.(d)} \frac{(1 + \sec \phi)}{\sec \phi} \times (1 - \cos \phi)$$

$$= (1 + \cos \phi)(1 - \cos \phi)$$

$$= (1 - \cos^2 \phi) = \sin^2 \phi$$

Sol.158.(c) $\sec C + \sin C \cdot \sec C$

$$= \sec C + \tan C = \sec 45^\circ + \tan 45^\circ$$

$$= \sqrt{2} + 1$$

$$\text{Sol.159.(b)} 4\sin \theta + 5\cos \theta = 3$$

$$4\cos \theta - 5\sin \theta = x$$

$$\dots \dots \dots$$

$$(4)^2 + (5)^2 = (3)^2 + (x)^2$$

$$x = \sqrt{16 + 25 - 9} = \sqrt{32} = 4\sqrt{2}$$

Sol.160.(d)

If $(A + B) = 90^\circ$, then $\tan A = \cot B$

Here,

$$(2\theta) + (\theta - 36^\circ) = 90^\circ$$

$$3\theta = 126^\circ \Rightarrow \theta = 42^\circ$$

$$\text{Sol.161.(a)} \cot \theta = \frac{2}{\sqrt{3}} \Rightarrow \tan \theta = \frac{\sqrt{3}}{2}$$

$$\text{Now, } 2\sqrt{3} \times \tan \theta + 1 = 2\sqrt{3} \times \frac{\sqrt{3}}{2} + 1$$

$$= 3 + 1 = 4$$

Sol.162.(c)

If $\sec \theta + \tan \theta = 5$ Eq. 1, Then

$$\sec \theta - \tan \theta = \frac{1}{5} \dots \text{Eq. 2}$$

Adding both the equation, we get

$$2\sec \theta = 5 + \frac{1}{5}$$

$$\Rightarrow \sec \theta = \frac{13}{5} = \frac{\text{Hypotenuse}}{\text{Base}}$$

$$\text{Perpendicular} = 12$$

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{12}{13}$$

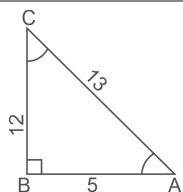
$$\text{Sol.163.(d)} \sin 2\theta = 2\sin \theta \cos \theta$$

$$= 2 \times \frac{6}{10} \times \frac{8}{10} = \frac{24}{25}$$

Sol.164.(b)

Pythagorean triplets:- (5,12,13)

$$\tan A = \frac{\text{Perpendicular}}{\text{Base}} = \frac{12}{5}$$



$$\operatorname{cosec} A = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{13}{12}$$

$$\sec A = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{13}{5}$$

$$\text{Sol.165.(a)} \cos \theta = \sin \theta \Rightarrow \theta = 45^\circ$$

$$\text{Now, } \sin^8 \theta + \cos^8 \theta = \left(\frac{1}{\sqrt{2}}\right)^8 + \left(\frac{1}{\sqrt{2}}\right)^8$$

$$= \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$\text{Sol.166.(d)} \frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z}$$

$$= \frac{3[(\sin^2 Z + \cos^2 Z)(\sin^2 Z - \cos^2 Z) + 1]}{\sin^2 Z} \because$$

$$(\sin^2 Z + \cos^2 Z = 1)$$

$$= \frac{3(\sin^2 Z - \cos^2 Z + 1)}{\sin^2 Z}$$

$$= \frac{3(\sin^2 Z + \sin^2 Z)}{\sin^2 Z} \because (1 - \cos^2 Z = \sin^2 Z)$$

$$= \frac{3 \times 2\sin^2 Z}{\sin^2 Z} = 6$$

Short trick : Put $Z = 90^\circ$

$$\frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z} = \frac{3(1 - 0 + 1)}{1} = 6$$

Sol.167.(a) According to the question,

$$\text{Let } \theta = 45^\circ$$

$$\tan \theta + \cot \theta = 2$$

$$1 + 1 = 2 \text{ (satisfied)}$$

Now,

$$2 \times 1^2 + 1^2 + 1^4 \times 1^4 = 4$$

Sol.168.(d) According to the question,

$$\sin 30^\circ = \frac{1}{2}$$

$$3 \sin A - 4 \sin^3 A = 3 \times \frac{1}{2} - 4 \times \left(\frac{1}{2}\right)^3 = 1$$

Sol.169.(c)

$$\sin 47^\circ \cos 33^\circ - \cos 43^\circ \sin 57^\circ$$

$$= \sin (90^\circ - 43^\circ) \cos 33^\circ - \cos 43^\circ$$

$$\sin (90^\circ - 33^\circ)$$

$$= \cos 43^\circ \cos 33^\circ - \cos 43^\circ \cos 33^\circ = 0$$

Sol.170.(c) $\cos 5\alpha = \sin \alpha$

$$\cos 5\alpha = \cos (90^\circ - \alpha)$$

$$5\alpha = 90^\circ - \alpha \Rightarrow \alpha = \frac{90^\circ}{6} = 15^\circ$$

$$\text{So, } \tan 2\alpha = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\text{Sol.171.(a)} \frac{1 + \sin A}{\cos A} + \frac{\cos A}{1 + \sin A}$$

$$= \frac{(1 + \sin A)^2 + (\cos A)^2}{(1 + \sin A)\cos A}$$

$$= \frac{1 + \sin^2 A + 2\sin A + \cos^2 A}{(1 + \sin A)\cos A}$$

$$= \frac{2 + 2\sin A}{(1 + \sin A)\cos A}$$

$$= \frac{2(1 + \sin A)}{(1 + \sin A)\cos A} = 2 \sec A$$

Sol.172.(b) $(1 + \tan^2 A)(1 + \cot^2 A)$

$$(\sec^2 A)(\operatorname{cosec}^2 A) = \frac{1}{\cos^2 A} \times \frac{1}{\sin^2 A}$$

$$= \frac{1}{\sin^2 A(1 - \sin^2 A)}$$

Sol.173.(c) According to the question,

$$\frac{\cos B}{\sin B} - \frac{\sin B}{\cos B} \times (\sin B \cdot \cos B)$$

$$= \frac{(\cos B)^2 - (\sin B)^2}{\sin B \cdot \cos B} \times (\sin B \cdot \cos B)$$

$$= 2\cos^2 B - 1$$

Sol.174.(d) Given:- $\cot^2 \theta - \operatorname{cosec}^2 \theta = ?$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

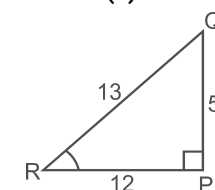
$$-(\cot^2 \theta - \operatorname{cosec}^2 \theta) = 1$$

$$\cot^2 \theta - \operatorname{cosec}^2 \theta = -1$$

$$\text{Sol.175.(b)} \tan A = \frac{\text{perpendicular}}{\text{base}} = \frac{3}{4}$$

$$\text{Hypotenuse} = \sqrt{(3)^2 + (4)^2} = 5$$

$$\text{Then, } \frac{6 \sin A}{1 - \sin A} = \frac{6 \times \frac{3}{5}}{1 - \frac{3}{5}} = 9$$

Sol.176.(b)

$$\tan Q = \frac{12}{5} \text{ and } \tan R = \frac{5}{12}$$

$$\text{So, } \tan Q - \tan R = \frac{12}{5} - \frac{5}{12}$$

$$= \frac{144 - 25}{60} = \frac{119}{60}$$

Sol.177.(d) According to the question,

$$(\sin^4 \theta + \cos^4 \theta) = (\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cdot \cos^2 \theta$$

$$\text{We know that } (\sin^2 \theta + \cos^2 \theta) = 1$$

$$1^2 - 2 \times \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

Sol.178.(d) According to the question,

$$\sqrt{\frac{1 + \cos P \times (1 + \cos P)}{1 - \cos P \times (1 + \cos P)}} = \frac{1 + \cos P}{\sin P}$$

$$= \operatorname{cosec} P + \cot P$$

Sol.179.(c) $\tan(50^\circ + \theta) - \cot(40^\circ - \theta)$

$$= \tan(50^\circ + \theta) - \cot[90^\circ - (50^\circ + \theta)] \{ \cot(90^\circ - \theta) = \tan \theta \}$$

$$= \tan(50^\circ + \theta) - \tan(50^\circ + \theta) = 0$$

Sol.180.(a) $(1 - \sin^2 A) \operatorname{cosec}^2 A$

$$= (\cos^2 A) \operatorname{cosec}^2 A = \frac{\cos^2 A}{\sin^2 A} = \cot^2 A$$

Sol.181.(d)

$$\begin{aligned}\tan 15^\circ &= 2 - \sqrt{3} \Rightarrow \cot 15^\circ = 2 + \sqrt{3} \\ \tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ \\ &= \tan 15^\circ \cot(90^\circ - 15^\circ) + \tan(90^\circ - 15^\circ) \cot 15^\circ \\ &= \tan 15^\circ \tan 15^\circ + \cot 15^\circ \cot 15^\circ \\ &= (\tan 15^\circ)^2 + (\cot 15^\circ)^2 \\ &= (2 - \sqrt{3})^2 + (2 + \sqrt{3})^2 \\ &= 4 + 3 - 4\sqrt{3} + 4 + 3 + 4\sqrt{3} = 14\end{aligned}$$

Sol.182.(a) $\sin(a+b) = 1 \Rightarrow (a+b) = 90^\circ$

$$\cos(a-b) = \frac{1}{2} \Rightarrow (a-b) = 60^\circ$$

$$a = \frac{90^\circ + 60^\circ}{2} = 75^\circ$$

Sol.183.(c) $\frac{3\sin\theta - \cos\theta}{\cos\theta + \sin\theta} = 1$

$$2\sin\theta = 2\cos\theta \Rightarrow \frac{\cos\theta}{\sin\theta} = \frac{2}{2} \Rightarrow \cot\theta = 1$$

Sol.184.(d) Formula used : $\cos(A-B)$

$$\begin{aligned}&= \cos A \cdot \cos B + \sin A \cdot \sin B \\ \cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos(54^\circ + A) \cdot \cos(54^\circ - A) \\ &\Rightarrow \cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos[90^\circ - (36^\circ - A)] \cdot \cos[90^\circ - (36^\circ + A)] \\ &\Rightarrow \cos(36^\circ + A) \cdot \cos(36^\circ - A) + \sin(36^\circ + A) \sin(36^\circ - A) \\ &\Rightarrow \cos[36^\circ + A - (36^\circ - A)] \Rightarrow \cos(2A)\end{aligned}$$

Sol.185.(b) $\sec^2 54^\circ - \cot^2 36^\circ + \frac{3}{2}$

$$\begin{aligned}\sin^2 37^\circ \times \sec^2 53^\circ + \frac{2}{\sqrt{3}} \tan 60^\circ \\ \Rightarrow \sec^2(90^\circ - 36^\circ) - \cot^2 36^\circ + \frac{3}{2} \\ \sin^2 37^\circ \times \sec^2(90^\circ - 37^\circ) + \frac{2}{\sqrt{3}} \tan 60^\circ \\ \Rightarrow \operatorname{cosec}^2 36^\circ - \cot^2 36^\circ + \frac{3}{2} \\ \sin^2 37^\circ \times \operatorname{cosec}^2 37^\circ + \frac{2}{\sqrt{3}} \tan 60^\circ \\ \Rightarrow 1 + \frac{3}{2} + \frac{2}{\sqrt{3}} \times \sqrt{3} \Rightarrow \frac{9}{2}\end{aligned}$$

Sol.186.(d)

$$\sin 23^\circ = \frac{a}{b} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\text{Base} = \sqrt{b^2 - a^2}$$

$$\sec 23^\circ - \sin 67^\circ = \sec 23^\circ - \cos 23^\circ$$

$$\begin{aligned}&= \frac{b}{\sqrt{b^2 - a^2}} - \frac{\sqrt{b^2 - a^2}}{b} \\ &= \frac{b^2 - b^2 + a^2}{b\sqrt{b^2 - a^2}} = \frac{a^2}{b\sqrt{b^2 - a^2}}\end{aligned}$$

Sol.187.(a) $\frac{\cos \beta}{\sec \alpha} = 15$ and $\frac{\sin \beta}{\sec \alpha} = 16$

$$\frac{\cos \beta}{\sec \alpha} \times \frac{\sec \alpha}{\sin \beta} = \frac{15}{16}$$

$$\Rightarrow \cot \beta = \frac{15}{16} = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\text{Hypotenuse} = \sqrt{225 + 256} = \sqrt{481}$$

$$\sin^2 \beta = \left(\frac{\text{Perpendicular}}{\text{hypotenuse}} \right)^2$$

$$= \left(\frac{16}{\sqrt{481}} \right)^2 = \frac{256}{481}$$

Sol.188.(d) $\sec^2 \theta - \tan^2 \theta = 1$

$$\sec^2 \theta + \tan^2 \theta = \frac{25}{18}$$

$$\text{Now, } \sec^4 \theta - \tan^4 \theta$$

$$\begin{aligned}&= (\sec^2 \theta + \tan^2 \theta) \times (\sec^2 \theta - \tan^2 \theta) \\ &= \frac{25}{18} \times 1 = \frac{25}{18}\end{aligned}$$

Sol.189.(c) $\frac{\sin^3 \alpha + \cos^3 \alpha}{\sin \alpha + \cos \alpha}$

$$\begin{aligned}&= \frac{(\sin \alpha + \cos \alpha)(\sin^2 \alpha + \cos^2 \alpha - \sin \alpha \cos \alpha)}{\sin \alpha + \cos \alpha} \\ &= 1 - \sin \alpha \cos \alpha\end{aligned}$$

Sol.190.(a) According to question,

$$\tan \frac{\pi}{6} + \sec \frac{\pi}{6} = x \Rightarrow \tan 30^\circ + \sec 30^\circ = x$$

$$\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \Rightarrow x = \sqrt{3}$$

Sol.191.(a) x lies in third quadrant In third quadrant only $\tan \theta$ and $\cot \theta$ are positive

$$\text{Now, } x = \cos^{-1}\left(-\frac{1}{2}\right) = 60^\circ$$

$$\text{Then, } \tan x = \tan(60^\circ) = \sqrt{3}$$

Sol.192.(c) According to the question,

$$\begin{aligned}\cos 75^\circ &= \cos(45^\circ + 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3}-1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4}\end{aligned}$$

Sol.193.(a) According to the question,

$$\begin{aligned}&\frac{\sin^4 \theta + \cos^4 \theta}{1 - 2\sin^2 \theta \cos^2 \theta} \\ &= \frac{(\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cos^2 \theta}{1 - 2\sin^2 \theta \cos^2 \theta} \\ &= \frac{1 - 2\sin^2 \theta \cos^2 \theta}{1 - 2\sin^2 \theta \cos^2 \theta} = 1\end{aligned}$$

Sol.194.(d) $\left(\frac{\sin(x-y)}{\sin(x+y)} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right)$

$$\begin{aligned}&= \left(\frac{\sin x \cos y - \cos x \sin y}{\sin x \cos y + \cos x \sin y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right) \\ &= \left(\frac{\sin x \cos y - \cos x \sin y}{\cos x \cos y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right) \\ &= \left(\frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right) \\ &= \left(\frac{\tan x - \tan y}{\tan x + \tan y} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right) = 1\end{aligned}$$

Sol.195.(c) $\sin\left(\frac{A}{2}\right) = \sqrt{\frac{(1-\cos A)}{2}}$

$$\text{And, } \cos\left(\frac{A}{2}\right) = \sqrt{\frac{(1+\cos A)}{2}}$$

$$\text{Given, } \tan \frac{A}{2} = x$$

$$= \sqrt{\frac{(1-\cos A)}{2}} \div \sqrt{\frac{(1+\cos A)}{2}}$$

$$= \sqrt{\frac{(1-\cos A)}{2}} \times \sqrt{\frac{2}{(1+\cos A)}}$$

$$= \sqrt{\frac{(1-\cos A)}{(1+\cos A)}}$$

$$\text{Formula :- } \tan \frac{A}{2} = \sqrt{\frac{(1-\cos A)}{(1+\cos A)}}$$

Sol.196.(d)

$$\left(\frac{1}{\sin \theta} + \frac{1}{\tan \theta} \right) \left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta} \right)$$

$$= \left(\frac{1}{\sin^2 \theta} - \frac{1}{\tan^2 \theta} \right) \Rightarrow (\operatorname{cosec}^2 \theta - \cot^2 \theta) = 1$$

Sol.197.(a) $\sin 2x + 2 \sin 4x + \sin 6x$

$$2 \sin \left(\frac{6x+2x}{2} \right) \cos \left(\frac{6x-2x}{2} \right) + 2 \sin 4x$$

$$= 2 \sin 4x \cos 2x + 2 \sin 4x$$

$$= 2 \sin 4x (\cos 2x + 1)$$

$$= 2 \sin 4x \{2 \cos^2 x - 1 + 1\}$$

$$= 2 \sin 4x (2 \cos^2 x) = 4 \cos^2 x \sin 4x$$

Sol.198.(b) In third quadrant, only value of $\tan \theta$ and $\cot \theta$ is positive.

$$\text{Given that, } \sin A = -\frac{3}{5}$$

$$= \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

Using pythagoras theorem,

$$H^2 = P^2 + B^2 \Rightarrow B^2 = 25 - 9 = 16$$

$$B = \sqrt{16} = \pm 4$$

$$\text{Now, } \sec A = \frac{\text{Hypotenuse}}{\text{Base}} = -\frac{5}{4}$$

Sol.199.(c) $\sin A = \cos B \Rightarrow A+B = 90^\circ$

$$\sin(5x - 25^\circ) = \cos(5y + 25^\circ)$$

$$5x - 25^\circ + 5y + 25^\circ = 90^\circ$$

$$5x + 5y = 90^\circ \Rightarrow x + y = \frac{90^\circ}{5} = 18^\circ$$

Sol.200.(a) $12(\sin^4 \theta + \cos^4 \theta) +$

$$18(\sin^6 \theta + \cos^6 \theta) + 78\sin^2 \theta \cos^2 \theta$$

$$\text{Put } \theta = 0^\circ$$

$$= 12(0+1) + 18(0+1) + 78 \times 0 \times 1 = 30$$

Sol.201.(c) Let $A = 45^\circ$ and $B = 30^\circ$

According to question,

$$\tan(45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$$

$$\tan 15^\circ = \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \times \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$= \frac{3+1-2\sqrt{3}}{3-1} = 2 - \sqrt{3}$$

Sol.202.(b) $\frac{\sin x - \cos x}{\sin x + \cos x} = \frac{2}{5}$

$$5 \sin x - 5 \cos x = 2 \sin x + 2 \cos x$$

$$3 \sin x = 7 \cos x$$

$$\frac{\cos x}{\sin x} = \frac{3}{7} \Rightarrow \cot x = \frac{3}{7} \Rightarrow \cot^2 x = \frac{9}{49}$$

$$\text{Now, } \frac{1 + \cot^2 x}{1 - \cot^2 x} = \frac{1 + \frac{9}{49}}{1 - \frac{9}{49}} = \frac{\frac{49+9}{49}}{\frac{49-9}{49}} = \frac{58}{40} \times \frac{49}{40} = \frac{58}{40} = 1.45$$

$$\begin{aligned} \text{Sol.203.(c)} (\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 \\ = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta \\ = 2(\sin^2\theta + \cos^2\theta) = 2 \times 1 = 2 \end{aligned}$$

$$\text{Sol.204.(c)} \left(\frac{\cos A}{1 - \sin A}\right) + \left(\frac{\cos A}{1 + \sin A}\right) = 4$$

$$\cos A \left(\frac{1 + \sin A + 1 - \sin A}{(1 - \sin A)(1 + \sin A)}\right) = 4$$

$$\cos A \left(\frac{2}{1 - \sin^2 A}\right) = 4$$

$$\cos A \left(\frac{2}{\cos^2 A}\right) = 4$$

$$\cos A = \frac{1}{2} \Rightarrow A = 60^\circ$$

$$\text{Sol.205.(a)} \cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

$$\cot 15^\circ = \cot(45^\circ - 30^\circ)$$

$$= \frac{\cot 45^\circ \cot 30^\circ + 1}{\cot 30^\circ - \cot 45^\circ}$$

$$= \frac{1 \times \sqrt{3} + 1}{\sqrt{3} - 1} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{4 + 2\sqrt{3}}{3 - 1} = 2 + \sqrt{3}$$

$$\text{Sol.206.(c)} P \cos \alpha = 3 \Rightarrow \sec \alpha = \frac{P}{3}$$

taking square both sides,

$$\sec^2 \alpha = \frac{P^2}{9} \quad \text{-----(1)}$$

$$\text{Now, } 4 \tan \alpha = Q \Rightarrow \tan \alpha = \frac{Q}{4}$$

taking square both sides

$$\tan^2 \alpha = \frac{Q^2}{16} \quad \text{-----(2)}$$

from equation (1) and (2), we get

$$\sec^2 \alpha - \tan^2 \alpha = \frac{P^2}{9} - \frac{Q^2}{16}$$

$$\therefore \frac{P^2}{9} - \frac{Q^2}{16} = 1$$

$$\text{Sol.207.(b)} \sin\theta + \sin^2\theta = 1$$

$$\sin\theta = 1 - \sin^2\theta$$

$$\sin\theta = \cos^2\theta \Rightarrow \sin^2\theta = \cos^4\theta$$

Now,

$$\cos^2\theta + \cos^4\theta = \cos^2\theta + \sin^2\theta = 1$$

$$\text{Sol.208.(a)} (a - b)^2 + (a + b)^2 = 2(a^2 + b^2)$$

$$(\cos x - \sin x)^2 + (\cos x + \sin x)^2$$

$$= 2(\cos^2 x + \sin^2 x) = 2$$

$$\text{Sol.209.(a)} \frac{\sec^2(90^\circ - \theta) - \cot^2\theta}{2(\sin^2 35^\circ + \sin^2 55^\circ)}$$

$$\frac{\operatorname{cosec}^2(\theta) - \cot^2\theta}{2(\sin^2 35^\circ + \cos^2 35^\circ)} = \frac{1}{2} = 0.5$$

$$\text{Sol.210.(c)} \operatorname{cosec} 2910^\circ = \operatorname{cosec} (360^\circ \times 8 + 30^\circ) = \operatorname{cosec} 30^\circ = 2$$

$$\sec 4260^\circ = \sec (360^\circ \times 11 + 300^\circ)$$

$$= \sec 300^\circ = \sec(360^\circ - 60^\circ) = \sec 60^\circ = 2$$

$$\tan 2565^\circ = \tan (360^\circ \times 7 + 45^\circ)$$

$$= \tan(45^\circ) = 1$$

$$\cot 1755^\circ = \cot (360^\circ \times 4 + 315^\circ)$$

$$= \cot(315^\circ) = \cot(270^\circ + 45^\circ)$$

$$= -\tan 45^\circ = -1$$

Now,

$$\operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ +$$

$$\cot 1755^\circ \Rightarrow 2 + 2 + 1 - 1 = 4$$

$$\text{Sol.211.(c)} \text{ For the maximum value of}$$

$$\sin^{12}\theta + \cos^{20}\theta, \text{ Put } \theta = 90^\circ,$$

$$\sin^{12}(90^\circ) + \cos^{20}(90^\circ) = 1 + 0 = 1$$

$$\text{Sol.212.(c)} \frac{2}{3} \tan^2 60^\circ + 3 \cos^2 30^\circ - 2$$

$$\sec^2 30^\circ - \frac{3}{4} \cot^2 60^\circ$$

$$= \frac{2}{3}(\sqrt{3})^2 + 3\left(\frac{\sqrt{3}}{2}\right)^2 - 2\left(\frac{2}{\sqrt{3}}\right)^2 - \frac{3}{4}\left(\frac{1}{\sqrt{3}}\right)^2$$

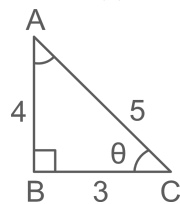
$$= 2 + \frac{9}{4} - \frac{8}{3} - \frac{1}{4} = 2 + 2 - \frac{8}{3}$$

$$= \frac{4}{3} = 1.33$$

$$\text{Sol.213.(d)} \sin(-405^\circ) = -\sin(360^\circ + 45^\circ)$$

$$= -\sin 45^\circ = \frac{-1}{\sqrt{2}}$$

$$\text{Sol.214.(a)}$$



From pythagoras theorem,

$$\text{Perpendicular} = \sqrt{25 - 9} = 4$$

$$\text{Now, } \tan \theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{AB}{BC} = \frac{4}{3}$$

$$\text{Sol.215.(a)} \cos 2\theta = (2\cos^2\theta - 1)$$

$$= (1 - 2\sin^2\theta)$$

$$\text{Now, } \frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \frac{\sin\theta(1 - 2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)} = \tan\theta$$

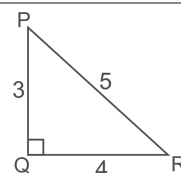
$$\text{Sol.216.(b)} \frac{1 + \sin t}{4 - 4 \sin t} - \frac{1 - \sin t}{4 + 4 \sin t}$$

$$= \frac{1}{4} \left[\frac{1 + \sin t}{(1 - \sin t)} - \frac{1 - \sin t}{(1 + \sin t)} \right]$$

$$= \frac{1}{4} \left[\frac{(1 + \sin t)^2 - (1 - \sin t)^2}{(1 - \sin t)(1 + \sin t)} \right]$$

$$= \frac{1}{4} \left[\frac{4 \sin t}{\cos^2 t} \right] = \frac{\sin t}{\cos t} \times \frac{1}{\cos t} = \tan t \sec t$$

$$\text{Sol.217.(c)}$$



$$\sin P = \frac{4}{5}, \sin Q = 90^\circ = 1 \text{ And, } \sin R = \frac{3}{5}$$

$$\text{Hence, } \sin P + \sin Q + \sin R$$

$$= \frac{4}{5} + 1 + \frac{3}{5} = \frac{12}{5}$$

$$\text{Sol.218.(c) Given, } \cot \theta = \frac{3}{2}$$

$$= \frac{\text{Base}}{\text{Perpendicular}}$$

By using pythagoras theorem, we get

$$\text{Hypotenuse} = \sqrt{13}$$

$$\text{Now, } \frac{\sqrt{13}\sin\theta - 3\tan\theta}{3\tan\theta + \sqrt{13}\cos\theta}$$

$$= \frac{\sqrt{13} \times \frac{2}{\sqrt{13}} - 3 \times \frac{2}{3}}{3 \times \frac{2}{3} + \sqrt{13} \times \frac{3}{\sqrt{13}}} = 0$$

$$\text{Sol.219.(a)}$$

$$\tan(45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{(1 + \tan 45^\circ \tan 30^\circ)}$$

$$= \frac{1 - \frac{1}{\sqrt{3}}}{(1 + 1 \times \frac{1}{\sqrt{3}})} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = 0.268$$

Short trick :-

$$\tan 15^\circ = (2 - \sqrt{3}) = 2 - 1.732 = 0.268$$

$$\text{Sol.220.(c) If } \operatorname{Cosec}\theta + \cot\theta = m,$$

$$\text{Then } \operatorname{Cosec}\theta - \cot\theta = \frac{1}{m}$$

$$\text{Now, } \frac{m^2 - 1}{m^2 + 1} = \frac{m - \frac{1}{m}}{m + \frac{1}{m}}$$

$$= \frac{\operatorname{Cosec}\theta + \cot\theta - (\operatorname{Cosec}\theta - \cot\theta)}{\operatorname{Cosec}\theta + \cot\theta + (\operatorname{Cosec}\theta - \cot\theta)}$$

$$= \frac{2\cot\theta}{2\operatorname{Cosec}\theta} = \cos\theta$$

$$\text{Sol.221.(c)}$$

$$\text{If } A + B = 90^\circ, \text{ then } \tan A \times \tan B = 1$$

$$\tan 5^\circ \tan 25^\circ \tan 45^\circ \tan 65^\circ \tan 85^\circ$$

$$= (\tan 5^\circ \times \tan 85^\circ) \times (\tan 25^\circ \times \tan 65^\circ) \times \tan 45^\circ = 1 \times 1 \times 1 = 1$$

$$\text{Sol.222.(c)} \tan x = \frac{7}{5}$$

$$\frac{9 \sin x - \frac{42}{5} \cos x}{15 \sin x + 21 \cos x} = \frac{9 \tan x - \frac{42}{5}}{15 \tan x + 21}$$

$$= \frac{9 \times \frac{7}{5} - \frac{42}{5}}{15 \times \frac{7}{5} + 21} = \frac{63 - 42}{105 + 105} = \frac{21}{210} = 0.1$$

$$\text{Sol.223.(d)} \sin\theta + \operatorname{cosec}\theta = \sqrt{5}$$

$$\sin\theta + \frac{1}{\sin\theta} = \sqrt{5} \text{ Then, } \sin^3\theta + \frac{1}{\sin^3\theta}$$

$$= (\sqrt{5})^3 - 3 \times \sqrt{5} = 2\sqrt{5}$$

$$\text{Sol.224.(c)}$$

$$\text{Given, } \cot \theta = \frac{b}{a} = \frac{\text{Base}}{\text{Perpendicular}}$$

By using pythagoras theorem,

$$\text{Hypotenuse} = \sqrt{b^2 + a^2}$$

$$\begin{aligned} \text{Now, } \frac{b \cos \theta - a \sin \theta}{b \cos \theta + a \sin \theta} &= \frac{\frac{b \times b}{\sqrt{b^2 + a^2}} - \frac{a \times a}{\sqrt{b^2 + a^2}}}{\frac{b \times b}{\sqrt{b^2 + a^2}} + \frac{a \times a}{\sqrt{b^2 + a^2}}} = \frac{b^2 - a^2}{b^2 + a^2} \end{aligned}$$

Sol.225.(c) Let $\theta = 45^\circ$

According to question,

$$\begin{aligned} \frac{1 + \sin^4 \theta + \cos^4 \theta}{\cos^2 \theta + \sin^4 \theta} &= \frac{1 + \sin^4 45^\circ + \cos^4 45^\circ}{\cos^2 45^\circ + \sin^4 45^\circ} \\ &= \frac{1 + \left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^4}{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^4} \\ &= \frac{1 + \frac{1}{4} + \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}} = \frac{\frac{6}{4}}{\frac{3}{4}} = 2 \end{aligned}$$

Sol.226.(c) $\operatorname{cosec}(\alpha + \beta)$

$$= \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$$

$\operatorname{cosec}(45^\circ + 30^\circ)$

$$= \frac{\sec 45^\circ \times \sec 30^\circ \times \operatorname{cosec} 45^\circ \times \operatorname{cosec} 30^\circ}{\sec 45^\circ \times \operatorname{cosec} 30^\circ + \operatorname{cosec} 45^\circ \times \sec 30^\circ}$$

$$\operatorname{cosec} 75^\circ = \frac{\sqrt{2} \times \frac{2}{\sqrt{3}} \times \sqrt{2} \times 2}{\sqrt{2} \times 2 + \sqrt{2} \times \frac{2}{\sqrt{3}}}$$

$$= \frac{\sqrt{2} \times \frac{2}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{2\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3} + 1}$$

$$= \frac{2\sqrt{2}}{\sqrt{3} + 1} \times \frac{\sqrt{3} - 1}{\sqrt{3} - 1}$$

$$= \sqrt{2}(\sqrt{3} - 1) = \sqrt{6} - \sqrt{2}$$

Sol.227.(d) $\cot A = 1 \Rightarrow A = 45^\circ$

$$\sin B = \frac{1}{\sqrt{2}} \Rightarrow B = 45^\circ$$

$$\text{Now, } \sin(A + B) - \cot(A + B) = 1 - 0 = 1$$

$$\text{Sol.228.(c)} \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$$

$$= \sqrt{\frac{(1 + \cos \theta) \times (1 + \cos \theta)}{(1 - \cos \theta) \times (1 + \cos \theta)}}$$

$$= \sqrt{\frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta}} = \sqrt{\frac{(1 + \cos \theta)^2}{\sin^2 \theta}}$$

$$= \frac{1 + \cos \theta}{\sin \theta} = \operatorname{cosec} \theta + \cot \theta$$

Sol.229.(b) $5\sqrt{3} \cos 60^\circ \tan 30^\circ - 3 \cos 0^\circ + 3 \cos^2 45^\circ + 2 \sin^2 60^\circ$

$$= 5\sqrt{3} \times \frac{1}{2} \times \frac{1}{\sqrt{3}} - 3 \times 1 + 3 \left(\frac{1}{\sqrt{2}}\right)^2 + 2 \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{5}{2} - 3 + 3 \left(\frac{1}{2}\right) + 2 \left(\frac{3}{4}\right)$$

$$= \frac{5}{2} - 3 + \left(\frac{3}{2}\right) + \left(\frac{3}{2}\right)$$

$$= \frac{5}{2} - 3 + 3 = \frac{5}{2} = 2.5$$

Sol.230.(d) Given, $\cos^2 \theta = \frac{3}{4}$

$$\Rightarrow \cos \theta = \sqrt{\frac{3}{4}} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$$

Now, $\sin(\theta + 30^\circ)$

$$= \sin(30^\circ + 30^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\text{Sol.231.(a)} \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$$

$$= \frac{\cos^2 \theta}{\sin \theta} = \cos \theta \cdot \cot \theta$$

Sol.232.(c) $\operatorname{cosec} 30^\circ - \cos 60^\circ$

$$= 2 - \frac{1}{2} = \frac{3}{2}$$

Sol.233.(a) Given,

$$\angle B = 90^\circ \Rightarrow \angle(A + C) = 90^\circ$$

Now, $\sin A \cos C + \cos A \sin C$

$$= \sin(A + C) \Rightarrow \sin 90^\circ = 1$$

$$\text{Sol.234.(c)} \frac{21 \cos A + 3 \sin A}{3 \cos A + 4 \sin A} = 2$$

$$= 21 \cos A + 3 \sin A = 6 \cos A + 8 \sin A$$

$$= 15 \cos A = 5 \sin A$$

$$\Rightarrow \cot A = \frac{5}{15} = \frac{1}{3}$$

$$\text{Sol.235.(d)} \text{ Given, } \cos x + \sec x = \frac{7}{2\sqrt{3}}$$

On squaring both side

$$\cos^2 x + \sec^2 x = \left(\frac{7}{2\sqrt{3}}\right)^2 - 2 = \frac{25}{12}$$

Sol.236.(c) $\tan 4384^\circ + \cot 6814^\circ$

$$\begin{aligned} &= \tan(360^\circ \times 12 + 64^\circ) + \cot(360^\circ \times 18 + 334^\circ) \\ &= \tan 64^\circ + \cot(270^\circ + 64^\circ) \\ &= \tan 64^\circ - \tan 64^\circ = 0 \end{aligned}$$

Sol.237.(a)

$$10\sqrt{2} \sin 45^\circ - 7\sqrt{2} \cos 45^\circ + 9 \tan 45^\circ$$

$$10 - 7 + 9 = 12$$

Sol.238.(d)

$$\begin{aligned} &\sin 600^\circ \cos 750^\circ + \sin 150^\circ \cos 240^\circ \\ &= \sin(180^\circ \times 3 + 60^\circ) \cos(180^\circ \times 4 + 30^\circ) \\ &+ \sin(180^\circ - 30^\circ) \cos(180^\circ + 60^\circ) \\ &= -\sin 60^\circ \cos 30^\circ - \sin 30^\circ \cos 60^\circ \\ &= -\sin(60^\circ + 30^\circ) = -1 \end{aligned}$$

Sol.239.(d) $\sin 2\theta = 2 \sin \theta \cos \theta$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\frac{\sin \theta}{1 + \cos \theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \tan\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{\sin \theta}{1 + \cos \theta}$$

Short trick: put $\theta = 30^\circ$

$$\text{Sol.240.(c)} \theta_1 + \theta_2 = \frac{\pi}{2}$$

$$\sin \theta_1 = \frac{1}{2} = \sin 30^\circ \Rightarrow \theta_2 = 60^\circ = \frac{\pi}{3}$$

Sol.241.(c)

$$3x = \sec A \Rightarrow 9x^2 = \sec^2 A \dots\dots\dots (1)$$

$$\frac{3}{x} = \tan A \Rightarrow \frac{9}{x^2} = \tan^2 A \dots\dots\dots (2)$$

Subtracting eqn.(2) from (1), we get

$$9x^2 - \frac{9}{x^2} = 1 \Rightarrow 9\left(x^2 - \frac{1}{x^2}\right) = 1$$

Sol.242.(d) $\tan^4 \theta + \tan^2 \theta = 1$

$$\tan^2 \theta (\tan^2 \theta + 1) = 1 \Rightarrow \tan^2 \theta \times \sec^2 \theta = 1$$

$$\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{1}{\cos^2 \theta} = 1 \Rightarrow \sin^2 \theta = \cos^4 \theta$$

Now, $11(\cos^4 \theta + \cos^2 \theta)$

$$= 11(\sin^2 \theta + \cos^2 \theta) = 11$$

$$\text{Sol.243.(d)} \frac{32}{\sec^2 \theta} - \frac{20}{1 + \tan^2 \theta} + 12 \sin^2 \theta$$

Put, $\theta = 0^\circ$ Then

$$\frac{32}{\sec^2 0^\circ} - \frac{20}{1 + \tan^2 0^\circ} + 12 \sin^2 0^\circ$$

$$= 32 - 20 + 0 = 12$$

$$\text{Sol.244.(b)} \sin \theta + \cos \theta = \frac{\sqrt{11}}{3}$$

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{11}{9}$$

$$2 \sin \theta \cos \theta = \frac{11}{9} - 1 = \frac{2}{9}$$

$$\sin \theta \cos \theta = \frac{1}{9}$$

$$\text{Now, } \tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = 1 \times \frac{9}{1} = 9$$

Sol.245.(b) If $\sec \theta + \tan \theta = a \dots\dots\dots \text{e.q. (1)}$

$$\text{Then } \sec \theta - \tan \theta = \frac{1}{a} \dots\dots\dots \text{e.q. (2)}$$

on solving e.q. (1) and e.q. (2), we get

$$\sec \theta = \frac{\left(a + \frac{1}{a}\right)}{2} = \frac{a^2 + 1}{2a}$$

Sol.246.(a)

$$\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots\dots\dots \cos 178^\circ \cdot \cos 179^\circ$$

$$\cos 1^\circ \cdot \cos 2^\circ \dots\dots \cos 90^\circ \dots\dots \cos 178^\circ \cdot \cos 179^\circ = 0$$

Sol.247.(c) $1 + \sin^2 \theta = a \cos^2 \theta$

$$\frac{1}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} = a$$

$$\sec^2 \theta + \tan^2 \theta = a$$

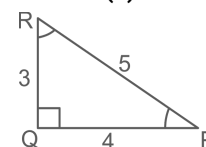
and $\sec^2 \theta - \tan^2 \theta = 1 \dots\dots \text{(identity)}$

Now, $\sec^4 \theta - \tan^4 \theta$

$$= (\sec^2 \theta + \tan^2 \theta)(\sec^2 \theta - \tan^2 \theta)$$

$$\sec^4 \theta - \tan^4 \theta = a \times 1 = a$$

Sol.248.(b)



Pythagoras triplets : - (3, 4, 5)

$$\sin P = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{3}{5}$$

$$\text{base} = 4 \text{ So } \angle Q = 90^\circ = \frac{\pi}{2}$$

Sol.249.(b) $\operatorname{Cosec} 41^\circ - \sec 49^\circ$
 $= \operatorname{Cosec} 41^\circ - \sec (90^\circ - 41^\circ)$
 $= \operatorname{Cosec} 41^\circ - \operatorname{Cosec} 41^\circ = 0$

Sol.250.(d) $\sin x + \sin^2 x = 1$

$$\sin x = \cos^2 x \Rightarrow \cos^4 x = \sin^2 x$$

Now,

$$\cos^8 x + 2\cos^6 x + \cos^4 x = (\cos^4 x + \cos^2 x)^2$$

$$= (\sin^2 x + \cos^2 x)^2 = 1$$

Sol.251.(a)

Formula :- $[\sec^2 \theta - \tan^2 \theta = 1]$

$$\Rightarrow (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

According to the question,

$$5 \times (\sec \theta - \tan \theta) = 1$$

$$\Rightarrow (\sec \theta - \tan \theta) = \frac{1}{5}$$

Sol.252.(d) $\left(\frac{1 + \sec^2 A}{1 + \cos^2 A}\right) \left(\frac{1 + \sin^2 A}{1 + \operatorname{cosec}^2 A}\right)$

$$= \left(\frac{1 + \frac{1}{\cos^2 A}}{1 + \cos^2 A}\right) \left(\frac{1 + \frac{1}{\operatorname{cosec}^2 A}}{1 + \operatorname{cosec}^2 A}\right)$$

$$\left(\frac{1}{\cos^2 A}\right) \left(\frac{1}{\operatorname{cosec}^2 A}\right) = \tan^2 A$$

Sol.253.(d) $\frac{\sqrt{1 - \sin A}}{\sqrt{1 + \sin A}}$

$$= \frac{\sqrt{1 - \sin A}}{\sqrt{1 + \sin A}} \times \frac{\sqrt{1 - \sin A}}{\sqrt{1 - \sin A}}$$

$$= \frac{(1 - \sin A)}{\sqrt{1 + \sin A} \sqrt{1 - \sin A}}$$

$$= \frac{(1 - \sin A)}{\sqrt{1 - \sin^2 A}} = \frac{(1 - \sin A)}{\cos A}$$

$$= \sec A - \tan A$$

Sol.254.(d)

$$(1 + \tan \theta - \sec \theta)(1 + \cot \theta + \operatorname{cosec} \theta)$$

$$= \left(1 + \frac{\sin \theta}{\cos \theta} - \frac{1}{\cos \theta}\right) \left(1 + \frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta}\right)$$

$$= \left(1 + \frac{\sin \theta - 1}{\cos \theta}\right) \left(1 + \frac{\cos \theta + 1}{\sin \theta}\right)$$

$$= \left(\frac{\cos \theta + \sin \theta - 1}{\cos \theta}\right) \left(\frac{\cos \theta + \sin \theta + 1}{\sin \theta}\right)$$

$$= \frac{(\cos \theta + \sin \theta)^2 - 1^2}{\sin \theta \cos \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta - 1}{\sin \theta \cos \theta}$$

$$= \frac{2\sin \theta \cos \theta}{\sin \theta \cos \theta} = 2$$

Sol.255.(d) $\frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \cdot \sec 40^\circ}$

$$= \frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \times \frac{1}{\cos 40^\circ}}$$

Let $A = 40^\circ$ then,

$$\frac{\cos(50^\circ + 40^\circ) - \sin(40^\circ - 40^\circ)}{1}$$

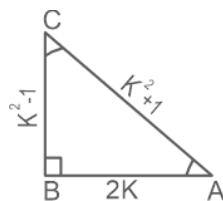
$$= \frac{\cos(90^\circ)}{1} = 0$$

Sol.256.(d) $\sec A + \tan A = K$ E.q.(1),

$$\sec A - \tan A = \frac{1}{K}$$
 E.q.(2)

On adding e.q.(1) and e.q.(2), we get

$$2 \sec A = \frac{K^2 + 1}{K} \Rightarrow \cos A = \frac{2K}{K^2 + 1}$$



$$\text{Now, } \sin A = \frac{K^2 - 1}{K^2 + 1}$$

Sol.257.(c) $4 \cot A = 5$

$$\Rightarrow \cot A = \frac{\text{base}}{\text{perpendicular}} = \frac{5}{4}$$

$$\text{Hypotenuse} = \sqrt{(5)^2 + (4)^2} = \sqrt{41}$$

So, $6 \sec A \tan A$

$$= 6 \times \frac{\sqrt{41}}{5} \times \frac{4}{5} = \frac{24\sqrt{41}}{25}$$

Sol.258.(b) $\frac{\operatorname{cosec} 90^\circ}{(1 - \cos^2 60^\circ)}$

$$= \frac{1}{(1 - \frac{1}{4})} = \frac{1}{(\frac{3}{4})} = \frac{4}{3}$$

Sol.259.(a) given that, $\theta = 45^\circ$

$$\text{Then, } 2 \operatorname{cosec}^2 \theta + 3 \sec^2 \theta = 2(\sqrt{2})^2 + 3$$

$$(\sqrt{2})^2 = 10$$

Sol.260.(c) According to the question,

$$\sin \theta = \sqrt{3} \cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = \sqrt{3} \Rightarrow \theta = 60^\circ$$

$$\text{So, } \frac{3 \sin^2 \theta + \cos \theta}{2 \cos \theta + 5} = \frac{3 \times (\frac{\sqrt{3}}{2})^2 + \frac{1}{2}}{2 \times \frac{1}{2} + 5}$$

$$= \frac{3 \times (\frac{3}{4}) + \frac{1}{2}}{(1 + 5)} = \frac{11}{24}$$

Sol.261.(a) $\sin(C + D) = \frac{\sqrt{3}}{2} \Rightarrow (C + D)$

$$= 60^\circ \text{-----(1)}$$

$$\text{And } \sec(C - D) = \frac{2}{\sqrt{3}} \Rightarrow (C - D)$$

$$= 30^\circ \text{-----(2)}$$

By the eq 1 and 2, $C = 45^\circ$ and $D = 15^\circ$

Sol.262.(d) $\tan 2A = \cot(46^\circ - A)$

$$\Rightarrow \cot(90^\circ - 2A) = \cot(46^\circ - A)$$

$$90^\circ - 2A = 46^\circ - A \Rightarrow A = 44^\circ$$

Sol.263.(d)

$$\sqrt{\frac{\operatorname{cosec} A + 1}{\operatorname{cosec} A - 1}} + \sqrt{\frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}}$$

$$= \frac{\sqrt{\operatorname{cosec} A + 1}}{\sqrt{\operatorname{cosec} A - 1}} + \frac{\sqrt{\operatorname{cosec} A - 1}}{\sqrt{\operatorname{cosec} A + 1}}$$

$$= \frac{(\operatorname{cosec} A + 1) + (\operatorname{cosec} A - 1)}{\sqrt{\operatorname{cosec}^2 A - 1}}$$

$$= \frac{(2 \operatorname{cosec} A)}{\sqrt{\cot^2 A}} = \frac{(2 \operatorname{cosec} A)}{\cot A}$$

$$= \frac{2(\frac{1}{\sin A})}{\frac{\cos A}{\sin A}} = 2 \sec A$$

Sol.264.(a) $\frac{1 + \sin A}{\cos^2 A} = \frac{1 + \sin A}{1^2 - \sin^2 A}$

$$= \frac{1 + \sin A}{(1 + \sin A)(1 - \sin A)} = \frac{1}{1 - \sin A}$$

Sol.265.(d) Putting $\theta = 45^\circ$

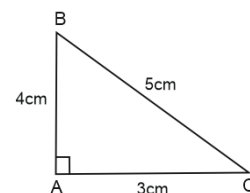
$$m = \sec \theta - \tan \theta = \sqrt{2} - 1$$

$$n = \operatorname{cosec} \theta + \cot \theta = \sqrt{2} + 1$$

Putting the value of m and n in $m + n(m - 1)$

$$\sqrt{2} - 1 + (\sqrt{2} + 1)(\sqrt{2} - 1 - 1) = -1$$

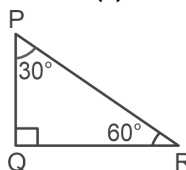
Sol.266.(d)



$$\cos B = \frac{4}{5} \text{ and } \cot C = \frac{3}{4}$$

$$\cos B + \cot C \Rightarrow \frac{4}{5} + \frac{3}{4} = \frac{31}{20}$$

Sol.267.(d)



$$\cos(30^\circ + 60^\circ) = \cos 90^\circ = 0$$

Sol.268.(b) $\sqrt{3} \tan^2 \theta - 4 \tan \theta + \sqrt{3} = 0$

$$= \sqrt{3} \tan^2 \theta - \tan \theta - 3 \tan \theta + \sqrt{3} = 0$$

$$= \tan \theta (\sqrt{3} \tan \theta - 1) - \sqrt{3} (\sqrt{3} \tan \theta - 1) = 0$$

$$= (\sqrt{3} \tan \theta - 1)(\tan \theta - \sqrt{3})$$

$$\Rightarrow \tan \theta = (\frac{1}{\sqrt{3}} \text{ and } \sqrt{3}) \text{ [here both value}$$

exists] $\Rightarrow \theta = 30^\circ$ and 60°

Then, by putting $\theta = 30^\circ$

$$\tan^2 \theta + \cot^2 \theta \Rightarrow \frac{1}{3} + 3 = \frac{10}{3}$$

Sol.269.(c)

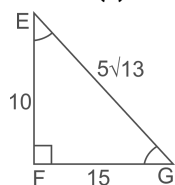
$$(\tan \theta + \cot \theta)^2 = \tan^2 \theta + \cot^2 \theta + 2$$

$$\Rightarrow \tan^2 \theta + \cot^2 \theta = (8)^2 - 2 = 62$$

Sol.270.(c) $6 \cot \theta = 5 \Rightarrow \cot \theta = \frac{5}{6}$

$$\text{Now, } \frac{(6 \cos \theta + \sin \theta)}{(6 \cos \theta - 4 \sin \theta)} = \frac{(6 \cot \theta + 1)}{(6 \cot \theta - 4)}$$

$$= \frac{(6 \times \frac{5}{6} + 1)}{(6 \times \frac{5}{6} - 4)} = 6$$

Sol.271.(a)

$$EG = \sqrt{10^2 + 15^2} = 5\sqrt{13}$$

$$\operatorname{cosec} G = \frac{\text{Hypotenuse}}{\text{Perpendicular}}$$

$$= \frac{5\sqrt{13}}{10} = \frac{\sqrt{13}}{2}$$

Sol.272.(b) $\operatorname{cosec} \theta - \sin \theta = 6$ Then, formula:- $(x + y)^2 = (x - y)^2 + 4xy$

$$\operatorname{cosec} \theta + \sin \theta = \sqrt{(6)^2 + 4} = \sqrt{40}$$

$$\text{Sol.273.(d)} \quad \sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$$

$$\sqrt{2} \sec^2 \theta - 2 \sec \theta - 2 \sec \theta + 2\sqrt{2} = 0$$

$$\sqrt{2} \sec \theta (\sec \theta - \sqrt{2}) - 2(\sec \theta - \sqrt{2}) = 0$$

$$\sec \theta = \sqrt{2}, \sec \theta = \frac{2}{\sqrt{2}} = \frac{\sqrt{2}}{1} = \frac{H}{B}$$

So, $P = 1$

$$\sin^2 \theta + \tan^2 \theta = \frac{1}{2} + 1 = \frac{3}{2}$$

Short trick :-

$$\sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$$

Put $\theta = 45^\circ$

$$\text{So value of } \sin^2 \theta + \tan^2 \theta = \frac{1}{2} + 1 = \frac{3}{2}$$

$$\text{Sol.274.(c)} \quad \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta}$$

$$= \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} \times \frac{\sin \theta + \cos \theta}{\sin \theta + \cos \theta} +$$

$$\frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} \times \frac{\sin \theta - \cos \theta}{\sin \theta - \cos \theta}$$

$$= \frac{(\sin \theta + \cos \theta)^2}{\sin^2 \theta - \cos^2 \theta} + \frac{(\sin \theta - \cos \theta)^2}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{2}{(\sin^2 \theta - \cos^2 \theta)}$$

$$\text{Sol.275.(d)} \quad \cot 3A = \tan (A - 36^\circ)$$

$$\Rightarrow \tan(90^\circ - 3A) = \tan (A - 36^\circ)$$

$$\Rightarrow 90^\circ - 3A = A - 36^\circ$$

$$\Rightarrow 4A = 126^\circ \Rightarrow A = 31.5^\circ$$

$$\text{Sol.276.(c)} \quad \frac{1 - \sin A}{\cos A} + \frac{\cos A}{1 - \sin A}$$

$$= \frac{(1 - \sin A)^2 + \cos^2 A}{(\cos A)(1 - \sin A)}$$

$$= \frac{(1)^2 + \sin^2 A - 2 \sin A + \cos^2 A}{(\cos A)(1 - \sin A)}$$

$$= \frac{2(1 - \sin A)}{(\cos A)(1 - \sin A)} = 2 \sec A$$

Sol.277.(b)

$$\cos (A + B) = 0 = \cos (90^\circ) \text{ and}$$

$$\sin (A - B) = 0 = \sin (0^\circ)$$

On comparing, we get

$$A + B = 90^\circ \text{ and } A - B = 0^\circ$$

$$\text{So, } A = 45^\circ \text{ and } B = 45^\circ$$

Sol.278.(b) $\sec x + \tan x = 5$

$$\text{Then, } \sec x - \tan x = \frac{1}{5}$$

$$\operatorname{cosec} y - \cot y = \frac{1}{3}$$

$$\text{then, } \operatorname{cosec} y + \cot y = 3$$

Now

$$(\sec x + \operatorname{cosec} y) - (\tan x - \cot y)$$

$$= (\sec x - \tan x + \operatorname{cosec} y + \cot y)$$

$$= \frac{1}{5} + 3 = \frac{16}{5} = 3.2$$

Sol.279.(c) $[\sec \theta = \operatorname{cosec}(90^\circ - \theta)]$

According to the question

$$3A = 90^\circ - A + 72^\circ$$

$$4A = 162^\circ \Rightarrow A = 40.5 \text{ degree}$$

Sol.280.(a)

$$\sin(x - y) = \frac{1}{2} = \sin 30^\circ \dots\dots \text{e.q. (1)}$$

$$\text{and } \cos(x + y) = \frac{1}{2} = \cos 60^\circ \dots\dots \text{e.q. (2)}$$

On solving e.q. (1) and e.q. (2), we get

$$x = 45^\circ \text{ and } y = 15^\circ$$

Now,

$$\sin x \cos x + 2 \sin^2 x + \cos^3 x \sec x$$

$$\sin 45^\circ \cdot \cos 45^\circ + 2 \sin^2 45^\circ + \cos^3 45^\circ$$

$$\sec 45^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + 2 \times \frac{1}{2} + \frac{1}{2\sqrt{2}} \times \sqrt{2} = 2$$

$$\text{Sol.281.(b)} \quad \frac{1 + \sin^2 45^\circ}{1 - \cos^2 45^\circ} = \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} = 3$$

Sol.282.(d) $\operatorname{cosec} \theta = \sec(90^\circ - \theta)$

$$= \sin \theta = \cos(90^\circ - \theta)$$

$$= \operatorname{cosec} 42^\circ = \sec 48^\circ$$

$$\Rightarrow \cos 32^\circ = \sin 58^\circ$$

According to the question,

$$\frac{3 \operatorname{cosec} 42^\circ}{\sec 48^\circ} - \frac{5 \cos 32^\circ}{\sin 58^\circ}$$

$$\frac{3 \operatorname{cosec} 42^\circ}{\operatorname{cosec} 42^\circ} - \frac{5 \cos 32^\circ}{\cos 32^\circ} = 3 - 5 \Rightarrow -2$$

Sol.283.(c) Put the value of $\theta = 45^\circ$,

$$A = a\sqrt{2} + b \text{ and } B = a + b\sqrt{2}$$

$$\text{Then, } A^2 = 2a^2 + b^2 + 2\sqrt{2}ab$$

$$B^2 = a^2 + 2b^2 + 2\sqrt{2}ab$$

$$A^2 - B^2 = 2a^2 + b^2 + 2\sqrt{2}ab - a^2 - 2b^2 - 2\sqrt{2}ab$$

$$ab \Rightarrow A^2 - B^2 = (a^2 - b^2)$$

Sol.284.(c) Let θ be 45°

$$\cos \theta + \sin \theta = \sqrt{2}$$

$$\cos 45^\circ + \sin 45^\circ = \sqrt{2}$$

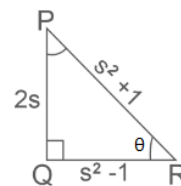
$$\text{LHS} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} = \text{RHS.}$$

$$\text{So, } \sec \theta \operatorname{cosec} \theta = \sec 45^\circ \operatorname{cosec} 45^\circ$$

$$= \sqrt{2} \times \sqrt{2} = 2$$

Sol.285.(a) Pythagoras triplets :-

$$(2s, s^2 + 1, s^2 - 1)$$



$$\operatorname{cosec} \theta + \cot \theta = s$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{s}$$

$$\operatorname{Cosec} \theta = \frac{s^2 + 1}{2s} \Rightarrow \sin \theta = \frac{2s}{s^2 + 1}$$

$$\text{So that, } \cos \theta = \frac{s^2 - 1}{s^2 + 1}$$

Sol.286.(c) $x = 30^\circ$ (put)

$$\sec x + \tan x = \sqrt{3}$$

$$\Rightarrow \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = \sqrt{3} \text{ (satisfied)}$$

According to the question,

$$\tan x = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Sol.287.(b)

Complementary angle of $\tan = \cot$

$$\text{Sol.288.(a)} \quad \sin^4 x + \sin^2 x = 1$$

$$\sin^4 x = 1 - \sin^2 x$$

$$\sin^4 x = \cos^2 x \text{ (divided by } \sin^2 x)$$

$$\sin^2 x = \cot^2 x$$

According to the question,

$$\cot^4 x + \cot^2 x = \sin^4 x + \sin^2 x = 1$$

Sol.289.(c) $\sec \theta + \tan \theta = 2$ ----- (eq. 1)

$$\text{As we know } \rightarrow \sec^2 \theta - \tan^2 \theta = 1$$

$$\text{Then, } (\sec \theta + \tan \theta) (\sec \theta - \tan \theta) = 1$$

$$\Rightarrow (\sec \theta - \tan \theta) = \frac{1}{2} \text{ ----- (eq. 2)}$$

By (eq. 1 and 2)

$$2 \sec \theta = 2 + \frac{1}{2}$$

$$\Rightarrow \sec \theta = \frac{\text{hypotenuse}}{\text{base}} = \frac{5}{4}$$

$$\text{then, perpendicular} = \sqrt{(5)^2 - (4)^2} = 3$$

$$\text{Therefore, } \cot \theta = \frac{\text{base}}{\text{perpendicular}} = \frac{4}{3}$$

Sol.290.(a) $\tan 10^\circ \tan 80^\circ + \tan 20^\circ \tan$

$$70^\circ + \tan 30^\circ \tan 60^\circ + \tan 40^\circ \tan 50^\circ$$

$$= \tan 10^\circ \tan(90^\circ - 10^\circ) + \tan 20^\circ \tan$$

$$(90^\circ - 20^\circ) + \tan 30^\circ \tan(90^\circ - 30^\circ) +$$

$$\tan 40^\circ \tan(90^\circ - 40^\circ)$$

$$= \tan 10^\circ \cot 10^\circ + \tan 20^\circ \cot 20^\circ + \tan$$

$$30^\circ \cot 30^\circ + \tan 40^\circ \cot 40^\circ$$

$$= 1 + 1 + 1 + 1 = 4$$

Sol.291.(b) $3 \operatorname{cosec} A = 7$

$$\operatorname{cosec} A = \frac{7}{3} = \frac{\text{Hypotenuse}}{\text{Perpendicular}}$$

$$\text{Now, } \cos A \tan A = \cos A \frac{\sin A}{\cos A} = \sin A$$

$$\Rightarrow \sin A = \frac{1}{\operatorname{cosec} A} = \frac{3}{7}$$

$$\text{Sol.292.(a)} \quad 2\sec^2\theta - 9\sec\theta + 7 = 0$$

$$\text{Put, } \theta = 0^\circ$$

$$2 \times 1 - 9 + 7 = 0 \dots (\text{satisfied})$$

$$\text{Now, } \cos^2\theta + \sec^2\theta = 1 + 1 = 2$$

$$\text{Sol.293.(d)} \quad \text{Let } \theta = 45^\circ$$

$$\frac{\cot\theta + \operatorname{cosec}\theta - 1}{\cot\theta - \operatorname{cosec}\theta + 1} = \frac{1 + \sqrt{2} - 1}{1 - \sqrt{2} + 1} = \frac{\sqrt{2}}{2 - \sqrt{2}}$$

$$\frac{\sqrt{2}}{\sqrt{2}(\sqrt{2} - 1)} \quad (\text{multiply by } (\sqrt{2} + 1) \text{ in numerator and denominator})$$

$$\frac{(\sqrt{2} + 1)}{1} = (\sqrt{2} + 1)$$

According to the question,

In option d, put $\theta = 45^\circ$

$$(\cot\theta + \operatorname{cosec}\theta) \Rightarrow (1 + \sqrt{2}) \text{ (satisfied)}$$

$$\text{Sol.294.(c)} \quad 6\sec\theta = 10$$

$$\sec\theta = \frac{10}{6} = \frac{5}{3} = \frac{\text{Hypotenuse}}{\text{Base}}$$

Using Pythagoras theorem;

$$\text{Perpendicular} = \sqrt{5^2 - 3^2} = 4,$$

$$\operatorname{cosec}\theta = \frac{5}{4} \text{ and } \cot\theta = \frac{3}{4}$$

$$\begin{aligned} & \frac{(5 \operatorname{cosec}\theta - 3 \cot\theta)}{(4 \cos\theta + 3 \sin\theta)} \\ &= \frac{(5 \times \frac{5}{4} - 3 \times \frac{3}{4})}{(4 \times \frac{3}{5} + 3 \times \frac{4}{5})} = \frac{(\frac{25}{4} - \frac{9}{4})}{(\frac{12}{5} + \frac{12}{5})} \\ &= \frac{(\frac{25-9}{4})}{(\frac{24}{5})} = \frac{20}{24} = \frac{5}{6} \end{aligned}$$

$$\text{Sol.295.(d)} \quad \sec 60^\circ + \tan 45^\circ - \sin^2 90^\circ = 2 + 1 - 1 = 2$$

$$\text{Sol.296.(d)} \quad \sin\theta + \cos\theta = \sqrt{2}$$

$$\text{Putting } \theta = 45^\circ$$

$$\sin 45^\circ + \cos 45^\circ = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$$

$$= \sqrt{2} \text{ (satisfied)}$$

$$\operatorname{cosec} 45^\circ + \cot 45^\circ = \sqrt{2} + 1$$

$$\text{Sol.297.(c)} \quad \cot(P + Q) = 0$$

$$\cot 90^\circ = 0 \Rightarrow P + Q = 90^\circ \text{-----e.q. (1)}$$

$$\tan(P - Q) = \frac{1}{\sqrt{3}}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow P - Q = 30^\circ \text{-----e.q. (2)}$$

$$\text{By adding e. q. (1) and e. q. (2); we get } \angle P = 60^\circ \text{ and } \angle Q = 30^\circ$$

$$\text{Sol.298.(b)}$$

$$\operatorname{cosec} A(1 - \cos A)(\operatorname{cosec} A + \cot A)$$

$$\Rightarrow \frac{1}{\sin A}(1 - \cos A)\left(\frac{1 + \cos A}{\sin A}\right)$$

$$\Rightarrow \frac{(1 - \cos^2 A)}{\sin^2 A} \Rightarrow \frac{(1 - \cos^2 A)}{(1 - \cos^2 A)} = 1$$

$$\text{Sol.299.(c)} \quad \frac{(2\cos^3 A - \cos A)}{(\sin A - 2\sin^3 A)} = 1$$

$$\Rightarrow \frac{\cos A(2\cos^2 A - 1)}{\sin A(1 - 2\sin^2 A)} = 1 \Rightarrow \frac{\cos A \cdot \cos 2A}{\sin A \cdot \cos 2A} = 1 \Rightarrow \cot A = 1 \Rightarrow A = 45^\circ$$

$$\text{Sol.300.(d)} \quad \frac{\cot 60^\circ - \cot 30^\circ}{1 + \cot 60^\circ \cot 30^\circ}$$

$$\frac{\frac{1}{\sqrt{3}} - \sqrt{3}}{1 + \frac{1}{\sqrt{3}} \times \sqrt{3}} = \frac{\frac{1-3}{\sqrt{3}}}{2} = -\frac{1}{\sqrt{3}}$$

$$\text{Sol.301.(b)} \quad A = \cot 30^\circ \tan 60^\circ + \cot 60^\circ \tan 30^\circ \Rightarrow A$$

$$= \sqrt{3} \times \sqrt{3} + \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} \Rightarrow A = \frac{10}{3}$$

$$\text{Sol.302.(c)}$$

$$\tan\theta + \sin\theta = A \text{ and } \tan\theta - \sin\theta = B$$

$$\text{Then, } A^2 - B^2 =$$

$$\begin{aligned} & (\tan\theta + \sin\theta)^2 - (\tan\theta - \sin\theta)^2 \\ & \Rightarrow \tan^2\theta + \sin^2\theta + 2\sin\theta\tan\theta - (\tan^2\theta \\ & \quad + \sin^2\theta - 2\sin\theta\tan\theta) \\ & = 4 \tan\theta \sin\theta \end{aligned}$$

$$\text{Sol.303.(c)}$$

$$(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2$$

$$\text{Putting the value of } \theta = 45^\circ$$

$$\left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2 = 9$$

$$\text{By putting value of } \theta = 45^\circ \text{ in the option (c) we get, } 7 + \tan^2\theta + \cot^2\theta = 7 + 1 + 1 = 9.$$

$$\text{Sol.304.(b)}$$

$$\tan^2\theta + \cot^2\theta - \sec^2\theta \cdot \operatorname{cosec}^2\theta$$

$$\text{Put } \theta = 45^\circ$$

$$\tan^2 45^\circ + \cot^2 45^\circ - \sec^2 45^\circ \cdot \operatorname{cosec}^2 45^\circ \Rightarrow 1 + 1 - 2 \times 2 \Rightarrow -2$$

$$\text{Sol.305.(c)}$$

$$\text{Formula :- } \sin(90^\circ - \theta) = \cos\theta$$

$$\begin{aligned} & \frac{\sin 30^\circ \sin 40^\circ \sin 50^\circ \sin 60^\circ}{\cos 30^\circ \cos 40^\circ \cos 50^\circ \cos 60^\circ} \\ &= \frac{\cos 60^\circ \cos 50^\circ \cos 40^\circ \cos 30^\circ}{\cos 30^\circ \cos 40^\circ \cos 50^\circ \cos 60^\circ} = 1 \end{aligned}$$

$$\text{Sol.306.(a) Formula :-}$$

$$\sin a - \sin b = 2 \sin\left(\frac{a-b}{2}\right) \cos\left(\frac{a+b}{2}\right)$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$\text{Sol.307.(c)} \quad \text{In third quadrant value of } \tan \theta \text{ and } \cot \theta \text{ is +ve.}$$

$$\tan 240^\circ = \tan(180^\circ + 60^\circ)$$

$$= \tan 60^\circ = \sqrt{3}$$

$$\text{Sol.308.(c)} \quad \tan^2\theta = 1 - a^2$$

$$\text{Adding 1 both side,}$$

$$1 + \tan^2\theta = 1 - a^2 + 1$$

$$\sec^2\theta = 2 - a^2$$

$$\sec\theta = \sqrt{2 - a^2} \dots\dots (\text{eq. 1})$$

$$\text{Now, } \sec\theta + \tan^3\theta \times \operatorname{Cosec}\theta$$

$$= \frac{1}{\cos\theta} + \frac{\sin^3\theta}{\cos^3\theta} \times \frac{1}{\sin\theta}$$

$$= \frac{\cos^2\theta + \sin^2\theta}{\cos^3\theta} = \sec^3\theta \dots\dots (\text{eq. 2})$$

$$\text{Putting the value of (eq. 1) in (eq. 2), we get, } \sec^3\theta = (2 - a^2)^{\frac{3}{2}}$$

$$\text{Sol.309.(a) Formula :-}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

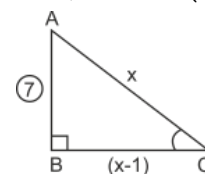
$$\sin(A + B) = \cos(A + B) \dots\dots (\text{given})$$

$$\text{Clearly value of } \sin\theta \text{ and } \cos\theta \text{ are equal at } 45^\circ, (A + B) = 45^\circ \Rightarrow A = 45^\circ - B$$

$$\text{Now, } \tan A = \tan(45^\circ - B)$$

$$= \frac{\tan 45^\circ - \tan B}{1 + \tan 45^\circ \tan B} = \frac{1 - \tan B}{1 + \tan B}$$

$$\text{Sol.310.(d)} \quad \text{Let, the length of side AC = x cm, then BC = (x - 1) cm}$$



Now, using pythagoras theorem :-

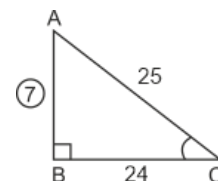
$$AC^2 = AB^2 + BC^2 \Rightarrow x^2 = 7^2 + (x - 1)^2$$

$$x^2 = 49 + x^2 + 1 - 2x$$

$$2x = 50 \Rightarrow x = 25 \text{ cm}$$

$$\text{Now, } \sin C = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{7}{25}$$

Short Trick:-



Using pythagoras triplets : (7, 24, 25)

$$AB = 7 \text{ cm}$$

$$\text{And } AC - BC = 25 - 24 = 1 \text{ cm} \dots\dots (\text{Satisfy the given condition})$$

$$\text{Then, } \sin C = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{7}{25}$$

$$\text{Sol.311.(b)} \quad \tan A + \cot A = 2 \dots\dots (\text{given})$$

On squaring both sides, we get,

$$\tan^2 A + \cot^2 A + 2 \tan A \cdot \cot A = 4$$

$$(\tan^2 A + \cot^2 A) = 2$$

$$\Rightarrow 2(\tan^2 A + \cot^2 A) = 4$$

Short Trick:-

$$\tan A + \cot A = 2 \dots\dots (\text{given})$$

Putting $A = 45^\circ$ equation satisfy, so

$$2(\tan^2 A + \cot^2 A) = 2(1 + 1) = 4$$

$$\text{Sol.312.(a) Formula :-}$$

$$\frac{(1 - \cos\theta)}{\sin\theta} = \frac{\sin\theta}{(1 + \cos\theta)}$$

$$\frac{(1 - \cos\theta)}{\sin\theta} = \frac{1}{5} \quad \dots(\text{given})$$

$$\text{Then, } \frac{(1 + \cos\theta)}{\sin\theta} = 5$$

$$\text{Sol.313.(b)} \quad \frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$$

$$= \frac{\cos A}{1 + \frac{\sin A}{\cos A}} - \frac{\sin A}{1 + \frac{\cos A}{\sin A}}$$

$$= \frac{\cos^2 A}{\cos A + \sin A} - \frac{\sin^2 A}{\sin A + \cos A}$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos A + \sin A}$$

$$= \frac{(\cos A + \sin A)(\cos A - \sin A)}{(\cos A + \sin A)}$$

$$= (\cos A - \sin A)$$

$$\text{Sol.314.(a)}$$

$$\frac{1}{1 + \cos(90^\circ - \theta)} + \frac{1}{1 - \cos(90^\circ - \theta)}$$

$$= \frac{1}{1 + \sin\theta} + \frac{1}{1 - \sin\theta}$$

$$= \frac{(1 - \sin\theta) + (1 + \sin\theta)}{1 - \sin^2\theta}$$

$$= \frac{2}{\cos^2\theta} \Rightarrow 2\sec^2\theta$$

$$\text{Sol.315.(d)} \text{ We know that,}$$

$$\cos^2\theta = (1 - \sin^2\theta)$$

$$\sin^2\theta + \cos^2\theta = 1 \dots\dots\text{eq. (1)}$$

$$\text{On squaring eq. (1), we get}$$

$$(\sin^2\theta + \cos^2\theta)^2 = (1)^2$$

$$\sin^4\theta + \cos^4\theta + 2\sin^2\theta\cos^2\theta = 1$$

$$\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$$

$$\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta(1 - \sin^2\theta)$$

$$\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta + 2\sin^4\theta$$

$$\text{Sol.316.(c)} \quad \frac{\cos x - \sqrt{3}\sin x}{2}$$

$$= \frac{1}{2} \times \cos x - \frac{\sqrt{3}}{2} \times \sin x$$

$$= \sin 30^\circ \cos x - \cos 30^\circ \sin x$$

$$= \sin(30^\circ - x) = \sin[90^\circ - (60^\circ + x)]$$

$$= \cos(60^\circ + x) = \cos\left(\frac{\pi}{3} + x\right)$$

$$\text{Sol.317.(b)}$$

$$\sqrt{\frac{1 + \cos A}{1 - \cos A}} = \sqrt{\frac{1 + \cos A}{1 - \cos A}} \times \frac{1 + \cos A}{1 + \cos A}$$

$$= \sqrt{\frac{(1 + \cos A)^2}{1 - \cos^2 A}} = \frac{1 + \cos A}{\sin A}$$

$$= \operatorname{cosec} A + \cot A$$

$$\text{Sol.318.(a)}$$

$$\sin\theta + \cos\theta = \frac{\sqrt{3} - 1}{2\sqrt{2}} \dots\dots\text{eq. (1)}$$

$$\text{Squaring eq. (1), we get}$$

$$\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$$

$$= \left(\frac{\sqrt{3} - 1}{2\sqrt{2}}\right)^2 = \frac{4 - 2\sqrt{3}}{8}$$

$$2\sin\theta\cos\theta = \frac{4 - 2\sqrt{3}}{8} - 1$$

$$= \frac{(4 - 2\sqrt{3}) - 8}{8} = \frac{-(2 + \sqrt{3})}{4}$$

$$\text{Now, } \tan\theta + \cot\theta = \frac{2}{2\sin\theta\cos\theta}$$

$$= \frac{-8}{(2 + \sqrt{3})} \times \frac{(2 - \sqrt{3})}{(2 - \sqrt{3})} = 8(\sqrt{3} - 2)$$

$$\text{Sol.319.(d)}$$

$$\text{If } A + B = 90^\circ, \text{ then } \sec A = \operatorname{cosec} B$$

$$\sec 2\theta = \operatorname{cosec}(\theta - 36^\circ)$$

$$\text{Now, } 2\theta + (\theta - 36^\circ) = 90^\circ \Rightarrow \theta = 42^\circ$$

$$\text{Sol.320.(a) Concepts:-}$$

$$\text{If } A + B = 90^\circ, \text{ then } \tan A = \cot B \text{ and } \sin A = \cos B$$

$$\cos^2 35^\circ + \cos 55^\circ \cdot \sin 35^\circ + \frac{\tan 34^\circ}{\cot 56^\circ}$$

$$= \cos^2 35^\circ + \sin^2 35^\circ + 1 \Rightarrow 1 + 1 = 2$$

$$\text{Sol.321.(d) Formula :-}$$

$$\tan 2x = \frac{2\tan x}{(1 - \tan^2 x)}$$

$$\cos x \cdot \operatorname{cosec} x - \sin x \cdot \sec x$$

$$= \cot x - \tan x = \frac{1}{\tan x} - \frac{\tan^2 x}{\tan x}$$

$$= \frac{2 \times (1 - \tan^2 x)}{2 \times \tan x} = \frac{2 \times 1}{\tan 2x} = 2\cot 2x$$

$$\text{Sol.322.(b)} \quad \cot^2 A - \cos^2 A$$

$$= \frac{\cos^2 A}{\sin^2 A} - \cos^2 A = \cos^2 A (\operatorname{cosec}^2 A - 1)$$

$$= \cos^2 A \cdot \cot^2 A$$

$$\text{Sol.323.(a)} \quad b \cdot \sin\theta = a$$

$$\Rightarrow \sin\theta = \frac{a}{b} = \frac{\text{Perpendicular}}{\text{hypotenuse}}$$

$$\text{by, using pythagoras theorem, we get}$$

$$\text{Base} = \sqrt{b^2 - a^2}$$

$$\text{Now,}$$

$$\sec\theta + \tan\theta = \frac{b}{\sqrt{b^2 - a^2}} + \frac{a}{\sqrt{b^2 - a^2}}$$

$$= \frac{b + a}{\sqrt{b^2 - a^2}} = \frac{\sqrt{b+a} \times \sqrt{b+a}}{\sqrt{b+a} \times \sqrt{b-a}} = \sqrt{\frac{b+a}{b-a}}$$

$$\text{Sol.324.(d)}$$

$$\text{If } A + B = 90^\circ, \text{ then } \cos A = \sin B$$

$$\frac{\cos 20^\circ}{\sin 70^\circ} + \frac{\cos\theta}{\sin(90^\circ - \theta)} = 1 + \frac{\cos\theta}{\cos\theta} = 2$$

$$\text{Sol.325.(b) Formula :-}$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$\sin 10^\circ - \frac{4}{3}\sin^3 10^\circ = \frac{3\sin 10^\circ - 4\sin^3 10^\circ}{3}$$

$$= \frac{\sin 30^\circ}{3} = \frac{1}{6}$$

$$\text{Sol.326.(c) Concepts:-}$$

$$(i) \cos(-\theta) = \cos\theta$$

$$(ii) \cos(2n\pi - \theta) = \cos\theta$$

$$\text{Now, } \cos\left(-\frac{17\pi}{3}\right) = \cos\left(\frac{17\pi}{3}\right)$$

$$= \cos\left(6\pi - \frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\text{Sol.327.(a) Formula :-}$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$(\sin A + \cos A)(1 - \sin A \cos A)$$

$$= (\sin A + \cos A)$$

$$(\sin^2 A + \cos^2 A - \sin A \cos A)$$

$$= \sin^3 A + \cos^3 A$$

$$\text{Sol.328.(d)} \quad \sec\theta - 2\cos\theta = \frac{7}{2}$$

$$= \sec\theta - \frac{2}{\sec\theta} = \frac{7}{2}$$

$$= \sec^2\theta - 2 = \frac{7}{2}\sec\theta$$

$$= 2\sec^2\theta - 7\sec\theta - 4 = 0$$

$$= 2\sec^2\theta - 8\sec\theta + \sec\theta - 4 = 0$$

$$= (\sec\theta - 4)(2\sec\theta + 1) = 0$$

$$= \sec\theta = 4$$

$$\text{Sol.329.(d)} \quad \sec 3\theta = \operatorname{cosec}(4\theta - 15^\circ)$$

$$\Rightarrow \operatorname{cosec}(90^\circ - 3\theta) = \operatorname{cosec}(4\theta - 15^\circ)$$

$$\Rightarrow 90^\circ - 3\theta = 4\theta - 15^\circ$$

$$\Rightarrow 7\theta = 90^\circ + 15^\circ$$

$$\Rightarrow 7\theta = 105^\circ \Rightarrow \theta = 15^\circ$$

$$\text{Then, } \tan 3\theta = \tan 45^\circ = 1$$

$$\text{Sol.330.(a)} \text{ In a right angled triangle, right angled at A,}$$

$$= \angle A = \angle B + \angle C = 90^\circ$$

$$= \angle A - \angle B - \angle C = 0^\circ$$

$$\tan(A - B - C) = \tan 0^\circ = 0$$

$$\tan(A - B - C)$$

$$= \frac{\tan A - \tan B - \tan C - \tan A \tan B \tan C}{1 + \tan A \tan B - \tan B \tan C + \tan C \tan A}$$

$$\tan A - \tan B - \tan C - \tan A \tan B \tan C = 0$$

$$\tan A - \tan B - \tan C = \tan A \tan B \tan C$$

$$\text{Sol.331.(d)}$$

$$1 + \tan^2\alpha = 4 + Q^2 \text{ (adding 1 both sides)}$$

$$\sec^2\alpha = 4 + Q^2$$

$$\cos\alpha = (4 + Q^2)^{-\frac{1}{2}}$$

$$\text{then, } \sec\alpha + \tan^3\alpha \operatorname{cosec}\alpha$$

$$= \frac{1}{\cos\alpha} + \frac{\sin^3\alpha}{\cos^3\alpha \times \sin\alpha}$$

$$= \frac{1}{\cos\alpha} + \frac{\sin^2\alpha}{\cos^2\alpha \times \cos\alpha}$$

$$= \frac{1 + \tan^2\alpha}{\cos\alpha} = \frac{(4 + Q^2)}{(4 + Q^2)^{-1/2}} = (4 + Q^2)^{\frac{3}{2}}$$

$$\text{Short Trick:- Put, } = 60^\circ, \text{ then,}$$

$$\tan^2 60^\circ = 3 + Q^2 \Rightarrow Q = 0$$

$$\text{Now,}$$

$$\sec\alpha + \tan^3\alpha \operatorname{cosec}\alpha = 2 + 3\sqrt{3} \times \frac{2}{\sqrt{3}} = 8$$

$$\text{Clearly, option (d) satisfy the given equation}$$

$$\Rightarrow (4 + Q^2)^{3/2} = (4 + 0^2)^{3/2}$$

$$= (2^2)^{3/2} = 2^3 = 8$$

Sol.332.(b) $1 + \sin\theta = m \cos\theta$

$$\frac{1 + \sin\theta}{\cos\theta} = m \Rightarrow \frac{(1 + \sin\theta)^2}{\cos^2\theta} = m^2$$

$$\frac{(1 + \sin\theta)^2}{(1 + \sin\theta)(1 - \sin\theta)} = m^2$$

$$\frac{(1 + \sin\theta)}{(1 - \sin\theta)} = m^2$$

By using componendo and dividendo ,

we get $\frac{(1 + \sin\theta + 1 - \sin\theta)}{(1 + \sin\theta - 1 + \sin\theta)} = \frac{m^2 + 1}{m^2 - 1}$

$$\frac{(2)}{(2\sin\theta)} = \frac{m^2 + 1}{m^2 - 1} \Rightarrow \sin\theta = \frac{m^2 - 1}{m^2 + 1}$$

Sol.333.(b) $\cos(30^\circ + \theta) - \sin(60^\circ - \theta)$

$$= \cos\{90^\circ - (60^\circ - \theta)\} - \sin(60^\circ - \theta)$$

$$= \sin(60^\circ - \theta) - \sin(60^\circ - \theta) = 0$$

Sol.334.(a) Put $\theta = 90^\circ$

$$5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19\sin^2\theta\cos^2\theta$$

$$= 5 \times 1 + 3 \times 1 + 19 \times 0 = 8$$

Sol.335.(c) $\tan\theta + \cot\theta = 12$,

Squaring both sides,

$$\tan^2\theta + \cot^2\theta + 2\tan\theta \cdot \cot\theta = 144$$

$$\tan^2\theta + \cot^2\theta = 144 - 2 = 142$$

Sol.336.(d) Formula: $\cos 2A = 1 - 2\sin^2A$

Sin Y = x then, $\cos 2Y = 1 - 2x^2$

Sol.337.(d) Formula:- $\tan(A + B + C)$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

In a right angled triangle, right angled at C,

$$= \angle C = \angle A + \angle B = 90^\circ$$

$$= \angle A + \angle B - \angle C = 0^\circ$$

$$\tan(A + B - C) = \tan 0^\circ = 0$$

$$\Rightarrow \tan(A + B - C)$$

$$= \frac{\tan A + \tan B - \tan C + \tan A \tan B \tan C}{1 - \tan A \tan B + \tan B \tan C + \tan C \tan A}$$

$$\Rightarrow \tan A + \tan B - \tan C + \tan A \tan B \tan C = 0$$

$$\Rightarrow \tan A + \tan B - \tan C = -\tan A \tan B \tan C$$

$$\Rightarrow \tan C - \tan A - \tan B = \tan A \tan B \tan C$$

Sol.338.(d) $x = 8(\sin\theta + \cos\theta)$ and $y = 9(\sin\theta - \cos\theta)$,

$$\Rightarrow \frac{x^2}{8^2} + \frac{y^2}{9^2} = \frac{64(\sin^2\theta + \cos^2\theta + 2\sin\theta \cdot \cos\theta)}{64}$$

$$+ \frac{81(\sin^2\theta + \cos^2\theta - 2\sin\theta \cdot \cos\theta)}{81}$$

$$= 1 + 2\sin\theta \cdot \cos\theta + 1 - 2\sin\theta \cdot \cos\theta = 2$$

Sol.339.(a) $\tan 6^\circ \times \tan 45^\circ \times \tan 84^\circ$

$$= \tan(90^\circ - 84^\circ) \times 1 \times \tan 84^\circ$$

$$= \cot 84^\circ \tan 84^\circ = 1$$

Sol.340.(b) $2 \cos 135^\circ \sin 15^\circ$

$$= [\sin(135^\circ + 15^\circ) - \sin(135^\circ - 15^\circ)]$$

$$= \sin 150^\circ - \sin 120^\circ$$

$$= \cos 60^\circ - \cos 30^\circ = \frac{1}{2} - \frac{\sqrt{3}}{2} = \frac{1 - \sqrt{3}}{2}$$

Sol.341.(c) $\sec x + \cos x =$

$$\sqrt{(\sec x - \cos x)^2 + 4 \times \sec x \cos x}$$

$$= \sqrt{4^2 + 4} = \sqrt{20} = 2\sqrt{5}$$

$$\frac{(1 + \cos^2 x)}{\cos x} = \frac{1}{\cos x} + \frac{\cos^2 x}{\cos x}$$

$$= \sec x + \cos x = 2\sqrt{5}$$

Sol.342.(c) $\tan(4\alpha - 50^\circ) = \cot(50^\circ - \alpha)$

$$4\alpha - 50^\circ + 50^\circ - \alpha = 90^\circ$$

$$3\alpha = 90^\circ \Rightarrow \alpha = \frac{90^\circ}{3} = 30^\circ$$

Sol.343.(c) $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$

$$= \frac{\sin A \sin B}{\cos A \cos B} + \frac{\cos x}{\cos A \cos B} = 1$$

$$= \frac{\sin A \sin B + \cos x}{\cos A \cos B} = 1$$

$$= \sin A \sin B + \cos x = \cos A \cos B$$

$$= \cos x = \cos A \cos B - \sin A \sin B$$

$$= \cos x = \cos(A + B) \Rightarrow x = A + B$$

Sol.344.(a) $\cos^3 60^\circ - \cos^3 240^\circ - \cos^3 360^\circ$

$$= \cos^3 60^\circ - \cos^3(180^\circ + 60^\circ) - 1$$

$$= \cos^3 60^\circ + \cos^3 60^\circ - 1$$

$$= \left(\frac{1}{2}\right)^3 \times 2 - 1 = \frac{1}{4} - 1 = \frac{-3}{4}$$

Sol.345.(c) $5\sin^2 A + 3\cos^2 A = 4$

$$\Rightarrow 2\sin^2 A + 3\sin^2 A + 3\cos^2 A = 4$$

$$\Rightarrow 2\sin^2 A + 3(\sin^2 A + \cos^2 A) = 4$$

$$\Rightarrow 2\sin^2 A + 3 = 4 \Rightarrow 2\sin^2 A = 1$$

$$\sin^2 A = \frac{1}{2} \Rightarrow \sin A = \frac{1}{\sqrt{2}}$$

$$\Rightarrow A = 45^\circ$$

$$\text{then, } \tan A = \tan 45^\circ = 1$$

Sol.346.(b) $\cos(A - B) = \frac{\sqrt{3}}{2}$

$$\cos(A - B) = \cos 30^\circ$$

$$A - B = 30^\circ \text{ ----- (1)}$$

$$\text{and, } \cos(A + B) = 0$$

$$\cos(A + B) = \cos 90^\circ$$

$$A + B = 90^\circ \text{ ----- (2)}$$

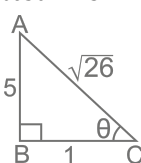
Adding eq. (1) and (2), we have

$$A = \frac{30^\circ + 90^\circ}{2} = \frac{120^\circ}{2} = 60^\circ$$

$$\text{Then } B = 90^\circ - 60^\circ = 30^\circ$$

Sol.347.(b) $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{3}{2}$

$$\frac{\sin\theta}{\cos\theta} = \frac{3+2}{3-2} = \frac{5}{1} \Rightarrow \tan\theta = \frac{5}{1}$$



$$\sin^4\theta - \cos^4\theta = \left(\frac{5}{\sqrt{26}}\right)^4 - \left(\frac{1}{\sqrt{26}}\right)^4$$

$$= \frac{625}{676} - \frac{1}{676} = \frac{624}{676} = \frac{12}{13}$$

Sol.348.(c)

$$\operatorname{cosec}^2\theta + \sec^2\theta = \frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}$$

$$= \frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta \cos^2\theta} = \frac{1}{\frac{2}{9}} = \frac{9}{2}$$

Sol.349.(c) Formula :- $(\cos 2x = 2\cos^2 x - 1)$

$$2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$$

$$\Rightarrow 2 \frac{\cos^2 x - \frac{1}{\cos^2 x}}{\frac{\sin^2 x}{\cos^2 x}} = 2 \frac{\cos^4 x - 1}{\sin^2 x}$$

$$= 2 \frac{(\cos^2 x - 1)(\cos^2 x + 1)}{1 - \cos^2 x}$$

$$= -2(\cos^2 x + 1) = -2\cos^2 x - 2$$

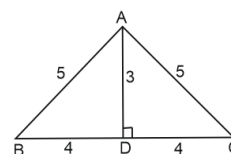
$$= -(\cos 2x + 1) - 2 = a + b \cos 2x$$

$$-3 - \cos 2x = a + b \cos 2x$$

Comparing both side, we get,

$$a = -3 \text{ and } b = -1$$

Sol.350.(b)



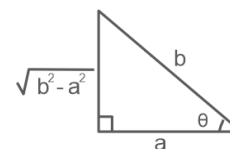
Since, $\triangle ABC$ is an isosceles triangle then, AD(median) divides BC in the ratio of 1 : 1.

By the given figure,

$$\tan C = \frac{3}{4}, \cot B = \frac{4}{3}$$

$$\text{So, } \tan C - \cot B = \frac{3}{4} - \frac{4}{3} = \frac{9 - 16}{12} = \frac{-7}{12}$$

Sol.351.(c) $b \cos\theta = a \Rightarrow \cos\theta = \frac{a}{b}$

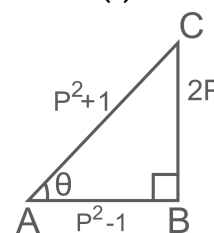


$$p = \sqrt{b^2 - a^2} \text{ (using pythagoras theorem)}$$

$$\operatorname{cosec}\theta + \cot\theta = \frac{b}{\sqrt{b^2 - a^2}} + \frac{a}{\sqrt{b^2 - a^2}}$$

$$= \frac{b + a}{(\sqrt{b - a})(\sqrt{b + a})} = \sqrt{\frac{b + a}{b - a}}$$

Sol.352.(a)



$$\operatorname{cosec}\theta + \cot\theta = p \text{ ----- (1)}$$

$$\operatorname{cosec}\theta - \cot\theta = \frac{1}{p} \text{ ----- (2)}$$

$$(\text{As } \operatorname{cosec}^2\theta - \cot^2\theta = 1)$$

Adding eq. (1) and (2) we get :

$$\operatorname{cosec} \theta = \frac{p^2 + 1}{2p} \text{ then, } \cot \theta = \frac{p^2 - 1}{2p}$$

$$\text{Now, } \frac{\frac{p^2 - 1}{2p}}{\frac{p^2 + 1}{2p}} = \frac{p^2 - 1}{p^2 + 1} = \frac{\cot \theta}{\operatorname{cosec} \theta} = \cos \theta$$

$$\text{Sol.353.(d)} \sin^2 \theta - 3 \sin \theta + 2 = 0$$

$$\sin^2 \theta - 2 \sin \theta - \sin \theta + 2 = 0$$

$$\sin \theta (\sin \theta - 2) - 1 (\sin \theta - 2) = 0$$

$$(\sin \theta - 1)(\sin \theta - 2) = 0 \Rightarrow \sin \theta = 1 \quad (\sin \theta \neq 2 \because -1 \leq \sin \theta \leq 1)$$

$$\text{So, } \sin \theta = 1 \quad (\text{As, } 0^\circ \leq \theta \leq 90^\circ) \Rightarrow \theta = 90^\circ$$

$$\text{Sol.354.(d)} \frac{1 + \sin \theta}{\cos \theta}$$

$$= \frac{(1 + \sin \theta)(1 - \sin \theta)}{\cos \theta (1 - \sin \theta)} = \frac{1 - \sin^2 \theta}{\cos \theta (1 - \sin \theta)}$$

$$= \frac{\cos^2 \theta}{\cos \theta (1 - \sin \theta)} = \frac{\cos \theta}{1 - \sin \theta}$$

$$\text{Sol.355.(b)} \tan 3\theta \cdot \tan 7\theta = 1$$

$$\tan 3\theta = \frac{1}{\tan 7\theta} = \cot 7\theta$$

$$\tan 3\theta = \tan (90^\circ - 7\theta)$$

$$3\theta = 90^\circ - 7\theta \Rightarrow 10\theta = 90^\circ$$

$$\Rightarrow \theta = \frac{90}{10} = 9^\circ$$

$$\text{So, } \cot 15\theta = \cot 135^\circ = -1$$

$$\text{Sol.356.(b)} 2 \sin \theta + 2 \sin^2 \theta = 2$$

$$\sin \theta + \sin^2 \theta = 1 \Rightarrow \sin \theta = 1 - \sin^2 \theta$$

$$\Rightarrow \sin \theta = \cos^2 \theta$$

$$\text{So, } 2 \cos^4 \theta + 2 \cos^2 \theta = 2(\cos^4 \theta + \cos^2 \theta)$$

$$= 2(\sin^2 \theta + \cos^2 \theta) = 2 \times 1 = 2$$

$$\text{Sol.357.(a)} \cos \theta + \cos^2 \theta = 1$$

$$\cos \theta = 1 - \cos^2 \theta \Rightarrow \cos \theta = \sin^2 \theta$$

$$\text{So, } \sqrt{\sin^4 \theta + \cos^2 \theta} = \sqrt{\cos^2 \theta + \cos^2 \theta}$$

$$= \sqrt{2 \cos^2 \theta} = \sqrt{2} \cos \theta$$

$$\text{Sol.358.(c)} \sec^2 A + \tan^2 A = 3$$

$$1 + \tan^2 A + \tan^2 A = 3$$

$$1 + 2 \tan^2 A = 3 \Rightarrow 2 \tan^2 A = 2$$

$$\tan^2 A = 1 \Rightarrow \frac{1}{\cot^2 A} = 1$$

$$\cot^2 A = 1 \Rightarrow \cot A = 1$$

$$\text{Sol.359.(b)} \operatorname{cosec} 15^\circ \sec 15^\circ$$

$$= \frac{1}{\sin 15^\circ \cos 15^\circ} = \frac{2}{2 \sin 15^\circ \cos 15^\circ}$$

$$= \frac{2}{\sin 30^\circ} = \frac{2}{\frac{1}{2}} = 4$$

$$\text{Sol.360.(d)} \text{ Putting } a = 45^\circ \text{ and } b = 15^\circ \text{ we have :}$$

$$\frac{\cos 30^\circ - \cos 60^\circ}{\cos 30^\circ + \cos 60^\circ} = \frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{\frac{\sqrt{3}}{2} + \frac{1}{2}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$= \frac{(\sqrt{3} - 1)(\sqrt{3} - 1)}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = \frac{3 + 1 - 2\sqrt{3}}{2}$$

$$= \frac{4 - 2\sqrt{3}}{2} = 2 - \sqrt{3}$$

$$\text{Sol.361.(d)} \frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A}$$

$$= \frac{\cos A}{1 - \frac{\sin A}{\cos A}} + \frac{\sin A}{1 - \frac{\cos A}{\sin A}}$$

$$= \frac{\cos^2 A}{\cos A - \sin A} - \frac{\sin^2 A}{\cos A - \sin A}$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos A - \sin A}$$

$$= \frac{(\cos A + \sin A)(\cos A - \sin A)}{\cos A - \sin A}$$

$$= \sin A + \cos A$$

$$\text{Sol.362.(a)} \sec \theta + \frac{1}{\cos \theta} = 2$$

$$\sec \theta + \sec \theta = 2$$

$$2 \sec \theta = 2 \Rightarrow \sec \theta = 1 = \sec 0^\circ \Rightarrow \theta = 0^\circ$$

$$\text{So, } \sec^{55} \theta + \frac{1}{\sec^{55} \theta} = \sec^{55} 0^\circ + \frac{1}{\sec^{55} 0^\circ}$$

$$= 1 + 1 = 2$$

$$\text{Sol.363.(c)} \tan 40^\circ = \cot 50^\circ = a$$

$$\text{So, } \frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ}$$

$$= \frac{\tan(360^\circ - 40^\circ) - \tan(360^\circ - 50^\circ)}{1 + \tan(360^\circ - 40^\circ) \tan(360^\circ - 50^\circ)}$$

$$= \frac{-\tan 40^\circ + \tan 50^\circ}{1 + \tan 40^\circ \tan 50^\circ}$$

$$= \frac{-a + \frac{1}{a}}{1 + a \times \frac{1}{a}} = \frac{1 - a^2}{2a}$$

$$\text{Sol.364.(a) Formula :-}$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan 2\alpha = \tan\{(\alpha + \beta) + (\alpha - \beta)\}$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \tan(\alpha - \beta)} = \frac{a + b}{1 - ab}$$

$$\text{Sol.365.(b) Short trick :}$$

$$\text{Putting } \theta = 45^\circ, \text{ we have,}$$

$$x = \frac{2 \sin 45^\circ}{1 + \cos 45^\circ + \sin 45^\circ} = \frac{\sqrt{2}}{1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}$$

$$= \frac{2}{\sqrt{2} + 2} = \frac{\sqrt{2}}{1 + \sqrt{2}}$$

$$\text{Now,}$$

$$\frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta} = \frac{1 - \cos 45^\circ + \sin 45^\circ}{1 + \sin 45^\circ}$$

$$= \frac{1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} = \frac{\sqrt{2} - 1 + 1}{\sqrt{2} + 1}$$

$$= \frac{\sqrt{2}}{\sqrt{2} + 1} = x$$

$$\text{Sol.366.(b)} \left\{ \left(\frac{\sec \theta - 1}{\sec \theta + 1} \right) \right\}^n = \operatorname{cosec} \theta - \cot \theta$$

$$\left\{ \frac{(\sec \theta - 1)(\sec \theta - 1)}{(\sec \theta + 1)(\sec \theta - 1)} \right\}^n = \operatorname{cosec} \theta - \cot \theta$$

$$\left\{ \frac{(\sec \theta - 1)}{\tan \theta} \right\}^{2n} = \operatorname{cosec} \theta - \cot \theta$$

$$\{\operatorname{cosec} \theta - \cot \theta\}^{2n} = \operatorname{cosec} \theta - \cot \theta$$

Comparing both side, we get

$$2n = 1 \Rightarrow n = \frac{1}{2} = 0.5$$

$$\text{Sol.367.(a)}$$

$$(2 \cos^2 \theta - 1) \left[\frac{1 + \tan \theta}{1 - \tan \theta} + \frac{1 - \tan \theta}{1 + \tan \theta} \right]$$

$$= \cos 2\theta \left[\frac{(1 + \tan \theta)^2 + (1 - \tan \theta)^2}{1 - \tan^2 \theta} \right]$$

$$(\because \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1)$$

$$= \cos 2\theta \left[\frac{2(1 + \tan^2 \theta)}{1 - \tan^2 \theta} \right]$$

$$= \cos 2\theta \left[\frac{2(\cos^2 \theta + \sin^2 \theta)}{\cos^2 \theta - \sin^2 \theta} \right]$$

$$= \cos 2\theta \times \frac{2}{\cos 2\theta} = 2$$

$$\text{Sol.368.(d)} \frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$$

Multiplying by 2 in numerator and denominator,

$$\frac{2 \sin 8\theta \cos \theta - 2 \sin 6\theta \cos 3\theta}{2 \cos 2\theta \cos \theta - 2 \sin 3\theta \sin 4\theta}$$

$$= \frac{\sin 9\theta + \sin 7\theta - (\sin 9\theta + \sin 3\theta)}{\cos 3\theta + \cos \theta - (\cos \theta - \cos 7\theta)}$$

$$= \frac{\sin 7\theta - \sin 3\theta}{\cos 3\theta + \cos 7\theta} = \frac{2 \cos 5\theta \sin 2\theta}{2 \cos 5\theta \cos 2\theta} = \tan 2\theta$$

$$\text{Sol.369.(a)}$$

$$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 1$$

Putting $A = 45^\circ$, we have :

$$(1 + \cot 45^\circ - \operatorname{cosec} 45^\circ)(1 + \tan 45^\circ + \sec 45^\circ) - 1$$

$$= (2 - \sqrt{2})(2 + \sqrt{2}) - 1 \Rightarrow 4 - 2 - 1 = 1$$

$$\text{Sol.370.(c)} \text{ As we know that,}$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\text{Then, } \cos A = \sqrt{1 - \left(\frac{1}{3}\right)^2} = \sqrt{\frac{8}{9}}$$

$$= -\frac{2\sqrt{2}}{3} \text{ and } \cos B = \sqrt{1 - \left(\frac{1}{5}\right)^2}$$

$$= \sqrt{\frac{24}{25}} = -\frac{2\sqrt{6}}{5}$$

($\because$ A and B are in second quadrant)

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$= \left(-\frac{2\sqrt{2}}{3}\right) \times \left(-\frac{2\sqrt{6}}{5}\right) + \frac{1}{3} \times \frac{1}{5}$$

$$= \frac{8\sqrt{3} + 1}{15}$$

$$\text{Sol.371.(a)}$$

$$\frac{\sin(A + B)}{\sin A \cos B} = \frac{\sin A \cos B + \cos A \sin B}{\sin A \cos B}$$

$$= 1 + \frac{\cos A \sin B}{\sin A \cos B} = 1 + \cot A \tan B$$

$$\text{Sol.372.(a)} \cos^2 45^\circ + \cos^2 135^\circ + \cos^2 225^\circ + \cos^2 315^\circ$$

$$= \left(\frac{1}{\sqrt{2}}\right)^2 + \cos^2(90^\circ + 45^\circ) + \cos^2(270^\circ - 45^\circ) + \cos^2(360^\circ - 45^\circ)$$

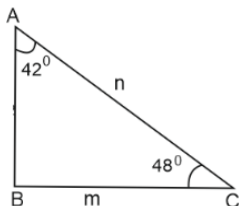
$$= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 2$$

Sol.373.(c) Formula used :

$$\cos 2\theta = 2\cos^2\theta - 1 \text{ or } 1 - 2\sin^2\theta$$

$$\text{Now, } \frac{2\cos^3\theta - \cos\theta}{\sin\theta - 2\sin^3\theta} = \frac{\cos\theta(2\cos^2\theta - 1)}{\sin\theta(1 - 2\sin^2\theta)} = \frac{\cos\theta \cdot \cos 2\theta}{\sin\theta \cdot \cos 2\theta} = \cot\theta$$

Sol.374.(d)



$$\sec 48^\circ = \frac{n}{m} \text{ and } \cot 42^\circ = \frac{\sqrt{n^2 - m^2}}{m}$$

$$\text{So, } \sec 48^\circ - \cot 42^\circ = \frac{n}{m} - \frac{\sqrt{n^2 - m^2}}{m} = \frac{n - \sqrt{n^2 - m^2}}{m}$$

$$\text{Sol.375.(c) } \sec^2\theta \sqrt{1 - \sin^2\theta} = \sec^2\theta \times \sqrt{\cos^2\theta} = \sec^2\theta \times \cos\theta = \sec\theta$$

Sol.376.(a) As we know, that

$$\begin{aligned} \operatorname{cosec}^2 A - \cot^2 A &= 1 \\ &= (\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1 \\ &= m \times (\operatorname{cosec} A - \cot A) = 1 \\ \Rightarrow (\operatorname{cosec} A - \cot A) &= \frac{1}{m} \end{aligned}$$

Sol.377.(c) $\cos\theta + \sec\theta = 2$

Putting $\theta = 0^\circ$, we get :

$$\cos 0^\circ + \sec 0^\circ = 2 \Rightarrow 1 + 1 = 2 \\ 2 = 2 \Rightarrow \text{LHS} = \text{RHS}$$

Now,

$$\sin^6\theta + \cos^6\theta = \sin^6 0^\circ + \cos^6 0^\circ = 0 + 1 = 1$$

Sol.378.(d) $(\operatorname{cosec} A + \cot A)(1 - \cos A)$

$$\begin{aligned} &= \left(\frac{1}{\sin A} + \frac{\cos A}{\sin A}\right)(1 - \cos A) \\ &= \frac{1 + \cos A}{\sin A} \times (1 - \cos A) \\ &= \frac{(1 - \cos^2 A)}{\sin A} = \frac{\sin^2 A}{\sin A} = \sin A \end{aligned}$$

Sol.379.(a) $\cos A = \sin^2 A$,

$$\begin{aligned} a \sin^{12} A + b \sin^{10} A + c \sin^8 A + \sin^6 A &= 1 \\ &= a \cos^6 A + b \cos^5 A + c \cos^4 A + \cos^3 A \\ &= 1 \dots \dots \dots (1) \end{aligned}$$

We know that, $(\cos A + \cos^2 A)^3 =$

$$\begin{aligned} &= \cos^3 A + \cos^6 A + 3\cos^3 A(\cos A + \cos^2 A) \\ &= \cos^6 A + 3\cos^5 A + 3\cos^4 A + \cos^3 A \\ &= 1 \dots \dots \dots (2) \end{aligned}$$

from eq. (2) with (1),

$$a = 1, b = 3 \text{ and } c = 3$$

$$\text{Then, } a + b + c = 7$$

Sol.380.(c) $4\cos\theta + 3\sin\theta = x \dots \dots \dots (1)$

$$4\sin\theta - 3\cos\theta = y \dots \dots \dots (2)$$

Squaring eq. (1) and (2), we get

$$\begin{aligned} 16\cos^2\theta + 9\sin^2\theta + 24\sin\theta \cdot \cos\theta &= x^2 \dots \dots \dots (3) \\ 16\sin^2\theta + 9\cos^2\theta - 24\sin\theta \cdot \cos\theta &= y^2 \dots \dots \dots (4) \end{aligned}$$

$$\begin{aligned} 16\sin^2\theta + 9\cos^2\theta - 24\sin\theta \cdot \cos\theta &= y^2 \dots \dots \dots (4) \\ \text{Adding eq. (3) and (4),} \end{aligned}$$

$$\begin{aligned} 16(\sin^2\theta + \cos^2\theta) + 9(\sin^2\theta + \cos^2\theta) &= x^2 + y^2 \\ &= x^2 + y^2 \end{aligned}$$

$$\text{then } x^2 + y^2 = 16 + 9 = 25$$

$$\begin{aligned} \text{Sol.381.(c) } \frac{\cos^4\theta - \sin^4\theta}{\sin^2\theta} &= \frac{(\cos^2\theta - \sin^2\theta)(\sin^2\theta + \cos^2\theta)}{\sin^2\theta} \\ &= \cot^2\theta - 1 \end{aligned}$$

Sol.382.(b)

$$\begin{aligned} \frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} + 1 &= \frac{\sin A \cot A - 1 - \sin A \cot A - 1}{\cot^2 A - \operatorname{cosec}^2 A} + 1 \\ &= \frac{-2}{-1} + 1 = 3 \end{aligned}$$

Sol.383.(b) Put $\theta = 45^\circ$

$$1 + \sin^2\theta - 3\sin\theta \cos\theta = 0$$

$$\Rightarrow 1 + \frac{1}{2} - \frac{3}{2} = 0$$

$$\text{Now, } \cot\theta = 1$$

$$\text{Sol.384.(b) } \cos^4\alpha - \sin^4\alpha = \frac{5}{6}$$

$$\Rightarrow (\cos^2\alpha - \sin^2\alpha)(\cos^2\alpha + \sin^2\alpha) = \frac{5}{6}$$

$$\Rightarrow \cos^2\alpha - (1 - \cos^2\alpha) = \frac{5}{6}$$

$$\Rightarrow \cos^2\alpha + \cos^2\alpha - 1 = \frac{5}{6}$$

$$\text{then, } 2\cos^2\alpha - 1 = \frac{5}{6}$$

Sol.385.(a)

$$\begin{aligned} \left[\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}\right] &= \frac{\cos^2\theta + \sin^2\theta}{\sin\theta \times \cos\theta} \\ &= \frac{1}{\sin\theta \times \cos\theta} = \operatorname{cosec}\theta \cdot \sec\theta \end{aligned}$$

Sol.386.(b)

$$\begin{aligned} \frac{\cos 50^\circ}{\sin 40^\circ} + \frac{3\operatorname{cosec} 80^\circ}{\sec 10^\circ} - 2\cos 50^\circ \cdot \operatorname{cosec} 40^\circ &= \frac{\sin 40^\circ}{\sin 40^\circ} + \frac{3\sec 10^\circ}{\sec 10^\circ} - 2\sin 40^\circ \cdot \operatorname{cosec} 40^\circ \\ &= 1 + 3 - 2(\sin 40^\circ \times \frac{1}{\sin 40^\circ}) \\ &= 1 + 3 - 2(1) = 4 - 2 = 2 \end{aligned}$$

Sol.387.(d) $\sqrt{3}\tan\theta = 3\sin\theta$

$$\Rightarrow \frac{\tan\theta}{\sin\theta} = \frac{3}{\sqrt{3}} = \sqrt{3} \Rightarrow \frac{1}{\cos\theta} = \sqrt{3}$$

$$\Rightarrow \cos^2\theta = \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

As per trigonometric identities, we have

$$\sin^2\theta + \cos^2\theta = 1$$

$$\Rightarrow \sin^2\theta = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{So, } \sin^2\theta - \cos^2\theta = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$

$$\begin{aligned} \text{Sol.388.(a) } \frac{1}{\tan\theta} + \tan\theta &= \frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta} \\ &= \frac{\cos^2\theta + \sin^2\theta}{\sin\theta \times \cos\theta} = \frac{1}{\sin\theta \times \cos\theta} \\ &= \frac{\sec\theta \times \operatorname{cosec}\theta}{1} = \sec\theta \times \operatorname{cosec}\theta \end{aligned}$$

Sol.389.(d) $\operatorname{cosec}(A - B) = 2$

$$\operatorname{cosec}(A - B) = \operatorname{cosec} 30^\circ$$

$$\Rightarrow A - B = 30^\circ \dots \dots \dots (1)$$

$$\cot(A + B) = \frac{1}{\sqrt{3}} \Rightarrow \cot(A + B) = \cot 60^\circ$$

$$\Rightarrow A + B = 60^\circ \dots \dots \dots (2)$$

Adding eqn(1) and (2) we get:

$$A = \frac{60^\circ + 30^\circ}{2} = \frac{90^\circ}{2} = 45^\circ$$

Sol.390.(a)

$$\begin{aligned} [(\sin\theta \tan\theta + \cos\theta)^2 - 1] &= \sin^2\theta \\ \tan^2\theta + \cos^2\theta + 2\sin\theta \cdot \tan\theta \cdot \cos\theta - 1 &= \sin^2\theta \\ &= \sin^2\theta \tan^2\theta + \cos^2\theta + 2\sin^2\theta - 1 \\ &= \sin^2\theta \tan^2\theta + \cos^2\theta + \sin^2\theta + \sin^2\theta - 1 \\ &= \sin^2\theta(1 + \tan^2\theta) + (\sin^2\theta + \cos^2\theta) - 1 \\ &= \sin^2\theta \cdot \sec^2\theta + 1 - 1 = \tan^2\theta \end{aligned}$$

Short Trick:- Put $\theta = 45^\circ$

$$[(\sin\theta \tan\theta + \cos\theta)^2 - 1]$$

$$\left[\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)^2 - 1\right] = 1$$

Only option (a) satisfies the given condition. $\Rightarrow \tan^2\theta = 1$

Sol.391.(b)

$$\begin{aligned} \sin^2 \frac{2\pi}{3} + \cos^2 \frac{5\pi}{6} - \tan^2 \frac{3\pi}{4} &= \sin^2 120^\circ + \cos^2 150^\circ - \tan^2 135^\circ \\ &= \left(-\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2 - 1 \\ &= \frac{3}{4} + \frac{3}{4} - 1 = \frac{6}{4} - 1 = \frac{1}{2} \end{aligned}$$

Sol.392.(a)

$$\begin{aligned} \frac{2\sin\theta}{\sqrt{4 - 4\sin^2\theta}} &= \frac{2\sin\theta}{\sqrt{4(1 - \sin^2\theta)}} \\ &= \frac{2\sin\theta}{\sqrt{4\cos^2\theta}} = \frac{2\sin\theta}{2\cos\theta} = \tan\theta \end{aligned}$$

Sol.393.(c)

$$\begin{aligned} \frac{1}{\cos x} \sqrt{\frac{\cos(\pi + x) \cos(-x)}{\sin(\pi - x) \cos\left(\frac{\pi}{2} + x\right)}} &= \frac{1}{\cos x} \sqrt{\frac{-\cos x \cos x}{\sin x (-\sin x)}} \\ &= \frac{1}{\cos x} \sqrt{\frac{-\cos x \cos x}{\sin x (-\sin x)}} \end{aligned}$$

$$= \frac{1}{\cos x} \times \frac{\cos x}{\sin x} = \frac{1}{\sin x} = \operatorname{cosec} x$$

Sol.394.(b) $\cos \theta = \sin(2\theta)$

$$\sin(90^\circ - \theta) = \sin(2\theta) \Rightarrow 90^\circ - \theta = 2\theta$$

$$\Rightarrow 90^\circ = 3\theta \Rightarrow \theta = 30^\circ$$

Now,

$$(\cos^4 30^\circ + \cos^3 30^\circ + \cos 30^\circ) + (\sin^4 30^\circ + \sin^3 30^\circ + \sin 30^\circ) + (\sin^2 30^\circ + \cos^2 30^\circ)$$

$$= \left(\frac{9}{16} + \frac{3\sqrt{3}}{8} + \frac{\sqrt{3}}{2}\right) + \left(\frac{1}{16} + \frac{1}{8} + \frac{1}{2}\right) + 1$$

$$= \left(\frac{9 + 14\sqrt{3}}{16}\right) + \frac{11}{16} + 1$$

$$= \frac{9 + 14\sqrt{3} + 11 + 16}{16} = \frac{36 + 14\sqrt{3}}{16}$$

$$= \frac{2(18 + 7\sqrt{3})}{16} = \frac{18 + 7\sqrt{3}}{8}$$

Sol.395.(c) Formula :- $\cot(90^\circ - \theta) = \tan \theta$

$$\tan 3\theta = \cot(\theta - 22^\circ)$$

$$\Rightarrow \cot(90^\circ - 3\theta) = \cot(\theta - 22^\circ)$$

$$\Rightarrow 90^\circ - 3\theta = \theta - 22^\circ$$

$$\Rightarrow 4\theta = 112^\circ \Rightarrow \theta = 28^\circ$$

Sol.396.(b)

$$\frac{x(1 - \operatorname{cosec}^2 30^\circ)}{1 + \operatorname{cosec}^2 30^\circ} = \left(\frac{1}{2}\right)^2 + (1 \times 4) - 2^2$$

$$\frac{x(1 - 4)}{1 + 4} = \frac{1}{4} + 4 - 4$$

$$x \times \frac{-3}{5} = \frac{1}{4} \Rightarrow x = \frac{1}{4} \times \frac{5}{-3} = -\frac{5}{12}$$

Sol.397.(a) Given that $14 \sec \theta = x$

$$\sec \theta - \cos \theta = 14$$

$$\sec \theta - \frac{1}{\sec \theta} = 14 \Rightarrow \frac{\sec^2 \theta - 1}{\sec \theta} = 14$$

[$14 \sec \theta = x$, given]

$$14 \sec \theta = \sec^2 \theta - 1 = \tan^2 \theta$$

$$14 \sec \theta = x = \tan^2 \theta$$

Sol.398.(d) $(1 + \tan 10^\circ)(1 + \tan 35^\circ)$

We know,

$$\text{If } A + B = 45^\circ \text{ or } 225^\circ \text{ then, } (1 + \tan A)$$

$$(1 + \tan B) = 2$$

$$(1 + \tan 10^\circ)(1 + \tan 35^\circ) = 2$$

Sol.399.(a) $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$

$$= \frac{2 \sin \left(\frac{4\theta + 2\theta}{2}\right) \cos \left(\frac{4\theta - 2\theta}{2}\right)}{2 \cos \left(\frac{4\theta + 2\theta}{2}\right) \cos \left(\frac{4\theta - 2\theta}{2}\right)}$$

$$= \frac{\sin 3\theta \cos \theta}{\cos 3\theta \cos \theta} = \frac{\sin 3\theta}{\cos 3\theta} = \tan 3\theta$$

Sol.400.(b) $\cos 2A = \sin 75^\circ$

$$\cos 2A = \sin(90^\circ - 15^\circ) = \cos 15^\circ$$

$$2A = 15^\circ \Rightarrow A = \frac{15^\circ}{2} = 7.5^\circ$$

So, smallest positive value of $A = 7.5^\circ$

Sol.401.(d) $\sin x - 3 \cos x = \sqrt{3} \cos x$

$$\sin x = \sqrt{3} \cos x + 3 \cos x$$

Dividing both side by $\cos x$. we get

$$\tan x = \sqrt{3} + 3$$

$$\cot x = \frac{1}{\sqrt{3} + 3} = \frac{\sqrt{3} - 3}{3 - 9}$$

$$= -\frac{\sqrt{3} - 3}{6} = \frac{3 - \sqrt{3}}{6}$$

Sol.402.(b) As per trigonometric identity,

$$\text{we have } \sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\text{Now, } (4 \sin^3 x - 3 \sin x + \sin 3x)$$

$$= 4 \sin^3 x - 3 \sin x + 3 \sin x - 4 \sin^3 x = 0$$

Sol.403.(c)

$$\cos^2(90^\circ - \theta) - \left[\frac{\cos(90^\circ - \theta) \cos \theta}{\cot \theta}\right]$$

$$= \sin^2 \theta - \left[\frac{\sin \theta \cos \theta}{\cot \theta}\right]$$

$$= \sin^2 \theta - [\sin \theta \cos \theta \times \frac{\sin \theta}{\cos \theta}]$$

$$\Rightarrow \sin^2 \theta - \sin^2 \theta = 0$$

Sol.404.(d) $\sin 2\theta \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}\right)$

$$\Rightarrow 2 \sin \theta \cos \theta \left(\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}\right)$$

$$\Rightarrow 2 \sin \theta \cos \theta \times \frac{1}{\sin \theta \cos \theta} = 2$$

Sol.405.(b) Formula used:-

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

Putting $A = 75^\circ$, $B = 15^\circ$. We have,

$$2 \times \cos 75^\circ \cos 15^\circ = \cos(75^\circ + 15^\circ)$$

$$+ \cos(75^\circ - 15^\circ)$$

$$2 \times \cos 75^\circ \cos 15^\circ = \cos 90^\circ + \cos 60^\circ$$

$$\cos 75^\circ \cos 15^\circ = \frac{1}{2} \left(0 + \frac{1}{2}\right)$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Sol.406.(c) $\operatorname{cosec} \theta + \cot \theta = 5 - (1)$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{5} - (2)$$

From equation (1) and (2)

$$\operatorname{Cosec} \theta = \frac{5 + \frac{1}{5}}{2} = \frac{1}{2} \left(5 + \frac{1}{5}\right)$$

Sol.407.(a) $1 + \tan \theta = \sqrt{3}$

$$\tan \theta = \sqrt{3} - 1$$

$$\sqrt{3} \cot \theta - 1 = \sqrt{3} \times \frac{1}{\sqrt{3} - 1} - 1$$

$$\Rightarrow \frac{\sqrt{3}(\sqrt{3} + 1)}{2} - 1 = \frac{3 + \sqrt{3} - 2}{2}$$

$$\Rightarrow \frac{\sqrt{3} + 1}{2}$$

Sol.408.(d) $\sin 2A = 2 \sin A \cos A$

$$100 (\sin 15^\circ \cos 15^\circ) = 100 \times \frac{\sin 30^\circ}{2}$$

$$= 100 \times \frac{1}{2} \times \frac{1}{2} = 25$$

Sol.409.(b) $\operatorname{cosec}^2 \theta + \cot^2 \theta = \frac{5}{3}$

$$\Rightarrow \frac{1}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{5}{3}$$

$$\Rightarrow \frac{1 + 1 - \sin^2 \theta}{\sin^2 \theta} = \frac{5}{3} \Rightarrow \frac{2 - \sin^2 \theta}{\sin^2 \theta} = \frac{5}{3}$$

$$\Rightarrow 6 - 3 \sin^2 \theta = 5 \sin^2 \theta$$

$$\Rightarrow 6 = 8 \sin^2 \theta \Rightarrow \sin^2 \theta = \frac{6}{8} = \frac{3}{4}$$

$$\Rightarrow \sin \theta = \frac{\sqrt{3}}{2} = \sin 60^\circ \Rightarrow \theta = 60^\circ$$

$$\cot 2\theta = \cot(2 \times 60^\circ) = \cot 120^\circ = -\frac{1}{\sqrt{3}}$$

Short trick:- Put $\theta = 60^\circ$

$$\operatorname{cosec}^2 \theta + \cot^2 \theta = \left(\frac{2}{\sqrt{3}}\right)^2 + \left(-\frac{1}{\sqrt{3}}\right)^2 = \frac{5}{3}$$

LHS = RHS

Now,

$$\cot 2\theta = \cot(2 \times 60^\circ) = \cot 120^\circ = -\frac{1}{\sqrt{3}}$$

Sol.410.(b) $(\operatorname{cosec} \theta - \cot \theta)^2$

$$= \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}\right)^2$$

$$= \frac{(1 - \cos \theta)^2}{\sin^2 \theta} = \frac{(1 - \cos \theta)(1 - \cos \theta)}{1 - \cos^2 \theta}$$

$$= \frac{(1 - \cos \theta)(1 - \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)} = \frac{(1 - \cos \theta)}{(1 + \cos \theta)}$$

Sol.411.(a) Put $x = 45^\circ$

$$\tan \frac{45^\circ}{2} = \frac{1 - \cos 45^\circ}{\sin 45^\circ}$$

$$\tan 22.5^\circ = \frac{1 - \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}$$

$$\Rightarrow \tan 22.5^\circ = \sqrt{2} - 1$$

Sol.412.(a)

$$\left(\frac{\sin 23^\circ \cos 67^\circ + \sec 52^\circ \sin 38^\circ}{\operatorname{cosec}^2 20^\circ - \tan^2 70^\circ}\right) +$$

$$\left(\frac{\cos 23^\circ \sin 67^\circ + \operatorname{cosec} 52^\circ \cos 38^\circ}{\operatorname{cosec}^2 20^\circ - \tan^2 70^\circ}\right)$$

$$= \left\{ \frac{\sin^2 23^\circ + \operatorname{cosec} 38^\circ \sin 38^\circ}{\operatorname{cosec}^2 20^\circ - \cot^2 20^\circ} \right.$$

$$+ \left. \frac{\cos^2 23^\circ + \sec 38^\circ \cos 38^\circ}{\operatorname{cosec}^2 20^\circ - \cot^2 20^\circ} \right\}$$

$$= \left\{ \frac{(\sin^2 23^\circ + \cos^2 23^\circ) + (\operatorname{cosec} 38^\circ \sin 38^\circ)}{(\operatorname{cosec}^2 20^\circ - \cot^2 20^\circ)} \right.$$

$$+ \left. \frac{(\sec 38^\circ \cos 38^\circ)}{(\operatorname{cosec}^2 20^\circ - \cot^2 20^\circ)} \right\}$$

$$= \frac{1 + 1 + 1}{1} = 3$$

Sol.413.(a) $4 \sin^2 \theta = 3(1 + \cos \theta)$

$$4 \sin^2 \theta = 3 + 3 \cos \theta$$

$$4(1 - \cos^2 \theta) = 3 + 3 \cos \theta$$

$$4 - 4 \cos^2 \theta = 3 + 3 \cos \theta$$

$$- 4 \cos^2 \theta - 3 \cos \theta + 1 = 0$$

$$4 \cos^2 \theta + 3 \cos \theta - 1 = 0$$

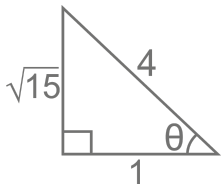
$$4 \cos^2 \theta + 4 \cos \theta - \cos \theta - 1 = 0$$

$$4 \cos \theta (\cos \theta + 1) - 1(\cos \theta + 1) = 0$$

$$(\cos \theta + 1)(4 \cos \theta - 1) = 0$$

$$\cos \theta = -1 \text{ not possible as } 0^\circ < \theta < 90^\circ, \text{ so}$$

$$\cos \theta = \frac{1}{4}$$



$$2\tan\theta + 4\sin\theta - \sec\theta$$

$$= 2\left(\frac{\sqrt{15}}{1}\right) + 4\left(\frac{\sqrt{15}}{4}\right) - \frac{4}{1}$$

$$= 2\sqrt{15} + \sqrt{15} - 4 = 3\sqrt{15} - 4$$

Sol.414.(c) $3\sec^2\theta + \tan\theta - 7 = 0$

$$3(1 + \tan^2\theta) + \tan\theta - 7 = 0$$

$$3 + 3\tan^2\theta + \tan\theta - 7 = 0$$

$$3\tan^2\theta + \tan\theta - 4 = 0$$

$$3\tan^2\theta + 4\tan\theta - 3\tan\theta - 4 = 0$$

$$3\tan^2\theta - 3\tan\theta + 4\tan\theta - 4 = 0$$

$$3\tan\theta(\tan\theta - 1) + 4(\tan\theta - 1) = 0$$

$$\tan\theta = -\frac{4}{3} \text{ not possible as } 0^\circ < \theta < 90^\circ, \tan\theta = 1, \text{ hence } \theta = 45^\circ$$

So, $\frac{2\sin\theta + 3\cos\theta}{\csc\theta + \sec\theta}$

$$= \frac{2\sin 45^\circ + 3\cos 45^\circ}{\csc 45^\circ + \sec 45^\circ}$$

$$= \frac{2\left(\frac{1}{\sqrt{2}}\right) + 3\left(\frac{1}{\sqrt{2}}\right)}{\sqrt{2} + \sqrt{2}} = \frac{\frac{5}{\sqrt{2}}}{2\sqrt{2}} = \frac{5}{4}$$

Alternate :- Put $\theta = 45^\circ$

$$3\sec^2\theta + \tan\theta - 7 = 3 \times (\sqrt{2})^2 + 1 - 7$$

$$= 6 + 1 - 7 = 0 \text{ (LHS = RHS)}$$

So, $\frac{2\sin\theta + 3\cos\theta}{\csc\theta + \sec\theta}$

$$= \frac{2\sin 45^\circ + 3\cos 45^\circ}{\csc 45^\circ + \sec 45^\circ}$$

$$= \frac{2\left(\frac{1}{\sqrt{2}}\right) + 3\left(\frac{1}{\sqrt{2}}\right)}{\sqrt{2} + \sqrt{2}} = \frac{\frac{5}{\sqrt{2}}}{2\sqrt{2}} = \frac{5}{4}$$

Sol.415.(d) Putting $\theta = 45^\circ$

$$1 + \sqrt{\frac{\cot\theta + \cos\theta}{\cot\theta - \cos\theta}}$$

$$= 1 + \sqrt{\frac{\cot 45^\circ + \cos 45^\circ}{\cot 45^\circ - \cos 45^\circ}}$$

$$= 1 + \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}} = 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}}$$

$$= 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$= 1 + \sqrt{\frac{(\sqrt{2} + 1)^2}{1}} = 1 + \sqrt{2} + 1 = 2 + \sqrt{2}$$

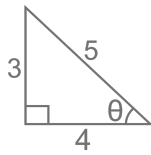
Putting $\theta = 45^\circ$ in options, we find that option (d) gets satisfied.

$$1 + \sec\theta + \tan\theta$$

$$= 1 + \sec 45^\circ + \tan 45^\circ$$

$$= 1 + \sqrt{2} + 1 \Rightarrow 2 + \sqrt{2}$$

Sol.416.(c) In this question, we will put the value of pythagoras triplet.



$$\left(\frac{1 - \cot\theta}{1 - \tan\theta}\right)^2 - 1 = \left(\frac{1 - \frac{4}{3}}{1 - \frac{3}{4}}\right)^2 - 1$$

$$\Rightarrow \left(\frac{-\frac{1}{3}}{\frac{1}{4}}\right)^2 - 1 = \left(\frac{-4}{3}\right)^2 - 1$$

$$\Rightarrow \frac{16}{9} - 1 = \frac{7}{9}$$

Putting given triplets in options, we find that option (c) gets satisfied.

$$\cot^2\theta - 1 = \left(\frac{4}{3}\right)^2 - 1 = \frac{16}{9} - 1 = \frac{7}{9}$$

Sol.417.(a) Let $\theta = 45^\circ$, now

$$(\cos^6\theta + \sin^6\theta - 1)$$

$$(\tan^2\theta + \cot^2\theta + 2) + 3$$

$$= \left[\left(\frac{1}{\sqrt{2}}\right)^6 + \left(\frac{1}{\sqrt{2}}\right)^6 - 1\right][1^2 + 1^2 + 2] + 3$$

$$= \left[\frac{1}{8} + \frac{1}{8} - 1\right][4] + 3 = -3 + 3 = 0$$

Sol.418.(c) $\sin^2\theta - \cos^2\theta - 3\sin\theta + 2 = 0$

$$\sin^2\theta - (1 - \sin^2\theta) - 3\sin\theta + 2 = 0$$

$$\sin^2\theta - 1 + \sin^2\theta - 3\sin\theta + 2 = 0$$

$$2\sin^2\theta - 3\sin\theta + 1 = 0$$

$$2\sin^2\theta - 2\sin\theta - \sin\theta + 1 = 0$$

$$2\sin\theta(\sin\theta - 1) - 1(\sin\theta - 1) = 0$$

$$(\sin\theta - 1)(2\sin\theta - 1) = 0$$

$$\sin\theta = 1 \text{ not possible, so } \sin\theta = \frac{1}{2},$$

$$\theta = 30^\circ \Rightarrow \text{So, } 1 + \sec\theta + \tan\theta$$

$$= 1 + \sec 30^\circ + \tan 30^\circ$$

$$= 1 + \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = 1 + \frac{3}{\sqrt{3}} = 1 + \sqrt{3}$$

Alternate :- put $\theta = 30^\circ, 60^\circ$ one by one, we find that $\theta = 30^\circ$ satisfies the given terms.

$$\sin^2\theta - \cos^2\theta - 3\sin\theta + 2$$

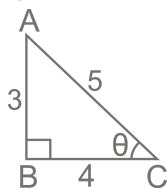
$$= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 - 3 \times \frac{1}{2} + 2$$

$$= \frac{1}{4} - \frac{3}{4} - \frac{3}{2} + 2 = 0 \text{ (LHS = RHS)}$$

$$\text{So, } 1 + \sec\theta + \tan\theta \Rightarrow 1 + \sec 30^\circ + \tan 30^\circ$$

$$= 1 + \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = 1 + \frac{3}{\sqrt{3}} = 1 + \sqrt{3}$$

Sol.419.(b) In this question, we will use pythagorean triplets of 3, 4, 5.



$$\sec\theta = \frac{5}{4}, \tan\theta = \frac{3}{4}, \sin\theta = \frac{3}{5} \text{ and}$$

$$\cos\theta = \frac{4}{5}$$

$$(\sec\theta - \tan\theta)^2 (1 + \sin\theta)^2 \div \cos^2\theta$$

$$= \left(\frac{5}{4} - \frac{3}{4}\right)^2 \left(1 + \frac{3}{5}\right)^2 \div \left(\frac{4}{5}\right)^2$$

$$= \left(\frac{1}{2}\right)^2 \times \left(\frac{8}{5}\right)^2 \div \left(\frac{4}{5}\right)^2$$

$$= \frac{1}{4} \times \frac{64}{25} \div \frac{16}{25} = \frac{1}{4} \times \frac{64}{25} \times \frac{25}{16} = 1$$

Sol.420.(c) $2\sin^2\theta + 3\cos\theta = 3$

$$2(1 - \cos^2\theta) + 3\cos\theta = 3$$

$$2 - 2\cos^2\theta + 3\cos\theta = 3$$

$$-2\cos^2\theta + 3\cos\theta - 1 = 0$$

$$2\cos^2\theta - 3\cos\theta + 1 = 0$$

$$2\cos^2\theta - 2\cos\theta - \cos\theta + 1 = 0$$

$$2\cos\theta(\cos\theta - 1) - 1(\cos\theta - 1) = 0$$

$$(\cos\theta - 1)(2\cos\theta - 1) = 0$$

$$\cos\theta = 1 \text{ not possible as } 0^\circ < \theta < 90^\circ,$$

$$\text{so } \cos\theta = \frac{1}{2}, \theta = 60^\circ$$

$$\text{So, } \sec^2\theta + \cot^2\theta = \sec^2 60^\circ + \cot^2 60^\circ$$

$$= (2)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 = 4 + \frac{1}{3} = 4\frac{1}{3}$$

Alternate method :- put $\theta = 60^\circ$

$$2\sin^2\theta + 3\cos\theta = 2 \times \left(\frac{\sqrt{3}}{2}\right)^2 + 3 \times \frac{1}{2}$$

$$= 2 \times \frac{3}{4} + \frac{3}{2} = \frac{3}{2} + \frac{3}{2} = 3$$

$$\text{LHS} = \text{RHS}$$

$$\text{So, } \sec^2\theta + \cot^2\theta = \sec^2 60^\circ + \cot^2 60^\circ$$

$$= (2)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 = 4 + \frac{1}{3} = 4\frac{1}{3}$$

Sol.421.(d) Put $A = 135^\circ$

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} - \sin A$$

$$= \frac{-\frac{1}{\sqrt{2}}}{1 - (-1)} + \frac{\frac{1}{\sqrt{2}}}{1 - (-1)} - \frac{1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

$$\text{Only } \cos A \text{ gives } -\frac{1}{\sqrt{2}} \text{ at } A = 135^\circ,$$

so $\cos A$ is answer

Sol.422.(d)

$$\csc^4 A (1 - \cos^4 A) - 2\cot^2 A - 1$$

Put $A = 45^\circ$ in equation because all option gives different values at $A = 45^\circ$

$$(\sqrt{2})^4 \left\{1 - \left(\frac{1}{\sqrt{2}}\right)^4\right\} - 2 \times (1)^2 - 1$$

$$= 4 \times \frac{3}{4} - 3 = 3 - 3 = 0$$

Sol.423.(d) $6\tan A(\tan A + 1) = 5 - \tan A$,

$$6\tan^2 A + 6\tan A + \tan A - 5 = 0$$

$$6\tan^2 A + 7\tan A - 5 = 0$$

$$6\tan^2 A + 10\tan A - 3\tan A - 5 = 0$$

$$2\tan A(3\tan A + 5) - 1(3\tan A + 5) = 0$$

$$(2\tan A - 1)(3\tan A + 5) = 0$$

$$\tan A = \frac{1}{2}, -\frac{5}{3}$$

$$\text{Taking } \tan A = \frac{1}{2} \dots (0 < A < \frac{\pi}{2})$$

$$\sin A = \frac{1}{\sqrt{5}}, \cos A = \frac{2}{\sqrt{5}}$$

$$\sin A + \cos A = \frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}} = \frac{3}{\sqrt{5}}$$

Sol.424.(a)

$$\left(\frac{1 - \cot \theta}{1 - \tan \theta}\right)^2 + 1 \Rightarrow \left(\frac{1 - \cot \theta}{1 - \frac{1}{\cot \theta}}\right)^2 + 1$$

$$= \left(\frac{1 - \cot \theta}{\cot \theta - 1}\right)^2 \times \cot^2 \theta + 1$$

$$= (-1)^2 \times \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$$

$$\Rightarrow \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$$

$$\text{Sol.425.(b)} \quad \frac{5 \cos^2 62^\circ + 5 \cos^2 28^\circ - 21}{7 \sin^2 35^\circ + 7 \sin^2 55^\circ + 1}$$

$$= \frac{5 \cos^2 62^\circ + 5 \sin^2 (90^\circ - 28^\circ) - 21}{7 \sin^2 35^\circ + 7 \cos^2 (90^\circ - 55^\circ) + 1}$$

$$= \frac{5 (\cos^2 62^\circ + \sin^2 62^\circ) - 21}{7 (\sin^2 35^\circ + \cos^2 35^\circ) + 1}$$

$$= \frac{5(1) - 21}{7(1) + 1} \Rightarrow \frac{-16}{8} = -2$$

$$\text{Sol.426.(b)} \quad \frac{\sin^2 \theta - 3 \sin \theta + 2}{\cos^2 \theta} = 1$$

$$\sin^2 \theta - 3 \sin \theta + 2 = \cos^2 \theta$$

$$\sin^2 \theta - 3 \sin \theta + 2 = 1 - \sin^2 \theta$$

$$2 \sin^2 \theta - 3 \sin \theta + 1 = 0$$

$$2 \sin^2 \theta - 2 \sin \theta - \sin \theta + 1 = 0$$

$$2 \sin \theta (\sin \theta - 1) - 1(\sin \theta - 1) = 0$$

$$(2 \sin \theta - 1)(\sin \theta - 1) = 0$$

$$\sin \theta = \frac{1}{2}, \theta = 30^\circ, 90^\circ$$

90° is not possible because $\cos \theta$ cannot be 0. $\Rightarrow \cos 2\theta - \sin 3\theta + \operatorname{cosec} 2\theta$

$$\cos 60^\circ - \sin 90^\circ + \operatorname{cosec} 60^\circ$$

$$= \frac{1}{2} - 1 + \frac{2}{\sqrt{3}} = \frac{-3 + 4\sqrt{3}}{6}$$

Sol.427.(a)

$$\frac{4 \tan^2 30^\circ + \sin^2 30^\circ \cos^2 45^\circ + \sec^2 48^\circ - \cot^2 42^\circ}{\cos 37^\circ \sin 53^\circ + \sin 37^\circ \cos 53^\circ + \tan 18^\circ \tan 72^\circ}$$

$$\frac{4 \times \frac{1}{3} + \frac{1}{4} \times \frac{1}{2} + \sec^2 48^\circ - \cot^2 42^\circ}{\cos 37^\circ \sin 53^\circ + \sin 37^\circ \cos 53^\circ + \tan 18^\circ \tan 72^\circ}$$

$$\frac{\frac{4}{3} + \frac{1}{8} + \sec^2 (90 - 42) - \cot^2 42^\circ}{\cos 37^\circ \sin (90 - 37) + \sin 37^\circ \cos (90 - 37) + \tan (90 - 72) \tan 72^\circ}$$

$$\frac{\frac{4}{3} + \frac{1}{8} + \operatorname{cosec}^2 42^\circ - \cot^2 42^\circ}{\cos 37^\circ \cos 37^\circ + \sin 37^\circ \sin 37^\circ + \cot 72^\circ \tan 72^\circ}$$

$$\frac{\frac{4}{3} + \frac{1}{8} + 1}{\cos^2 37^\circ + \sin^2 37^\circ + 1}$$

$$= \frac{\frac{4}{3} + \frac{1}{8} + 1}{1 + 1} = \frac{\frac{59}{24}}{2} = \frac{59}{48}$$

Sol.428.(d)

$$\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$$

$$= \frac{1 - \sin^2 \theta + \cos \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}}}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}$$

$$= \frac{\cos^2 \theta + \cos \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta \times \sin^2 \theta}}}{\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \times \sin \theta}}$$

$$= \frac{\cos \theta (1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} \times \frac{\frac{1}{\cos \theta \times \sin \theta}}{\frac{1}{\cos \theta \times \sin \theta}}$$

$$= \frac{\cos \theta}{\sin \theta} = \cot \theta$$

Sol.429.(d)

$$1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$$

$$1 + \tan^2 \theta + \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$$

$$\sec^2 \theta + \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$$

$$\sec^2 \theta + \sec^2 \theta - 1 + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$$

$$\frac{2 \sec^2 \theta - 1 + 2 \sin \theta \sec^2 \theta}{1} = \frac{a}{b}$$

Using componendo and dividendo, we get

$$\frac{2 \sec^2 \theta - 1 + 2 \sin \theta \sec^2 \theta + 1}{2 \sec^2 \theta - 1 + 2 \sin \theta \sec^2 \theta - 1} = \frac{a + b}{a - b}$$

$$\text{So, } \frac{a + b}{a - b} = \frac{2 \sec^2 \theta + 2 \sin \theta \sec^2 \theta}{2 \sec^2 \theta - 2 + 2 \sin \theta \sec^2 \theta}$$

$$= \frac{\frac{2}{\cos^2 \theta} + \frac{2 \sin \theta}{\cos^2 \theta}}{\frac{2}{\cos^2 \theta} - 2 + \frac{2 \sin \theta}{\cos^2 \theta}}$$

$$= \frac{2 + 2 \sin \theta}{2 - 2 \cos^2 \theta + 2 \sin \theta}$$

$$\Rightarrow \frac{2 + 2 \sin \theta}{2 \sin^2 \theta + 2 \sin \theta} = \frac{2(1 + \sin \theta)}{2 \sin \theta (1 + \sin \theta)}$$

$$\Rightarrow \frac{1}{\sin \theta} = \operatorname{cosec} \theta$$

Short trick :- Put $\theta = 30^\circ$

$$1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta$$

$$= 1 + 2 \times \left(\frac{1}{\sqrt{3}}\right)^2 + 2 \times \frac{1}{2} \times \left(\frac{2}{\sqrt{3}}\right)^2$$

$$= 1 + \frac{2}{3} + \frac{4}{3} = 3 = \frac{a}{b}$$

Here $a = 3$ and $b = 1$

$$\frac{a + b}{a - b} = \frac{3 + 1}{3 - 1} = 2$$

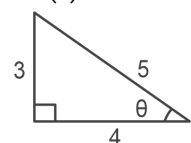
now, checking options,

$$\operatorname{cosec} 30^\circ = 2$$

Sol.430.(d)

$$\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2}$$

.....(1)



$$\text{Put } \tan \theta = \frac{3}{4}, \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5} \text{ and}$$

so on.

Putting respective values in (1), we get

$$\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2} = \frac{(1 + \frac{5}{4} \times \frac{5}{3})^2 (\frac{5}{4} - \frac{3}{4})^2 (1 + \frac{3}{5})}{(\frac{3}{5} + \frac{5}{4})^2 + (\frac{4}{5} + \frac{5}{3})^2}$$

$$\begin{aligned} &= \frac{(1 + \frac{25}{12})^2 \times (\frac{1}{4}) \times (\frac{8}{5})}{(\frac{12 + 25}{20})^2 + (\frac{12 + 25}{15})^2} \\ &= \frac{\frac{1}{12 \times 12} \times \frac{1}{4} \times \frac{8}{5}}{\frac{1}{400} + \frac{1}{225}} = \frac{\frac{1}{12 \times 6} \times \frac{1}{5}}{\frac{625}{400 \times 225}} = \frac{2}{5} \end{aligned}$$

Among all the given options, only option (d) satisfies the above obtained value.

$$\begin{aligned} \text{Sol.431.(c)} & \left(\frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta \right) \div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta) \\ &= \left(\frac{\sin^3 \theta}{\cos^3 \theta \times \sec^2 \theta} + \frac{\cos^3 \theta}{\sin^3 \theta \times \operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta \right) \div (\sec^2 \theta + \operatorname{cosec}^2 \theta) \\ &= \left(\frac{\sin^3 \theta}{\cos \theta} + \frac{\cos^3 \theta}{\sin \theta} + 2 \sin \theta \cos \theta \right) \div \left(\frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} \right) \\ &= \left(\frac{\sin^4 \theta + \cos^4 \theta + 2 \sin^2 \theta \cos^2 \theta}{\sin \theta \cos \theta} \right) \div \left(\frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \theta \cos^2 \theta} \right) \\ &= \frac{(\sin^2 \theta + \cos^2 \theta)^2}{\sin \theta \cos \theta} \div \left(\frac{1}{\sin^2 \theta \cos^2 \theta} \right) \\ &= \frac{1}{\sin \theta \cos \theta} \div \left(\frac{1}{\sin^2 \theta \cos^2 \theta} \right) = \sin \theta \cos \theta \end{aligned}$$

Sol.432.(c) $3 \tan \theta = 2\sqrt{3} \sin \theta$

$$\frac{3 \sin \theta}{\cos \theta} = 2\sqrt{3} \sin \theta \Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$$

So, $\theta = 30^\circ$

$$\frac{\operatorname{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2 \theta + \tan^2 2\theta} = \frac{\operatorname{cosec}^2 60^\circ + \cot^2 60^\circ}{\sin^2 30^\circ + \tan^2 60^\circ}$$

$$= \frac{\frac{4}{3} + \frac{1}{3}}{\frac{1}{4} + 3} = \frac{\frac{5}{3}}{\frac{13}{4}} = \frac{20}{39}$$

$$\text{Sol.433.(d)} \quad \frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 2}$$

Putting $\theta = 45^\circ$ in above expression, we get

$$\begin{aligned} & \frac{\tan^6 45^\circ - \sec^6 45^\circ + 3 \sec^2 45^\circ \tan^2 45^\circ}{\tan^2 45^\circ + \cot^2 45^\circ + 2} \\ &= \frac{(1)^6 - (\sqrt{2})^6 + 3(\sqrt{2})^2 (1)^2}{(1)^2 + (1)^2 + 2} \\ &= \frac{1 - 8 + 6}{4} = \frac{-1}{4} \end{aligned}$$

If we put $\theta = 45^\circ$ in given options, then only option (d) satisfies the above value.

Sol.434.(b)

$$\begin{aligned} & \frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2} \\ &= \frac{\cos^2 \theta - \sin^2 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} + 1)^2 - \cot^2 - 2} \\ &= \frac{\cos^2 \theta + \sin^2 \theta + 3}{\operatorname{cosec}^2 \theta + 1 + 2 \operatorname{cosec} \theta - \cot^2 \theta - 2} \end{aligned}$$

$$= \frac{1+3}{1+1+2\operatorname{cosec}\theta-2} = \frac{4}{2\operatorname{cosec}\theta}$$

$$= 2\sin\theta$$

Sol.435.(d)

$$\frac{(1-\sin\theta+\cos\theta)^2(1-\cos\theta)\sec^3\theta\operatorname{cosec}^2\theta}{(\sec\theta-\tan\theta)(\tan\theta+\cot\theta)}$$

Putting $\theta = 45^\circ$,

$$= \frac{(1-\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}})^2(1-\frac{1}{\sqrt{2}})(\sqrt{2})^3(\sqrt{2})^2}{(\sqrt{2}-1)(1+1)}$$

$$= \frac{(1)^2(\frac{\sqrt{2}-1}{\sqrt{2}})4\sqrt{2}}{(\sqrt{2}-1)(1+1)} = \frac{4}{2} = 2$$

Put $\theta = 45^\circ$ in options

So, option (d) satisfies.

$$2\tan 45^\circ = 2$$

$$\text{Sol.436.(d)} \quad \frac{\sin^2\theta}{\cos^2\theta-3\cos\theta+2} = 1$$

$$\Rightarrow \sin^2\theta = \cos^2\theta - 3\cos\theta + 2$$

$$\Rightarrow 1 - \cos^2\theta = \cos^2\theta - 3\cos\theta + 2$$

$$\Rightarrow 2\cos^2\theta - 3\cos\theta + 1 = 0$$

$$\Rightarrow 2\cos^2\theta - 2\cos\theta - \cos\theta + 1 = 0$$

$$\Rightarrow 2\cos\theta(\cos\theta - 1) - 1(\cos\theta - 1) = 0$$

$$\Rightarrow (2\cos\theta - 1)(\cos\theta - 1) = 0$$

$$\text{For } \cos\theta = \frac{1}{2}, \theta = 60^\circ \text{ and for}$$

$$\cos\theta = 1, \theta = 0^\circ,$$

as value of θ lies in first quadrant which is not possible in case of $\cos\theta = 1$,

so value of $\cos\theta = \frac{1}{2}$ and $\theta = 60^\circ$ is admissible and correct.

$$\text{Putting } \theta = 60^\circ, \text{ in } \frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta}$$

$$\text{we get, } \frac{\tan^2\frac{60^\circ}{2} + \sin^2\frac{60^\circ}{2}}{\tan 60^\circ + \sin 60^\circ}$$

$$= \frac{\tan^2 30^\circ + \sin^2 30^\circ}{\tan 60^\circ + \sin 60^\circ}$$

$$= \frac{(\frac{1}{\sqrt{3}})^2 + (\frac{1}{2})^2}{\sqrt{3} + \frac{\sqrt{3}}{2}} = \frac{\frac{1}{3} + \frac{1}{4}}{\frac{3\sqrt{3}}{2}} = \frac{\frac{7}{12}}{\frac{3\sqrt{3}}{2}}$$

$$= \frac{14}{36\sqrt{3}} = \frac{7}{18\sqrt{3}} = \frac{7}{18\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

(multiplying numerator and denominator

$$\text{by } \sqrt{3}) = \frac{7\sqrt{3}}{54}$$

Sol.437.(b)

$$\frac{(1-2\sin^2\theta\cos^2\theta)(\cot\theta+1)\cos\theta}{(\sin^4\theta+\cos^4\theta)(1+\tan\theta)\operatorname{cosec}\theta} - 1$$

Put $\theta = 45^\circ$

$$\frac{\{(1-2(\frac{1}{\sqrt{2}})^2(\frac{1}{\sqrt{2}})^2)\}(1+1)\frac{1}{\sqrt{2}}}{[(\frac{1}{\sqrt{2}})^4+(\frac{1}{\sqrt{2}})^4](1+1)\sqrt{2}} - 1$$

$$= \frac{(1-\frac{1}{2})(1+1)\frac{1}{\sqrt{2}}}{[\frac{1}{4}+\frac{1}{4}]2\sqrt{2}} - 1 = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2} \times 2\sqrt{2}} - 1$$

$$= \frac{1}{\sqrt{2} \times \sqrt{2}} - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$$

on putting $\theta = 45^\circ$ in given options, then only option (b) satisfies.

$$\Rightarrow -\sin^2\theta = -(\frac{1}{\sqrt{2}})^2 = -\frac{1}{2}$$

Sol.438.(b) As, $(\cos^2\theta + \sin^2\theta) = 1$

So,

$$(\cos^2x + \sin^2x) + (\cos^22x + \sin^22x) +$$

$$(\cos^23x + \sin^23x) + \dots +$$

$$(\cos^230x + \sin^230x)$$

$$= 1 + 1 + 1 + \dots + 1 = 1 \times 30 = 30$$

$$\text{So, required average} = \frac{30}{60} = 0.5$$

Sol.439.(b)

$$= \frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta-1} + \frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta+1} - \tan^2\theta$$

$$= \frac{\operatorname{cosec}^2\theta + \operatorname{cosec}\theta + \operatorname{cosec}^2\theta - \operatorname{cosec}\theta}{\operatorname{cosec}^2\theta - 1} - \tan^2\theta$$

$$= \frac{2\operatorname{cosec}^2\theta}{\operatorname{cosec}^2\theta - 1} - \tan^2\theta = \frac{2\operatorname{cosec}^2\theta}{\cot^2\theta} - \tan^2\theta$$

$$= 2\sec^2\theta - \tan^2\theta = \sec^2\theta + \sec^2\theta - \tan^2\theta$$

$$= \sec^2\theta + 1$$

Sol.440.(a) $2\cos^2\theta = 3\sin\theta$

$$2(1 - \sin^2\theta) = 3\sin\theta \Rightarrow 2 - 2\sin^2\theta = 3\sin\theta$$

$$2\sin^2\theta + 3\sin\theta - 2 = 0 \Rightarrow \sin\theta = -2 \text{ or } 0.5$$

$$\text{So, } \theta = 30^\circ$$

$$(\sec^2\theta - \tan^2\theta + \cos^2\theta)$$

$$\text{Putting } \theta = 30^\circ \Rightarrow \frac{4}{3} - \frac{1}{3} + \frac{3}{4} = \frac{7}{4}$$

Sol.441.(c) $3\tan\theta = 2\sqrt{3}\sin\theta$

$$\sec\theta = \frac{2}{\sqrt{3}}, \text{ So } \theta = 30^\circ$$

$$2\sin^2\theta - 3\cos^23\theta = 2 \times \frac{1}{4} - 3 \times 0 = \frac{1}{2}$$

Sol.442.(a) $\frac{\tan(45^\circ - \alpha)}{\cot(45^\circ + \alpha)}$

$$= \frac{(\cos 19^\circ + \sin 71^\circ)(\sec 19^\circ + \operatorname{cosec} 71^\circ)}{\tan 12^\circ \cdot \tan 24^\circ \cdot \tan 66^\circ \cdot \tan 78^\circ}$$

$$\text{Let } \alpha = 0^\circ \Rightarrow \frac{\tan(45^\circ)}{\cot(45^\circ)}$$

$$= \frac{(\cos 19^\circ + \cos 19^\circ)(\sec 19^\circ + \sec 19^\circ)}{\tan 12^\circ \cdot \tan 24^\circ \cdot \cot 24^\circ \cot 12^\circ}$$

$$1 - \frac{2\cos 19^\circ \cdot 2\sec 19^\circ}{1} \Rightarrow 1 - \frac{2 \times 2}{1}$$

$$\Rightarrow 1 - 4 = -3$$

Sol.443.(d) $1 + 2\tan^2\theta + 2\sin\theta\sec^2\theta$

$$= \sec^2\theta + \tan^2\theta + 2\sin\theta \frac{\sec\theta}{\cos\theta}$$

$$= \sec^2\theta + \tan^2\theta + 2\sec\theta\tan\theta$$

$$= (\sec\theta + \tan\theta)^2$$

$$= \left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right)^2 = \left(\frac{1+\sin\theta}{\cos\theta}\right)^2$$

$$= \frac{(1+\sin\theta)^2}{\cos^2\theta} = \frac{(1+\sin\theta)^2}{(1+\sin\theta)(1-\sin\theta)}$$

$$= \frac{1+\sin\theta}{1-\sin\theta}$$

Sol.444.(d) $2\sin(3x - 15)^\circ = 1$

$$\sin(3x - 15)^\circ = \frac{1}{2} = 30^\circ$$

$$3x - 15^\circ = 30^\circ \text{ so, } x = 15^\circ,$$

$$\Rightarrow \cos^2(2x + 15)^\circ + \cot^2(x + 15)^\circ$$

$$\Rightarrow \cos^2 45^\circ + \cot^2 30^\circ = \frac{1}{2} + 3 = \frac{7}{2}$$

Sol.445.(b) $\sin\left(\frac{2A+B}{2}\right) = \frac{\sqrt{3}}{2} = \sin 60^\circ$

$$\Rightarrow 2A + B = 120^\circ \dots (\text{eq. 1})$$

$$\cos\left(\frac{2A-B}{2}\right) = \frac{\sqrt{3}}{2} = \cos 30^\circ$$

$$\Rightarrow 2A - B = 60^\circ \dots (\text{eq. 2})$$

On solving eq. (1) and (2) we get,

$$A = 45^\circ \text{ and } B = 30^\circ$$

$$\sin[3(A - B)] = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

Sol.446.(c)

$$(\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta}) \left(\frac{\sin\theta(1+\cos\theta)}{1+\cos\theta-\sin^2\theta} \right)$$

$$\left(\sqrt{\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}} \right) \left(\frac{\sin\theta(1+\cos\theta)}{1+\cos\theta-1+\cos^2\theta} \right)$$

$$= \frac{1}{\sin\theta \cdot \cos\theta} \times \frac{\sin\theta}{\cos\theta} = \sec^2\theta$$

Sol.447.(d) $\frac{1}{\cos\theta} + \frac{1}{\tan\theta - \sec\theta}$

$$= \frac{1}{\cos\theta} - \frac{(\tan\theta - \sec\theta)(\tan\theta + \sec\theta)}{\tan\theta - \sec\theta}$$

$$= \sec\theta - \tan\theta - \sec\theta = -\tan\theta$$

Sol.448.(b) $4\sin^2(2x - 10)^\circ = 3$

$$\sin^2(2x - 10)^\circ = 3/4$$

$$\sin(2x - 10)^\circ = \frac{\sqrt{3}}{2}$$

$$(2x - 10)^\circ = 60^\circ \Rightarrow x = 35^\circ$$

$$\frac{\sin^4(x-5)^\circ + \cos^4(x-5)^\circ}{1 - 2\sin^2(3x-15)^\circ \cos^2(3x-15)^\circ}$$

$$\Rightarrow \frac{\sin^4 30^\circ + \cos^4 30^\circ}{1 - 2\sin^2 90^\circ \cos^2 90^\circ}$$

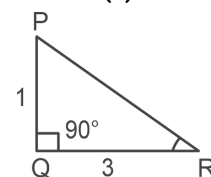
$$= \frac{(\frac{1}{2})^4 + (\frac{\sqrt{3}}{2})^4}{1 - 2 \times 1 \times 0} = \frac{\frac{1}{16} + \frac{9}{16}}{1} = \frac{5}{8}$$

Sol.449.(d) $\frac{\sqrt{2}\tan(60^\circ - \theta)\tan(30^\circ + \theta)}{\sin^2(45^\circ + \theta) + \sin^2(45^\circ - \theta)}$

$$\text{Put } \theta = 0^\circ \Rightarrow \frac{\sqrt{2}\tan(60^\circ)\tan(30^\circ)}{\sin^2(45^\circ) + \sin^2(45^\circ)}$$

$$\Rightarrow \frac{\sqrt{2} \times \sqrt{3} \times \frac{1}{\sqrt{3}}}{\frac{1}{2} + \frac{1}{2}} = \sqrt{2}$$

Sol.450.(d)



$$PR = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$\frac{\sec P(\cos R + \sin P)}{\operatorname{cosec} R(\sin R - \operatorname{cosec} P)}$$

$$= \frac{\frac{\sqrt{10}}{1} \left(\frac{3}{\sqrt{10}} + \frac{3}{\sqrt{10}} \right)}{\frac{\sqrt{10}}{1} \left(\frac{1}{\sqrt{10}} - \frac{\sqrt{10}}{3} \right)} = \frac{\frac{6\sqrt{10}}{\sqrt{10}}}{\frac{3-10}{3}} = \frac{6}{\frac{-7}{3}} = -\frac{18}{7}$$

Sol.451.(a)

$$\begin{aligned} & (\sec\theta + \tan\theta)^2 + \frac{1 + \operatorname{cosec}\theta}{1 - \operatorname{cosec}\theta} \\ &= (\sec\theta + \tan\theta)^2 + \frac{(1 + \operatorname{cosec}\theta)^2}{1 - \operatorname{cosec}^2\theta} \\ &\Rightarrow \sec^2\theta + \tan^2\theta + 2\sec\theta \times \tan\theta + \frac{1 + \operatorname{cosec}^2\theta + 2\operatorname{cosec}\theta}{- \cot^2\theta} \\ &\Rightarrow \sec^2\theta + \tan^2\theta + 2\sec\theta \times \tan\theta - \tan^2\theta - \sec^2\theta - 2\sec\theta \times \tan\theta = 0 \end{aligned}$$

$$\text{Sol.452.(b)} \sin^6\theta + \cos^6\theta = \frac{1}{3}$$

$$(\sin^2\theta + \cos^2\theta) [(\sin^2\theta + \cos^2\theta)^2 - 3\sin^2\theta \cdot \cos^2\theta] = \frac{1}{3}$$

$$[1 - 3\sin^2\theta \cdot \cos^2\theta] = \frac{1}{3}$$

$$\Rightarrow 3\sin^2\theta \cdot \cos^2\theta = \frac{2}{3}$$

$$\sin^2\theta \cdot \cos^2\theta = \frac{2}{9} \Rightarrow \sin\theta \cdot \cos\theta = \frac{\sqrt{2}}{3}$$

$$\text{Sol.453.(d)} 2\cos^2\theta - 5\cos\theta + 2 = 0$$

$$\text{Put } \theta = 60^\circ$$

$$= 2\cos^2 60^\circ - 5\cos 60^\circ + 2 = 0$$

$$= 2 \times \frac{1}{4} - \frac{5}{2} + 2 = 0 \Rightarrow 0 = 0,$$

satisfy then

$$\begin{aligned} (\sec\theta + \tan\theta) &= \sec 60^\circ + \tan 60^\circ \\ &= 2 + \sqrt{3} \end{aligned}$$

Sol.454.(c)

$$\begin{aligned} & \frac{\tan 13^\circ \tan 36^\circ \tan 45^\circ \tan 54^\circ \tan 77^\circ}{2 \sec^2 60^\circ (\sin^2 60^\circ - 3\cos 60^\circ + 2)} \\ &= \frac{\tan 13^\circ \tan 36^\circ \tan 45^\circ \cot 36^\circ \cot 13^\circ}{2 \sec^2 60^\circ (\sin^2 60^\circ - 3\cos 60^\circ + 2)} \\ &= \frac{1}{8 \left(\frac{3}{4} - \frac{3}{2} + 2 \right)} = \frac{1}{10} \end{aligned}$$

$$\begin{aligned} \text{Sol.455.(a)} & \sec^4\theta(1 - \sin^4\theta) - 2\tan^2\theta \\ &= \sec^4\theta \cos^2\theta(1 + \sin^2\theta) - 2\tan^2\theta \\ &= \sec^2\theta + \sec^2\theta \cdot \sin^2\theta - \tan^2\theta - \tan^2\theta \\ &= 1 + 0 = 1 \end{aligned}$$

$$\text{Sol.456.(a)} \sin^2\theta - \cos^2\theta - 3\sin\theta + 2 = 0$$

$$\text{Put } \theta = 30^\circ \Rightarrow \frac{1}{4} - \frac{3}{4} - \frac{3}{2} + 2 = 0$$

$$\Rightarrow \frac{9}{4} - \frac{9}{4} = 0$$

 $\Rightarrow 0 = 0$ (condition satisfied). So,

$$\frac{1}{\sqrt{\sec\theta - \tan\theta}} = \frac{1}{\sqrt{\frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}}}} = \sqrt[4]{3}$$

$$\text{Sol.457.(c)} \sqrt{\frac{\sec^2\theta + \operatorname{cosec}^2\theta}{\tan^2\theta - \sin^2\theta}}$$

$$\begin{aligned} &= \sqrt{\frac{\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}}{\sin^2\theta \left(\frac{1}{\cos^2\theta} - 1 \right)}} \\ &= \sqrt{\frac{\frac{1}{\sin^2\theta \cdot \cos^2\theta}}{\frac{\sin^2\theta \cdot \cos^2\theta}{\sin^2\theta \cdot \sin^2\theta}}} = \sqrt{\operatorname{cosec}^6\theta} = \operatorname{cosec}^3\theta \end{aligned}$$

Short trick :

$$\text{Put } \theta = 30^\circ \Rightarrow \sqrt{\frac{\sec^2 30^\circ + \operatorname{cosec}^2 30^\circ}{\tan^2 30^\circ - \sin^2 30^\circ}} = 8$$

$$\text{Check option (c)} \Rightarrow \operatorname{Cosec}^3 30^\circ = 8$$

Sol.458.(c)

$$\frac{1}{1 - \sin\theta} + \frac{1}{1 + \sin\theta} = 4 \sec\theta$$

$$\frac{2}{1 - \sin^2\theta} = 4 \sec\theta \Rightarrow \frac{2}{\cos^2\theta} = \frac{4}{\cos\theta}$$

$$\sec\theta = 2, \theta = 60^\circ, \text{ then}$$

$$\cot 60^\circ + \operatorname{cosec} 60^\circ = \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} = \sqrt{3}$$

$$\text{Sol.459.(c)} \sin^2\theta = 2 \sin\theta - 1,$$

$$\sin^2\theta - 2 \sin\theta + 1 = 0 \Rightarrow (\sin\theta - 1)^2 = 0$$

$$\sin\theta = 1 \text{ so, } \theta = 90^\circ$$

$$\frac{1 + \operatorname{cosec}\theta}{1 - \cos\theta} = \frac{1 + 1}{1 - 0} = 2$$

Sol.460.(b)

$$\sin^2 59^\circ + \frac{1}{\sec^2 31^\circ} - \frac{1}{\sin^2 59^\circ \operatorname{cosec}^2 59^\circ}$$

$$\Rightarrow \cos^2 31^\circ + \frac{1}{x^2} - \frac{1}{1}$$

$$\Rightarrow \frac{1}{x^2} + \frac{1}{x^2} - 1 = \frac{2}{x^2} - 1 = \frac{2 - x^2}{x^2}$$

$$\text{Sol.461.(a)} 2 \tan x + 3 \cot x = 5$$

Squaring both side:

$$4 \tan^2 x + 9 \cot^2 x + 2 \times 2 \tan x \times 3 \cot x = 25$$

$$4 \tan^2 x + 9 \cot^2 x + 12 = 25$$

$$4 \tan^2 x + 9 \cot^2 x = 25 - 12 = 13$$

Alternate method: Put $x = 45^\circ$

$$\text{Sol.462.(d)} 2 \cos\theta + \sec\theta - 2\sqrt{2} = 0$$

$$\text{Put } \theta = 45^\circ$$

$$2 \cos 45^\circ + \sec 45^\circ - 2\sqrt{2} = 0$$

$$= 2 \frac{1}{\sqrt{2}} + \sqrt{2} - 2\sqrt{2} = 0$$

$$= \sqrt{2} + \sqrt{2} - 2\sqrt{2} = 0 \Rightarrow 0 = 0, \text{ satisfy}$$

Sol.463.(d) As we know that

$$\cot\theta = \frac{B}{P} = \frac{\sqrt{2} + 1}{1}$$

Applying pythagoras theorem

$$H^2 = B^2 + P^2 = 2 + 1 + 2\sqrt{2} + 1$$

$$= 4 + 2\sqrt{2}$$

$$\operatorname{Cosec}\theta \sec\theta = \frac{H^2}{B \times P} = \frac{4 + 2\sqrt{2}}{\sqrt{2} + 1} = 2\sqrt{2}$$

Sol.464.(b) Put $\theta = 30^\circ$ in

$$\frac{(1 - \sin\theta)(\sec\theta + \tan\theta) \tan\theta}{(\sec\theta + \tan\theta + 1)(\cot\theta - \operatorname{cosec}\theta + 1)}$$

$$\begin{aligned} &= \frac{(1 - \frac{1}{2}) \left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right) \frac{1}{\sqrt{3}}}{\left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} + 1 \right) (\sqrt{3} - 2 + 1)} \\ &= \frac{\left(\frac{1}{2} \right) \left(\frac{3}{\sqrt{3}} \right) \frac{1}{\sqrt{3}}}{\left(\frac{3}{\sqrt{3}} + 1 \right) (\sqrt{3} - 1)} \\ &= \frac{\left(\frac{1}{2} \right)}{(\sqrt{3} + 1) (\sqrt{3} - 1)} = \frac{1}{4} = \frac{\sin\theta}{2} \end{aligned}$$

$$\text{Sol.465.(d)} \frac{3\sqrt{3} \sec\theta + 4 \tan\theta}{3 \tan\theta + \sqrt{3} \sec\theta} = 2$$

$$3\sqrt{3} \sec\theta + 4 \tan\theta = 2 \times (3 \tan\theta + \sqrt{3} \sec\theta)$$

$$3\sqrt{3} \sec\theta + 4 \tan\theta = 6 \tan\theta + 2\sqrt{3} \sec\theta$$

$$\sqrt{3} \sec\theta = 2 \tan\theta$$

$$\sin\theta = \frac{\sqrt{3}}{2}, \theta = 60^\circ \Rightarrow \cos 60^\circ = \frac{1}{2}$$

Sol.466.(c)

$$4(\sin^4 60^\circ + \cos^4 30^\circ) - (\tan^2 45^\circ - \cot^2 30^\circ) + \cos^2 45^\circ - \operatorname{cosec}^2 45^\circ + \sec^2 60^\circ$$

$$= 4 \left(2 \times \frac{9}{16} \right) - (1 - 3) + \frac{1}{2} - 2 + 4$$

$$= \frac{9}{2} + 2 + \frac{1}{2} + 2 = 5 + 4 = 9$$

Number 7 should be subtracted to get 2.

Sol.467.(a) $Y = \tan 35^\circ$

$$Y = \cot 55^\circ (\tan A = \cot B, \text{ if } A + B = 90^\circ)$$

$$\text{Now, } 2 \tan 55^\circ + \cot 55^\circ$$

$$= \frac{2 + \cot^2 55^\circ}{\cot 55^\circ} = \frac{2 + Y^2}{Y}$$

$$\text{Sol.468.(d)} \tan\theta + \cot\theta = 4$$

Squaring both sides

$$\tan^2\theta + \cot^2\theta = 14 \quad \dots(1)$$

$$\text{Also, } \tan\theta + \cot\theta = 4$$

$$\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = 4 \Rightarrow \operatorname{cosec}\theta \sec\theta = 4$$

$$\operatorname{cosec}^2\theta \sec^2\theta = 16 \quad \dots(2)$$

On putting value from eq. 1 and 2

$$\frac{3(\tan^2\theta + \cot^2\theta)}{(2 \operatorname{cosec}^2\theta \sec^2\theta - 4)} = \frac{3 \times 14}{2 \times 16 - 4}$$

$$= \frac{42}{28} = 3 : 2$$

$$\text{Sol.469.(c)} 3(\sec^2\theta + \tan^2\theta) = 5$$

$$\theta = 30^\circ \text{ (by hit and trial)} \Rightarrow \operatorname{Cosec} 30^\circ = 2$$

Sol.470.(c)

$$(\cos^2\theta - 1)(2 \sec^2\theta) + \sec^2\theta + 2 \tan^2\theta = 2$$

$$\Rightarrow (2 - 2 \sec^2\theta) + \sec^2\theta + 2 \tan^2\theta = 2$$

$$\Rightarrow -2 \tan^2\theta + \sec^2\theta + 2 \tan^2\theta = 2$$

$$\Rightarrow \sec^2\theta = 2 \Rightarrow \theta = 45^\circ$$

$$\frac{(\sec\theta + \sin\theta)}{(\operatorname{cosec}\theta - \cos\theta)} = \frac{\sqrt{2} + \frac{1}{\sqrt{2}}}{\sqrt{2} - \frac{1}{\sqrt{2}}} = 3$$

Sol.471.(a)

$$\frac{(1 + \cos\theta)(\operatorname{cosec}\theta - \cot\theta) \sec\theta}{\sin\theta(1 - \sin\theta)(\sec\theta + \tan\theta)}$$

$$= \frac{(1 + \cos\theta)(1 - \cos\theta) \frac{1}{\sin\theta\cos\theta}}{\sin\theta(1 - \sin\theta)(1 + \sin\theta) \frac{1}{\cos\theta}}$$

$$= \frac{\frac{\sin\theta}{\cos\theta}}{\sin\theta\cos\theta} = \sec^2\theta$$

Sol.472.(d) $\sin\theta(2\sin\theta + 3) = 2$

$\theta = 30^\circ$ (by hit and trial)

$$\sec^2\theta + \cot^2\theta - \cos^2\theta$$

$$= \sec^2 30^\circ + \cot^2 30^\circ - \cos^2 30^\circ$$

$$= \frac{4}{3} + 3 - \frac{3}{4} = \frac{43}{12}$$

Sol.473.(b) $\sin\theta = \frac{2\sqrt{ab}}{a+b}$

Let $a = 2$, $b = 1$

Then $\sin\theta = \frac{2\sqrt{2}}{3}$, $\cos\theta = \frac{1}{3}$

Now, $\frac{\cos\theta + 1}{\cos\theta - 1} = \frac{\frac{1}{3} + 1}{\frac{1}{3} - 1} = \frac{-2}{1}$

When the value of a and b are put in all options, we see that option (b) is the correct answer.

Sol.474.(c) $\frac{3\cos^2 27^\circ - 5 + 3\cos^2 63^\circ}{\tan^2 32^\circ + 4 - \operatorname{cosec}^2 58^\circ}$

$$+ \sin 35^\circ \cos 55^\circ + \cos 35^\circ \sin 55^\circ$$

We know that $\sin A = \cos B$, if

$A + B = 90^\circ$

So, $\frac{3\cos^2 27^\circ - 5 + 3\sin^2 27^\circ}{\cot^2 58^\circ + 4 - \operatorname{cosec}^2 58^\circ}$

$$+ \sin 35^\circ \sin 35^\circ + \cos 35^\circ \cos 35^\circ$$

$$= \frac{3-5}{-1+4} + 1 = \frac{1}{3}$$

Sol.475.(b) $2\cos^2\theta = 3(1 - \sin\theta)$

Put $\theta = 30^\circ$

$$\Rightarrow 2\cos^2 30^\circ = 3(1 - \sin 30^\circ) \Rightarrow \frac{3}{2} = \frac{3}{2}$$

$\theta = 30^\circ$ satisfies the given equation.

$$(\tan 2\theta + \operatorname{cosec} 3\theta - \sec 2\theta) = \sqrt{3} + 1 - 2$$

$$= \sqrt{3} - 1$$

Sol.476.(b) $\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$

$$\frac{2 - \sin^2\theta}{1 - \cot^2\theta} + \sec^2\theta + \operatorname{cosec}\theta$$

$$= \frac{2 - \frac{1}{4}}{1 - 3} + \left(\frac{4}{3} + 2\right)$$

$$= \frac{7}{-8} + \frac{10}{3} = \frac{-21 + 80}{24} = \frac{59}{24}$$

Sol.477.(a)

$$\sec^2\theta + (1 - \sqrt{3})\tan\theta - (1 + \sqrt{3}) = 0,$$

$$1 + \tan^2\theta + \tan\theta - \sqrt{3}\tan\theta - 1 - \sqrt{3} = 0$$

$$\tan\theta(\tan\theta + 1) - \sqrt{3}(\tan\theta + 1) = 0$$

$$(\tan\theta + 1)(\tan\theta - \sqrt{3}) = 0$$

$$\tan\theta = -1, \sqrt{3} \Rightarrow \theta = 60^\circ$$

Sol.478.(b)

$$\operatorname{cosec}^2\theta(\cos\theta - 1)(1 + \cos\theta) = k,$$

$$\operatorname{cosec}^2\theta\{-(1 - \cos\theta)(1 + \cos\theta)\}$$

$$- \operatorname{cosec}^2\theta \times \sin^2\theta = k \Rightarrow k = -1$$

Sol.479.(a)

$$\frac{\cot\theta + \cos\theta}{\cot\theta - \cos\theta} = \frac{k+1}{1-k}$$

(By using componendo and dividendo)

$$\frac{\cot\theta}{\cos\theta} = \frac{2}{2k} \Rightarrow k = \sin\theta$$

Sol.480.(c) $\cos^2\theta - \sin^2\theta - 3\cos\theta + 2 = 0,$

Put $\theta = 60^\circ$

$$\cos^2 60^\circ - \sin^2 60^\circ - 3\cos 60^\circ + 2 = 0,$$

$$\frac{1}{4} - \frac{3}{4} - \frac{3}{2} + 2 = 0$$

$$\Rightarrow 0 = 0 \text{ (LHS = RHS)}$$

$$\sec\theta - \cos\theta = 2 - \frac{1}{2} = \frac{3}{2}$$

Sol.481.(b) $\frac{\operatorname{cosec}^2\theta}{\operatorname{cosec}^2\theta - \cot^2\theta} = \frac{13}{4}$

$$\Rightarrow \operatorname{cosec}^2\theta = \frac{13}{4} (\operatorname{cosec}^2\theta - \cot^2\theta = 1)$$

$$\sin\theta = \frac{2}{\sqrt{13}} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

By pythagoras theorem

$$\text{Base} = \sqrt{\text{Hypotenuse}^2 - \text{Perpendicular}^2}$$

$$\Rightarrow \text{Base} = 3$$

$$\cos\theta = \frac{3}{\sqrt{13}}, \sec\theta = \frac{\sqrt{13}}{3},$$

$$\tan\theta = \frac{2}{3}, \cot\theta = \frac{3}{2}$$

$$\frac{52\cos^2\theta - 9\tan^2\theta}{18\sec^2\theta + 8\cot^2\theta} = \frac{52 \times \frac{9}{13} - 9 \times \frac{4}{9}}{18 \times \frac{13}{9} + 8 \times \frac{9}{4}}$$

$$= \frac{36-4}{26+18} = \frac{32}{44} = \frac{8}{11}$$

Sol.482.(a) $\frac{3(\cot^2 46^\circ - \sec^2 44^\circ)}{2(\sin^2 28^\circ + \sin^2 62^\circ)} +$

$$\frac{2\cos^2 60^\circ \tan^2 33^\circ \tan^2 57^\circ}{\sec^2(90^\circ - \theta) - \cot^2\theta}$$

$$\Rightarrow \frac{3(\tan^2 44^\circ - \sec^2 44^\circ)}{2(\sin^2 28^\circ + \cos^2 28^\circ)} +$$

$$\frac{2\cos^2 60^\circ \tan^2 33^\circ \cot^2 33^\circ}{\operatorname{cosec}^2\theta - \cot^2\theta} \Rightarrow \frac{-3}{2} + \frac{1}{2} = -1$$

Sol.483.(b) $3\sec\theta + 4\cos\theta = 4\sqrt{3},$

Put $\theta = 30^\circ$

$$3 \times \frac{2}{\sqrt{3}} + 4 \times \frac{\sqrt{3}}{2} = 4\sqrt{3} \text{ (Equation satisfies)}$$

$$(1 - \sin\theta + \cos\theta) = (1 - \sin 30^\circ +$$

$$\cos 30^\circ) = \frac{1 + \sqrt{3}}{2}$$

Sol.484.(d) $\frac{5\cot\theta + \sqrt{3}\operatorname{cosec}\theta}{2\sqrt{3}\operatorname{cosec}\theta + 3\cot\theta} = 1$

$$\Rightarrow 5\cos\theta + \sqrt{3} = 2\sqrt{3} + 3\cos\theta$$

$$2\cos\theta = \sqrt{3} \Rightarrow \cos\theta = \frac{\sqrt{3}}{2}, \theta = 30^\circ$$

$$\Rightarrow \frac{\frac{7}{2}\cot^2\theta - \frac{3}{4}\operatorname{cosec}^2\theta}{4\sin^2\theta + \frac{3}{2}\tan^2\theta}$$

$$= \frac{\frac{7}{2} \times 3 - \frac{3}{4} \times 4}{4 \times \frac{1}{4} + \frac{3}{2} \times \frac{1}{3}} = 5$$

Sol.485.(c) $(1 + \cot^2\theta) + (1 + (\cot^2\theta)^{-1})$

$$\Rightarrow \operatorname{cosec}^2\theta + (1 + \tan^2\theta)$$

$$\Rightarrow \operatorname{cosec}^2\theta + \sec^2\theta \Rightarrow \frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}$$

$$\Rightarrow \operatorname{cosec}^2\theta \cdot \sec^2\theta$$

$$k = \operatorname{cosec}^2\theta \cdot \sec^2\theta \Rightarrow \sqrt{k} = \operatorname{cosec}\theta \cdot \sec\theta$$

Sol.486.(c) $\frac{\sec\theta \operatorname{cosec}\theta}{2 + \tan^2\theta + \cot^2\theta}$

$$= \frac{\sec\theta \operatorname{cosec}\theta}{\sec^2\theta + \operatorname{cosec}^2\theta} = \frac{\sec\theta \operatorname{cosec}\theta}{\frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}}$$

$$= \sec\theta \cdot \operatorname{cosec}\theta \sin^2\theta \cos^2\theta = \sin\theta \cos\theta$$

Sol.487.(c) $\cot A + \operatorname{cosec} A = 2 \text{ ---- (1)}$

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$(\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$$

$$(\operatorname{cosec} A - \cot A) = \frac{1}{2} \text{ ---- (2)}$$

Adding (1) and (2)

$$2\operatorname{cosec} A = \frac{5}{2} \Rightarrow \operatorname{cosec} A = \frac{5}{4}$$

$$\sin A = \frac{4}{5}; \tan A = \frac{4}{3}$$

$$\frac{9\tan A + 16\operatorname{cosec} A}{5\sin A + 3\tan A} = \frac{9 \times \frac{4}{3} + 16 \times \frac{5}{4}}{5 \times \frac{4}{5} + 3 \times \frac{4}{3}} = 4$$

Sol.488.(c) $\cot^2\theta + \cot^4\theta = 2$

$$\cot^2\theta(1 + \cot^2\theta) = 2$$

$$\cot^2\theta = 2\sin^2\theta \Rightarrow \cot^4\theta = 4\sin^4\theta$$

$$2\sin^4\theta + \sin^2\theta = \frac{\cot^4\theta}{2} + \frac{\cot^2\theta}{2}$$

$$= \frac{\cot^2\theta + \cot^4\theta}{2} = 1$$

Short-Trick: put $\theta = 45^\circ$

Sol.489.(c) $\sin\alpha + \operatorname{cosec}\alpha = \tan\frac{\pi}{3}$

$$\sin\alpha + \operatorname{cosec}\alpha = \sqrt{3}$$

$$\sin^3\alpha + \operatorname{cosec}^3\alpha = a^3 - 3a$$

$$= (\sqrt{3})^3 - 3(\sqrt{3}) = 0$$

Sol.490.(c)

$$\operatorname{cosec}(58^\circ + \theta) - \sec(32^\circ - \theta)$$

$$+ \sin 15^\circ \cdot \sin 35^\circ \cdot \sec 55^\circ \cdot \sin 30^\circ \cdot \sec 75^\circ$$

$$\sec(32^\circ - \theta) - \sec(32^\circ - \theta) + \frac{1}{2} = 0.5$$

Sol.491.(c) $\frac{\sin^2 52^\circ + 2 + \sin^2 38^\circ}{4\cos^2 43^\circ - 5 + 4\cos^2 47^\circ}$

$$\frac{\sin^2 52^\circ + 2 + \cos^2 52^\circ}{4\cos^2 43^\circ - 5 + 4\sin^2 43^\circ}$$

$$\frac{1+2}{4\cos^2 43^\circ + 4\sin^2 43^\circ - 5} = \frac{3}{-1} = -3$$

Sol.492.(d) $4(\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ)$
 $- \cos 90^\circ + y \times \tan^2 66^\circ \times \tan^2 24^\circ = \frac{y}{2},$

We know $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$

And $\tan A \tan B = 1$ if $(A + B = 90^\circ)$

So, $4(1 + \cot^2 57^\circ - \cot^2 57^\circ) - \cos 90^\circ +$

$y \times 1 = \frac{y}{2} \Rightarrow 4 + y = \frac{y}{2}$

$\Rightarrow \frac{y}{2} = -4 \Rightarrow Y = -8$

Sol.493.(a)

$$\frac{\sin \theta [(1 - \tan \theta) \tan \theta + \sec^2 \theta]}{(1 - \sin \theta) \tan \theta (1 + \tan \theta) (\sec \theta + \tan \theta)}$$

$$\frac{\sin \theta [\tan \theta - \tan^2 \theta + \sec^2 \theta]}{(\sec \theta - \tan \theta) \sin \theta (1 + \tan \theta) (\sec \theta + \tan \theta)}$$

$$\Rightarrow \frac{\sin \theta (1 + \tan \theta)}{\sin \theta (1 + \tan \theta)} = 1$$

Sol.494.(a) Solving separately

$$\left(\frac{1}{\cos \theta} - \frac{1}{\sin \theta} \right) = \sec \theta - \operatorname{cosec} \theta$$

We know that $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

$$\frac{1}{\operatorname{cosec} \theta - \cot \theta} = \operatorname{cosec} \theta + \cot \theta$$

Similarly $\frac{1}{\sec \theta + \tan \theta} = \sec \theta - \tan \theta$

Now combining

$$\sec \theta - \operatorname{cosec} \theta + \operatorname{cosec} \theta + \cot \theta - \sec \theta + \tan \theta = \cot \theta + \tan \theta$$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$$

$$= \frac{1}{\sin \theta \cos \theta} = \sec \theta \operatorname{cosec} \theta$$

Short trick :- put $\theta = 45^\circ$

$$\left(\frac{1}{\cos \theta} - \frac{1}{\sin \theta} \right) + \frac{1}{\operatorname{cosec} \theta - \cot \theta}$$

$$- \frac{1}{\sec \theta + \tan \theta} = 0 + \frac{1}{\sqrt{2} - 1} - \frac{1}{\sqrt{2} + 1}$$

$$= \sqrt{2} + 1 - (\sqrt{2} - 1)$$

$$= \sqrt{2} + 1 - \sqrt{2} + 1 = 2$$

Put $\theta = 45^\circ$ in options, option (a) satisfy.

$$\sec \theta \operatorname{cosec} \theta = \sqrt{2} \times \sqrt{2} = 2$$

Sol.495.(c) $\sec \theta = \frac{a}{b} = \frac{\text{hypotenuse}}{\text{base}}$

Perpendicular = $\sqrt{a^2 - b^2}$

$$\tan \theta = \frac{\sqrt{a^2 - b^2}}{b}, \sin \theta = \frac{\sqrt{a^2 - b^2}}{a}$$

$$\frac{1 - \tan^2 \theta}{2 - \sin^2 \theta} = \frac{1 - \left(\frac{a^2 - b^2}{b^2} \right)}{2 - \left(\frac{a^2 - b^2}{a^2} \right)}$$

$$= \frac{a^2(2b^2 - a^2)}{b^2(a^2 + b^2)}$$

Sol.496.(a)

$$\frac{\sec^2 \theta (1 + \tan^2 \theta + 1 + \cot^2 \theta) \div (\sin^2 \theta - \tan^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)(1 + \cot^2 \theta)^2}$$

$$\frac{\sec^2 \theta (\sec^2 \theta + \operatorname{cosec}^2 \theta)}{(\sin^2 \theta - \tan^2 \theta)(\operatorname{cosec}^2 \theta + \sec^2 \theta)(\operatorname{cosec}^2 \theta)^2}$$

$$\frac{\sec^2 \theta}{(\sin^2 \theta - \tan^2 \theta)(\operatorname{cosec}^2 \theta)^2}$$

Put value = 45° or (change each and every function into $\sin \theta$ and $\cos \theta$)

$$\frac{2}{-\frac{1}{2} \times 4} = -1$$

Sol.497.(d)

$$\frac{2 \sin^2 38^\circ \sec^2 52^\circ + \cos 64^\circ \sin 26^\circ + \sin^2 64^\circ}{\tan^2 23^\circ + \cot^2 23^\circ - \sec^2 67^\circ - \operatorname{cosec}^2 67^\circ}$$

$$\frac{2 \sin^2 38^\circ \frac{1}{\sin^2 38^\circ} + \cos 64^\circ \cos 64^\circ + \sin^2 64^\circ}{\tan^2 23^\circ + \cot^2 23^\circ - \operatorname{cosec}^2 23^\circ - \sec^2 23^\circ}$$

$$= \frac{2 + \cos^2 64^\circ + \sin^2 64^\circ}{-2} = \frac{2 + 1}{-2} = -\frac{3}{2}$$

Sol.498.(c)

$$\frac{\cos^6 \theta + \sin^6 \theta + 3 \sin^2 \theta \cos^2 \theta \times 1}{\operatorname{cosec} \theta \sec \theta (\sin \theta + \cos \theta - 1)(\sin \theta + \cos \theta + 1)}$$

$$\frac{\cos^6 \theta + \sin^6 \theta + 3 \sin^2 \theta \cos^2 \theta \times (\sin^2 \theta + \cos^2 \theta)}{\operatorname{cosec} \theta \sec \theta (\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta - 1)}$$

$$\frac{(\sin^2 \theta + \cos^2 \theta)^3}{\operatorname{cosec} \theta \sec \theta (2 \sin \theta \cos \theta)}$$

$$\frac{(1)^3}{\frac{1}{\sin \theta \cos \theta} (2 \sin \theta \cos \theta)} = \frac{1}{2}$$

Sol.499.(c) $\frac{\sin^2 \theta}{\cos \theta (1 + \cos \theta)} + \frac{1 + \cos \theta}{\cos \theta}$

$$= \frac{\sin^2 \theta + (1 + \cos \theta)^2}{\cos \theta (1 + \cos \theta)}$$

$$= \frac{\sin^2 \theta + (1 + \cos^2 \theta + 2 \cos \theta)}{\cos \theta (1 + \cos \theta)}$$

$$= \frac{1 + 1 + 2 \cos \theta}{\cos \theta (1 + \cos \theta)}$$

$$= \frac{2(1 + \cos \theta)}{\cos \theta (1 + \cos \theta)} = \frac{2}{\cos \theta} = 2 \sec \theta$$

Sol.500.(a)

$$\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$$

Put $\theta = 45^\circ$

$$= \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1}{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1} \times \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}}$$

$$= \frac{\sqrt{2} - 1}{1} \times \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$= \sqrt{2} - 1 \times \sqrt{2} + 1 = 2 - 1 = 1$$

Sol.501.(a) Short-Trick : Put $A = 45^\circ$

$$\frac{\sqrt{2}(\sqrt{2} + 1)(1 - \frac{1}{\sqrt{2}})}{\frac{1}{2}}$$

$$(\sqrt{2} + 1)(\sqrt{2} - 1) \times 2 = 2$$

Now, go through option A

$$\sec^2 A = \sec^2 45^\circ = (\sqrt{2})^2 = 2 \text{ (which satisfies)}$$

Sol.502.(a)

$$(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$$

$$= k + \tan^2 \theta + \cot^2 \theta,$$

$$\Rightarrow \sin^2 \theta + \operatorname{cosec}^2 \theta + 2 \sin \theta \operatorname{cosec} \theta + \cos^2 \theta$$

$$+ \sec^2 \theta + 2 \cos \theta \sec \theta$$

$$= k + \tan^2 \theta + \cot^2 \theta$$

$$\Rightarrow 5 + 1 + \tan^2 \theta + 1 + \cot^2 \theta = \tan^2 \theta + \cot^2 \theta + k \Rightarrow K = 7$$

Short trick:- Put $\theta = 45^\circ$

$$(\sin \theta + \operatorname{cosec} \theta)^2 +$$

$$(\cos \theta + \sec \theta)^2 = k + \tan^2 \theta + \cot^2 \theta$$

$$\Rightarrow \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right)^2 + \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right)^2 = k + 1 + 1$$

$$\Rightarrow \left(\frac{3}{\sqrt{2}} \right)^2 + \left(\frac{3}{\sqrt{2}} \right)^2 = k + 2$$

$$\Rightarrow 9 = k + 2 \Rightarrow K = 7$$

Sol.503.(a) $3 \sin x + 4 \cos x = 2$

$$3 \cos x - 4 \sin x = y \text{ (let)}$$

Squaring and adding both equations

$$9 + 16 = 4 + y^2 \Rightarrow y = \sqrt{21}$$

Sol.504.(a) Minimum value of 8

$$\operatorname{cosec}^2 \theta + 25 \sin^2 \theta = 2\sqrt{25 \times 8} = 20\sqrt{2}$$

Sol.505.(b)

Put $\theta = 0^\circ$, then $A = 2 - 3 = -1$

$$\cos \alpha = \sqrt{\frac{3 + A}{5 + A}}$$

Put value of A, we get

$$\cos \alpha = \frac{1}{\sqrt{2}} \Rightarrow \alpha = 45^\circ$$

$$\text{So, } 3\alpha = 3 \times 45^\circ = 135^\circ$$

Sol.506.(d) $117 \cos^2 A + 129 \sin^2 A = 120$

$$\Rightarrow 117 \cos^2 A + 117 \sin^2 A + 12 \sin^2 A = 120$$

$$\Rightarrow 117 + 12 \sin^2 A = 120 \Rightarrow 12 \sin^2 A = 3$$

$$\Rightarrow \sin^2 A = \frac{1}{4}$$

$$170 \cos^2 B + 158 \sin^2 B = 161$$

$$\Rightarrow 12 \cos^2 B + 158 \cos^2 B + 158 \sin^2 B = 161$$

$$\Rightarrow 12 \cos^2 B + 158 = 161$$

$$\Rightarrow 12 \cos^2 B = 3 \Rightarrow \cos^2 B = \frac{1}{4}$$

$$\operatorname{cosec}^2 A \sec^2 B = \frac{1}{\sin^2 A \cos^2 B} = 4 \times 4 = 16$$

Sol.507.(a) Put $\theta = 30^\circ$

$$= \frac{\sqrt{3}}{1 + 2} + \frac{1 + 2}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

Put the value of θ , in option (a) which satisfies.

Sol.508.(c) $\frac{A \tan 62^\circ \sec 28^\circ \cot 38^\circ}{\operatorname{Cosec} 62^\circ \tan 11^\circ} = 1$

$$\Rightarrow A = \frac{\operatorname{Cosec} 62^\circ \tan 11^\circ}{\tan 62^\circ \sec 28^\circ \cot 38^\circ}$$

$$\Rightarrow A = \frac{\sec 28^\circ \cot 79^\circ}{\tan 62^\circ \sec 28^\circ \cot 38^\circ}$$

$$\Rightarrow A = \frac{\tan 28^\circ \tan 38^\circ}{\tan 79^\circ}$$

Sol.509.(d) $\frac{\cos x}{1 + \sin x} + \frac{1 + \sin x}{\cos x}$

$$= \frac{\cos^2 x + (1 + \sin x)^2}{(1 + \sin x)\cos x} = \frac{2 + 2\sin x}{(1 + \sin x)\cos x}$$

$$= \frac{2(1 + \sin x)}{(1 + \sin x)\cos x} = 2\sec x$$

Sol.510.(c) $2\sin^2 x + \sqrt{3}\cos x + 1 = 0$

Go through option: (c) $\Rightarrow \frac{5\pi}{6} = 150^\circ$

$$2\sin^2 150^\circ + \sqrt{3}\cos 150^\circ + 1 = 0$$

$$\Rightarrow 2 \times \left(\frac{1}{2}\right)^2 + \sqrt{3} \times \left(-\frac{\sqrt{3}}{2}\right) + 1 = 0$$

$$\Rightarrow \frac{1}{2} - \frac{3}{2} + 1 = 0 \text{ (which satisfies)}$$

Sol.511.(d) $\tan(63^\circ - \theta) - \cot(27^\circ + \theta) +$

$$\frac{\operatorname{cosec}^2 70^\circ - \tan^2 20^\circ}{\sec^2 37^\circ - \cot^2 53^\circ}$$

$$= \tan(63^\circ - \theta) - \tan(63^\circ - \theta) +$$

$$\frac{\operatorname{cosec}^2 70^\circ - \tan^2 20^\circ}{\sec^2 37^\circ - \cot^2 53^\circ}$$

$$= 0 + \frac{\sec^2 20^\circ - \tan^2 20^\circ}{\operatorname{cosec}^2 53^\circ - \cot^2 53^\circ} = 0 + 1 = 1$$

Sol.512.(b) $x = \operatorname{cosec} A + \cos A$

Put $A = 45^\circ \Rightarrow x = \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}}$

$$y = \operatorname{cosec} A - \cos A = \sqrt{2} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$x + y = \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{2}} = 2\sqrt{2}$$

$$x - y = \frac{3}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \sqrt{2}$$

So,

$$\Rightarrow \left(\frac{2}{x+y}\right)^2 + \left(\frac{x-y}{2}\right)^2 - 1$$

$$= \left(\frac{2}{2\sqrt{2}}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2 - 1 = 0$$

Sol.513.(c) $3b \operatorname{cosec} \theta = a \sec \theta$

$$b = \frac{a \sec \theta}{3 \operatorname{cosec} \theta} \Rightarrow 3a \sec \theta - b \operatorname{cosec} \theta = 8$$

$$3a \sec \theta - \frac{a \sec \theta}{3 \operatorname{cosec} \theta} \times \operatorname{cosec} \theta = 8$$

$$\Rightarrow 8a \sec \theta = 24 \Rightarrow a \sec \theta = 3 \Rightarrow a = 3 \cos \theta$$

$$\Rightarrow 3b \operatorname{cosec} \theta = a \sec \theta$$

$$\Rightarrow 3b \operatorname{cosec} \theta = 3 \Rightarrow b \operatorname{cosec} \theta = 1 \Rightarrow b = \sin \theta$$

$$9b^2 + a^2 = 9(\sin^2 \theta) + (3\cos \theta)^2$$

$$= 9\sin^2 \theta + 9\cos^2 \theta = 9$$

Sol.514.(b) $\sqrt{3}\cos \theta = \sin \theta,$

$$\tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$$

$$\frac{4\sin^2 \theta - 5\cos \theta}{3\cos \theta + 1} = \frac{4 \times \frac{3}{4} - \frac{5}{2}}{\frac{3}{2} + 1} = \frac{1}{5}$$

Sol.515.(b) if $\tan 2\theta \cdot \tan 3\theta = 1$

It means, $2\theta + 3\theta = 90^\circ \Rightarrow \theta = 18^\circ$

Sol.516.(b) if $\sin A = \cos B$, then

$$A + B = 90^\circ$$

Therefore, $7x + 11x = 90^\circ \Rightarrow x = 5^\circ$

$$\tan 9x = \tan 45^\circ = 1$$

Sol.517.(d) $\sin(\theta + 30^\circ) = \frac{3}{\sqrt{12}} = \frac{\sqrt{3}}{2}$

$$\sin(\theta + 30^\circ) = \sin 60^\circ \Rightarrow \theta = 30^\circ$$

Sol.518.(d) $(\sin \theta + \cos \theta)^2 = 2$

$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cdot \cos \theta = 2$$

$$\Rightarrow 1 + 2 \sin \theta \cdot \cos \theta = 2$$

$$\Rightarrow 2 \sin \theta \cdot \cos \theta = 1 \Rightarrow \sin \theta \cdot \cos \theta = \frac{1}{2}$$

take, $\theta = \frac{\pi}{4}, \sin \frac{\pi}{4} \cdot \cos \frac{\pi}{4} = \frac{1}{2}$

Alternate method:- $(\sin \theta + \cos \theta)^2 = 2$

$$(\sin \theta + \cos \theta) = \sqrt{2}$$

Now checking through options, $\theta = 45^\circ$ satisfies the given condition.

$$(\sin 45^\circ + \cos 45^\circ)$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} \text{ (LHS = RHS)}$$

Sol.519.(b)

Given: $a \sin A + b \cos A = c$ (i)

Let, $a \cos A - b \sin A = x$ (ii)

On squaring and adding both equations; we get:-

$$a^2 + b^2 = c^2 + x^2 \Rightarrow x = \sqrt{a^2 + b^2 - c^2}$$

Sol.520.(b) $\frac{1 - \tan A}{1 + \tan A} =$

$$\frac{\tan 3^\circ \tan 15^\circ \tan 30^\circ \tan 75^\circ \tan 87^\circ}{\tan 27^\circ \tan 39^\circ \tan 51^\circ \tan 60^\circ \tan 63^\circ}$$

$$\Rightarrow \frac{1 - \tan A}{1 + \tan A}$$

$$= \frac{\tan 3^\circ \tan 15^\circ \tan 30^\circ \cot 15^\circ \cot 3^\circ}{\tan 27^\circ \tan 39^\circ \cot 39^\circ \tan 60^\circ \cot 27^\circ}$$

$$\Rightarrow \frac{1 - \tan A}{1 + \tan A} = \frac{\tan 30^\circ}{\tan 60^\circ}$$

$$\Rightarrow \frac{\frac{1}{\tan A} - \frac{\tan A}{\tan A}}{\frac{1}{\tan A} + \frac{\tan A}{\tan A}} = \frac{\frac{1}{\sqrt{3}}}{\sqrt{3}}$$

$$\Rightarrow \frac{\cot A - 1}{\cot A + 1} = \frac{1}{3}$$

$$\Rightarrow 3 \cot A - 3 = \cot A + 1$$

$$\Rightarrow 2 \cot A = 4 \Rightarrow \cot A = 2$$

Sol.521.(a) $\frac{\sqrt{\operatorname{Cosec} x - 1}}{\sqrt{\operatorname{Cosec} x + 1}}$

$$\Rightarrow \frac{\sqrt{\operatorname{Cosec} x - 1}}{\sqrt{\operatorname{Cosec} x + 1}} \times \frac{\sqrt{\operatorname{Cosec} x - 1}}{\sqrt{\operatorname{Cosec} x - 1}}$$

$$\Rightarrow \frac{\operatorname{Cosec} x - 1}{\sqrt{\operatorname{Cosec}^2 x - 1}}$$

$$\Rightarrow \frac{\operatorname{Cosec} x - 1}{\cot x} = \frac{\operatorname{cosec} x}{\cot x} - \frac{1}{\cot x}$$

$$\Rightarrow \sec x - \tan x$$

Sol.522.(b)

$$x \cos \theta - y \sin \theta = 2$$

$$x \sin \theta + y \cos \theta = 4$$

$$x^2 + y^2 = 2^2 + 4^2 \Rightarrow x^2 + y^2 = 20$$

Sol.523.(b) We know that, if $x + y = 45^\circ$

then $(1 + \tan x)(1 + \tan y) = 2$

Here $A + B = 45^\circ$ Then,

$$2(1 + \tan A)(1 + \tan B) = 4$$

Sol.524.(b) Put $\theta = 5^\circ$ in given equation:
 $\operatorname{cosec}(85^\circ + \theta) - \sec(5^\circ - \theta) - \tan(55^\circ + \theta) + \cot(35^\circ - \theta)$

We get : $\operatorname{Cosec} 90^\circ - \sec 0^\circ - \tan 60^\circ + \cot 30^\circ$

$$\Rightarrow 1 - 1 - \sqrt{3} + \sqrt{3} = 0$$

Sol.525.(d)

$$3\sec^2 \theta \tan^2 \theta + \tan^6 \theta - \sec^6 \theta = ?$$

We know that: $\sec^2 \theta - \tan^2 \theta = 1$

Cubing both sides we get:

$$\sec^6 \theta - \tan^6 \theta - 3\sec^2 \theta \tan^2 \theta = 1$$

Thus,

$$3\sec^2 \theta \tan^2 \theta + \tan^6 \theta - \sec^6 \theta = -1$$

Sol.526.(c)

put $\theta = 45^\circ$ in $\frac{\tan^2 \theta - \sin^2 \theta}{2 + \tan^2 \theta + \cot^2 \theta}$

$$= \frac{\tan^2 45^\circ - \sin^2 45^\circ}{2 + \tan^2 45^\circ + \cot^2 45^\circ} = \frac{1}{8}$$

Only option (c) satisfies.

Sol.527.(d) $x \cos A - y \sin A = 1$ and $x \sin A + y \cos A = 4$

Squaring both sides and adding,

$$\Rightarrow (x^2 + y^2)(\sin^2 A + \cos^2 A) = (1 + 16) = 17$$

$$\Rightarrow 17(x^2 + y^2) = 289$$

Sol.528.(a) $(\cos^2 \theta - 1)(1 + \tan^2 \theta) +$

$$2 \tan^2 \theta = 1, (-\sin^2 \theta)(\sec^2 \theta) + 2 \tan^2 \theta = 1$$

$$-\tan^2 \theta + 2 \tan^2 \theta = 1 \Rightarrow \tan^2 \theta = 1 \Rightarrow \theta = 45^\circ$$

Sol.529.(b) In the third quadrant $\tan A$ and $\cot A$ are positive.

$$\tan A = \frac{21}{20} \Rightarrow \frac{5 \sin A - 2 \cos A}{4 \cos A - \frac{5}{7} \sin A};$$

Divide numerator and denominator by $\cos A$.

$$\text{We get, } \frac{5 \tan A - 2}{4 - \frac{5}{7} \tan A} = 1$$

On putting the value of $\tan A$

$$\Rightarrow \frac{5 \times \frac{21}{20} - 2}{4 - \frac{5}{7} \times \frac{21}{20}} = 1$$

Sol.530.(c) $4 \left[\frac{(1 - \sec A)^2 + (1 + \sec A)^2}{1 + \sec^2 A} \right]$

$$= 4 \left[\frac{(1 + \sec^2 A - 2 \sec A) + (1 + \sec^2 A + 2 \sec A)}{1 + \sec^2 A} \right] = 8$$

Sol.531.(d) $\frac{\sin A + \cos A}{\cos A} = \frac{17}{12}$

$$\Rightarrow \tan A + 1 = \frac{17}{12} \Rightarrow \tan A = \frac{5}{12}$$

$$\Rightarrow \cos A = \frac{12}{13} \text{ and } \sin A = \frac{5}{13}$$

$$\frac{1 - \cos A}{\sin A} = \frac{13 - 12}{5} = \frac{1}{5}$$

Sol.532.(b) $6 \tan \theta - 5\sqrt{3} \sec \theta + 12 \cot \theta = 0$

let $\theta = 60^\circ$

$$\Rightarrow 6 \tan 60^\circ - 5\sqrt{3} \sec 60^\circ + 12 \cot 60^\circ = 0$$

$$\Rightarrow 6\sqrt{3} - 5\sqrt{3} \times 2 + 12 \times \frac{1}{\sqrt{3}} = 0$$

$\Rightarrow 0 = 0$ (condition satisfied). put $\theta = 60^\circ$,

$$\operatorname{Cosec}\theta + \sec\theta = \frac{2}{\sqrt{3}} + 2 = \frac{2}{3}(3 + \sqrt{3})$$

Sol.533.(b) $[(\operatorname{cosec}A + \cot A + 1)][(\operatorname{cosec}A - \cot A + 1)] - 2\operatorname{cosec}A$
 $= \operatorname{cosec}^2A - \cot^2A + \operatorname{cosec}A + \cot A + \operatorname{cosec}A - \cot A + 1 - 2\operatorname{cosec}A$
 $\Rightarrow 1 + 2\operatorname{cosec}A + 1 - 2\operatorname{cosec}A = 2$

Sol.534.(b) Given that: $\frac{1}{\operatorname{Cosec}\theta + 1} + \frac{1}{\operatorname{Cosec}\theta - 1} = 2\sec\theta$, $0^\circ < \theta < 90^\circ$
 $\Rightarrow \frac{\operatorname{cosec}\theta - 1 + \operatorname{cosec}\theta + 1}{\operatorname{cosec}^2\theta - 1} = 2\sec\theta$
 $\Rightarrow \frac{2\operatorname{cosec}\theta}{\cot^2\theta} = 2\sec\theta$
 $\Rightarrow \tan\theta = 1 \Rightarrow \theta = 45^\circ$
 Then, $\frac{\tan\theta + 2\sec\theta}{\operatorname{cosec}\theta} = \frac{\tan 45^\circ + 2\sec 45^\circ}{\operatorname{cosec} 45^\circ}$
 $= \frac{1 + 2\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2} + 4}{2}$

Sol.535.(d) Given that:

$$\frac{(\cos\theta + \sin\theta)}{(\cos\theta - \sin\theta)} = \frac{(\sqrt{3} + 1)}{(\sqrt{3} - 1)}$$

$$\Rightarrow \frac{(\cos\theta + \sin\theta)}{(\cos\theta - \sin\theta)} = \frac{\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)}$$

Thus, $\theta = 30^\circ \Rightarrow \sec 30^\circ = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$

Sol.536.(a) Put $A = 90^\circ$

$$\frac{(2\sin A)(1 + \sin A)}{1 + \sin A + \cos A}$$

$$= \frac{(2\sin 90^\circ)(1 + \sin 90^\circ)}{1 + \sin 90^\circ + \cos 90^\circ}$$

$$= \frac{2(1)(1 + 1)}{1 + 1 + 0} = 2$$

Now go through options,

Choose option (a)

$$1 + \sin A - \cos A = 1 + \sin 90^\circ - \cos 90^\circ$$

$$= 1 + 1 - 0 = 2$$

Condition satisfied, hence option A is the correct answer.

Note: We can check other options also but no option will satisfy the condition.

Sol.537.(d) $\frac{(1 + \cos\theta)^2 + \sin^2\theta}{(\operatorname{cosec}^2\theta - 1)\sin^2\theta}$
 $= \frac{1 + \cos^2\theta + 2\cos\theta + \sin^2\theta}{\cot^2\theta \cdot \sin^2\theta}$
 $= \frac{1 + 1 + 2\cos\theta}{\frac{\cos^2\theta}{\sin^2\theta} \cdot \sin^2\theta} = \frac{2 + 2\cos\theta}{\cos^2\theta}$
 $= \frac{2}{\cos^2\theta} + \frac{2}{\cos\theta} = 2\sec\theta(1 + \sec\theta)$

Sol.538.(a) $\sqrt{\frac{\cot\theta + \cos\theta}{\cot\theta - \cos\theta}}$

$$= \sqrt{\frac{\frac{\cos\theta}{\sin\theta} + \cos\theta}{\frac{\cos\theta}{\sin\theta} - \cos\theta}} = \sqrt{\frac{(1 + \sin\theta)\cos\theta}{(1 - \sin\theta)\cos\theta}}$$

$$= \sqrt{\frac{(1 + \sin\theta)(1 + \sin\theta)}{(1 - \sin\theta)(1 + \sin\theta)}}$$

$$= \sqrt{\frac{(1 + \sin\theta)^2}{(1 - \sin^2\theta)}} = \frac{(1 + \sin\theta)}{\cos\theta}$$

$$= \sec\theta + \tan\theta$$

Sol.539.(b) $\frac{1 + \sin\theta}{1 - \sin\theta} = \frac{p^2}{q^2}$
 $\Rightarrow \frac{1 + \sin\theta}{1 - \sin\theta} \times \frac{1 + \sin\theta}{1 + \sin\theta} = \frac{p^2}{q^2}$
 $\Rightarrow \frac{(1 + \sin\theta)^2}{1 - \sin^2\theta} = \frac{p^2}{q^2}$
 $\Rightarrow \frac{1 + \sin^2\theta + 2\sin\theta}{\cos^2\theta} = \frac{p^2}{q^2}$
 $\Rightarrow \sec^2\theta + \tan^2\theta + 2\sec\theta \cdot \tan\theta = \frac{p^2}{q^2}$

$$\Rightarrow (\sec\theta + \tan\theta)^2 = \frac{p^2}{q^2}$$

$$\Rightarrow \sec\theta + \tan\theta = \frac{p}{q} \dots (1)$$

$$\Rightarrow \sec\theta - \tan\theta = \frac{q}{p} \dots (2) \dots$$

{If $\sec\theta + \tan\theta = k$, then $\sec\theta - \tan\theta = \frac{1}{k}$ }

Add (1) and (2)

$$= 2\sec\theta = \frac{p}{q} + \frac{q}{p} \Rightarrow \sec\theta = \frac{1}{2}\left(\frac{p}{q} + \frac{q}{p}\right)$$

Sol.540.(d)

$$\frac{\sec\theta(1 - \sin\theta)(\sin\theta + \cos\theta)(\sec\theta + \tan\theta)}{\sin\theta(1 + \tan\theta) + \cos\theta(1 + \cot\theta)}$$

$$\Rightarrow \frac{\frac{1}{\cos\theta}(1 - \sin\theta)(\sin\theta + \cos\theta)\left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right)}{\sin\theta\left(1 + \frac{\sin\theta}{\cos\theta}\right) + \cos\theta\left(1 + \frac{\cos\theta}{\sin\theta}\right)}$$

$$\Rightarrow \frac{\frac{1}{\cos\theta}(1 - \sin\theta)(\sin\theta + \cos\theta)\left(\frac{1 + \sin\theta}{\cos\theta}\right)}{\sin\theta\left(\frac{\cos\theta + \sin\theta}{\cos\theta}\right) + \cos\theta\left(\frac{\sin\theta + \cos\theta}{\sin\theta}\right)}$$

$$\Rightarrow \frac{\frac{1}{\cos^2\theta}(1 - \sin^2\theta)(\sin\theta + \cos\theta)}{\left\{\left(\frac{\sin\theta}{\cos\theta}\right) + \left(\frac{\cos\theta}{\sin\theta}\right)\right\}(\sin\theta + \cos\theta)}$$

$$\Rightarrow \frac{\frac{1}{\cos^2\theta} \times \cos^2\theta}{\left(\frac{\sin^2\theta + \cos^2\theta}{\sin\theta \cdot \cos\theta}\right)} = \sin\theta \cdot \cos\theta$$

Sol.541.(b)

$$(\sec\theta - \tan\theta)^2(1 + \sin\theta)^2 \div \sin^2\theta$$

$$= \left(\frac{1}{\cos\theta} - \frac{\sin\theta}{\cos\theta}\right)^2(1 + \sin\theta)^2 \div \sin^2\theta$$

$$= \left(\frac{1 - \sin\theta}{\cos\theta}\right)^2(1 + \sin\theta)^2 \div \sin^2\theta$$

$$= \frac{1}{\cos^2\theta} [(1 + \sin\theta)(1 - \sin\theta)]^2 \div \sin^2\theta$$

$$\sin^2\theta = \frac{1}{\cos^2\theta} [(1 - \sin^2\theta)]^2 \div \sin^2\theta$$

$$= \frac{1}{\cos^2\theta} [\cos^2\theta]^2 \div \sin^2\theta = \cot^2\theta$$

$$= \frac{1}{\sin^2\theta} [\cos^2\theta] = \cot^2\theta$$

Sol.542.(d)

$$\sqrt{\frac{\operatorname{cosec}\theta - \cot\theta}{\operatorname{cosec}\theta + \cot\theta}} \div \frac{\sin\theta}{1 + \cos\theta}$$

$$\Rightarrow \sqrt{\frac{\operatorname{cosec}\theta - \cot\theta}{\operatorname{cosec}\theta + \cot\theta}} \times \frac{\operatorname{cosec}\theta - \cot\theta}{\operatorname{cosec}\theta - \cot\theta}$$

$$\div \frac{\sin\theta}{1 + \cos\theta}$$

$$\Rightarrow \sqrt{\frac{(\operatorname{cosec}\theta - \cot\theta)^2}{\operatorname{cosec}^2\theta - \cot^2\theta}} \div \frac{\sin\theta}{1 + \cos\theta}$$

$$\Rightarrow \operatorname{cosec}\theta - \cot\theta \div \frac{\sin\theta}{1 + \cos\theta}$$

$$\Rightarrow \frac{1}{\sin\theta} - \frac{\cos\theta}{\sin\theta} \div \frac{\sin\theta}{1 + \cos\theta}$$

$$\Rightarrow \frac{1 - \cos\theta}{\sin\theta} \times \frac{1 + \cos\theta}{\sin\theta} = 1$$

$$\Rightarrow (1 - \cos^2\theta = \sin^2\theta)$$

Sol.543.(b) Given, $\sec\theta + \tan\theta = p$ (1)

$$\Rightarrow \sec\theta - \tan\theta = \frac{1}{p} \dots (2)$$

$$\text{Add (1) and (2)} \Rightarrow 2\sec\theta = p + \frac{1}{p}$$

$$\text{Put } p = 2, \Rightarrow \sec\theta = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{5}{4}$$

$$\Rightarrow \operatorname{cosec}\theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{5}{3}$$

$$\frac{\operatorname{cosec}\theta + 1}{\operatorname{cosec}\theta - 1} = \frac{\frac{5}{3} + 1}{\frac{5}{3} - 1} = \frac{\frac{8}{3}}{\frac{2}{3}} = 4$$

Going through options

$$\text{Chose option b} \Rightarrow p^2 = 2^2 = 4$$

Condition satisfied.

We can put this value in any option; no other option will satisfy the condition. Hence option b is the correct option.

Sol.544.(b) $\frac{\sec^2\theta}{\operatorname{cosec}^2\theta} + \frac{\operatorname{cosec}^2\theta}{\sec^2\theta} - (\sec^2\theta + \operatorname{cosec}^2\theta)$
 $= \frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\sin^2\theta} - (\sec^2\theta + \operatorname{cosec}^2\theta)$
 $= \tan^2\theta + \cot^2\theta - (\sec^2\theta + \operatorname{cosec}^2\theta)$
 $= -(\sec^2\theta - \tan^2\theta) - (\operatorname{cosec}^2\theta - \cot^2\theta)$
 $= -2$

Sol.545.(a)

$$\frac{(\sin\theta - \cos\theta)(1 + \tan\theta + \cot\theta)}{1 + \sin\theta\cos\theta}$$

$$= \frac{(\sin\theta - \cos\theta)\left(1 + \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}\right)}{1 + \sin\theta\cos\theta}$$

$$= \frac{(\sin\theta - \cos\theta)\frac{\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}}{1 + \sin\theta\cos\theta}$$

$$= \frac{(\sin\theta - \cos\theta)(\sin\theta\cos\theta + 1)}{(1 + \sin\theta\cos\theta)\sin\theta\cos\theta}$$

$$= \frac{\sin\theta}{\sin\theta\cos\theta} - \frac{\cos\theta}{\sin\theta\cos\theta}$$

$$= \sec\theta - \operatorname{cosec}\theta$$

$$\text{Sol.546.(a)} \quad \frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \frac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \frac{1+1+2\cos\theta}{\sin\theta(1+\cos\theta)} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ$$

$$(\tan\theta + \sec\theta)^{-1} \Rightarrow \frac{1}{(\tan\theta + \sec\theta)}$$

$$\Rightarrow \frac{1}{(\tan 60^\circ + \sec 60^\circ)}$$

$$\Rightarrow \frac{1}{\sqrt{3}+2} \times \frac{\sqrt{3}-2}{\sqrt{3}-2} = 2 - \sqrt{3}$$

Sol.547.(d) Short-Trick :

$$\cot\theta = \frac{1}{\sqrt{3}}, \text{ clearly } \theta = 60^\circ$$

$$\frac{2 - \sin^2\theta}{1 - \cos^2\theta} + \operatorname{cosec}^2\theta + \sec\theta$$

$$= \frac{2 - (\frac{\sqrt{3}}{2})^2}{1 - (\frac{1}{2})^2} + (\frac{2}{\sqrt{3}})^2 + 2$$

$$= \frac{5}{3} + \frac{4}{3} + 2 = 5$$

Sol.548.(b) Given, Cosec $31^\circ = x$

$$\sin 31^\circ = \frac{1}{x} \Rightarrow \cos^2 31^\circ = \frac{x^2 - 1}{x^2}$$

$$\Rightarrow \sin^2 59^\circ + \frac{1}{\operatorname{cosec}^2 31^\circ} + \tan^2 59^\circ -$$

$$\frac{1}{\sin^2 59^\circ \operatorname{cosec}^2 59^\circ}$$

$$\Rightarrow \sin^2 59^\circ + \sin^2 31^\circ + \tan^2 (90 - 31) - 1$$

$$\Rightarrow 1 + \tan^2 (90 - 31) - 1$$

$$\Rightarrow \cot^2 31^\circ = \frac{\cos^2 31^\circ}{\sin^2 31^\circ}$$

$$\Rightarrow \frac{x^2 - 1}{x^2} \times \frac{x^2}{1} = x^2 - 1$$

Sol.549.(c) Short-Trick : Put $\theta = 30^\circ$

$$\sin\theta \cdot \sec^2\theta \Rightarrow \frac{1}{2} \times (\frac{2}{\sqrt{3}})^2 = \frac{2}{3}$$

Condition satisfied. Clearly $\theta = 30^\circ$

$$(\tan^2\theta + \cos^2\theta) \Rightarrow (\frac{1}{3})^2 + (\frac{\sqrt{3}}{2})^2$$

$$= \frac{1}{3} + \frac{3}{4} = \frac{13}{12}$$

Sol.550.(b) Given, $\cos x = \frac{-\sqrt{3}}{2}$

$$\cos^2 x = (\frac{-\sqrt{3}}{2})^2 = \frac{3}{4}$$

$$\text{And } \sin^2 x = 1 - \cos^2 x = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\text{Now, } 4\cot^2 x - 3\operatorname{cosec}^2 x$$

$$= 4(\frac{\cos^2 x}{\sin^2 x}) - 3(\frac{1}{\sin^2 x})$$

$$= 4(\frac{\frac{3}{4}}{\frac{1}{4}}) - 3(\frac{1}{\frac{1}{4}}) = 12 - 12 = 0$$

Sol.551.(d) Given, $5\sin\theta + 12\cos\theta = 13$

Trick :

Here, 5, 12 and 13 are triplets. In such questions coefficient of $\sin\theta$ will be perpendicular and coefficient of $\cos\theta$ will be base and the coefficient on the right side of sign "=" will be hypotenuse.

$$\Rightarrow \tan\theta = \frac{\text{perpendicular}}{\text{base}} = \frac{5}{12}$$

Sol.552.(c)

Short-Trick : $\sin\theta + \operatorname{cosec}\theta = 2$

Put $\theta = 90^\circ$

$$\sin\theta + \operatorname{cosec}\theta = 2 \Rightarrow 1 + 1 = 2$$

$$\Rightarrow 2 = 2 \text{ condition satisfied}$$

$$\sin^{153}\theta + \operatorname{cosec}^{253}\theta = 1^{153} + 1^{253} = 2$$

Sol.553.(c) Put $\theta = 45^\circ$

$$\frac{2 + \tan^2\theta + \cot^2\theta}{\sec\theta \cdot \operatorname{cosec}\theta} \Rightarrow \frac{2 + 1 + 1}{\sqrt{2} \cdot \sqrt{2}} = 2$$

Going through the options only option (c) satisfies the given condition

$$\sec\theta \cdot \operatorname{cosec}\theta = \sqrt{2} \cdot \sqrt{2} = 2$$

Sol.554.(d)

$$\text{Given, } \cos\theta = \frac{2p}{1+p^2} = \frac{\text{Base}}{\text{Hypotenuse}}$$

So, Perpendicular,

$$P = \sqrt{(1+p^2)^2 - 4p^2} = 1 - p^2$$

$$\text{Therefore, } \tan\theta = \frac{1-p^2}{2p}$$

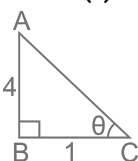
Sol.555.(a) Put $\theta = 90^\circ$

$$\cos 90^\circ = 0; \sin 90^\circ = 1$$

$$\sqrt{3}\cos\theta + \sin\theta = \sqrt{3}(0) + 1 = 1$$

condition satisfied

Sol.556.(c) $\sin\theta = 4\cos\theta \Rightarrow \tan\theta = 4$



In the given figure

$$\tan\theta = \frac{AB}{BC} = \frac{4}{1}; AC^2$$

$$= 4^2 + 1^2 \Rightarrow AC = \sqrt{17}$$

$$\Rightarrow \sin\theta \cos\theta = \frac{AB}{AC} \times \frac{BC}{AC}$$

$$= \frac{4}{\sqrt{17}} \times \frac{1}{\sqrt{17}} = \frac{4}{17}$$

Sol.557.(b) $\sin^2 30^\circ \cos^2 45^\circ + 4\tan^2 30^\circ$

$$+ \frac{1}{2} \sin^2 90^\circ + 2\cos 90^\circ$$

$$\Rightarrow (\frac{1}{2})^2 \times (\frac{1}{\sqrt{2}})^2 + 4 \times (\frac{1}{\sqrt{3}})^2 + \frac{1}{2} \times$$

$$(1)^2 + 2 \times 0 \Rightarrow \frac{1}{8} + \frac{4}{3} + \frac{1}{2} = \frac{47}{24}$$

$$\text{Sol.558.(a)} \quad \frac{3\sin\theta - 4\cos\theta}{3\sin\theta + 4\cos\theta}$$

Divide by $\cos\theta$,

$$\Rightarrow \frac{3\tan\theta - 4}{3\tan\theta + 4} \Rightarrow \frac{3(\frac{2}{3}) - 4}{3(\frac{2}{3}) + 4} = \frac{-1}{3}$$

Sol.559.(a)

$$\frac{\sin 44^\circ}{\cos 46^\circ} + \sin^2 60^\circ - \cos^2 45^\circ + \sec 60^\circ$$

$$= 1 + (\frac{\sqrt{3}}{2})^2 - (\frac{1}{\sqrt{2}})^2 + 2$$

$$= 1 + \frac{3}{4} - \frac{1}{2} + 2 = \frac{4+3-2+8}{4} = \frac{13}{4}$$

Sol.560.(b) $\sin(A+B) = \sin 60^\circ$

$$\Rightarrow A+B = 60^\circ \text{--- (i)}$$

$$\tan(A-B) = \tan 30^\circ$$

$$\Rightarrow A-B = 30^\circ \text{--- (ii)}$$

On solving eq (i) and (ii), we get

$$A = 45^\circ \text{ and } B = 15^\circ$$

$$\text{Therefore, } 2A + 3B = 90^\circ + 45^\circ = 135^\circ$$